BARNES & NOBLE

3173 CINEMA POINT,
COLORADO SPRINGS, CO 80922

WJCA

Designer

299 MARKET STREET
SUITE 340
SADDLE BROOK, NJ 07663

CUSTOM DESIGN AND ENGINEERING

Mechanical, Electrical & Plumbing Consultant

153 EAST MAIN STREET
ROCKAWAY, NJ 07866

DRAWING INDEX

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SHEET NUMBER	SHEET TITLE	ISSUED FOR REVIEW 04/25/2025				REVISION DATE AND NO.						
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1 2

BARNES & NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J. COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3
08-8-25	BD COMMENTS	4

These plans are an instrument of service and the property of the Architect, infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
LICENSED ARCHITECT

LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

Cover Sheet

Drawing Number:

A-000

DIVISION 01 GENERAL REQUIREMENTS

A. GENERAL CONDITIONS OF THE CONTRACT:
1. GENERAL CONDITIONS OF THE CONTRACT, STANDARD FORM A-201, CURRENT EDITION, OF THE AMERICAN INSTITUTE OF ARCHITECTS ARE HEREBY MADE PART OF THIS SPECIFICATION THE SAME AS IF BOUND HEREIN.
2. THE GENERAL CONDITIONS, INCLUDING MODIFICATIONS AND SPECIAL CONDITIONS HEREIN SHALL BECOME A PART OF THE CONTRACT AND SHALL APPLY TO THE GENERAL CONTRACTOR AND ALL SUBCONTRACTORS.
3. THE GENERAL REQUIREMENTS & GENERAL CONDITIONS ARE HEREBY MADE A PART OF AND SHALL APPLY TO THE WORK OF EVERY SECTION OR SUB-SECTION OF THESE SPECIFICATIONS.
4. COORDINATE ALL DEVICE LOCATIONS WITH EXISTING CONDITIONS AND WITH PROJECT DESIGN INTENT - CLARIFY INTENT WITH TENANT'S REPRESENTATIVE. ALL LIFE SAFETY DEVICE LOCATIONS ARE TO BE REVIEWED WITH TENANT'S REPRESENTATIVE FOR DESIGN INTENT, BUT FINAL LOCATIONS ARE TO BE COORDINATED WITH INSPECTOR OR OTHER AUTHORITIES HAVING JURISDICTION.

B. SCOPE:
1. BE RESPONSIBLE FOR A COMPLETE AND OPERATIONAL TENANT SPACE INCLUDING, BUT NOT LIMITED TO ALL FINISHES, MECHANICAL, ELECTRICAL, PLUMBING, AND FIRE PROTECTION.
2. WORK OF THE CONTRACT INCLUDES FURNISHING OF ALL LABOR, MATERIALS, APPLIANCES, TRANSPORTATION AND SERVICES REQUIRED FOR SATISFACTORY CONSTRUCTION OF THIS PROJECT.
3. VERIFY EXISTING CONDITIONS THAT AFFECT ALL ALTERATIONS ADJUSTMENTS, ETC. TO DELIVER A COMPLETED PROJECT THAT CONFORMS TO THE CONTRACT DRAWINGS, SPECIFICATIONS AND CODE COMPLIANCE.
4. VERIFY SETBACKS, AND/OR EASEMENTS, EXISTING UTILITIES, AND ANY UNDERGROUND OR SURFACE STRUCTURE OR CONDITIONS PRIOR TO COMMENCING THE WORK.
5. **ALL REQUESTS FOR INFORMATION MUST BE MADE IN WRITING TO THE DESIGNATED TENANT REPRESENTATIVE. WRITTEN INQUIRIES SHOULD REASONABLY IDENTIFY THE INFORMATION NEEDED AND SHOULD INCLUDE THE NAME, ADDRESS, PHONE NUMBER, OR OTHER MEANS TO CONTACT SENDER REGARDING THE REQUEST.**

C. SCHEDULING AND COORDINATION:
1. OBTAIN A COPY OF THE TENANT CONTRACTOR GUIDELINES HANDBOOK AND COMPLY WITH THE REQUIREMENTS. NOTE IN PARTICULAR THE INSURANCE REQUIREMENTS, UTILITY LOCATIONS, CONSTRUCTION PROCEDURES, TEMPORARY BARRICADE WALLS, CONSTRUCTION MATERIAL STORAGE, HAUL OFF AND PERMIT INFORMATION.
2. COORDINATE WORK WITH WORK OF OTHER TRADES, THE LANDLORD'S FORCES, OTHER TENANT CONTRACTORS AND TENANT STORE OPERATIONS IN ORDER TO AVOID CONFLICTS AND EXPEDITE THE COMPLETION OF THE PROJECT.
3. FURNISH LABOR, MATERIALS, EQUIPMENT AND APPLIANCES NECESSARY TO COMPLETE THE INSTALLATION OF ALL ITEMS FURNISHED BY OTHERS.
4. COORDINATE ALL DEVICE LOCATIONS WITH EXISTING CONDITIONS AND WITH PROJECT DESIGN INTENT - CLARIFY INTENT WITH TENANT'S REPRESENTATIVE. ALL LIFE SAFETY DEVICE LOCATIONS ARE TO BE REVIEWED WITH TENANT'S REPRESENTATIVE FOR DESIGN INTENT, BUT FINAL LOCATIONS ARE TO BE COORDINATED WITH INSPECTOR OR OTHER AUTHORITIES HAVING JURISDICTION.
5. REPORT ALL CONFLICTS IN THE CONSTRUCTION DRAWINGS TO THE TENANT REPRESENTATIVE IMMEDIATELY. - NO CREDIT FOR VISIBLE SITE CONDITIONS
6. VISIT THE SITE AND VERIFY ALL EXISTING CONDITIONS. BRING ANY DEVIATIONS TO THE ATTENTION OF THE TENANT'S PROJECT MANAGER.
7. ANY WORK REQUIRED TO BE COMPLETED DURING NON-BUSINESS HOURS SHALL BE INCLUDED IN THE BID.
8. COORDINATE AND SCHEDULE ALL NON-BUSINESS HOUR WORK WITH LOCAL JURISDICTION AND/OR BUILDING MANAGER
9. REFER TO THE RESPONSIBILITY SCHEDULE TO CLARIFY RESPONSIBILITIES FOR SPECIFIED WORK.

D. SUPERVISION:
1. PROVIDE A FULL TIME, QUALIFIED SUPERVISOR ON THE JOB SITE.
2. PROVIDE A JOB SITE TELEPHONE AND FAX MACHINE DURING THE CONSTRUCTION PERIOD. NOTIFY TENANT'S REPRESENTATIVE IMMEDIATELY OF CONTACT NUMBERS.

E. PERMITS:
1. OBTAIN AND PAY FOR ALL PERMITS AND INSPECTIONS REQUIRED, UNLESS SPECIFIED OTHERWISE.

F. UTILITIES:
1. PROVIDE ALL TEMPORARY UTILITIES (INCLUDING, BUT NOT LIMITED TO, WATER, SEWERAGE, ELECTRICAL POWER, FUEL AND TELEPHONES) AND THE COSTS ASSOCIATED WITH THESE.
2. PROVIDE THE TENANT'S REPRESENTATIVE WITH ALL UTILITY COMPANY NAMES, NUMBERS, CONTACTS AND ACCOUNT NUMBERS (OWNER WILL TRANSFER ACCOUNTS TO THEIR NAME AT TURNOVER).

G. TESTS AND INSPECTIONS:
1. ALL TESTS AND INSPECTIONS, OF ANY KIND, REQUIRED, AND WHICH INVOLVE EXPENSE, SHALL BE PAID BY THE CONTRACTOR AT NO ADDITIONAL EXPENSE.

H. PROTECTION OF WORK IN PLACE:
1. ALL WORK IN PLACE SHALL BE ENCLOSED WITH ADEQUATE PROTECTION. PERMANENT OPENINGS USED AS THOROUGHFARES FOR THE INTRODUCTION OF WORK AND MATERIALS TO THE STRUCTURE SHALL HAVE HEADS, JAMBS AND SILLS WELL BLOCKED AND BOARDED.
2. PROTECT ALL NEW WORK, FURNISHINGS, AND EQUIPMENT AT ALL TIMES. ANY DAMAGE THAT OCCURS AS A RESULT OF CONSTRUCTION IS TO BE REPAIRED TO LIKE NEW CONDITION.
3. ERECT A TEMPORARY BARRICADE AT THE STOREFRONT ONLY IF REQUIRED BY THE LOCAL JURISDICTION AND/OR BUILDING OWNER.

I. GUARANTEE:
1. A WRITTEN GUARANTEE SHALL BE PROVIDED ON ALL LABOR AND MATERIALS FOR A PERIOD OF NO LESS THAN ONE YEAR FROM THE DATE OF THE SUBSTANTIAL COMPLETION OF THE PROJECT.
2. THE TENANT'S REPRESENTATIVE SIGNATURE AND DATE OF ACCEPTANCE IS REQUIRED IN ORDER TO ACTIVATE THIS GUARANTEE. THIS GUARANTEE SHALL COVER ALL OF THE GENERAL AND SUBCONTRACTORS' WORK.
3. ALL DEFECTS DISCOVERED DURING THIS GUARANTEE PERIOD SHALL BE REPAIRED TO THE OWNER/TENANT'S SATISFACTION AT CONTRACTOR'S EXPENSE AT NO COST TO THE OWNER/TENANT.
4. FURNISH THE FOLLOWING UPON COMPLETION OF WORK:
   a. CERTIFICATE OF OCCUPANCY.
   b. GENERAL CONTRACTOR'S AFFIDAVIT THAT THE PROJECT WAS BUILT PER THE CONSTRUCTION DOCUMENTS.
   c. GENERAL CONTRACTOR'S RELEASE OF WAIVER OF LIEN.
   d. AN EXECUTED AND ACKNOWLEDGED RELEASE OF MECHANIC'S LIENS WITH RESPECT TO DEMISED PREMISES EXECUTED BY THE TENANT'S GENERAL CONTRACTOR AND EVERY SUB CONTRACTOR AND SUPPLIER OF LABOR AND MATERIALS ENGAGED IN TENANT'S WORK.
   e. ELECTRICAL INSPECTION CERTIFICATE
   f. UNDERWRITER'S APPROVAL CERTIFICATE FOR THE ELECTRICAL WORK.
   g. HVAC WARRANTY.
   h. OPERATION MANUALS FOR ALL MECHANICAL AND ELECTRICAL EQUIPMENT.
   i. ROOFING SYSTEM - SUBMIT COPIES OF THE MANUFACTURER'S FINAL INSPECTION PRIOR TO THE ISSUANCE OF THE MANUFACTURER'S WARRANTY.

J. PAYMENT AND WAIVER OF LIENS:
1. PAYMENTS MADE FOR THIS PROJECT ARE TO BE ALLOCATED FOR THIS PROJECT AND ONLY FOR PAYMENTS TO SUBCONTRACTORS AND MATERIAL SUPPLIERS, AND AGENTS AUTHORIZED BY SPECIFIC CONTRACTORS TO PERFORM WORK OR SUPPLY MATERIALS TO THIS PROJECT.
2. SUBMIT PARTIAL WAIVERS OF LIEN FOR WORK INCLUDING SUBCONTRACTORS AND MATERIAL SUPPLIERS.
3. THE FINAL 10% RETENTION ON THIS PROJECT MUST INCLUDE SIGNED AND EXECUTED WAIVERS OF LIEN INDICATING 100% COMPLETION AND 100% PAYMENT IN FULL BY ALL SUBCONTRACTORS AND MATERIAL SUPPLIERS.
4. PROVIDE A "SWORN STATEMENT" LISTING ALL SUBCONTRACTORS AND CONTRACT VALUE.
5. PROVIDE A LIST OF ALL SUBCONTRACTORS, INCLUDING ADDRESS AND PHONE NUMBER TO THE TENANT'S REPRESENTATIVE.
6. REFER TO THE CONTRACT FOR PAYMENT TERMS.

K. TRASH REMOVAL:
1. PROVIDE FOR REMOVAL OF ALL TRASH, RUBBISH AND SURPLUS MATERIALS RESULTING FROM CONSTRUCTION OF THE PROJECT.

L. CLEAN UP AT TURNOVER:
1. REMOVE ALL PROTECTIVE COVERINGS AND ALL CONSTRUCTION MATERIALS AS DIRECTED BY THE TENANT'S REPRESENTATIVE.
2. HVAC FILTERS TO BE CHANGED FOLLOWING THE CLEANING.
3. THE PROJECT IS TO BE CLEAN AND READY FOR OCCUPANCY

M. THE TERM "OR EQUAL":
1. WHEREVER THE TERM "OR EQUAL" IS USED, IT SHALL MEAN EQUAL PRODUCT AS APPROVED IN WRITING BY THE OWNER'S REPRESENTATIVE, NO EXCEPTIONS.

N. QUALITY OF WORK:
1. ALL WORK SHALL BE PERFORMED IN A SAFE AND WORKMANLIKE MANNER IN ACCORDANCE WITH ACCEPTED CONSTRUCTION STANDARDS AND O.S.H.A. REQUIREMENTS.
2. ALL CONSTRUCTION SHALL BE IN STRICT ACCORDANCE WITH THE PLANS UNLESS A VARIANCE IS APPROVED BY THE OWNER/TENANT AND ALL AUTHORITIES HAVING JURISDICTION.
3. ALL EQUIPMENT AND MATERIALS UNDER THE CONTRACT ARE TO BE NEW UNLESS OTHERWISE SPECIFIED IN THE PLANS.

O. CODE COMPLIANCE:
1. VERIFY ALL WORK IS IN COMPLIANCE WITH ALL CODES, ORDINANCES, AND REGULATIONS OF ALL CONSTITUTED PUBLIC AUTHORITIES HAVING JURISDICTION. THIS INCLUDES BUT IS NOT LIMITED TO LOCAL BUILDING CODE REQUIREMENTS, FIRE DEPARTMENT REGULATIONS, UTILITY COMPANY REQUIREMENTS, O.S.H.A., AND THE BEST TRADE PRACTICES.

P. DIMENSIONS:
1. UNLESS OTHERWISE NOTED, ALL DIMENSIONS ARE MEASURED FROM FINISHED FACE TO FINISHED FACE OF MATERIALS, OR LEASE LINE.
2. VERIFY ALL PROPOSED DIMENSIONS WITH FIELD CONDITIONS. ADVISE THE TENANT'S REPRESENTATIVE OF ANY DISCREPANCY BETWEEN THE DRAWINGS AND ACTUAL JOB CONDITIONS BEFORE BEGINNING THE WORK. FAILURE TO DO SO SHALL PLACE FULL RESPONSIBILITY FOR ANY ERRORS UPON THE CONTRACTOR.
3. ALL DIMENSIONS SHOWN SHALL BE CONSIDERED APPROXIMATE.
4. ALL ANGLES SHALL BE 90 OR 45 DEGREES UNLESS OTHERWISE NOTED IN THE CONSTRUCTION DOCUMENTS.
5. WRITTEN DIMENSIONS TAKE PRECEDENCE OVER SCALE. DO NOT SCALE.

Q. SHOP DRAWINGS & SUBMITTALS:
1. SUBMIT THE NUMBER OF OPAQUE REPRODUCTIONS WHICH CONTRACTOR REQUIRES, PLUS TWO COPIES, TO THE TENANT'S REPRESENTATIVE FOR APPROVAL.
2. SUBMITTALS MUST BE SCHEDULED SO AS NOT TO IMPEDE THE PROGRESS OF THE WORK.
3. MARK EACH COPY TO IDENTIFY APPLICABLE PRODUCTS, MODELS, OPTIONS, AND OTHER DATA. SUPPLEMENT MANUFACTURER'S STANDARD DATA TO PROVIDE INFORMATION SPECIFIC TO THIS PROJECT.
4. INDICATE SPECIAL UTILITY AND ELECTRICAL CHARACTERISTICS, UTILITY CONNECTION REQUIREMENTS, AND LOCATION OF UTILITY OUTLETS FOR SERVICE FOR FUNCTIONAL EQUIPMENT AND APPLIANCES. 5. IDENTIFY VARIATIONS FROM CONTRACT DOCUMENTS AND PRODUCT OR SYSTEM
   6. LIMITATIONS WHICH MAY AFFECT THE CONSTRUCTION PROGRESS.
   7. PROVIDE SHOP DRAWINGS AND SUBMITTALS FOR ITEMS INDICATED HEREIN, PER THE CONSTRUCTION DOCUMENTS, OR UPON REQUEST BY THE OWNER.
   8. DIRECT ALL SUBMITTALS TO THE TENANT'S REPRESENTATIVE.

R. SAMPLES:
1. SUBMIT THREE COPIES OF EACH SAMPLE TO THE OWNER'S REPRESENTATIVE FOR APPROVAL.
2. INCLUDE IDENTIFICATION ON EACH SAMPLE WITH FULL PROJECT INFORMATION.
3. PROVIDE SAMPLES FOR ITEMS INDICATED HEREIN, PER THE CONSTRUCTION DOCUMENTS, OR UPON REQUEST BY THE OWNER.

S. INSURANCE:
1. SECURE, PAY FOR AND MAINTAIN, DURING CONSTRUCTION OF LEASED PREMISES, ALL OF THE INSURANCE POLICIES REQUIRED AND IN AMOUNTS AS SET FORTH HEREIN. WORK SHALL NOT COMMENCE UNTIL ALL REQUIRED INSURANCE HAS BEEN OBTAINED AND CERTIFICATES OF SUCH INSURANCE HAVE BEEN DELIVERED TO THE OWNER & BUILDING OWNER
2. BUILDING OWNER MUST BE NAMED AS "ADDITIONAL INSURED". CERTIFICATE OF INSURANCE SHALL COVER WORK PERFORMED BY CONTRACTOR, EACH SUBCONTRACTOR AND ALL MAJOR SUPPLIERS.
3. THE MINIMUM COVERAGE AND LIMITS OF LIABILITY ARE AS FOLLOWS:
   a. WORKMEN'S COMPENSATION INSURANCE INCLUDING PAYMENT TO THE EMPLOYEES OF THE CONTRACTOR, COMPENSATION PROVIDED BY THE WORKMEN'S COMPENSATION LAWS OF THE STATE IN WHICH THE WORK IS BEING PERFORMED, WITH LIMITS OF NOT LESS THAN $1,000,000.
   b. CONTRACTOR'S PUBLIC LIABILITY INSURANCE, HAVING LIMITS OF NOT LESS THAN $1,000,000.
   c. CONTRACTOR'S PROTECTIVE LIABILITY INSURANCE, HAVING LIMITS OF NOT LESS THAN $1,000,000.
   d. GENERAL THIRD PARTY PROPERTY DAMAGE INSURANCE, HAVING LIMITS OF NOT LESS THAN $1,000,000.

T. INDEMNIFICATION:
1. TO THE FULLEST EXTENT PERMITTED BY LAW, INDEMNIFY AND HOLD HARMLESS THE OWNER AND ARCHITECT AND THEIR AGENTS AND EMPLOYEES FROM AND AGAINST ALL CLAIMS, DAMAGES, LOSSES AND EXPENSES, INCLUDING BUT NOT LIMITED TO, ATTORNEY'S FEES ARISING OUT OF OR RESULTING FROM THE PERFORMANCE OF THE WORK, PROVIDED THAT ANY SUCH CLAIM, DAMAGE, LOSS OR EXPENSE IS ATTRIBUTABLE TO BODILY INJURY, SICKNESS, DISEASE OR DEATH, OR TO INJURY OR TO DESTRUCTION OF TANGIBLE PROPERTY (OTHER THAN THE WORK ITSELF) INCLUDING THE LOSS OF USE RESULTING THEREFROM.

USE OF CONSTRUCTION DOCUMENTS & COMPUTER-AIDED CONTRACT DESIGN DOCUMENTS:
1. DRAWINGS ARE TO BE USED SOLELY FOR THE PROJECT INDICATED.
2. ARCHITECT WILL RETAIN OWNERSHIP OF THE DOCUMENTS AND MAY ALLOW A SEPARATE LICENSEE AGREEMENT.
3. IN THE EVENT THAT THE GENERAL CONTRACTOR OR ANY SUBCONTRACTORS REQUEST COMPUTER AIDED BACKGROUND DRAWINGS COPIES OF THESE DRAWINGS MAY BE PROVIDED SUBJECT TO THE FOLLOWING CONDITIONS:
   a. A SEPARATE AGREEMENT TITLED AIA C106-2007 MUST BE SIGNED BY THE CONTRACTOR/SUBCONTRACTOR BEFORE THE TRANSFER OF THE FILES.

V. MISSING/DAMAGED FREIGHT (OWNER/TENANT FURNISHED MATERIALS):
1. OPEN BOXES/CARTONS WITH OBVIOUS DAMAGE
2. INSPECT FREIGHT BEFORE SIGNING THE FREIGHT BILL, NOTE SPECIFICALLY THE ITEM(S) MISSING/DAMAGED.
3. AFTER THE PROBLEM IS NOTED ON THE FREIGHT BILL, HAVE CARRIER'S DRIVER ALSO SIGN IN ORDER TO VERIFY THAT BOTH PARTIES AGREE TO THE EXTENT OF THE DAMAGE.
4. WHEN DAMAGE IS DISCOVERED UPON UNPACKING THE FREIGHT, INSPECT ALL PACKING FOR SIGNS OF DAMAGE OR ABUSE. THE FREIGHT BILL IS TO BE NOTED WITH THE SPECIFIED DAMAGE.
5. CONTACT THE TENANT'S REPRESENTATIVE WITHIN 48 HOURS OF TAKING RECEIPT OF MATERIALS TO INFORM THEM OF ANY LOSS. FAX ALL REQUIRED DOCUMENTATION TO TENANT'S REPRESENTATIVE.
6. IF CONTRACTOR FAILS TO FOLLOW THIS PROCEDURE, CONTRACTOR SHALL BE RESPONSIBLE FOR ALL COSTS RELATED TO THE REPLACEMENT OF MISSING/ DAMAGED OWNER FURNISHED MATERIALS.

DIVISION 02 EXISTING CONDITIONS

A. DEMOLITION:
1. VISIT THE SITE PRIOR TO BID SUBMISSION AND VERIFY ALL EXISTING CONDITIONS (I.E., OVERALL DIMENSIONS, CONDITION OF CONCRETE SLAB, LOCATION OF SERVICES, STRUCTURE, CLEARANCES, ETC.) AGAINST THE PROPOSED WORK. REPORT ANY DISCREPANCIES, OMISSIONS OR OBSTRUCTIONS THAT WILL AFFECT THE LAYOUT OR DESIGN IMMEDIATELY TO THE TENANT REPRESENTATIVE.
2. ALL CONDITIONS AND DIMENSIONS SHOWN ARE FOR REFERENCE ONLY AND MUST BE FIELD VERIFIED.
3. UPON COMPLETION OF DEMOLITION, ALL CONDITIONS AND DIMENSIONS ARE TO BE VERIFIED AGAIN FOR VARIANCES AGAINST THE PROPOSED WORK. REPORT ANY DISCREPANCIES IMMEDIATELY TO THE TENANT REPRESENTATIVE.
4. THIS PORTION OF THE WORK INCLUDES ALL LABOR, MATERIALS AND SERVICES NECESSARY FOR AND REASONABLY INCIDENTAL TO COMPLETE ALL DISMANTLING, DEMOLITION, AND SALVAGE AS SHOWN AND NOTED ON THE DRAWINGS AND AS SPECIFIED HEREIN.
5. PROTECT ALL AREAS NOT INCLUDED IN THE WORK. ANY DAMAGE THAT OCCURS AS A RESULT OF THE WORK SHALL BE REPAIRED TO A LIKE NEW CONDITION AT G.C. EXPENSE.
6. NO ALTERATION'S TO THE EXISTING STRUCTURAL MEMBERS ARE ALLOWED UNLESS SPECIFICALLY NOTED.
7. REFER TO ARCHITECTURAL AND MEP DRAWINGS FOR SCOPE OF NEW WORK.
8. MAINTAIN AND/OR REPAIR ALL FIRE RATED ASSEMBLIES AND SEPARATIONS (PARTITION, FLOOR, CEILING, OR BETWEEN ADJACENT SPACES ETC.) REQUIRED BY CODE OR REGULATION.
9. NOTIFY THE TENANT REPRESENTATIVE OF ANY UNFORESEEN HAZARD OF ANY KIND BEFORE DEMOLITION IS COMPLETE. THE REMOVAL AND/OR HANDLING OF HAZARDOUS MATERIALS (E.G. - ACM.) IS NOT PART OF THIS WORK. IF ANY DISTURBANCE OF HAZARDOUS MATERIAL IS ANTICIPATED BY THIS WORK, AUTHORITIES MUST BE NOTIFIED AND PRESCRIBED PROCEDURES MUST BE FOLLOWED.
10. DISCONNECTION OF SERVICES:
   a. EXISTING SERVICES THAT WILL BE DISRUPTED BY CONSTRUCTION WORK SHALL BE RE-ROUTED AS NECESSARY TO PROVIDE TEMPORARY CONTINUATION OF SERVICE.
   b. DISCONNECTED SERVICES SHALL BE CAPPED AND SEALED IN AN APPROVED MANNER.
   c. ANY DISCONNECTED SERVICES INTERFERING WITH THE NEW CONSTRUCTION SHALL BE REMOVED ONLY TO THE EXTENT THAT THEY DO NOT INTERFERE WITH NEW CONSTRUCTION.
   d. SPRINKLER SHUT DOWN SHALL BE COORDINATED WITH THE BUILDING MANAGEMENT PRIOR TO EXECUTION OF WORK.
11. ABANDONED ELECTRICAL WIRING:
   a. CONDUITS AND OTHER UTILITIES ENCOUNTERED IN THE PROGRESS OF THE DEMOLITION THAT ARE TO BE ABANDONED SHALL BE DISCONNECTED AND THE ENDS PLUGGED IN AN APPROVED MANNER. ANY ABANDONED UTILITIES INTERFERING WITH WORK UNDER THIS CONTRACT SHALL BE REMOVED ONLY TO THE EXTENT THAT THEY DO NOT INTERFERE WITH NEW CONSTRUCTION.
12. SALVAGED MATERIALS AND EQUIPMENT:
   a. REFER TO THE PLANS FOR ITEMS TO BE SALVAGED. CONSULT WITH THE TENANT REPRESENTATIVE PRIOR TO THE DISPOSAL OF MATERIALS AND EQUIPMENT.
   b. PACKAGE, PROTECT AND STORE SALVAGED ITEMS FOR FUTURE USE.
13. DISPOSAL OF NON-SALVAGED MATERIALS AND EQUIPMENT:
   a. ALL MATERIALS AND EQUIPMENT NOT SCHEDULED TO BE SALVAGED, INCLUDING DEBRIS, SHALL BECOME THE PROPERTY OF THE CONTRACTOR AND SHALL BE DISPOSED OF IN A LEGAL MANNER.
   b. LOCATION OF DISPOSAL AREA AND LENGTH OF HAUL SHALL BE CONTRACTOR'S RESPONSIBILITY.

DIVISION 03 CONCRETE

A. PATCHING (WHEN REQUIRED)
1. INSTALL MINIMUM 3000 PSI CONCRETE FILLER AND PATCHING WHERE EXISTING SLAB HAS BEEN REMOVED FOR UNDER SLAB CONDUITS, PLUMBING, ETC.

B. CONCRETE SLAB
1. INSTALL MINIMUM 3000 PSI CONCRETE WITH 6" X 6" - W4 X W4 WWF, AT MID LEVEL OVER 6 MIL VAPOR BARRIER (AT SLABS ON GRADE CONDITIONS OR ADHERE TO LOCAL CODE REQUIREMENTS). INSTALL CONTROL JOINTS AT MAXIMUM 300 SF.

DIVISION 04 MASONRY

A. NO WORK UNDER THIS SECTION UNLESS REQUIRED FOR PROPER CUTTING AND REPAIR OF BUILDING SHELL. IN THIS EVENT ALL WORK SHALL BE PERFORMED TO THE HIGHEST LEVELS OF QUALITY AND CRAFTSMANSHIP AND SHALL BE COORDINATED WITH LANDLORD'S REPRESENTATIVE.

DIVISION 05 METALS

A. METAL FRAMING:
1. LIGHT GAUGE METAL FRAMING:
   a. ALL STUD FRAMING MEMBERS, RUNNER TRACKS, BRIDGING AND NON-LOAD BEARING STUDS SHALL BE OF TYPE AND SIZE INDICATED ON DRAWINGS.
   b. PROVIDE ALL FASTENERS, SCREWS, BOLTS, CLIPS, ANCHORS, ANGLES, BRIDGES, BRACKETS, SLEEVES, ETC. AS REQUIRED.
   c. MATERIALS AND INSTALLATION SHALL BE IN STRICT CONFORMANCE WITH MANUFACTURER'S RECOMMENDATIONS.
2. STRUCTURAL STEEL, AND/OR HEAVY GAUGE METAL FRAMING:
   a. ALL STRUCTURAL STEEL/ HEAVY GAUGE METAL FRAMING SHALL BE OF TYPE AND SIZE INDICATED ON DRAWINGS.
   b. INSTALL FRAMING, CONNECTION PLATES, SCREWS, BOLTS, ETC. AS SHOWN IN PLANS AND DETAILS.
   c. MATERIALS AND INSTALLATION SHALL BE IN STRICT CONFORMANCE WITH MANUFACTURER'S RECOMMENDATIONS.
3. GALVANIZED INSULATION:
   a. INSTALL GALVANIC INSULATION BETWEEN DISSIMILAR METALS.
4. WELDING:
   a. PERFORM ALL WELDING IN CONFORMANCE WITH AWS STANDARDS.

B. PROTECTIVE CORNER GUARDS
1. INSTALL METAL CORNER GUARDS (2"X2"X12") AT ALL OUTSIDE CORNERS OR AS SPECIFIED IN THE PLANS. SPACKLE AND SAND SMOOTH.

C. REVEALS
1. INSTALL METAL REVEALS: SIZE AND LENGTH AS INDICATED ON PLANS. PROVIDE BLOCKING OR BACKING AS REQUIRED TO SECURE IN PLACE.
2. PAINT OR FINISH PER PLANS.

D. SUSPENSION SYSTEM
1. PROVIDE AND INSTALL UNISTRUT SPACE FRAME GRID SYSTEM PER MANUFACTURERS SPECIFICATIONS AS REQUIRED FOR LIGHTING, CEILING ACCESSORIES OR MECHANICAL SYSTEMS SUSPENSION.
2. SUPPORT FROM STRUCTURE WITH THREADED RODS AND NUTS AS REQUIRED OR DETAILED.
3. CONTACT UNISTRUT CORPORATION 800-468-9510.

DIVISION 06 WOOD, PLASTICS & COMPOSITES

A. CARPENTRY & BLOCKING:
1. BEFORE USING WOOD FRAMING OR BLOCKING OF ANY TYPE, CONFIRM CODE RESTRICTIONS. IF USE OF WOOD IS PERMITTED BY REQUIREMENTS, ADHERE TO THE FOLLOWING GUIDELINES:
   a. USE NEW LUMBER BEARING GRADE AND TRADEMARK OF ASSOCIATION UNDER WHICH IT WAS PRODUCED.
   b. LAY-OUT, CUT, FIT AND ERECT FRAMING FOR ROUGH AND FINISHED WORK. INSTALL BLOCKING, NAILERS, FURRING AND OTHER ROUGH AND FINISHED MATERIALS AS REQUIRED. BRACE, PLUMB, AND LEVEL MEMBERS IN TRUE ALIGNMENT AND RIGIDLY SECURE IN PLACE WITH SUFFICIENT NAILS, SPIKES, SCREWS AND BOLTS AS NECESSARY.
   c. FURNISH AND INSTALL ROUGH AND FINISHED PLYWOOD, FURRING SUPPORTS, ETC. FURNISH AND INSTALL ALL ROUGH HARDWARE, ANCHORS, BOLTS, ETC. AS REQUIRED FOR CARPENTRY WORK.
   d. ALL EXISTING OR NEW WOOD SHALL BE FIRE RETARDANT AND MUST BE IDENTIFIED WITH A UL LABEL CERTIFYING THIS CLASSIFICATION.
   e. WORKMANSHIP AS PER THE ARCHITECTURAL WOODWORK INSTITUTE'S ARCHITECTURAL WOODWORK QUALITY STANDARDS.

B. MILLWORK:
1. RECEIVE AND INSTALL ALL OWNER PROVIDED FINISHED MILLWORK CABINETS, PANELING, DOORS AND TRIMWORK, ETC. AS SHOWN ON DRAWINGS OR AS DIRECTED.
2. CUT, FIT, AND INSTALL ALL MILLWORK AND SET PLUM AND TRUE IN ACCORDANCE WITH ASSEMBLY/INSTALLATION DRAWINGS PROVIDED.
3. BACK CUT FLAT TRIM TO PREVENT WARPING. DRILL WHERE NECESSARY TO PREVENT SPLITTING.
4. CONFIRM BLOCKING REQUIREMENTS WITH VENDORS PRIOR TO CLOSING UP ALL ASSEMBLIES.

DIVISION 07 THERMAL & MOISTURE PROTECTION

A. VAPOR BARRIER:
1. PROVIDE AND INSTALL 6 MIL POLYETHYLENE BELOW ALL REQUIRED PATCH AREAS AT SLAB ON GRADE CONDITIONS AND 4 MIL POLYETHYLENE AT WALLS THAT RECEIVE BATT INSULATION.

B. ELASTOMERIC CRACK BRIDGING:
1. PROVIDE AND INSTALL ELASTOMERIC CRACK BRIDGING AS MANUFACTURED BY NATIONAL APPLIED CONSTRUCTION PRODUCTS, INC. 800-633-4672 OR APPROVED EQUAL.

C. BATT INSULATION:
1. PROVIDE AND INSTALL MINIMUM R-13 FRICTION FIT, FULL BATT INSULATION AS MANUFACTURED BY OWENS-CORNING FIBERGLASS OR EQUAL WHERE INDICATED IN THE DRAWINGS.

D. ACOUSTICAL INSULATION:
1. PROVIDE AND INSTALL ASTM C665, UNFACED, Ø.32 K-VALUE, WIDTHS TO COORDINATE WITH FRAMING FOR FRICTION FILL, THICKNESS SHOWN OR IF NOT SHOWN, FILL CAVITY.

E. WATERPROOF MEMBRANE:
1. PROVIDE AND INSTALL A (M) 30 MIL (MIN.) WATERPROOF MEMBRANE.
2. EXTEND A MINIMUM OF 4" UP VERTICAL SURFACES IN TOILET ROOMS OR AS REQUIRED BY LOCAL JURISDICTION.

F. SEALANTS:
1. PROVIDE AND INSTALL THE SEALANTS REQUIRED IN CONJUNCTION WITH ROOFING, WATERPROOFING, FIREPROOFING, FIRESTOPPING, DOOR FRAMES & CARPENTRY, ETC.
2. COLOR TO APPROXIMATE ADJACENT SURFACES, UNLESS DIRECTED OTHERWISE.

G. FIRESTOPPING:
1. PROVIDE AND INSTALL UL TESTED FIRESTOPPING ASSEMBLIES AS ARE APPLICABLE FOR EACH CONDITION ENCOUNTERED. IN THE ABSENCE OF TESTED ASSEMBLIES, PROVIDE FIRESTOPPING ASSEMBLIES AS RECOMMENDED BY THE FIRESTOPPING SYSTEM MANUFACTURER'S STAFF ENGINEERS AND AS APPROVED BY AUTHORITIES HAVING JURISDICTION.
2. PROVIDE FIRESTOPPING MATERIALS OF THE THICKNESS REQUIRED TO MAINTAIN THE FIRE RATINGS OF THE ASSEMBLIES INTO WHICH THE MATERIALS ARE INSTALLED.
3. STRICTLY MATCH UL TESTED ASSEMBLIES.
4. INSTALL FIRESTOPPING WITHOUT GAPS AND VOIDS OF ANY KIND.
5. INSPECT FIRESTOPPING WORK AND REPAIR OR REPLACE WORK WHICH HAS BEEN DAMAGED, DISTURBED, OR REMOVED BEFORE FIRESTOPPING IS CONCEALED OR ENCLOSED.
6. PROVIDE ANCHORAGE ACCESSORIES AND OTHER COMPONENTS AS NEEDED TO PROVIDE COMPLETE, EFFECTIVE FIRESTOPPING SYSTEMS COMPLYING WITH UL TESTED ASSEMBLIES.

H. FIRE CAULKING:
1. ALL SLEEVES, OPENINGS, ETC. THROUGH FIRE RATED WALLS AND FLOORS SHALL BE FIRE SEALED WITH CALCIUM SILICATE, SILICONE "RTV" FOAM, "3M" FIRE RATED SEALANTS OR EQUAL BY HILTI AFTER INSTALLATION SO AS TO RETAIN THE FIRE RATING.

I. FIREPROOFING:
1. PROVIDE AND INSTALL UL TESTED FIREPROOFING ASSEMBLIES AS ARE APPLICABLE FOR EACH CONDITION ENCOUNTERED AND AS APPROVED BY AUTHORITIES HAVING JURISDICTION.
2. PROVIDE FIREPROOFING MATERIALS OF THE THICKNESS AND DENSITY REQUIRED TO ACHIEVE THE REQUIRED FIRE RATINGS.
3. STRICTLY MATCH UL TESTED ASSEMBLIES.
4. INSPECT EXISTING FIREPROOFING WORK AND REPAIR OR REPLACE WORK WHICH HAS BEEN DAMAGED, DISTURBED, OR REMOVED.
5. PROVIDE ANCHORAGE ACCESSORIES AND OTHER COMPONENTS AS NEEDED TO PROVIDE COMPLETE, EFFECTIVE FIREPROOFING SYSTEMS COMPLYING WITH UL TESTED ASSEMBLIES.

J. EXPANSION JOINTS/EXPANSION JOINT COVERS
1. PROVIDE EXPANSION JOINTS, AS REQUIRED FOR APPLICATION WITH ELASTOMERIC FILLER TO MATCH ADJACENT FINISH AS MANUFACTURED BY BALCO INC.
2. INSTALL PER PLANS AND MANUFACTURER'S RECOMMENDATIONS.

DIVISION 08 OPENINGS

A. DOORS AND HARDWARE:
1. FURNISH AND INSTALL ALL DOORS, FRAMES, HARDWARE AND RELATED PARTS AND MATERIALS AS INDICATED ON DRAWINGS AND RESPONSIBILITY SCHEDULE.
2. HOLLOW METAL FRAMES: KD, MINIMUM 16 GA. SIZE, SHAPE AND PARTITION TYPE PER DOOR SCHEDULE.
3. WOOD DOORS: FLUSH TYPE DOORS, EDGE BANDING AND VENEERS SHALL BE APPEARANCE GRADE READY FOR PAINT.
4. LABELED DOORS: PROVIDE UL LABEL DOORS AND FRAMES PER APPLICABLE STATE AND LOCAL BUILDING CODES, WHERE REQUIRED.
5. FABRICATE AND ERECT TRUE, PLUMB, AND SQUARE AS SPECIFIED AND DETAILED AND AS PER MANUFACTURER'S RECOMMENDATIONS.
6. SEE DOOR SCHEDULE FOR SPECIFIC INFORMATION AND HARDWARE.

B. STOREFRONT CLOSURE:
1. INSTALLATION AND SUPPORT OF ROLL-UP GRILLE OR SECURITY GATE MUST COMPLY WITH LOC

GENERAL NOTES

1. AN APPROVED SET OF PLANS (BUILDING, FIRE, SPRINKLER, FIRE ALARM, ETC.) SHALL BE ON THE JOB SITE DURING CONSTRUCTION. NO INSPECTIONS WILL BE MADE WITHOUT COMPLIANCE WITH THE ABOVE.
2. ALL METAL STUD PARTITIONS SHALL BE CONSTRUCTED OF METAL STUDS AT 16" O.C. MAXIMUM. SEE PLANS FOR EXCEPTIONS.
3. MAINTAIN A TELEPHONE AND FAX MACHINE WITH WORKING PHONE LINES THROUGHOUT THE PROGRESS OF THE WORK.
4. DRAWINGS ARE NOT TO BE SCALED FOR INFORMATION.
5. CONTRACTOR SHALL PROVIDE FIRE EXTINGUISHERS AND CABINETS WITH REQUIRED SIGNAGE AS DIRECTED BY THE LOCAL FIRE DEPARTMENT. ARCHITECT SHALL SELECT/APPROVE TYPE AND LOCATION.
6. CONTRACTOR SHALL PROVIDE CONCEALED BLOCKING AS REQUIRED TO SUPPORT/SECURE WALL-HUNG FIXTURES, WALL STANDARDS, BATH ACCESSORIES, ARCHITECTURAL AND MILLWORK ITEMS, ETC.
7. CONTRACTOR'S INSTALLATION OF ALL UTILITIES AND SUBSEQUENT HOOKUP SHALL BE AS NECESSARY TO ASSURE PERFORMANCE WITH UTILITY COMPANY.
8. CONTRACTOR SHALL PROVIDE ACCESS PANELS AS REQUIRED BY PLUMBING, AIR CONDITIONING AND OTHER TRADES, AND AS REQUIRED BY CODES. LOCATIONS TO BE APPROVED BY ARCHITECT.
9. FIRE DAMPERS ARE REQUIRED IF DUCTS PENETRATE THE CEILING OF A FIRE-RESISTIVE FLOOR-CEILING ASSEMBLY, OR PENETRATE FIRE-RATED CORRIDOR WALLS. PROVIDE AS REQUIRED.
10. SIGN CONTRACTOR TO OBTAIN SEPARATE APPROVALS AND PERMITS FROM APPLICABLE GOVERNMENT AGENCIES FOR SIGNS.
11. CONTRACTOR SHALL BE RESPONSIBLE FOR SCHEDULING AND OBTAINING ALL INSPECTIONS AND TESTS REQUIRED TO IMPLEMENT THE PLANS AND SPECIFICATIONS. TESTING AND INSPECTION COSTS BY GENERAL CONTRACTOR.
12. THE CONTRACTOR SHALL VERIFY ALL EXISTING CONDITIONS PRIOR TO COMMENCEMENT OF WORK AND REPORT ANY DISCREPANCIES TO ARCHITECT.
13. ALL NEW WORK TO BE SQUARE AND TRUE TO THE "REFERENCE LINES", AS SHOWN ON THE PLANS. THE EXISTING STRUCTURE SHOULD NOT BE ASSUMED TO BE SQUARE OR TRUE AND SHOULD NOT BE USED AS A REFERENCE OR CONTROL GUIDELINE FOR LAYOUT UNLESS OTHERWISE NOTED.
14. ALL WORK PERFORMED BY THE GENERAL CONTRACTOR AND HIS SUBCONTRACTORS SHALL CONFORM TO THE REQUIREMENTS OF MUNICIPAL, LOCAL AND STATE LAWS AND ALL OTHER GOVERNING AGENCIES HAVING JURISDICTION, WHETHER OR NOT SPECIFIED HEREIN.
15. THE CONTRACTOR SHALL ERECT AND MAINTAIN AS REQUIRED BY EXISTING FIELD CONDITIONS, THROUGHOUT THE ENTIRE PROGRESS OF THE WORK, ALL SAFEGUARDS AND BARRICADES FOR SAFETY PER O.S.H.A. STANDARDS INCLUDING (BUT NOT LIMITED TO) POSTING WARNING SIGNS, ENFORCING SAFETY REGULATIONS, AND PROTECTION OF PROPERTY.
16. UNLESS OTHERWISE NOTED ALL MATERIALS SHALL BE CONSIDERED NEW AND ARE TO BE PROVIDED BY THE GENERAL CONTRACTOR.
17. THE CONTRACTOR SHALL COORDINATE WORK BETWEEN ALL TRADES ON JOB SITE INCLUDING BARNES & NOBLE'S VENDORS.
18. G.C. SHALL COORDINATE ALL BUILDING TRADES TO PROVIDE ROUTING AND LOCATION FOR H.V.A.C., PLUMBING, ELECTRICAL, FIRE PROTECTION, STRUCTURAL, ETC. TO PERMIT EXACT LOCATION OF ALL ARCHITECTURAL ELEMENTS. ANY CONFLICTS OR DISCREPANCIES SHALL BE BROUGHT TO THE ATTENTION OF THE ARCHITECT IN WRITING IMMEDIATELY BEFORE PROCEEDING WITH THE WORK.
19. REFER TO PLUMBING AND ELECTRICAL DRAWINGS FOR SCOPE OF ALL UNDERGROUND TRENCHING WORK. GENERAL CONTRACTOR SHALL CUT, PATCH AND REPAIR CONCRETE SLAB AS REQUIRED.
20. ALL WOOD BLOCKING, SHEATHING, GROUNDS, SHIMS, ETC. TO BE FIRE-RETARDANT TREATED WOOD. (TYPICAL UNLESS OTHERWISE NOTED)
21. COORDINATE ALL TELEPHONE WORK WITH OWNER'S REPRESENTATIVE AND LOCAL TELEPHONE COMPANY BEFORE STARTING WORK.
22. BEFORE CORE DRILLING STARTS, NOTIFY THE BUILDING MANAGEMENT AND TENANTS OF SUCH WORK. MAKE ALL ARRANGEMENTS AND SCHEDULES OF WORK WITH SAME. PROTECT ALL WALLS, FLOORS, CEILINGS, ETC., THAT ARE ADJACENT TO THE CORE DRILLING OPERATIONS.
23. FOR WOODWORK INSTALLATION, CHECK AND VERIFY ALL DIMENSIONS AND CONDITIONS AT THE JOB SITE PRIOR TO STARTING WORK. PROVIDE SHOP DRAWINGS TO ARCHITECT FOR APPROVAL OF ALL MILLWORK.
24. ALL FINISHED WORK SHALL BE CONSTRUCTED IN THE MOST CAREFUL AND THOROUGH MANNER IN ACCORDANCE WITH DETAILS, AFTER SHOP DRAWINGS AND THE BEST METHODS KNOWN IN THE INDUSTRY. ALL EXPOSED STRUCTURES SHALL BE FREE FROM MARKS, BLEMISHES, OR DEFECTS OF ANY KIND. PROVIDE ALL CONNECTIONS FOR ELECTRICAL FIXTURES.
25. ALL SURFACES TO BE PAINTED OR TO RECEIVE WALL COVERING SHALL PRESENT A TRUE, SMOOTH, LEVEL FINISH AND SHALL BE CLEANED FREE FROM ALL GRIT, GREASE, DIRT AND LOOSE MATERIALS AND ALL SCRATCHES, CRACKS, OPEN JOINTS, HOLES AND OTHER DEFECTS, SHALL BE PROPERLY CUT OUT, SPACKLED AND BROUGHT TO A SMOOTH EVEN FINISH AND GIVEN A PRIME COAT AS SPECIFIED FOR THE AFFECTED WORK.
26. PROVIDE AND MAINTAIN ADEQUATE PROTECTION FOR ALL FLOORING WORK FOR THE DURATION OF CONSTRUCTION.
27. FOR COMPLETION, ALL WORK SHALL BE CLEANED, REMOVING ALL SPOTS OF ADHESIVE, SURFACE STAINS AND ALL SCRATCHES.
28. SUBMIT TO THE ARCHITECT, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES FOR REVIEW AND APPROVAL AS LISTED BELOW BUT NOT LIMITED TO:
    - MECHANICAL DUCT WORK LAYOUT
    - MECHANICAL LOUVERS, DIFFUSERS, FIRE DAMPERS, DUCT DETECTORS, THERMOSTATS, ROOF TOP UNITS, ETC.
    - SPRINKLER SYSTEM DESIGN WITH HYDRAULIC CALCULATIONS
    - STEEL STAIRS, LANDINGS AND RAILINGS
    - TOILET PARTITIONS AND ACCESSORIES
    - ENTRY DOORS, SIDELIGHTS AND TRANSOMS
    - LIGHTING FIXTURES
    - PAINT SAMPLES
    - EXTERIOR SIGNAGE (GENERAL CONTRACTOR TO COORDINATE)
    - ELEVATOR AND ESCALATOR DRAWINGS WHERE APPLICABLE
    - ELECTRICAL PANELS AND TRANSFORMERS
    - MILLWORK AND CABINETRY
    - SAMPLES OF ALL WOOD FINISHES
    - ORNAMENTAL METALS (STEEL RAILINGS, ETC.)
    - CERAMIC TILE
    - ALUMINUM STORE FRONT AND GLAZING
    - METAL LOCKERS
    - STOCK ROOM SHELVING
    - DOORS, FRAMES AND HARDWARE
    - PLUMBING FIXTURES
    - EXTERIOR COLD ROLLED METAL STUDS & FRAMING
    - EXTERIOR AWNING (SIGNED AND SEALED)
    - VERTICAL/INCLINED TRANSPORTATION (PASSENGER CIRCULATION)
    - VERTICAL/INCLINED TRANSPORTATION (MATERIAL LOADING)
29. GENERAL CONTRACTOR SHALL NOTE THAT, UNLESS OTHERWISE AGREED IN WRITING WITH THE ARCHITECT, SHOULD GENERAL CONTRACTOR AND/OR HIS SUBCONTRACTOR(S) COMMENCE FABRICATION AND/OR INSTALLATION ON ANY ITEM LISTED ABOVE PRIOR TO SHOP DRAWING SUBMITTAL AND APPROVAL BY THE ARCHITECT, GENERAL CONTRACTOR SHALL ASSUME ALL RESPONSIBILITY FOR THAT SPECIFIC ITEM (MATERIAL AND LABOR COSTS) AND, SHOULD THE ARCHITECT DETERMINE THAT MODIFICATIONS ARE NECESSARY TO ENABLE THE ITEM OR SYSTEM TO PERFORM TO DESIGN SPECIFICATIONS, THE GENERAL CONTRACTOR SHALL PERFORM OR HAVE PERFORMED THE MODIFICATIONS NECESSARY AT HIS OWN COST.
30. THE GENERAL CONTRACTOR MUST SUBMIT THE FOLLOWING ITEMS TO THE BARNES AND NOBLE DEVELOPING MANAGER AND ARCHITECT ALONG WITH FINAL CONSTRUCTION INVOICE PRIOR TO FINAL PAYMENT.

    GENERAL CONTRACTOR PROVIDES:
    - GC CERTIFICATION (100% COMPLETE AND 1 YEAR WARRANTY)
    - CERTIFICATE OF OCCUPANCY
    - AS-BUILT DRAWINGS (DIGITAL FORMAT)
    - ALL BUILDING INSPECTION SIGN OFF
    - COPY OF BUILDING PERMIT
    - CERTIFICATE OF INSURANCE FROM CONTRACTOR
    - LIEN RELEASE (FINAL UNCONDITIONAL RELEASES 2 WEEKS AFTER FINAL PAYMENT)
    - SUBCONTRACTOR LIST
    - MANUFACTURER'S AND SUBCONTRACTOR'S GUARANTEES AND WARRANTIES
    - AIR BALANCE REPORT
    - OWNERS MANUAL OF EQUIPMENT
    - COPY OF GC CONTRACT

    ALONG WITH FINAL CONSTRUCTION INVOICE PRIOR TO FINAL PAYMENT.
31. GENERAL CONTRACTOR WILL BE FULLY RESPONSIBLE FOR ALL COSTS ASSOCIATED WITH OBTAINING PERMITS, APPROVALS, IMPACT FEES, TIE-IN FEES, ETC. AS REQUIRED BY ALL APPROVING AGENCIES AND LANDLORDS.
32. GENERAL CONTRACTOR SHALL PREPARE AND SUBMIT TO FIRE DEPARTMENT AND OTHER REQUIRED APPROVING AGENCIES THE FOLLOWING. FIRE ALARM AND SPRINKLER DRAWINGS SIGNED, SEALED AND PREPARED BY A LICENSED ENGINEER FOR REVIEW AND APPROVAL PRIOR TO OBTAINING PERMITS. REFER TO ADDITIONAL NOTES ON SPRINKLER DRAWINGS AND FIRE ALARM DRAWINGS FOR FURTHER CLARIFICATIONS.
33. GENERAL CONTRACTOR TO VERIFY THAT MAXIMUM HEIGHT FOR ALL STAIR RISERS SHALL BE 7" AND MINIMUM STAIR TREAD WIDTH TO BE 11".
34. GENERAL CONTRACTOR/ELECTRICIAN IS RESPONSIBLE FOR PROVIDING KROY TYPED LABELS INDICATING PANEL AND CIRCUIT NUMBER FOR ALL OUTLETS AND JUNCTION BOXES.

INTERPRETATIONS AND CONFLICTS

1. BRING ANY DISCREPANCIES BETWEEN DIFFERENT DRAWINGS, BETWEEN THE DRAWINGS AND FIELD CONDITIONS OR BETWEEN THE DRAWINGS AND SPECIFICATIONS, OR ANY APPARENT OMISSIONS, TO THE ATTENTION OF THE ARCHITECT/ENGINEER PRIOR TO SUBMISSION OF ANY BID. AFTER AWARD OF CONTRACT, THE INTERPRETATION OF ANY CONFLICT WILL BE MADE BY THE ARCHITECT AND SHALL BE ACCEPTED AS FINAL.
2. THE FAILURE TO QUESTION ANY CONTROVERSIAL ITEM WILL CONSTITUTE ACCEPTANCE BY THE BIDDER, WHO SHALL EXECUTE SAME TO THE SATISFACTION OF THE ARCHITECT/ENGINEER AFTER BEING AWARDED THE CONTRACT.
3. SUBMIT DETAILS OF ANY DESIRED DEPARTURES FROM THE CONTRACT DOCUMENTS AS MAY BE DEEMED NECESSARY, AND THE REASONS THEREFORE, WITHIN 15 DAYS AFTER AWARD OF THE CONTRACT. MAKE NO DEPARTURES WITHOUT WRITTEN APPROVAL.
4. IF MENTION HAS BEEN OMITTED PERTAINING TO DETAILS, ITEMS OR RELATED ACCESSORIES REQUIRED FOR THE COMPLETION OF ANY ELECTRICAL, PLUMBING, HVAC OR OTHER SYSTEM, INCLUDE SUCH ITEMS AND ACCESSORIES IN THE CONTRACT WITHOUT ADDITIONAL CHARGE.
5. AFTER THE CONTRACT IS AWARDED, CLAIMS BASED ON INSUFFICIENT DATA OR INCORRECTLY ASSUMED CONDITIONS, OR CLAIMS BASED ON MISUNDERSTANDING THE NATURE OR CHARACTER OF THE WORK OR THE CONDITIONS UNDER WHICH IT MUST BE PERFORMED, WILL NOT BE RECOGNIZED.
6. PRIOR TO SUBMITTING THE BID, RESOLVE ANY APPARENT CONFLICTS IN REGARD TO LOCATIONS, INTERFERENCE AND JURISDICTION.
7. THE CONTRACTOR SHALL NOT INTERPRET THIS PROJECT TO BE A "MINIMUM CODE DESIGN", OR AN "INDUSTRY STANDARD DESIGN", WHERE MATERIALS OR METHODS SHOWN ON THE DRAWINGS, OR SPECIFIED HEREIN, INDICATE A PREMIUM MATERIAL OR METHOD, THE CONTRACTOR IS EXPECTED TO PROVIDE SAME, AND PROPOSE ALTERNATE METHODS OR MATERIALS AS A SIDE ITEM. EXCUSES SUCH AS "IT MEETS NEC REQUIREMENTS" OR "THIS IS THE INDUSTRY STANDARD" WILL NOT BE RECOGNIZED AS JUSTIFICATION FOR ACCEPTANCE OF ITEMS NOT CONFORMING TO THE INTENDED DESIGN.

BUILDING CODE ANALYSIS

PROJECT LOCATION BARNES @ NOBLE
3173 CINEMA POINT,
COLORADO SPRINGS, CO 80922

DESCRIPTION WJCA PROJECT NO. – 33288

ARCHITECTURAL- INTERIOR TENANT FIT-OUT WITHIN AN EXISTING SPACE IN AN EXISTING BUILDING. NO CHANGE TO USE OR OCCUPANCY. NO EXTERIOR WORK.

MEP SCOPE- OVERALL, RENOVATIONS OF EXISTING BUILDING TO B&N. EXISTING SERVICES TO REMAIN.

NUMBER OF STORIES: 1

TENANT SQUARE FOOTAGE: 17,660 SQ.FT.

CONSTRUCTION CLASSIFICATION TYPE 2B
1 STORY

OCCUPANCY GROUP GROUP M (MERCANTILE), GROUP B (BUSINESS), GROUP A-2 (ASSEMBLY) 1

BUILDING DEPARTMENT PIKES PEAK REGIONAL BUILDING DEPARTMENT
2880 INTERNATIONAL CIRCLE,
COLORADO SPRINGS, CO 80910.

ALL CONSTRUCTION SHALL COMPLY WITH THE FOLLOWING CODES AND AMENDMENTS PER THEIR ADOPTING ORDINANCES.

2023 REGIONAL BUILDING CODE

INTERNATIONAL BUILDING CODE 2021

INTERNATIONAL PLUMBING CODE 2021

NFPA 70, 2023

INTERNATIONAL MECHANICAL CODE 2021

2010 ADA STANDARDS

COLORADO ACCESSIBILITY CODE A117.1, 2017

INTERNATIONAL ENERGY CONSERVATION CODE 2021

NOTE: THE ONLY ENERGY PROVISION THAT APPLIES TO THIS APPLICATION BASED ON SCOPE AND SIZE IS REGARDING THE INTERIOR LIGHTING AND POWER COMPLIANCE. THIS IS AN INTERIOR TENANT FIT-OUT WITHIN AN EXISTING BUILDING WITHIN ITS EXISTING ENVELOPE, THEREFORE WE ARE EXEMPT FROM THE ENVELOPE'S ENERGY CALCULATIONS.

FIRE ALARM SYSTEM (FILED UNDER SEPARATE APPLICATION)

FIRE SUPPRESSION SYSTEM (FILED UNDER SEPARATE APPLICATION)

CHAPTER 2 GENERAL DEFINITIONS

NET FLOOR AREA:
THE ACTUAL OCCUPIED AREA NOT INCLUDING UNOCCUPIED ACCESSORY AREAS SUCH AS CORRIDORS, STAIRWAYS, TOILET ROOMS, MECHANICAL ROOMS AND CLOSETS.

CHAPTER 10 MEANS OF EGRESS
SECTION 1004 DESIGN OCCUPANT LOAD

Table 1004.5 MAXIMUM FLOOR AREA ALLOWANCES PER OCCUPANT
USE GROUP M 60 SF GROSS
BUSINESS AREAS 150 SF GROSS

PROPOSED OCCUPANCY LOAD

	Area (S.F.)	Occupant Load	Occupants
GENERAL SALES AREA	12,437	60	207
STOCK ROOM	1,925	60	32
MANAGER'S OFFICE	133	150	1
RESTROOM AREA	163	60	3
CORRIDOR	228	60	4
JANITOR CLOSET	42	60	1
STORAGE	445	60	8
BREAK ROOM	282	150	2
CAFE/CAFE WORK ROOM	786	200	4
CAFE SEATING AREA	1,219	15 NET	81
TOTAL	17,660		343 2

SECTION 1005 MEANS OF EGRESS SIZING (SEE EGRESS PLAN ON A-003)

SECTION 1005.3.2 OTHER EGRESS COMPONENTS. CAPACITY FACTOR OF 0.2" PER OCCUPANT.

DOORS, CORRIDORS = 0.2/OCCUPANT
USE GROUP M, WITH SPRINKLER SYSTEM
TOTAL REQUIRED: 343 OCCUPANTS X 0.2 INCHES = 68.6 INCHES 2

EXIT #1 FRONT ENTRY SWING DOOR: 6'-0" = 72" WIDTH
EXIT #2 SIDE ENTRY SWING DOOR: 4'-0" = 48" WIDTH
EXIT #3 REAR ENTRY SWING DOOR: 6'-0" = 72" WIDTH 2

TOTAL PROVIDED: 192 INCHES > 68.6 INCHES OK.

SECTION 1007.1 MINIMUM NUMBER OF EXITS REQUIRED:
1-500 OCCUPANTS = 2 EXITS (TABLE 1006.3.2)

NUMBER OF EXITS PROVIDED: 3

SECTION 1007.1.1 TWO EXITS OR EXIT ACCESS DOORWAYS- WHERE TWO EXITS, EXIT ACCESS DOORWAYS, EXIT ACCESS STAIRWAYS OR RAMPS, OR ANY COMBINATION THEREOF, ARE REQUIRED FROM ANY PORTION OF THE EXIT ACCESS, THEY SHALL BE PLACED A DISTANCE APART EQUAL TO NOT LESS THAN ONE-HALF OF THE LENGTH OF THE MAXIMUM OVERALL DIAGONAL DIMENSION OF THE BUILDING OR AREA TO BE SERVED MEASURED IN A STRAIGHT LINE BETWEEN THEM. INTERLOCKING OR SCISSOR STAIRWAYS SHALL BE COUNTED AS ONE EXIT STAIRWAY.

EXCEPTION 2 WHERE A BUILDING IS EQUIPPED THROUGHOUT WITH AN AUTOMATIC SPRINKLER SYSTEM IN ACCORDANCE WITH SECTION 903.3.1.1 OR 903.3.1.2, THE SEPARATION DISTANCE SHALL BE NOT LESS THAN ONE-THIRD OF THE LENGTH OF THE MAXIMUM OVERALL DIAGONAL DIMENSION OF THE AREA SERVED.

BUILDING DIAGONAL = 202'-11" / 3 = 67'-8"
ACTUAL REMOTENESS = 167'-0" (EXIT 1 TO EXIT 2)

SECTION 1017.2 EXIT ACCESS AND TRAVEL DISTANCE.
OCCUPANCY M WITH SPRINKLER SYSTEM = 250'
(NOTE: SEE A-003 FOR EGRESS PATHS)

SECTION 1020.4 TWO EXITS OR EXIT ACCESS DOORWAYS:
EXCEPTION 2 WHERE MORE THAN ONE EXIT OR EXIT ACCESS DOORWAY IS REQUIRED, THE EXIT ACCESS SHALL BE ARRANGED SUCH THAT THERE ARE NO DEAD ENDS IN CORRIDORS MORE THAN 50 FEET IN LENGTH. (WHERE THE BUILDING IS EQUIPPED WITH AUTOMATIC SPRINKLER SYSTEM)

PLUMBING CODE ANALYSIS CHAPTER 4 MINIMUM NUMBER OF PLUMBING FIXTURES (TABLE 403.1)
BUILDING TYPE OR OCCUPANCY : MERCANTILE
MERCANTILE: 343/2 = 172 MALE & 172 FEMALE 2

WATER CLOSETS REQUIRED: 1 PER 500 OCCUPANTS
WATER CLOSETS PROVIDED: 2

URINALS PROVIDED: 0

LAVATORIES REQUIRED: 1 PER 750 OCCUPANTS
LAVATORIES PROVIDED: 2

DRINKING FOUNTAINS REQUIRED: 1 PER 1000 OCCUPANTS
DRINKING FOUNTAINS PROVIDED: 1 HIGH & LOW TYPE

SERVICE SINK REQUIRED: 1 SERVICE SINK
SERVICE SINK PROVIDED: 2

SYMBOL LEGEND

ROOM NUMBER IDENTIFICATION — ROOM DESCRIPTION / NO.

DOOR NUMBER IDENTIFICATION — 06

INTERIOR/EXTERIOR ELEVATIONS — 1 / 3 A-400 4 / 2

WALL SECTION/DETAILS — 1 / A-410

FINISH IDENTIFICATION — CT-1

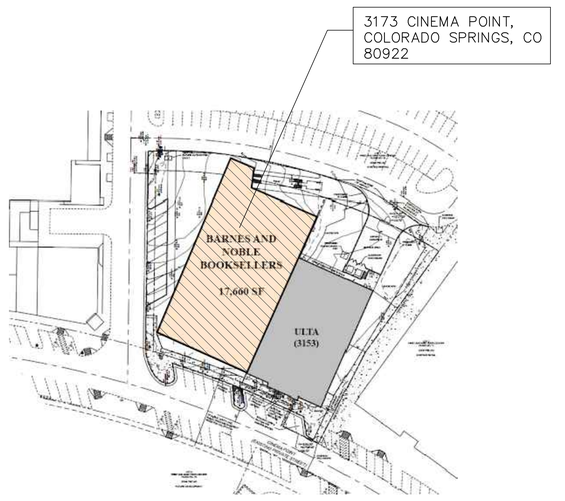
KEY NOTES — 1

TYPICAL DETAIL DESIGNATION — DETAIL NUMBER → 1; SHEET WHERE DETAIL IS SHOWN → A-410

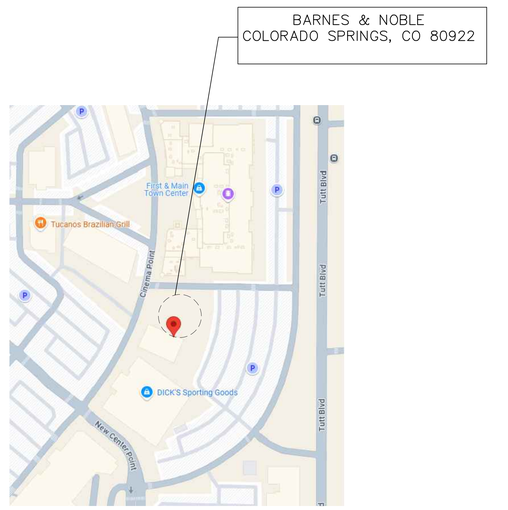
ABBREVIATIONS

A.C.T.	ACOUSTICAL CEILING TILE	MIN.	MINIMUM
A.F.F.	ABOVE FINISH FLOOR	M.O.	MASONRY OPENING
BLK	BLOCK	N.I.C.	NOT IN CONTRACT
BLK'G	BLOCKING	N/A	NOT APPLICABLE
CLG.	CEILING	NO.	NUMBER
C.T.	CERAMIC TILE	O.C.	ON CENTER
CONC.	CONCRETE	OPN'G	OPENING
CONT.	CONTINUOUS	PLAM.	PLASTIC LAMINATE
C.J.	CONTROL JOINT	PL.	PLATE
DN.	DOWN	PLY'D	PLYWOOD
DIA.	DIAMETER	RAD.	RADIUS
DWG.	DRAWING	R.C.P.	REFLECTED CEILING PLAN
E.W.C.	ELECTRIC WATER COOLER	REQ'D	REQUIRED
EL.	ELEVATION	STL.	STEEL
EXIST.	EXISTING	SUSP. CLG.	SUSPENDED CEILING
EXP.	EXPANSION	T.O.M.	TOP OF MASONRY
F.P.	FIREPROOF	T.O.S.	TOP OF STEEL
F.R.	FIRE RETARDANT	TYP.	TYPICAL
F.B.O.	FURNISHED BY OWNER	V.C.T.	VINYL COMPOSITE TILE
FIN.	FINISHED)	V.I.F.	VERIFY IN FIELD
GA.	GAUGE	W/	WITH
G.C.	GENERAL CONTRACTOR	WD.	WOOD
GALV.	GALVANIZED	W/O	WITHOUT
GL.	GLASS		
H.M.	HOLLOW METAL		
H.P.	HIGH POINT		
I.B.C.	INSTALLED BY CONTRACTOR		
JT.	JOINT		
LAM.	LAMINATE		
LAV.	LAVATORY		
L.P.	LOW POINT		
MAX.	MAXIMUM		
MD.	MOLDING		
MTL.	METAL		

KEY PLAN



VICINITY MAP



BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
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06-12-25	LL AND BD COMMENTS	2
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These plans are an instrument of service and the property of the Architect. Infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

STATE OF COLORADO — WILLIAM J. COMMER — LICENSED ARCHITECT

LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

General Notes, Legend, Code Analysis, Plot Plan

Drawing Number:

A-001

VENDOR LIST

VENDOR	NEW STORE RESPONSIBILITY	CONTACT INFORMATION
SINAGE		
Ruggles Signs	Exterior Signage / Illuminated Signage	93 Industry Drive | PO BOX 349 Versailles, KY 40383 Contacts: Tim Cambron Email: tim@rugglessign.com Phone: 859-879-1199 ext. 104 Farriah Hardin Email: farriah@rugglessign.com Phone: 859-879-1199 ext. 163
Color Image, Inc	Coming Soon Graphics / Receiving Room Decal	9148 King St Franklin Park, IL 60131 Contacts: Kyle O'Donnell Email:kyle@colorimage.com Phone: 847-226-5645
LOW VOLTAGE		
MRF Tek	LV Wiring and Installation	8342 Milky Way Zionsville PA 18092 Contacts: Angel Veloz Email: aveloz2580@gmail.com Phone: 484-538-3249
TDX	LV Wiring and Installation	5735 W Old Shakopee Rd #100 Bloomington, MN 55437 Contacts: Michael Matt Email: mmatt@tdxtech.com Phone: 952-912-9281
TST Network Corp.	LV Wiring and Installation / Cell Booster / CCTV	Contacts: Frank Stantos Email: frank.santos@tstcorporation.com Phone: 347-755-0393
LightStat	Lighting / HVAC Controls	22 W West Hill Rd Barkhamsted, CT 06063 Contacts: Kevin Ellis Email: k.ellis@lightstat.com Phone: 800-292-2444 ext. 277 Karen Simons Email: k.simons@lightstat.com Phone: 860.738.4111 Ext: 279
Mood Media	Music	Contacts: Eve DeForest Email: eve.deforest@moodmedia.com Phone: 512.788.4103
PSI	CCTV	598 Hillsboro Technology Drive Deerfield Beach, FL 33441 Contacts: Andrew Creta Email: andy.creta@prosegur.com Phone: 845-325-9455
Global Security Services	EAS	Contacts: Chris Saporito Email: chris.saporito@global-security-solutions.com Phone:
Checkpoint	EAS	101 Wolf Drive Thorofare NJ, 08086 Contacts: Allan Beam Email: ABeam@checkpt.com Phone: (937) 522 5118 Jerry O'Leary Email: joleary@checkpt.com Phone: (1)- 800- 257- 5540 X 2311
Scarsdale	Burglar Alarm / Fire Alarm Monitoring	Contact: Ramiro Martinez Email: rmartinez@scarsdalesecurity.com Phone: 914-722-2389
FLOORING / TILE		
Associated Flooring	Flooring- Carpet and Tile Wall - Tile	245 Portofino Drive North Venice, FL 34275 Contacts: Richard Goodman Email: rgoodman@assocint.com Phone: 917-328-1462 Robert Odgis Email:bodgis@assocint.com Phone: (973) 376 1111
RHI Inc.	Tile and Carpet Installation (Flooring and Wall)	7397 Washington Blvd. Suite 106 Elkridge, MD 21075 Contact: Marty Hilliard Email: mhilliard@rhi-flooring.com Phone: 443-677-8408
LIGHTING		
Specialty Lighting	Lighting	74 Pickering Street Portland, CT 06480 Contacts: Meghan Donooghue Email: meghan.donoghue@sslighting.com Phone: (860)767-0110 x232 Paul O'Donnell Email: paul.odonnell@sslighting.com Phone: (203) 499-7030

VENDOR	NEW STORE RESPONSIBILITY	CONTACT INFORMATION
CAFE EQUIPMENT		
DWM	Water Filtration	Contacts: Dovie Wood Email:dwood@dwminc.com Phone:888-396-9111, x215
Wasserstrom	Café Equipment	4500 E Broad St Columbus OH 43213 Contacts: Jim Miller Email: jimmiller@wasserstrom.com Phone: (800) 999-9277 X 8592
Woodin	Café Millwork (Cabinets and Countertops)	280 Gannett Drive South Portland, ME 04106 Contacts: Troy Fredericks Email: troy@woodinco.com Phone: 207-671-9551
Melitta	Espresso Machines	2150 Point Blvd., Suite 200 Elgin, IL 60123 Contacts: Karie Van Sistine Email: karie.vansistine@melitta.com Phone: (847) 717-8887
SUPPORT AREA FIXTURES		
PIPP	Receiving Room Shelving	2966 Wilson Dr. NW Walker MI 49534 Contacts: Cassandra Beach Email: cbeach@pippmobile.com Phone: 616-988-4060
mmi Storage Solutions	Receiving Room Shelving	45 Turner Drive, Suite 2A Middletown, NY 10941 Contacts: Paul Ruksenas Email: pr@mobilemediastorage.com Phone: 216-849-1469
SALES FLOOR FIXTURES		
Nicewood	Sales Floor Fixtures / Café Wood Paneling	9001 Westmont Drive Toano, VA 23168 Contacts: John Finney Email: jfinney@davidnicebuilders.com Phone: 757-566-2156 Jacob Oberding Email: joberding@davidnicebuilders.com Phone: 757.566.2156 Kelsi Foster Email: kfoster@nicewood.com Phone: 757.566.2156
PCL/Ardent	Sales Floor Fixtures / Cafe Furniture	81 Legget ST. East Hartford, CT 06108 Contacts: Donald Budnick Email: dbudnick@ardentdisplays.com Phone: 860.214.2233 Brenda Winter Email: bwinter@pclfixtures.net Phone: 860-904-6906 Larry Cahill Email: lcahill@ardentdisplays.com Phone: 972-691-5603
Visual Creations	Sales Floor Fixtures / Merchandise Accessories	500 Narragansett Park Drive Pawtucket, RI 02861 Contacts: Kathleen Rohrman Email: krohrman@vci-displays.com Phone: 401-862-7462
SAMA	Sales Floor Fixtures / Café Furniture / Merchandise Accessories	20 Sand Park Road Cedar Grove, NJ 07009 Contacts: Mark Wolfberg Email: mlwolfberg@samaplastics.com Phone: (973) 239-7200
IMG	n/a	
TinkerTin	n/a	
FURNITURE		
Jasper Chair	Café Seating / Sales Floor Seating	534 E 8TH STREET JASPER, IN 47546 Contacts: David Laforest Email: dlaforest@jasperchair.com Phone: 401.578.7382
CF Group	Kids Ottomans / Café Table Bases & Legs	Contacts: Steve Cohen Email: scohen@mycfgroup.com Phone: 636-448-3465

VENDOR	NEW STORE RESPONSIBILITY	CONTACT INFORMATION
FIXTURE INSTALLATION / CAF INSTALLATION		
Patterson McClain	Café Installation / Millwork Installation	P.O. Box 369 Cary, NC 27512 Contacts: Greg Morf Email: gmorf@pattersonmcclain.com Phone: (919) 380-0772
D&S Millwork	Millwork Installation	Contacts; Mike Gish Email: gmorf@pattersonmcclain.com Phone: 541-654-2380
Nicon Construction	Millwork Installation	1004 Ionia Avenue Staten Island, NY 10309 Contacts: Angelo Nicastro Email: angelonicastro@aol.com Phone: 917 567 1694
Sunrise Installations	Millwork Installation	19737 Canary Way Queen Creek AZ 85242 Contacts: Lee Tomczak Email: lee.tomczak@gmail.com Phone: 562-355-3303
Installation Solutions	Millwork Installation	11660 Theatre Drive N Suite 250 Champlin, MN 55316 Contacts: Issac Bindert Email: ibindert@installationsolutionsinc.com Phone: 763.315.0492

CONTACT LIST

LANDLORD CONTACT

Bob Cope
Norwood Development Group
111 South Tejon Street - Suite 222
Colorado Springs, CO 80903
Cell: 719-330-7272
Office:719-593-2615
Email: bcope@norwood.dev

TENANT COORDINATOR'S NAME

Matt Dockter
Email: mdockter@norwood.dev

TENANT REPRESNETATIVE

Robert Richardson
Barnes & Noble Booksellers, Inc.
33 East 17th/ Street
New York, NY 10003
E: richardsonebn.com
P: 646.984.5561

LANDLORD NOTES:

1. TENANT MUST USE LANDLORD ROOFING CONTRACTOR FOR ALL ROOF WORK. WEATHERCRAFT : 719-382-8511

RESPONSIBILITY SCHEDULE

NOTES:

1. ALL ITEMS SUPPLIED BY BARNES & NOBLE SHALL BE UNLOADED, COUNTED, INVENTORIED, COORDINATED, STORED, AND SUPERVISED BY THE GENERAL CONTRACTOR. THE GENERAL CONTRACTOR WILL ASSUME FULL RESPONSIBILITY FOR ALL BARNES & NOBLE SUPPLIED MATERIALS / ITEMS.
2. ITEMS NOT LISTED ABOVE, BUT INCLUDED IN THE CONSTRUCTION DRAWINGS AND/OR SPECIFICATIONS DO NOT RELIEVE THE GENERAL CONTRACTOR FROM RESPONSIBILITY FOR THE SUPPLY AND INSTALLATION OF THOSE ITEMS.

Responsibility Schedule

BARNES & NOBLE COLORADO SPRINGS, CO

Date:04/25/25

SCOPE OF WORK	Existing	Furnish: Landlord	Furnish: Tenant	Furnish: Tenant Contractor	Install: Landlord	Install: Tenant	Install: Tenant Contractor	Not Applicable	REMARKS
DIVISION 01-GENERAL REQUIREMENTS									
PERMITS, FEES, INSURANCE CERTIFICATES, ETC.				X					SIGN PERMIT BY TENANTS SIGN CONTRACTOR
BARRICADE								X	
TEMP WINDOW GRAPHICS			X			X			
TEMP UTILITIES & EQUIPMENT								X	
RECEIVING & UNLOADING TENANT PROVIDED MATERIALS				X					
FINAL PROFESSIONAL CLEANUP & TRASH REMOVAL				X			X		DAILY SWEEP CLEANING DURING CONSTRUCTION 2 CLEANS PRIOR TO OPS TURNOVER. ONE BEFORE PUNCH AND WHITE GLOVE BEFORE OPS TURNOVER
FINAL JOB SITE PLANS				X			X		
CERTIFICATE OF OCCUPANCY				X					
DIVISION 02-EXISTING CONDITIONS									
DEMOLITION & HAUL OFF				X			X		
DIVISION 03-CONCRETE									
CONCRETE SLAB	X								
CONCRETE SLAB TRENCHING, PATCHING, LEVELING				X			X		
DIVISION 05-METALS									
STRUCTURAL STEEL	X								
STRUCTURAL & LIGHT-GAUGE METAL FRAMING		X		X			X		PERIMETER STUDS, INSULATION, VAPOR BARRIER FILBY LL
ALUMINUM STOREFRONT FRAMING	X								
DEMISING WALL STUDS	X	X							CMU EXTERIOR WALL ADJACENT TO ULTA BUILDING
DIVISION 06-WOODS, PLASTICS & COMPOSITES									
ROUGH FRAMING, BLOCKING AND FURRING				X			X		
FINISH CARPENTRY			X				X		MISC WOOD TRIM BOTH CARPENTRY BY GC, MILLWORK PACKAGE BY BN MILLWORKER
DIVISION 07-THERMAL & MOISTURE PROTECTION									
WATERPROOFMEMBRANE & SEALANTS				X			X		
BATT INSULATION	X								
SOUND BATTING				X			X		
ROOFING PENETRATIONS / FLASHINGS / CURBS				X			X		FOR CELL BOOSTER AND CAFE VENTS
FIRE PROOFING / STOPPING	X								
SEALANTS & CAULKING				X			X		
EXPANSION JOINTS								X	
DIVISION 08-OPENINGS									
DOORS, FRAMES & HARDWARE	X			X			X		
REAR DOOR AND HARDWARE	X			X			X		SEE DOOR SCHEDULE
ACCESS PANELS				X			X		
STOREFRONT DOOR AND DOOR HARDWARE	X			X			X		VERIFY CYLINDER HOUSING WILL ACCOMODATE 7 PIN CORE
VESTIBULE				X			X		
STOREFRONT GLAZING SYSTEM	X								
KEYS AND CORES			X				X		
DIVISION 09-FINISHES									
GYPSUM BOARD, DEMISING PARTITIONS	X			X			X		PATCH/ REPAIR AS REQUIRED
GYPSUM BOARD				X			X		
ACOUSTICAL CEILING SYSTEM				X			X		
GROUT				X			X		
PORCELAIN FLOORING			X				X		
WALK OFF MATT AND TRAY			X				X		
VINYL FLOORING				X			X		
CARPET AND GLUE			X				X		
WALL TILE			X				X		
VINYL BASE			X				X		
TRANSITION STRIPS & REDUCERS			X				X		
WALL BASE - WOOD			X				X		
PAINT/PAINTING & ACCESSORIES				X			X		1 PRIMER, 2 COATS FINISH MIN.
REVEALS								X	
DIVISION 10-SPECIALTIES									
STOREFRONT SIGN				X			X		GC MUST INITIATE AND COOR. WITH B&N VENDOR
SIGNAGE FOR STORE FIXTURES				X			X		
BLADE SIGN								X	
CAFE SIGNAGE - ILLUMINATED				X			X		GC TO PROVIDE POWER, BLOCKING, AND STRUT CHANNEL
BABY CHANGING SIGNAGE				X					
TOILET ROOM MIRRORS							X		
BLOCKING/POWER FOR SIGNAGE				X			X		
TOILET ROOM ACCESSORIES			X				X		GC TO COOR. WITH B&N VENDOR
INTERIOR GRAPHICS			X			X			
TOILET ROOM SIGNS			X				X		
SEISMIC BRACING				X			X		
ELEVATORS	X								
FIRE EXTINGUISHER(S) AND CABINETS				X			X		COORDINATE WITH A.H.J. ON LOCATIONS
DIVISION 12-FURNISHINGS									
DISPLAY FIXTURES AND TABLES			X			X			SEE FIXTURE/FURNITURE SCHEDULE
CASHWRAP			X			X			
RECEIVING TABLE WALL SHELVING			X				X		
RECEIVING TABLE			X						
JANITOR'S CLOSET SHELVING AND HARDWARE				X			X		
NON SALES SHELVING			X			X			
BREAKROOM APPLIANCES			X				X		REFRIDGERATOR FURNISHED BY BN
SAFES			X				X		
BULLETIN BOARDS			X			X			BN VENDOR INSTALLS
BREAKROOM LOCKERS			X				X		GC INSTALLS
BREAKROOM CABINETS			X				X		NICEWOOD FURNISHES. GC INSTALLS
OFFICE SHELVING AND COUNTERTOP			X				X		NICEWOOD FURNISHES. GC INSTALLS
OFFICE FURNISHINGS			X				X		
CAFRTS			X				X		
SALES PERIMETER WALL SYSTEM			X			X			BN MILLWORKER INSTALLS
DIVISION 21-FIRE SUPPRESSION SPRINKLER SYSTEMS									
SPRINKLER STUB		X							
SPRINKLER MAIN		X							
SPRINKLER GRID MODIFICATIONS				X			X		INCLUDING PERMITTING AND DRAWINGS
DIVISION 22 PLUMBING									
PLUMBING STUBS (SAN, DCW, VENT)	X			X			X		
WATER FILTRATION SYSTEM			X				X		
BREAKROOM FAUCET				X			X		
PLUMBING WORK & FIXTURES				X			X		SEE PLUMBING SHEETS
GAS SERVICE LINE								X	
WATER METER	X								
PAPER TOWEL DISPENSER			X				X		
SOAP DISPENSERS			X				X		
CAFE PAPER TOWEL DISPENSER			X				X		
MOP HOLDER			X				X		
DIVISION 23-HVAC									
MECHANICAL EQUIPMENT	X			X			X		SEE MECHANICAL DRAWINGS
HVAC PIPING/REFRIGERANT LINES							X		
OUTSIDE AIR ACCESS AND DUCTWORK				X			X		
RELIEF AIR ACCESS AND DUCTWORK							X		
DUCTWORK & AIR DEVICES	X			X			X		
FIRE DAMPERS							X		
TEMPERATURE CONTROL SYSTEM			X				X		LIGHTSTAT
DUCT SMOKE DETECTORS							X		
SMOKE EVAC & CONTROLS							X		
TOILET EXHAUST FAN, DUCTWORK, ROOF CAP				X			X		
AIR BALANCE REPORT				X					CERTIFIED REPORT BY INDEPENDENT AGENCY REQUIRED.
DIVISION 26-ELECTRICAL / DIVISION 27-COMMUNICATIONS / DIVISION 28-ELECTRONIC SAFETY & SECURITY									
MAIN SWITCHBREAKER IN LL'S ELECTRICAL ROOM	X								
TENANT ELECTRICAL METER	X								
MAIN SERVICE CONDUIT	X								
MAIN SERVICE WIRING	X								
TELEPHONE CONDUIT & PULL WIRE TO SPACE	X								
TELEPHONE CONDUIT & PULL WIRE FROM STUB TO BOARD				X			X		
TELEPHONE WIRE FROM LL ROOM TO TENANT BOARD								X	
TENANT ELECTRICAL PANELS (SWITCHGEAR)	X			X			X		SEE ELECTRICAL DRAWINGS
TENANT ELECTRICAL TRANSFORMERS								X	
HVAC POWER CONNECTION AND DISCONNECT		X		X	X		X		LL IS RESPONSIBLE FOR PROVIDING UNITS IS GOOD WORKING CONDITION
CONDUIT, WIRING & DEVICES				X			X		
LIGHTING FIXTURES /LAMPS			X				X		DELIVERY BY BN MDS - ANY CHANGES TO GO THROUGH BN
LIGHTING AND SIGNAGE CONTROLS			X				X		
TELEPHONE SYSTEM			X			X			
PHONE AND DATA CONDUIT			X			X			
PHONE AND DATA CABLING			X			X			
SOUND SYSTEM WIRING			X			X			
SOUND SYSTEM AND SPEAKERS			X			X			MOOD MEDIA
FIRE ALARM	X			X			X		INCLUDING PERMITTING AND DRAWINGS
SECURITY SYSTEM			X			X			GC TO PROVIDE CONDUIT PULL STRINGS AND J-BOXES
COMPUTERS			X			X			
LOW VOLTAGE PERMITS				X					
LOW VOLTAGE EMPLOYEE BUZZER			X			X			
LOW VOLTAGE DELIVERY BUZZER				X			X		
CCTV			X			X			
POINT OF SALE (POS)			X			X			

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

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Professional Seal:

LIC.# : 305353
EXP. DATE : 10/31/25

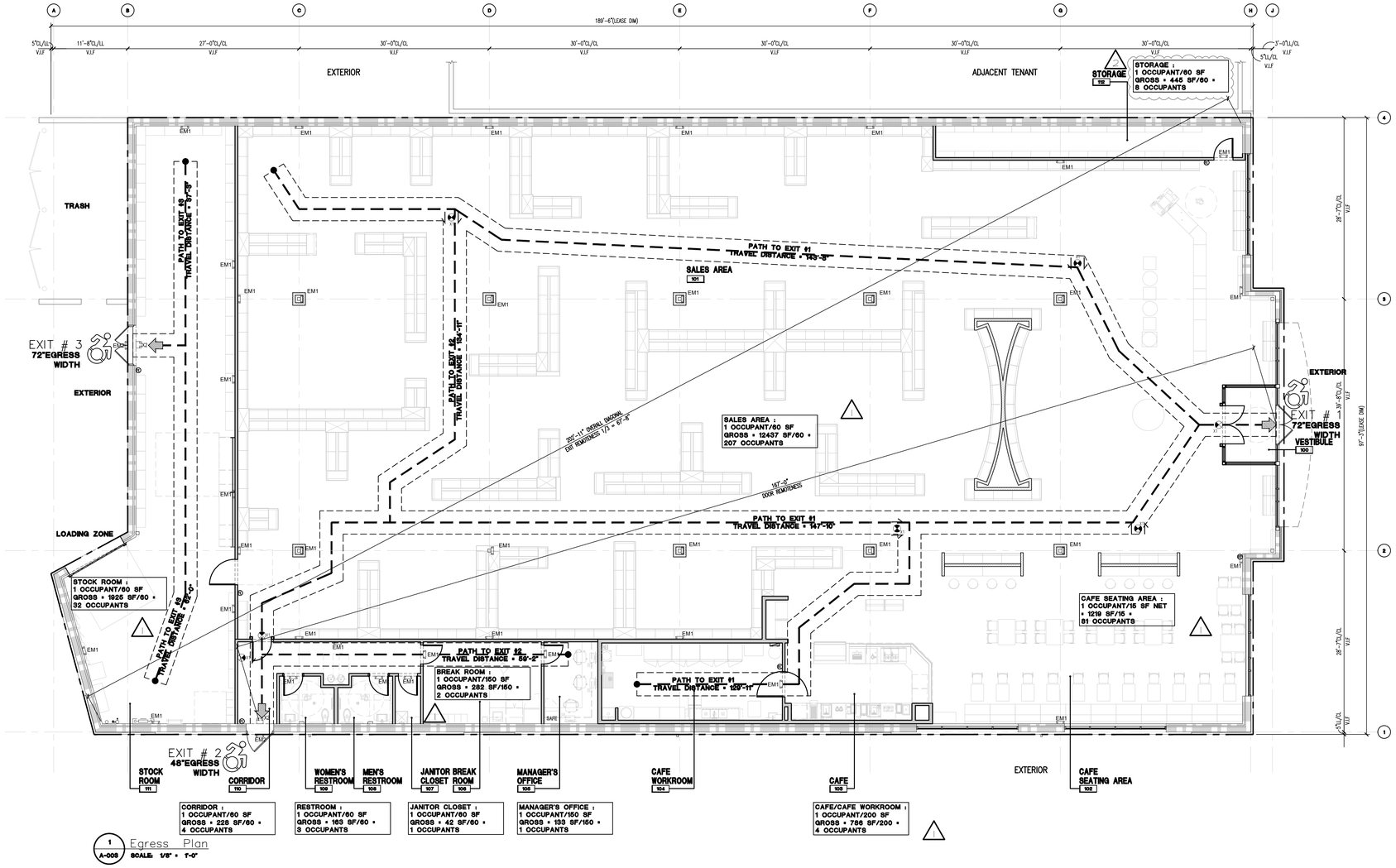
Drawing Description:

Vendor List, Responsibility Schedule and Contacts

Drawing Number:

A-002

NOTE:
SEE BUILDING CODE ANALYSIS, PROPOSED OCCUPANCY LOAD AND EGRESS ANALYSIS ON A-001



1 A-003 Egress Plan
SCALE: 1/8" = 1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

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Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
LICENSED ARCHITECT

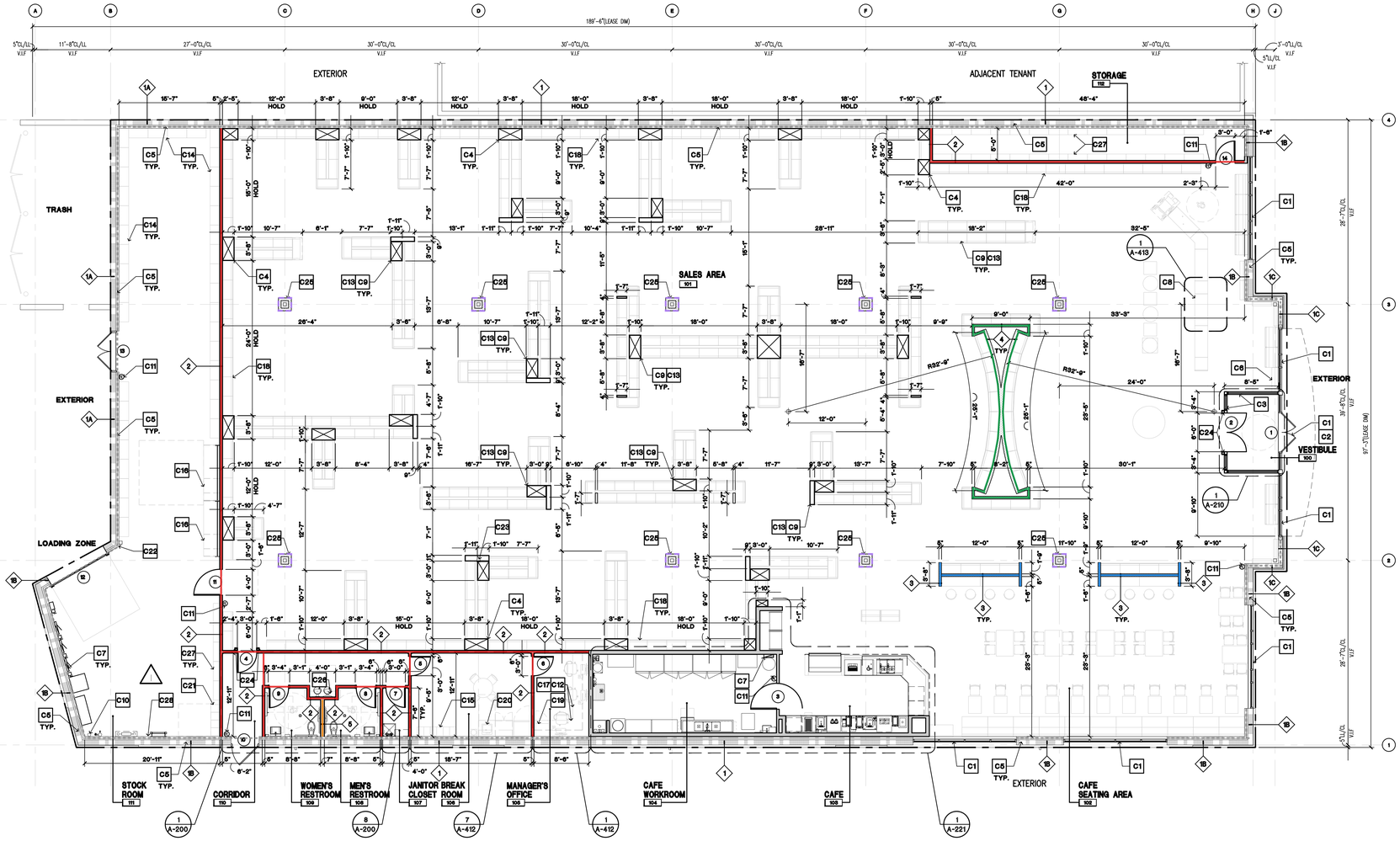
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

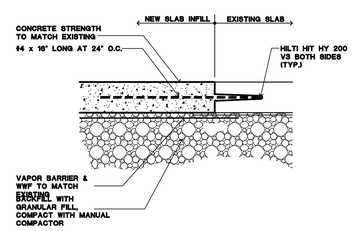
Egress Plan

Drawing Number:

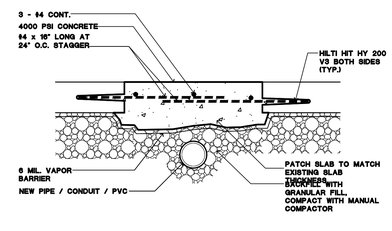
A-003



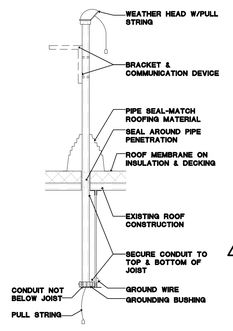
1 A-101 Construction Plan
SCALE: 1/8" = 1'-0"



2 A-101 Concrete Slab Infill/ Patch Detail
SCALE: 1 1-2" = 1'-0"



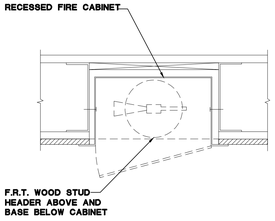
NOTES:
1. GENERAL CONTRACTOR SHALL VERIFY THAT CONCRETE SLAB DOES NOT HAVE REINFORCING BARS AND IT IS A SLAB ON GRADE SYSTEM PRIOR TO CUTTING OR CHOPPING.
2. DUE TO POTENTIAL UNEVEN SPOTS WITHIN THE COMPACTED SUBGRADE, FIBER MESH IS NOT



ROOF PENETRATION SPECS:
1. 1 1/2" CONDUIT (WITH PULL STRING & WEATHER HEAD BY ELECT. CONTRACTOR.
2. INSTALL WEATHER HEAD 60" ABOVE ROOF DECK.
3. SECURE CONDUIT TO STRUCTURE. CONDUIT SHALL NOT PROTRUDE BELOW JOIST.
4. WEATHER PROOFING BY ROOFING CONTRACTOR.
6. GROUND CONDUIT TO BUILDING STEEL (GROUNDING BUSHING) (COPPER WIRE 10 AWG. BY ELECT. CONTRACTOR.)

GUIDELINES:
1. WHEN THBB IS ON GROUND LEVEL, ROOF PENETRATION SHALL BE MAX. OF 10' LEFT OR RIGHT OR IN FRONT OF THE CABINET.
2. WHEN THBB IS ON MEZZANINE, ROOF PENETRATION SHALL BE A MAX. OF 10' LEFT OR RIGHT IN FRONT OF THE CABINET.
3. LOCATE ANTENNA A MIN. OF 48" FROM ROOF DEVICES & EQUIPMENT.
4. INSTALL PRIOR TP SCHEDULED MOVE-IN DATE.

3 A-101 Roof Penetration Detail
SCALE: 3/4" = 1'-0"



4 A-101 Fire Extinguisher Detail
SCALE: 3/8" = 1'-0"

GENERAL NOTES

1. G.C. TO UTILIZE LANDLORD'S ROOFING CONTRACTOR WHEN APPLICABLE FOR ALL REPAIRS AND MAINTAIN CURRENT ROOFING WARRANTY.
2. G.C. TO CHOP, TRENCH, OR CORE AS REQUIRED, FILL AND PATCH CONCRETE SLAB TO ACCOMMODATE ALL UNDERGROUND UTILITIES, SUCH AS SANITARY LINES, STORM LINES, DOMESTIC WATER LINES, ELECTRICAL CONDUIT AND SPRINKLER MAINS. G.C. TO VERIFY IN FIELD IF SLAB IS STRUCTURAL. IF SO DO NOT COMMENCE ANY CUTTING UNTIL APPROVALS ARE OBTAINED BY ARCHITECT AND LANDLORD.
3. ALL ITEMS TO BE SUPPLIED AND INSTALLED BY GENERAL CONTRACTOR UNLESS SPECIFICALLY LISTED "BY OTHERS", "BY LANDLORD", OR "BY SHELL CONTRACTOR".
4. THE TERMS G.C., S&N G.C., GENERAL CONTRACTOR, ETC. SHALL REFER TO THE BARNES & NOBLE GENERAL CONTRACTOR.
5. THE TERMS "LANDLORD, SHELL CONTRACTOR, AND L.L. SHALL REFER TO THE LANDLORD AND / OR HIS SHELL CONTRACTOR, UNLESS BARNES & NOBLE IS RESPONSIBLE FOR SHELL CONSTRUCTION IN CONJUNCTION WITH INTERIOR FIT UP.
6. GENERAL CONTRACTOR WILL BE RESPONSIBLE FOR ALL COSTS ASSOCIATED WITH OBTAINING PERMITS, FILING FEES, IMPACT FEES, TIE-IN FEES, ETC. AS REQUIRED BY ALL APPROVING AGENCIES AND LANDLORD.
7. ALL MECHANICAL, PLUMBING, AND ELECTRICAL EQUIPMENTS BE PROPERLY SUPPORTED FROM STEEL STRUCTURE, NOT ROOF DECK.
8. AS PER LANDLORD SPECIFICATIONS ALL WALLS IN WET AREAS AND/ OR BEHIND PLUMBING FIXTURES SHALL BE CONSTRUCTED OF MOISTURE RESISTANT DRYWALL OR CEMENT BOAR WITH THE BOTTOM OF THE WALL CAULKED AT FLOOR LEVEL TO PREVENT MOISTURE INFILTRATION INTO BASE BUILDING ELEMENTS. WATERPROOFING TO BE PROVIDED BELLOW NEW TILE IN WET AREAS

PROVIDE 3-1/2" SOUND ATTENUATION BLANKETS WITHIN THE ENCLOSING WALLS AND CEILINGS OF THE FOLLOWING ROOMS.

ROOM	CEILING	WALLS
103 - CAFE		●
104 - CAFE WORKROOM		●
105 - MANAGER'S OFFICE		●
106 - BREAK ROOM		●
107 - JANITOR CLOSET		●
108 - MEN'S RESTROOM		●
109 - WOMEN'S RESTROOM		●
110 - CORRIDOR		●
111 - STOCK ROOM		●
112 - STORAGE		●

CONSTRUCTION LEGEND

DESIGNATION	DESCRIPTION
	EXISTING EXTERIOR WALL
	EXISTING PARTITION
	NEW NON-BEARING PARTITION
	PARTIAL HEIGHT GYP. BD. PARTITION

CONSTRUCTION PLAN KEY NOTES

- C1 STOREFRONT SYSTEM. ENTRANCE DOORS TO BE REMAIN. PROTECT DURING ALL PHASES OF CONSTRUCTION.
- C2 STOREFRONT DOORS SEE DOOR SCHEDULE FOR MORE INFORMATION.
- C3 NEW VESTIBULE
- C4 3/4" THICK FILLER PANEL, HEIGHT TO MATCH SHELVING UNITS
- C5 DEMISING PARTITION TO BE PATCHED AND REPAIRED TO MAINTAIN EXISTING FIRE RATING. PREPARE WALLS FOR NEW FINISH. ADD GWB/FURRING AS REQUIRED.
- C6 ILLUMINATED CAFE PEDESTAL SIGN
- C7 ELECTRICAL PANEL BOARDS. SEE ELECTRICAL PLANS FOR FURTHER INFORMATION.
- C8 CASHWRAP FIXTURE. POWER AND DATA CONDUIT STUBBED UP INTO FIXTURE BASE. DO FINAL CONNECTION TO DEVICES - SEE POWER PLAN FOR FURTHER INFORMATION
- C9 DISPLAY SHELVING UNITS. SEE FIXTURE PLAN FOR FURTHER INFORMATION
- C10 "AS BUILT" PLAN TUBE HOLDER IN STOCK ROOM. SEE DETAIL #5/A-413
- C11 FIRE EXTINGUISHER AND CABINET. LOCATIONS AND QUANTITIES SHOWN SHALL BE APPROVED BY FIRE MARSHALL. MIN. 1 PER 3000 SF). CABINETS TO BE MOUNTED AT A MAXIMUM HEIGHT OF 48" TO TOP. CABINETS BY J.L. INDUSTRIES, (800) 654-8077, AT SALES AREA, PROVIDE COSMOPOLITAN SERIES, FULLY RECESSED FULL GLASS DOOR, STAINLESS STEEL FINISH FOR DOOR AND TRIM. TUB TO MATCH ADJACENT FINISH, AT WORK ROOM, PROVIDE AMBASSADOR SERIES 1017F1D IN STEEL WITH WHITE EPOXY FINISH. FIRE EXTINGUISHER (MIN. 2A10BC). PROVIDE FIRE RATED CABINETS FOR LOCATIONS IN RATED WALLS (TYP).
- C12 WALL SHELVING AND COUNTER TOP AT OFFICE. SEE TYPICAL DETAIL #12/A-412. PROVIDE F.R.T BLOCKING AS NEEDED
- C13 ELECTRICAL CONDUIT STUBBED UP THROUGH SLAB FROM NEAREST WALL. SEE POWER PLANS FOR FURTHER INFORMATION.
- C14 1'-6" DEEP X 8'-0" WIDE X 8'-0" HIGH STEEL SHELVING UNITS FASTENED TO PERIMETER WALLS, SEE DETAIL#7/A-413
- C15 LOCKERS
- C16 RECEIVING TABLE, BLOCKING AND WALL MOUNTED ADJUSTABLE SHELVING. PROVIDE POWER AND (2) GROMMETS. SEE TYPICAL DETAIL #8/A-413.
- C17 VOLUME CONTROL IN MANAGER'S OFFICE & B.O.H. MOUNTED AT 48" A.F.F. RESTROOM CORRIDOR V.C. TO BE LOCATED IN THE BREAK ROOM. COORDINATE REQUIREMENTS WITH THE POWER PLAN
- C18 SALES FLOOR PERIMETER FIXTURES. COORDINATE WITH DETAIL MILLWORK DRAWINGS.
- C19 SAFE
- C20 KITCHENETTE & REFRIGERATOR, SEE DETAIL #9/A-412. PROVIDE F.R.T. BLOCKING AS NEEDED
- C21 MUSIC SYSTEM RACK AND POS SERVER RACK/ CCTV SYSTEM. COORDINATE REQUIREMENTS WITH POWER PLAN.
- C22 STAINLESS STEEL CORNER GUARDS AT 48" A.F.F. FOR ALL OUTSIDE GYPSUM BOARD CORNERS AT STOCKROOM WALLS.
- C23 EXISTING COLUMN TO REMAIN EXPOSED. COORDINATE WITH THE FINISH PLAN. VERIFY EXACT LOCATION ON SITE. SEE POWER PLAN FOR POWER TO MILLWORK.
- C24 NEW EAS SYSTEM
- C25 EX. COLUMN. NEW 24"x24" GWB SURROUND OVER 3 5/8" METAL FRAMING FROM SLAB TO 8'-0" AFF PAINT COLUMN PER FINISH PLAN
- C26 HIGH-LOW DRINKING FOUNTAIN.
- C27 1' DEEP X 8'-0" WIDE X 8'-0" HIGH STEEL SHELVING UNITS FASTENED TO PERIMETER WALLS, SEE DETAIL#7/A-413
- C28 G.C. TO COORDINATE ALL WATER METERING, LADDER, HATCH, PLUMBING FILTRATION WITH TENANT AND LANDLORD PRIOR TO INSTALLATION.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BD, LANDLORD & CLIENT REVIEW	3
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	1

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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

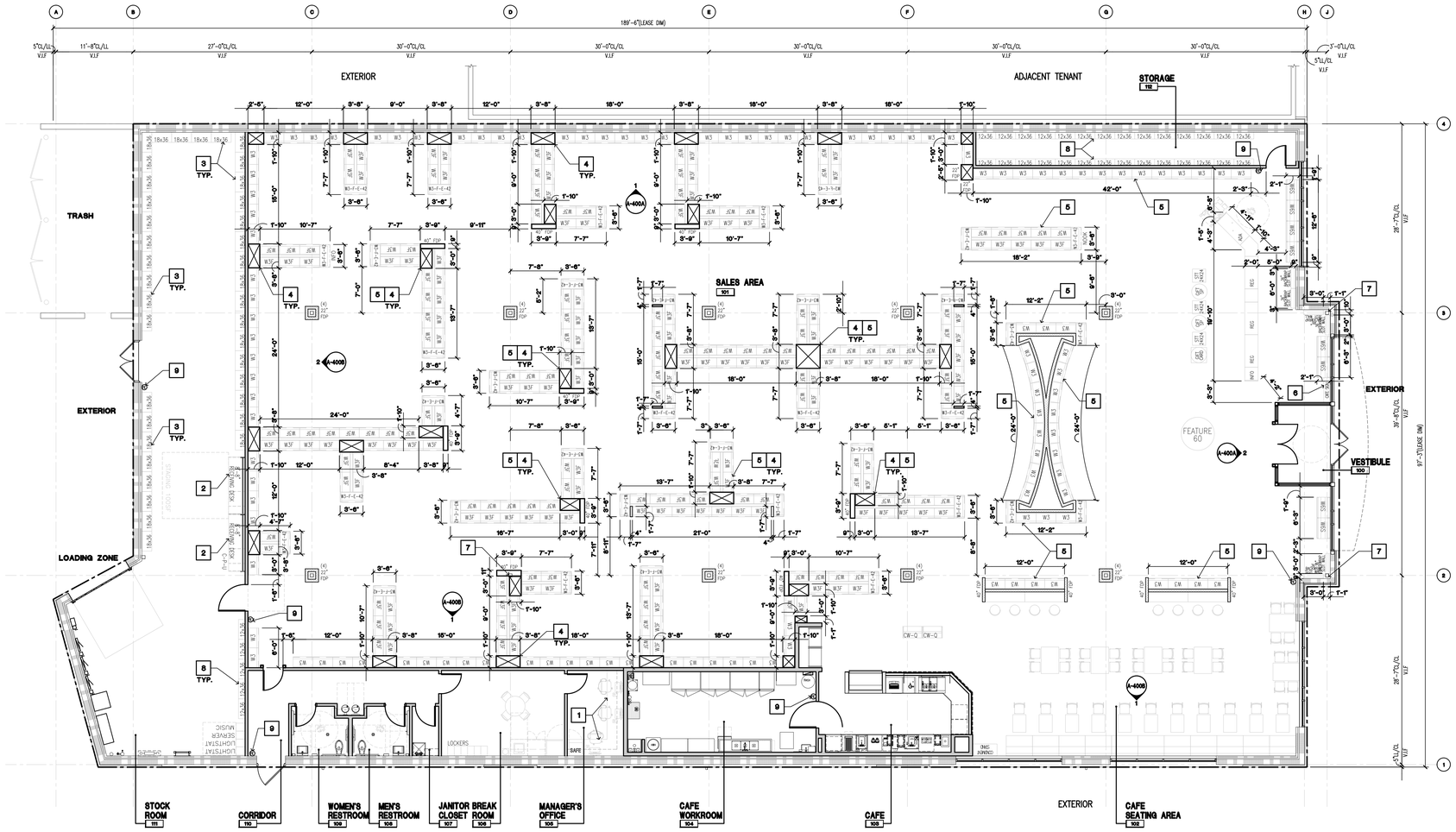
Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
CORPORATION
LICENSED ARCHITECT

LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:
Construction Plan, Legend, and Notes

Drawing Number:
A-101



1 A-111 Fixture Plan
SCALE: 1/8" = 1'-0"

NOTE:
THE MINIMUM CLEAR AISLE WIDTH SHALL BE 44" WHEN MERCHANDISE OR OTHER SIMILAR OBSTRUCTIONS ARE PLACED ON BOTH SIDES OF THE AISLE AND A MINIMUM OF 36" WHEN PLACED ON ONE SIDE OF THE AISLE.

RETAIL FIXTURE SCHEDULE

DESCRIPTION	
W2	24" x 95" HIGH PERIMETER WALL UNIT
W3	36" x 95" HIGH PERIMETER WALL UNIT
W3-F	36" x 95" HIGH FLOOR UNIT
W3-F-E	36" x 95" HIGH FLOOR END UNIT
W3-F-E-42	36" x 95" HIGH FLOOR END UNIT
W6	36" x 60" HIGH FLOOR UNIT
W73	36" x 73" HIGH WINDOW BAY UNIT
W-77	36" x 77" HIGH WINDOW BAY UNIT
FF-9	40" x 77" HIGH WINDOW DOUBLE SLATWALL UNIT
48" BW-SW	48" BACK WRAP SLOT WALL
36" BW-SW	36" BACK WRAP SLOT WALL
T8	31" x 31" TABLE
T10	42" x 42" TABLE
T12	47" ROUND TABLE
FEATURE	60" ROUND FEATURE TABLE
FAB DIS DOUBLE	FABRIC DISPLAY - DOUBLE SIDED
FAB DIS SINGLE	FABRIC DISPLAY - SINGLE SIDED
DONKEY SM	DONKEY FIXTURE - SMALL
DONKEY LG	DONKEY FIXTURE - LARGE
STS 24	SINGLE TITLE SQUARE TABLE
STR 24	SINGLE TITLE ROUND TABLE
PLUSH	PLUSH FLOOR FIXTURE
SCHL EICH	VENDOR SUPPLIED
POPS	POP! FIXTURE
PICT BKS	PICTURE BOOKS
CGW GIFT	CGW GIFT FIXTURE
GIFT SP	GIFT SPINNER
CGW SHC	CGW, SPECIALTY, HOBBIES AND COLLECTIBLES
CGW DVD	CGW DVD FIXTURE
CGW CDS	CGW CD FIXTURE
CGW JUV	CGW KIDS FIXTURE
VINYL	VINYL FIXTURE
TRAIN	TRAIN FIXTURE
LEGO	KIDS ACTIVITY TABLE
QUE	QUE LINE FIXTURES

FIXTURE PLAN KEY NOTES

KEY	DESCRIPTION
1	TABLE WITH SHELVING REFER TO TYPICAL DETAIL 12/A-412. PROVIDE F.R.T. PLYWOOD BLOCKING AS NEEDED
2	RECEIVING TABLE BLOCKING AND WALL MOUNTED ADJUSTABLE SHELVING POWER AND GROMMETS (2). (SEE DETAIL 16/A-413)
3	1'-6" DEEP X 8'-0" WIDE X 8'-0" HIGH STEEL SHELVING UNITS FASTENED TO PERIMETER WALLS. SEE DETAIL17/A-413
4	98"HT FILLER PANEL, TYP.
5	POWER AND DATA CONDUITS STUBBED UP FROM FLOOR OR ADJACENT WALL. COORDINATE WITH ELECTRICAL POWER PLAN
6	ILLUMINATED PEDESTAL SIGN, SEE SHOP DRAWINGS.
7	POWER FOR MILLWORK LIGHTING TO BE RUN DOWN BACK OF COLUMNS
8	1' DEEP X 8'-0" WIDE X 8'-0" HIGH STEEL SHELVING UNITS FASTENED TO PERIMETER WALLS. SEE DETAIL17/A-413
9	FIRE EXTINGUISHER AND CABINET. LOCATIONS AND QUANTITIES SHOWN SHALL BE APPROVED BY FIRE MARSHALL. (MIN. 1 PER 3000 SF). CABINETS TO BE MOUNTED AT A MAXIMUM HEIGHT OF 48" TO TOP. CABINETS BY J.L. INDUSTRIES, (800) 554-6077. AT SALES AREA, PROVIDE COSMOPOLITAN SERIES, FULLY RECESSED FULL GLASS DOOR, STAINLESS STEEL FINISH FOR DOOR AND TRIM. TUB TO MATCH ADJACENT FINISH. AT WORK ROOM, PROVIDE AMBASSADOR SERIES 1017F10 IN STEEL WITH WHITE EPOXY PAINT FINISH, FIRE EXTINGUISHER (MIN. 5A10BC). PROVIDE FIRE RATED CABINETS FOR LOCATIONS IN RATED PARTITIONS.

FIXTURE PLAN GENERAL NOTES

NOTE:
ALL FREE-STANDING FIXTURES TO HAVE 36" MINIMUM WIDTH FOR ALL ACCESS AISLES AS WELL AS HANDICAP ACCESSIBILITY.

NOTE:
CONTRACTOR TO VERIFY ALL QUANTITIES AND MUST NOTIFY B&N OF ANY DISCREPANCIES.

BARNES&NOBLE

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BUILDING VALUE SINCE 1994

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Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
LICENSED ARCHITECT

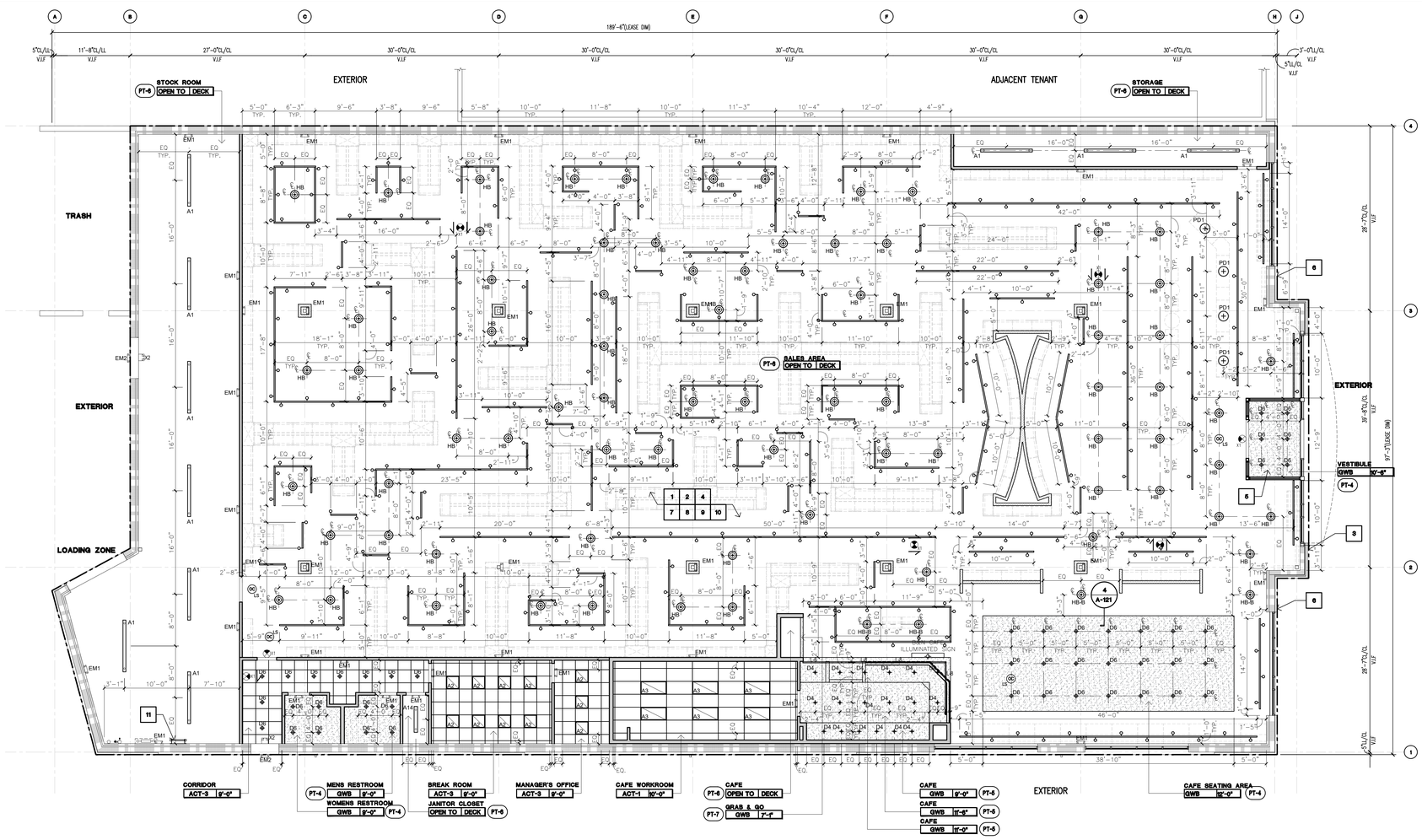
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

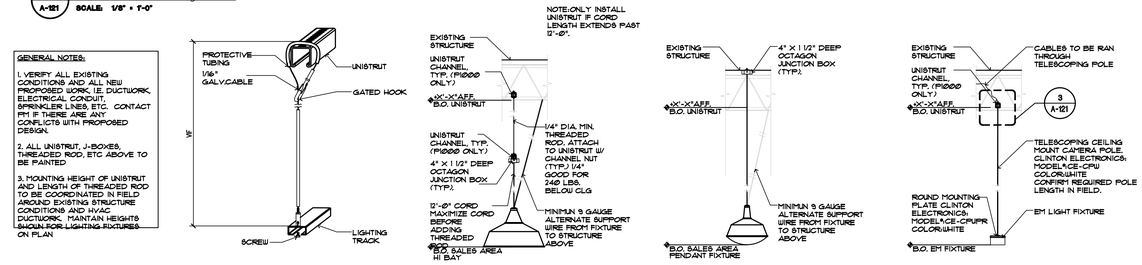
Fixture Plan

Drawing Number:

A-111



1 A-121 Reflected Ceiling Plan
SCALE: 1/8" = 1'-0"

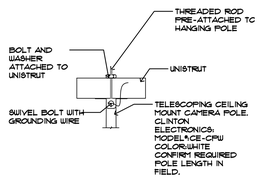


A DETAIL AT TRACK LIGHT FIXTURE

B DETAIL AT SALES AREA HI-BAY FIXTURE

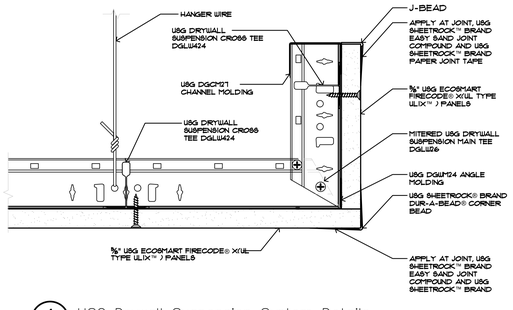
C DETAIL AT CASH WRAP PENDANT LIGHT FIXTURE

D DETAIL AT EM LIGHT FIXTURE AND LIGHTSTAT SALES DROPS



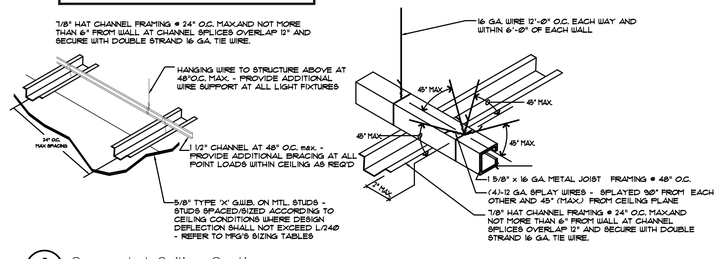
3 A-121 Mounting Detail At Pole
SCALE: 8" = 1'-0"

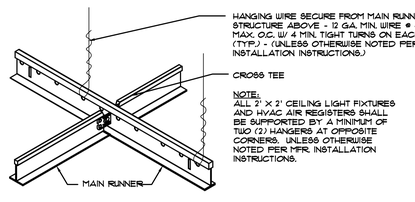
2 A-121 Hanging Light Detail
SCALE: 1/2" = 1'-0"



TYPICAL CEILING CONSTRUCTION OPTIONS:
THE CONTRACTOR HAS THE OPTION TO EMPLOY OTHER USG STANDARD FRAMING METHODS NOT SHOWN IN THE BELOW DETAIL WHEN CONSTRUCTING CEILING ELEMENTS. CONTRACTOR SHALL BE RESPONSIBLE FOR THE CEILING SYSTEM'S PERFORMANCE (L/240 MAX.) AND MUST RECEIVE THE ARCHITECT'S OR TENANT'S REVIEW OF ANY ALTERNATE FRAMING SYSTEMS PRIOR TO BID SUBMISSION.

NOTE: ALL 6PLAY WIRES SHALL BE TAUT & TIED @ BOTH ENDS W/ MIN. OF 3 TURNS IN 1" OF RUN
SECURE SUPPORT WIRES TO STRUCTURAL STEEL MEMBERS ABOVE - NOT TO ROOF DECK





4 A-121 USG Drywall Suspension System Details
SCALE: 6" = 1'-0"

5 A-121 Suspended Ceiling Section
SCALE: N.T.S

CEILING NOTES

1. USE ONLY 12 GAUGE STEEL WIRE FOR ALL SUSPENDED CEILING SYSTEMS AT 48" O.C. WRAP 3 FULL TIMES. SEE A-420 DRAWINGS REGARDING CEILING BRACING REQUIREMENTS.
2. REFER TO MECHANICAL DRAWINGS FOR LOCATION OF ALL FIRE DAMPERS AND DIFFUSERS.
3. G.C. TO PROVIDE FIRESTOPPING/DRAFTSTOPPING ABOVE CEILING AS REQUIRED BY CODE.
4. LOCATE RECESSED DOWN LIGHTS CENTERED IN TILES U.O.N.
5. G.C. TO PAINT ALL SUPPLY / RETURN GRILLES, AND LIGHTING TRIM SET IN DRYWALL CEILINGS AND SOFFITS SAME COLOR TO MATCH ADJACENT DRYWALL SURFACE. SEMI-GLOSS ENAMEL.
6. G.C. TO PAINT ALL SUPPLY / RETURN GRILLES, AND LIGHTING TRIM SET IN A.C.T. CEILINGS OTHER THAN WHITE TO MATCH COLOR OF A.C.T.
7. FOR EMERGENCY LIGHT FIXTURE LOCATIONS REFER TO ELECTRICAL ENGINEER'S DRAWINGS
8. ALL CEILING HEIGHTS SHALL BE REFERENCED FROM FLOOR SLAB ELEVATION 0'-0"
9. ALL LIGHTING FIXTURES SPECIFIED WITHIN THE CAFE SERVICE AREA AS WELL AS THE CAFE WORK ROOM SHALL BE LENSED AS REQUIRED BY THE LOCAL HEALTH DEPARTMENT.
10. GENERAL CONTRACTOR TO OBTAIN LIGHTING FIXTURE CUT SHEETS FROM LIGHTING SUPPLIER AND COORDINATE ALL MOUNTING DETAILS AND INSTALLATION PROCEDURES WITH MANUFACTURER.
11. ALL CEILING GRID SUPPORTS ATTACHED TO BAR JOISTS SHALL BE ATTACHED TO THE TOP CHORD OR THE BOTTOM CHORD AT A PANEL POINT ONLY WITH THE APPROVAL FROM THE BUILDING STRUCTURAL ENGINEER.

CEILING KEY NOTES

1. NEW SUSPENDED HIGH BAY FIXTURES THROUGHOUT THE SALES FLOOR.
2. NEW TRACK LIGHTING THROUGHOUT THE SALES FLOOR
3. EXTERIOR SIGNAGE MOUNTED TO FASCIA. REFER TO VENDOR SHOP DRAWINGS AND ELECTRICAL PLANS FOR FURTHER INFORMATION. PROVIDE POWER AS REQ'D
4. MECHANICAL DUCTS - REFER TO MECHANICAL PLANS FOR NEW SCOPE OF WORK.
5. NEW GWB CEILING AT ENTRY VESTIBULE
6. VERIFY ALL EXTERIOR LIGHTING IS IN GOOD WORKING ORDER, REPLACE IF NEEDED
7. UNDERSIDE OF DECK, STRUCTURE AND DUCTWORK TO BE PAINTED
8. ALL NEW EMERGENCY LIGHT FIXTURES THROUGHOUT WALL MOUNTED OR SUSPENDED FROM CEILING
9. ADJUST ANY CEILING DEVICES DUCTWORK, ETC. IF THERE IS CONFLICT WITH NEW LIGHT FIXTURE LOCATIONS.
10. MODIFY SPRINKLER HEAD LOCATION AS REQUIRED
11. ROOF HATCH ACCESS LADDER

LIGHT FIXTURE SCHEDULE (LIGHT CONCEPT)

SYMBOL	TAG	LOAD (VAE)	DESCRIPTION	AREA
	HB	15	HIGH BAY HALOGEN LIGHTINGS NUVO SF76-283, L16A19N1530K FINISH : WHITE	SUSPENDED IN SALES FLOOR AT 8'-0" A.F.F.
	HB-B	15	HIGH BAY HALOGEN LIGHTINGS NUVO SF76-283, L16A19N1530K FINISH : BLACK	SUSPENDED ABOVE CAFE SEATING AREA AT 8'-0" A.F.F.
	LW-01	22	NEW TRACK HEADS MANUFACTURER: ATTAIN LED TRACK HEAD HP MODEL: AT-26-36-95-24-WH-J	TRACK MOUNTED IN SALES AREA
	LWT-01 TRACK	-	NEW PENDANT MOUNTED MANUFACTURER: CONTECH SINGLE CIRCUIT LINE VOLTAGE TRACK VARIOUS LENGTHS- WHITE UNISTRUT AND HANGING HARDWARE	SUSPENDED IN SALES FLOOR AT 12'-0" A.F.F.
	A1	68	BOH LIGHTING MANUFACTURER: LITHONIA LIGHTING MODEL: TZLIN LED L96 10000LM FST MVOLT 50K 80CRI PLR E7W WH	SUSPENDED AT 12'-0" A.F.F. IN STOCK ROOM
A2	A2	-	OFFICE AND BREAK ROOM LIGHTING MANUFACTURER: DT BY TCP MODEL:DTF2UZD5835K	LAY-IN CEILING
A3	A3	46	ATTAIN 2 FT. X 4 FT. INTEGRATED LED 4000 LUMENS 4000K 120V COMMERCIAL GRADE RECESSED TROFFER W/ PRISMATIC LENSES	LAY-IN CEILING
	A14	34	MANUFACTURER: LITHONIA LIGHTING MODEL#: 2LIN LED 8TRIPLIGHT L48 5000LM F8T MVOLT 30K 80CRI PLR E7W WH	SURFACE MOUNTED TO CEILING OR SUSPENDED AT 12FT A.F.F.
X1	X1	3.8	THERMOPLASTIC LED EXIT SIGN MANUFACTURER: BEST LIGHTING MODEL: EZXTEU-2RW-EM	SUSPENDED OR WALL MOUNTED AT 8'-0" A.F.F., MIN. B.O. SIGN
X2	X2	5	LED EXIT SIGN AND EMERGENCY LIGHT, THERMOPLASTIC COMBO MANUFACTURER: BEST LIGHTING MODEL: LEDXTEU-2-R-W-NC	SUSPENDED OR WALL MOUNTED AT 8'-0" A.F.F., MIN. B.O. FIXTURE
EM1	EM1	5	EMERGENCY LIGHT MANUFACTURER: ASTRALITE MODEL: EU-8-LED	SUSPENDED OR WALL MOUNTED AT 10'-0" A.F.F. MIN. B.O. FIXTURE (SALES AREA) OR AS HIGH AS POSSIBLE (NON-SALES)
EM2	EM2	2	EXTERIOR EMERGENCY LIGHT MANUFACTURER: BEST LIGHTING MODEL: LED HIGH LUMEN THERMOPLASTIC HPEM	WALL MOUNTED AT 8'-0" A.F.F. MIN. B.O. FIXTURE
	D4	12	4" DIAMETER LED DOWNLIGHT MANUFACTURER: ATTAIN LED MODEL: AT-DL4-12-35-WH	RECESSED IN CAFE CEILING
	D6	32	6" DIAMETER LED DOWNLIGHT MANUFACTURER: MAXI LUME MODEL: MS-LED-2000L-DIM10-120-MD-35K-90/ L607 SHZ WH	RECESSED IN RESTROOM CEILING
	PD1	60	22" DIAMETER LED PENDANT MANUFACTURER: WAYFAIR MODEL: METAIRIE 1 LIGHT SINGLE DOME PENDANT	SUSPENDED ABOVE CASHWRAP AT 7'-0" AFF FROM J-BOX AT DECK
	L-8	80	LED TUBE PENDANT ADJUSTABLE MANUFACTURER: CORONET LED MODEL: 8ft. CRDS-8-30-HIGH-UNV-DB-BLK -AC-SD	SUSPENDED ABOVE COUNTER AT 9'-0" AFF
	L-6	60	8ft. CRDS-6-30-HIGH-UNV-DB-BLK -AC-SD	
	L-4	40	4ft. CRDS-4-30-HIGH-UNV-DB-BLK -AC-SD	
	L-2	20	2ft. CRDS-2-30-HIGH-UNV-DB-BLK -AC-SD	
OCLS	OCCUPANCY SENSOR (LIGHTSTAT)			DECK MOUNTED J-BOX WITH 3/4" CONDUIT DROP 10'-8" AFF

BARNES&NOBLE

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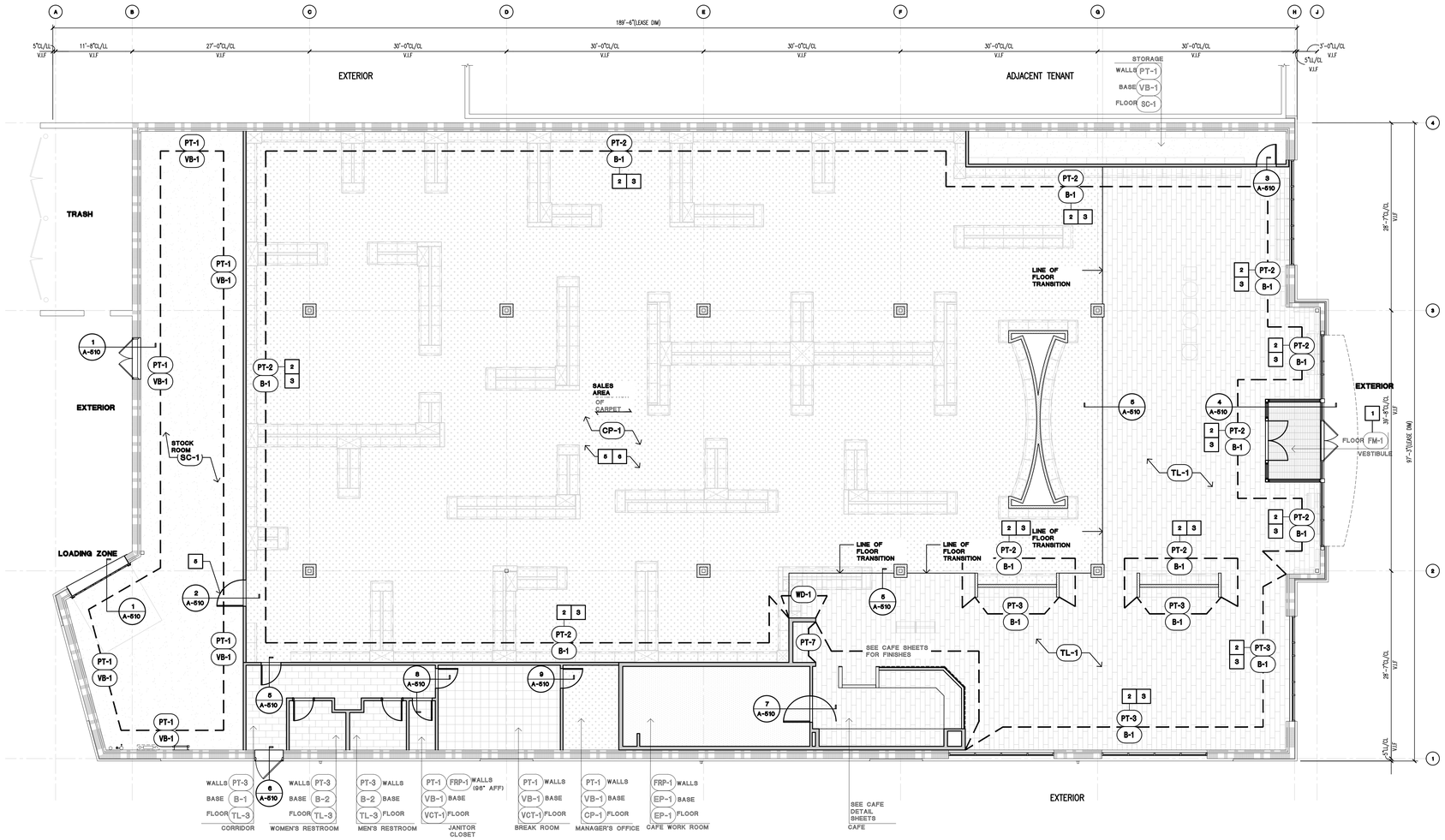
Professional Seal:
STATE OF COLORADO WILLIAM J COMMER LICENSED ARCHITECT

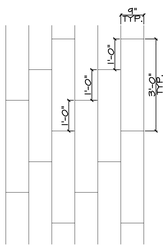
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:
Reflected Ceiling Plan
Legend, Notes and
Ceiling Details

Drawing Number:
A-121



1 A-131 Finish Plan
SCALE 1/8" = 1'-0"



2 A-131 Tile Detail
SCALE 1/2" = 1'-0"

FLOOR FINISH LEGEND

REFER TO SHEET A-510 FOR ALL FINISHES, SPECIFICATIONS, AND SCHEDULES (WALLS, FLOORS, CEILINGS, ETC)

SYMBOL	DESCRIPTION
TL-#	CERAMIC FLOOR TILE (SEE FINISH SCHEDULEBEE NOTE #10)
VCT-#	VINYL COMPOSITE TILE (SEE FINISH SCHEDULE)
SC-#	SEALED CONCRETE FLR SLAB (SEE FINISH SCHEDULE)
EP-#	POURED EPOXY FLOORING (SEE FINISH SCHEDULE)
VB-#	VINYL BASE
FM-#	VESTIBULE FLOOR MAT (SEE FINISH SCHEDULE)
CP-#	FIELD CARPET (SEE FINISH SCHEDULE)
B-#	WOOD BASE

FINISH PLAN KEY NOTES

1. NEW ENTRY MAT AT SALES
2. PAINT WALLS AND COLUMNS PT-2 FROM 7'-6" TO UNDER SIDE OF DECK WHERE THERE ARE DISPLAY WALL UNITS.
3. WOOD BASE IN SALES FLOOR AT WALLS WITHOUT FIXTURES.
4. NOT USED.
5. PAINT ALL DOORS SEMI GLOSS-COLOR TO MATCH THE ADJACENT WALL.
6. PREP AND PAINT MILLWORK FILLER PANELS TYP.

FINISH NOTES

1. ALL INTERIOR FINISHES, SPECIFIED WALL COVERINGS AND CEILING MATERIALS, SHALL HAVE A FLAME SPREAD RATING NOT TO EXCEED 75 AND SMOKE-DEVELOPED INDEX NOT TO EXCEED 450.

2. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR PROPER PREPARATION OF ALL NEW AND EXISTING SURFACES IN A SATISFACTORY MANNER TO RECEIVE NEW FINISHES. THIS INCLUDES THE DEMOLITION AND REMOVAL OF NECESSARY ITEMS. TOUCH-UP AND/OR REFINISH OF SURFACES DAMAGED BY SUBSEQUENT WORK SHALL BE THE RESPONSIBILITY OF THE GENERAL CONTRACTOR. ALL WORK SHALL BE PERFORMED IN CONFORMANCE WITH THE MANUFACTURERS RECOMMENDED INSTALLATION METHODS. THE GENERAL CONTRACTOR SHALL PREPARE THE EXISTING FLOOR PRIOR TO THE APPLICATION OF FINISH FLOORING. THE EXISTING CONCRETE SLAB MUST BE SMOOTH AND LEVEL WITH A TOLERANCE OF ONE-EIGHT INCH PER FOOT. LATEX CEMENT PATCHING COMPOUND SHALL BE UTILIZED (NO ASPHALT BASED COMPOUNDS).

3. G.C. TO PROVIDE GROUT SAMPLE FOR APPROVAL AT ALL TILE LOCATIONS, ARCHITECT TO APPROVE PRIOR TO COMPLETE INSTALLATION. GROUT SPECIFIED DOES NOT REQUIRE SEALING.

4. ALL JOINTS IN GYPSUM BOARD WALLS SHALL BE FINISHED WITH PAPER TAPE 2" WIDE AND THREE COATS OF VINYL, DRY OR PREMIXED JOINT COMPOUND. ALL OUTSIDE CORNERS SHALL BE FINISHED WITH METAL CORNER BEADS, TAPED AND SPACKLED. ALL AREAS TO BE PAINTED SHALL BE SANDED SMOOTH, JUST PRIOR TO APPLICATION OF FIRST COAT OF PAINT. WIPE SANDED SURFACES WITH DAMP CLOTH IN ORDER TO LAY FLAT ANY NAP WHICH MAY HAVE FORMED IN SANDING.

GENERAL NOTES

1. G.C. TO PROVIDE WATERPROOFING AS SPECIFIED FOR THE FOLLOWING CONDITIONS:
   A. AT RESTROOM SLAB-ON-GRADE CONDITIONS, G.C. TO PROVIDE WATERPROOFING MEMBRANE 12" UP WALL AND 12" ON TO FLOOR.
      LATICRETE 9235 OR MAPEI MAPELASTIC HPG OR EQUAL AS APPROVED BY LANDLORD. MUST BE FLOOD TESTED AND APPROVED BY ON-SITE LANDLORD REPRESENTATIVE PRIOR TO THE INSTALLATION OF NEW FLOORING).
2. REFER THE FINISH LEGEND ON SHEET A-510 FOR MORE INFORMATION ON FINISHES.
3. AT THE BACK WORKROOM AND JANITOR'S CLOSET PROVIDE CONTINUOUS SEALANT JOINT AT ALL FRP AND STAINLESS STEEL INTERSECTIONS AT TILE BASE, CEILINGS, DOOR FRAMES, PLUMBING/GAS/ELECTRICAL ROUGH-INS, AND MISCELLANEOUS PENETRATIONS.
4. CAULK ALL JOINTS BETWEEN WALL TILE AND DOOR FRAMES, COLOR TO MATCH GROUT, U.N.O. INSTALL CLEAR SEALANT JOINTS WHERE WINDOWS ABUT A WOOD/METAL FINISH, U.N.O. INSTALL COLORED SEALANT JOINTS WHERE WINDOWS ABUT AN INTERIOR WALL SURFACE, COLOR TO MATCH STOREFRONT, U.N.O.
5. TILE CONTRACTOR TO ENSURE THAT FLOOR TILE IS FLUSH W/ COVED BASE TILE PER HEALTH DEPARTMENT AT CAFE AND TOILET ROOMS.
6. G.C. TO ENSURE THAT THE FLOOR SLAB IS AT AN ACCEPTABLE TOLERANCE LEVEL TO MINIMIZE THE LIPPAGE FOR THE REQUIRED FLOOR FINISH INSTALLATION, REFER TO FINISH SPECIFICATIONS FOR MORE INFORMATION, G.C. TO INFORM ARCHITECT OF DISCREPANCIES.
7. FLOOR TILES TO BE NON-SLIP WITH A MIN. SLIP COEFFICIENT OF FRICTION OF 0.6 OR BETTER.
8. ALL MATERIAL TRANSITIONS TO MEET A.D.A. GUIDELINES.
9. MAXIMUM FLOOR SLOPE IN ANY DIRECTION IS 1 IN 20.
10. ALL INTERIOR WOOD BLOCKING TO BE FIRE TREATED (F.T.), U.N.O., V.I.F.
11. TILE NOTES (PER TCNA HANDBOOK 2011):
    A. FOR TILE SIZES 15" OR SMALLER, MORTAR BED TO BE NO GREATER THAN ¼" THICK
    B. FOR TILE SIZES GREATER THAN 15", MORTAR BED TO BE NO GREATER THAN ⅜" THICK
12. ALL FLOOR PENETRATIONS TO BE WATERPROOFED, U.N.O.
13. G.C. TO PROVIDE LATICRETE EPOXY GROUT AT ALL TILE LOCATIONS. DESIGNER TO APPROVE TYPE AND COLOR.

WATERPROOFING FOR TOILETS

PROVIDE WATERPROOFING MEMBRANE THROUGHOUT EXTENTS:

TOILET ROOMS:ENTIRE EXTENTS OF FLOOR AND A MIN. OF 12" UP PERIMETER WALLS. MUST USE LATICRETE 9235 OR EQUAL AND REFER TO MANUFACTURER SPECIFICATIONS FOR INSTALLATION.

FLOOR PENETRATIONS: MUST PATCH AND REPAIR ALL EXISTING WATERPROOFING AT
ALL FLOOR SINKS, FLOOR DRAINS, AND CORE DRILLS.

COORDINATE WITH LANDLORD'S EXISTING CONDITIONS FOR ALL PATCH AND REPAIR

FLOOR FINISH REQUIREMENTS

1. GENERAL CONTRACTOR TO PREPARE FLOOR AS REQUIRED (SCRAPING, LATEXING ETC.) TO RECEIVE SPECIFIED LEVEL FLOOR FINISHES.
2. REFER TO FLOORING TRANSITION DETAILS FOR MORE INFORMATION.
3. ALL CERAMIC SANITARY COVE BASE MATERIAL MUST BE SET INTEGRAL WITH FLOOR TILE. DO NOT SET THE BASE ON TOP OF FLOOR TILE. THIS IS REQUIRED IN THE TOILET ROOMS, JANITOR'S CLOSET, KITCHENS, AND SERVICE AREA.
4. WHEREVER APPLICATION OF TILE FLOORING, PRETREAT CRACKS, COLD JOINTS, CONTROL JOINTS WITH ANTI-FRACTURE MEMBRANE (LATICRETE OR EQUAL).
5. ALL WORK ROOM WALLS TO RECEIVE 2X2 FULL HEIGHT STAINLESS STEEL CORNER GUARDS WITH CHAMFERED AND SILICONE EDGES.

DECK/STRUCTURAL FINISHING

DESIGN INTENT IS TO PAINT ALL STRUCTURAL ELEMENTS, DUCTWORK, DECKING, PIPING, ETC

ALL CONDUIT TO BE PAINTED. COORDINATE SEQUENCING OF PAINTING SO ALL ELEMENTS ARE PAINTED

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

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07-23-25	BD COMMENTS	3

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Professional Seal:

STATE OF COLORADO
WILLIAM J COMMER
REGISTERED ARCHITECT

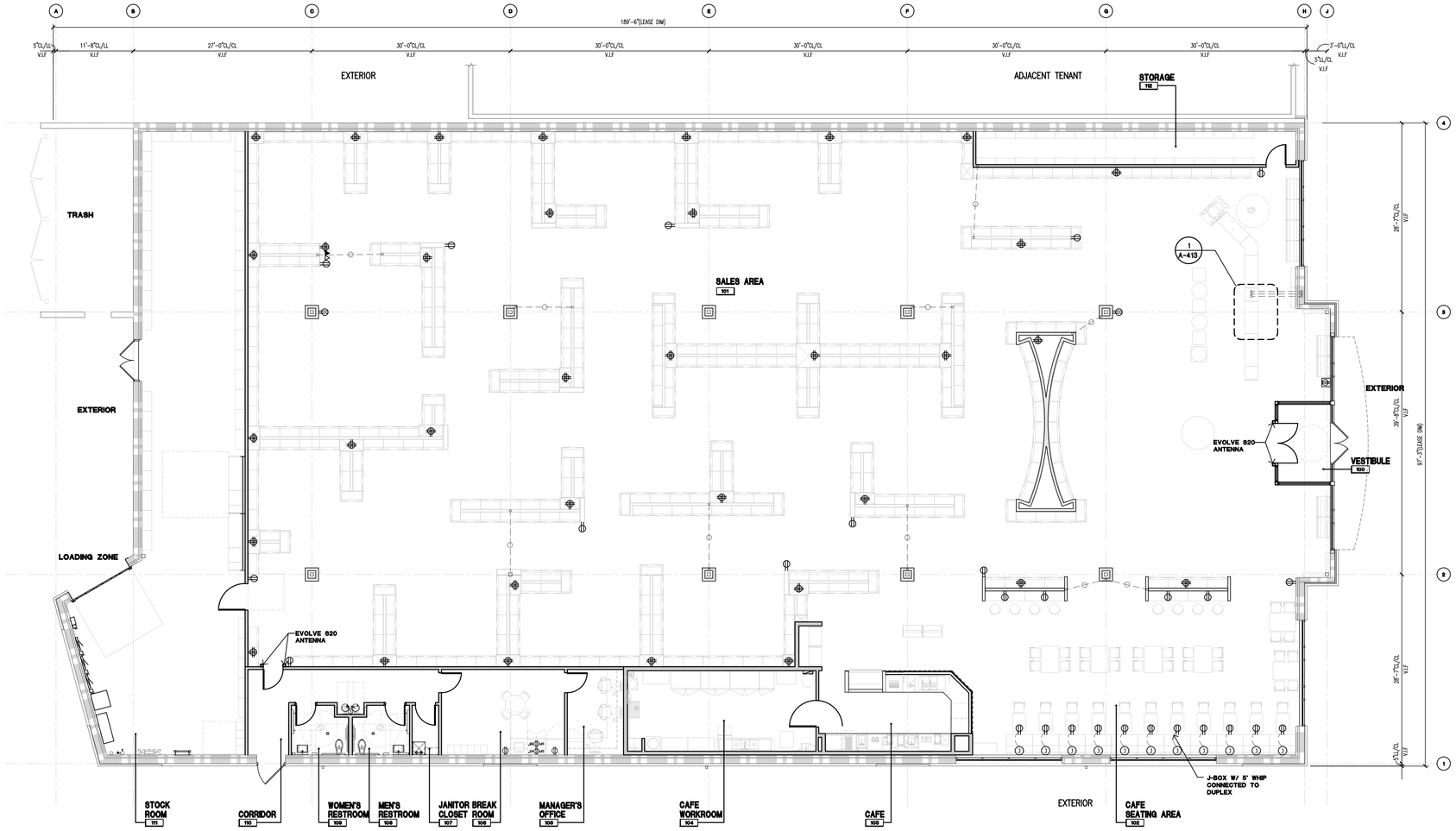
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

Floor Finish Plan, Legend and Notes

Drawing Number:

A-131



1 A-151 Power And Signal Plan

SCALE: 1/8" = 1'-0"

ELECTRICAL SYMBOL LEGEND

SYMBOL	DESCRIPTION
	CONDUIT WITH WIRING RUN CONCEALED IN OR ABOVE CEILING OR WALL, OR RUN EXPOSED IN UNFINISHED AREAS.
	CONDUIT WITH WIRING RUN CONCEALED BELOW FLOOR.
	PANELBOARD SEE ELECTRICAL ONE-LINE DIAGRAM.
	DISCONNECT SWITCH - TYPE & RATING AS SHOWN ON PLANS.
	20A - 120V/277V SINGLE POLE TOGGLE SWITCH MOUNTED 48" AFF TO TOP OF BOX, UNLESS NOTED OTHERWISE. IF APPLICABLE, LOWER CASE SUBSCRIPT "x" = KEYS SWITCH TO FIXTURES BEING CONTROLLED (TYPICAL OF ALL SWITCH SYMBOLS)
	LINE VOLTAGE PASSIVE INFRARED WALL SWITCH OCCUPANCY SENSOR MOUNTED 48"AFF TO TOP OF BOX, UNLESS NOTED OTHERWISE (OCC = DUAL RELAY). MANUFACTURER SHALL BE WATTSTOPPER, HUBBELL, SENSOR SWITCH, COOPER CONTROLS, OR LUTRON)
	LOW VOLTAGE CEILING MOUNTED OCCUPANCY SENSOR. LOCATE AS DIRECTED PER MANUFACTURER AND PROVIDE STANDARD VS. EXTENDED COVERAGE RANGE AS DIRECTED BY MANUFACTURER. REFER TO TYPICAL LOW VOLTAGE OCCUPANCY SENSOR WIRING DIAGRAM ON SHEET E101. (1 = DUAL TECHNOLOGY 360 COVERAGE PATTERN, 2 = DUAL TECHNOLOGY CORNER MOUNTED WITH WIDE VIEW COVERAGE PATTERN, 3 = PIR NARROW HALLWAY COVERAGE PATTERN). MANUFACTURER SHALL BE WATTSTOPPER, HUBBELL, SENSOR SWITCH, COOPER CONTROLS, OR LUTRON)
	GROUNDING TYPE SIMPLEX RECEPTACLE
	GROUNDING TYPE DUPLEX RECEPTACLE
	20A - 125V GROUNDING TYPE QUADRAPLEX RECEPTACLE
	GROUND FAULT INTERRUPTING TYPE DUPLEX RECEPTACLE,
	GROUND FAULT INTERRUPTING TYPE DUPLEX RECEPTACLE MOUNTED
	GROUNDING TYPE DUPLEX RECEPTACLE WITH (2) USB PORTS MOUNTED
	GROUNDING TYPE QUADRAPLEX RECEPTACLE WITH (4) USB PORTS MOUNTED 18" AFF, UNLESS NOTED OTHERWISE
	COMPLETE FLOOR BOX SYSTEM WHICH INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING: SINGLE GANG, CAST IRON, FULLY ADJUSTABLE FLOOR BOX RECESSED IN CONCRETE FLOOR SLAB
	COMPLETE FLOOR BOX SYSTEM WHICH INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING: 2-GANG, CAST IRON, FULLY ADJUSTABLE MULTI-SERVICE FLOOR BOX RECESSED IN CONCRETE FLOOR SLAB
	STANDARD GANG BOX REFER TO TERMINAL LIST BELOW FOR INDIVIDUAL DEVICES: T TELEPHONE, D DATA, P DUPLEX, DP DEDICATED DUPLEX WITH ISOLATED GROUND, SP SPARE CONDUIT (RAN UP TO CEILING), JB J-BOX WITH GENERAL POWER, JP J-BOX POWER FROM LIGHTING CONTACTOR, Q QUAD
	DOOR BELL ASSEMBLY
	CHIME ASSEMBLY
	PUSH-BUTTON ASSEMBLY
	JUNCTION BOX
	TELEPHONE OUTLET
	DATA OUTLET MOUNTED
	COMBINATION TELEPHONE/DATA
	PLUG MOLD

NOTE:NOT ALL SYMBOLS MAY APPLY TO PROJECT.

BARNES&NOBLE

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WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
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3173 CINEMA POINT,
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80922

Project Number 33288
Store Number 3566

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Professional Seal:

STATE OF COLORADO
WILLIAM J COMMER
305353
LICENSED ARCHITECT

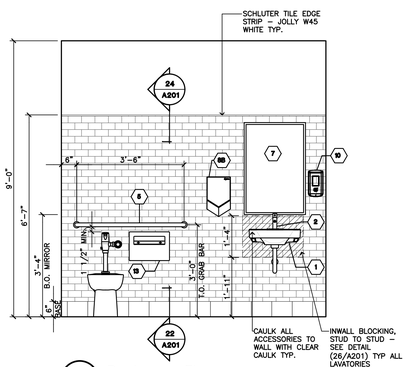
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

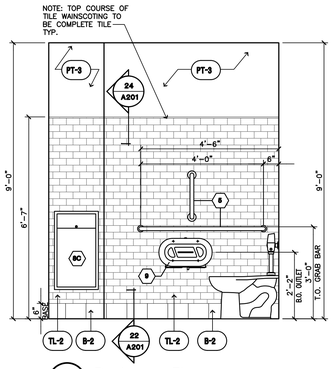
Power And Signal Plan

Drawing Number:

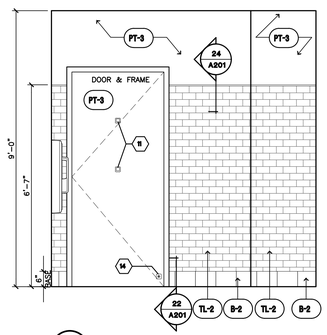
A-151



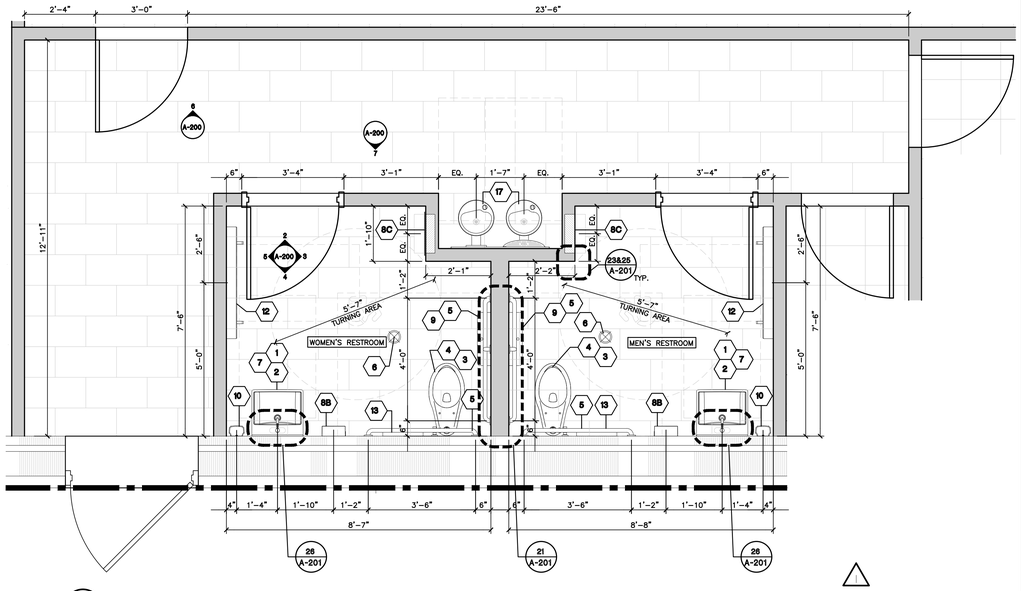
4 A-200 Restroom Elev.
SCALE: 1/2"=1'-0"



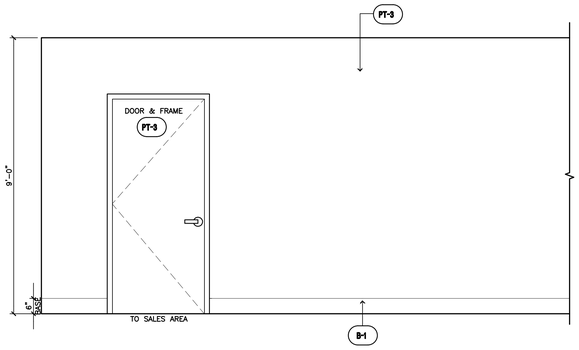
8 A-200 Restroom Elev.
SCALE: 1/2"=1'-0"



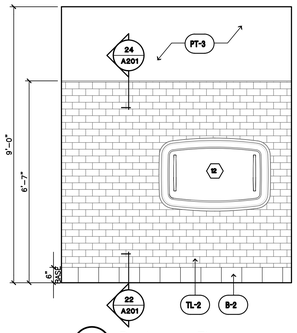
2 A-200 Restroom Elev.
SCALE: 1/2"=1'-0"



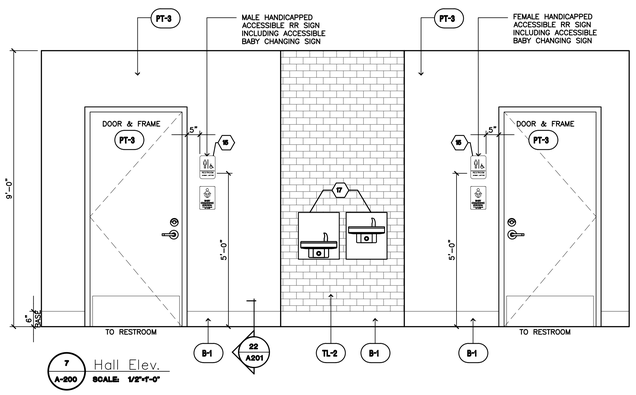
1 A-200 Enlarged Rest Room and Corridor Plan
SCALE: 1/2"=1'-0"



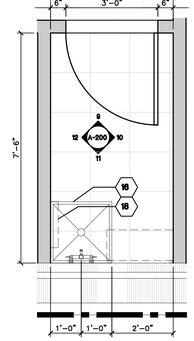
6 A-200 Hall Elev.
SCALE: 1/2"=1'-0"



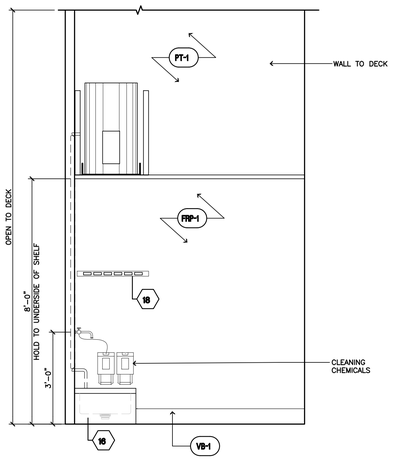
5 A-200 Restroom Elev.
SCALE: 1/2"=1'-0"



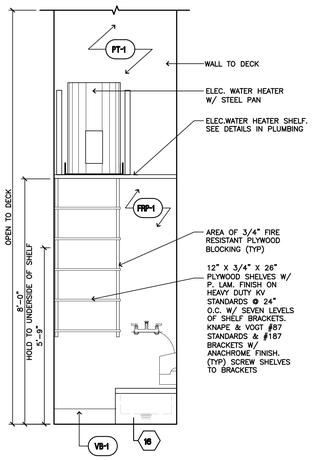
7 A-200 Hall Elev.
SCALE: 1/2"=1'-0"



8 A-200 Enlarged Janitors Closet Plan
SCALE: 1/2"=1'-0"



12 A-200 Janitor's Closet Elev.
SCALE: 1/2"=1'-0"



11 A-200 Janitor's Closet Elev.
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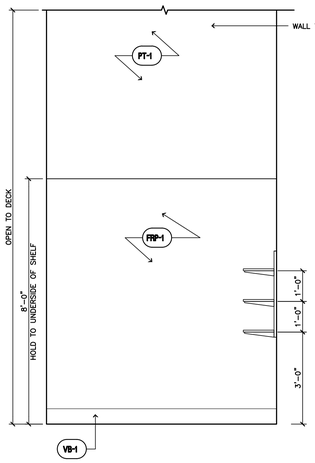
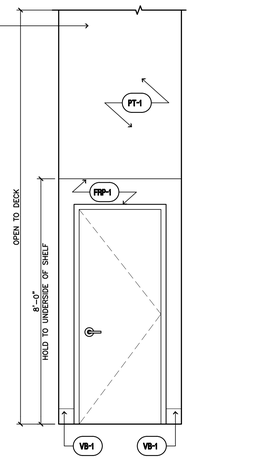


10 A-200 Janitor's Closet Elev.
SCALE: 1/2"=1'-0"



9 A-200 Janitor's Closet Elev.
SCALE: 1/2"=1'-0"

TOILET ROOM FIXTURE AND ACCESSORY SCHEDULE

FIXT. No.	DESCRIPTION	QTY.	MAKE	MODEL #	REMARKS
1	LAVATORY	2	AMERICAN STANDARD	9024.001EC	A.D.A., WALL MOUNTED
2	FAUCET	2	AMERICAN STANDARD	2064.101	
3	WATER CLOSET	2	AMERICAN STANDARD	2467.016	A.D.A., FLOOR MOUNT, 1.6 GPF CADET SUITE
4	FLUSHOMETER	2	AMERICAN STANDARD	6065.111	COMPATIBLE WITH 1.1 GPF HET
5	GRAB BARS 42 48" 18"	2 2 2	BOBRICK	B-6806 x42 x48 x18	ABOVE WATER CLOSET SIDE WALL VERTICAL GRAB BAR
6	FLOOR DRAIN	2	–	–	SEE PLUMBING DRAWINGS
7	MIRROR	2	BOBRICK	B-290 24x36	WELDED FRAME MIRROR
8B	HAND DRYER	2	DYSON	AIRBLADE V SPRAYED NICKEL	–
8C	WASTE RECEPTACLE	2	BOBRICK	B-3644	RECESSED
9	TOILET PAPER HOLDER	2	BOBRICK	B-2892	SURFACE MOUNTED TWIN JUMBO-ROLL TOILET TISSUE DISPENSER
10	SOAP DISPENSER	2	GOJO INDUSTRIES	PURELL CS6	PROVIDED BY BN
11	BAG/COAT HOOK	4	BOBRICK	B-6717	MOUNT 46" AND 66" AFF CENTERED ON BACK OF DOOR
12	BABY CHANGING STATION	2 0	KOALA KARE	KB-300-01-GRAY KB-301-01-GRAY	HORIZONTAL SURFACE MOUNTED BABY CHANGING STATION VERTICAL SURFACE MOUNTED BABY CHANGING STATION
13	TOILET SEAT COVER DISPENSER	2	BOBRICK	B-4221	SURFACE MOUNTED
14	DOOR BUMPER	2	BOBRICK	B-687	MOUNT MIN 4"x4" FROM DOOR LOWER CORNER STRIKE POINT
15	RESTROOM SIGN	2	AMERA PRODUCTS	GREY 4405	PROVIDED BY BN
16	MOP SINK W/RACK AND FAUCETS	1	A&J WASHROOM	UJ12A	MOP SINK – SEE PLUMBING DRAWINGS
17	DRINKING FOUNTAIN	1	OASIS	MMRSL	RADII, BI-LEVEL MODULAR FOUNTAIN, A.D.A. INCLUDING CANE APRON
18	MOP HOLDER	1	–	–	PROVIDED BY BN
19	URINAL	0	AMERICAN STANDARD	6515001	–
20	PARTITION	0	BOBRICK	1180	TRADITIONAL GRAY – DURALINE SERIES (CLASS A)

BARNES&NOBLE

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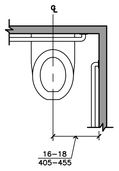
Professional Seal:

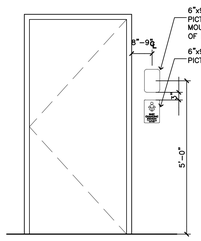
STATE OF COLORADO
WILLIAM J COMMER
LICENSED ARCHITECT

LIC.# : 305353
EXP. DATE : 10/31/25

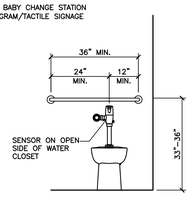
Drawing Description:
Enlarged Toilet Room, Elevations, Details and Schedule

Drawing Number:
A-200

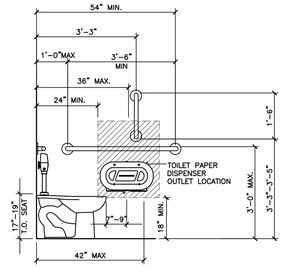




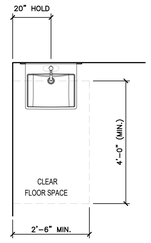
DOOR IDENTIFICATION SYMBOLS



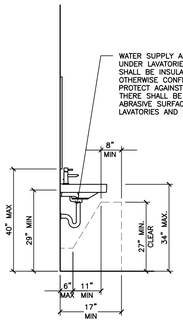
GRAB BARS / SEAT COVER DISPENSER (ADA)

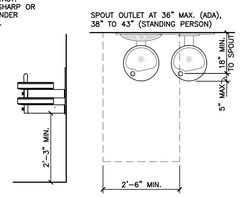


TOILET (FLOOR MOUNTED)

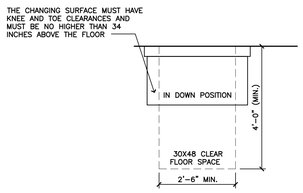


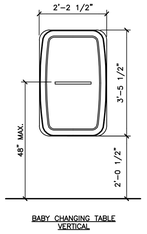
LAVATORY



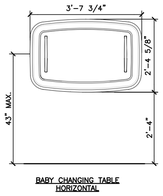


DRINKING FOUNTAIN (CANTILEVERED)

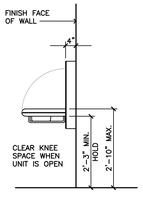




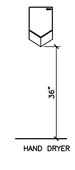
BABY CHANGING TABLE VERTICAL



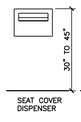
BABY CHANGING TABLE HORIZONTAL



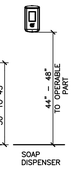
PAPER TOWEL DISPENSER



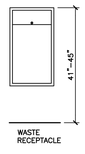
HAND DRYER



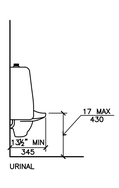
SEAT COVER DISPENSER



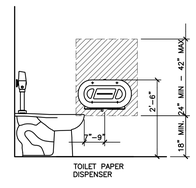
SOAP DISPENSER



WASTE RECEPTACLE

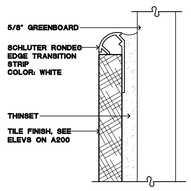


URINAL

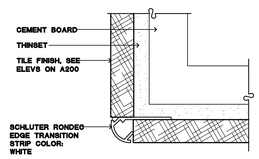


TOILET PAPER DISPENSER

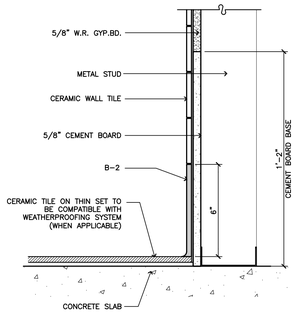
20 A-201 Accessibility Mounting Standards
SCALE: NOT TO SCALE



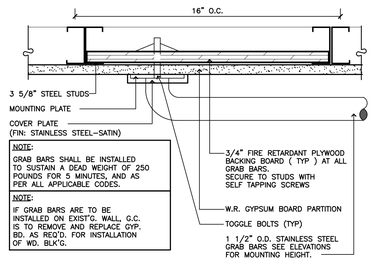
24 A-201 Detail – Tile Corner Edge
SCALE: FULL SIZE



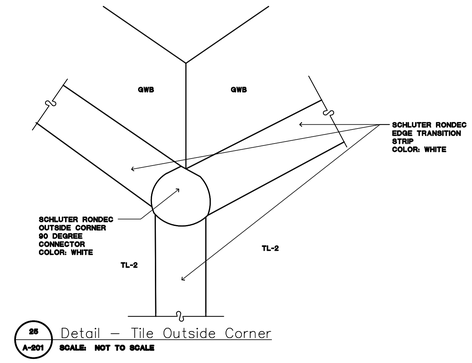
29 A-201 Detail – Tile Corner Edge
SCALE: FULL SIZE



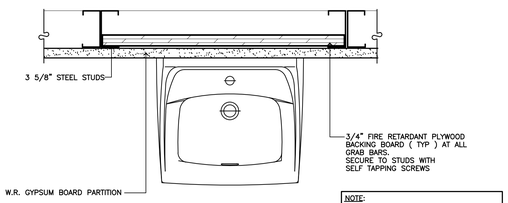
22 A-201 Tile Base Detail
SCALE: 3"=1'-0"



21 A-201 Detail at Grab Bar
SCALE: 3"=1'-0"



25 A-201 Detail – Tile Outside Corner
SCALE: NOT TO SCALE



29 A-201 Detail at Lavatory
SCALE: 3"=1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

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Professional Seal:

STATE OF COLORADO · WILLIAM J COMMER · LICENSED ARCHITECT

LIC.# : 305353
EXP. DATE : 10/31/25

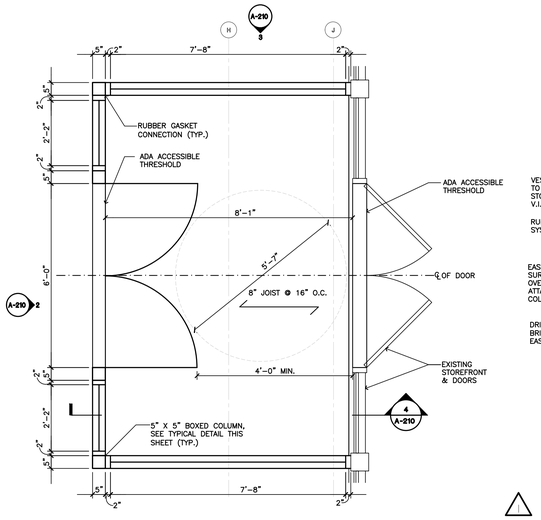
Drawing Description:

Toilet Room Details

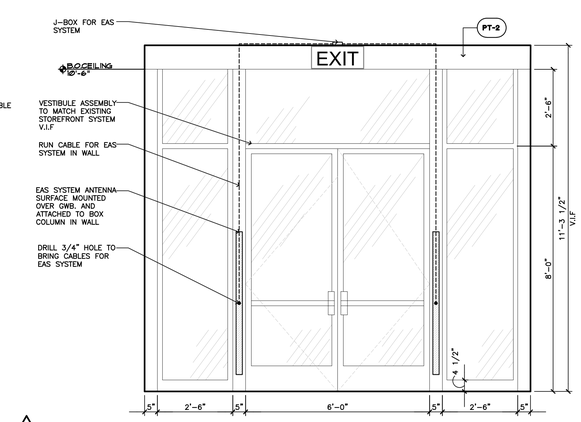
Drawing Number:

A-201

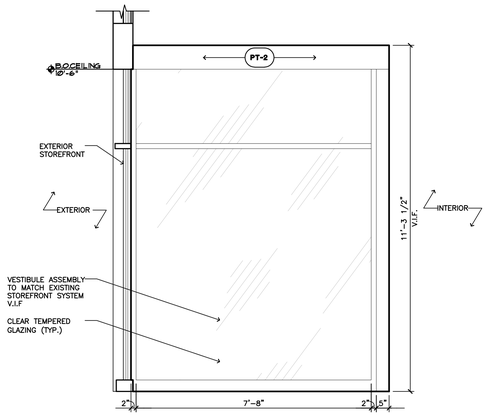
NOTES:
- VERIFY ALL EXISTING STOREFRONT IN FIELD.



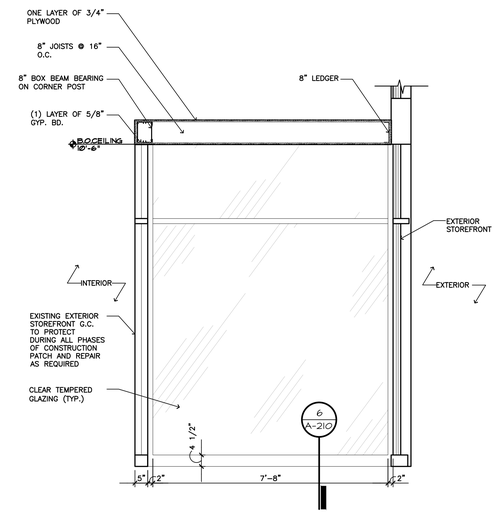
1 A-210 Vestibule Floor Plan
SCALE: 1/2"=1'-0"



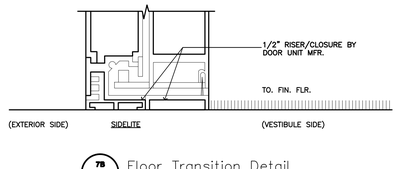
2 A-210 Vestibule Interior Door Elevation
SCALE: 1/2"=1'-0"



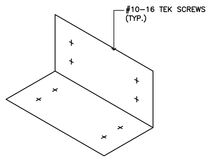
3 A-210 Vestibule Side Elevation
SCALE: 1/2"=1'-0"

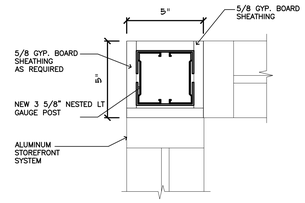


4 A-210 Vestibule Section
SCALE: 1/2"=1'-0"

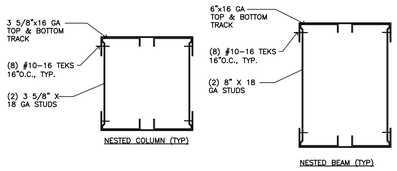


7B A-210 Floor Transition Detail
SCALE: 3"=1'-0"

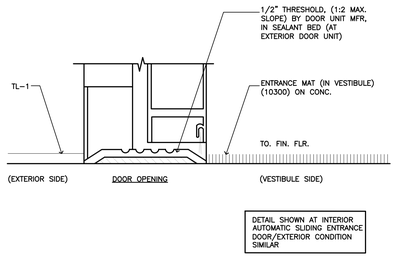




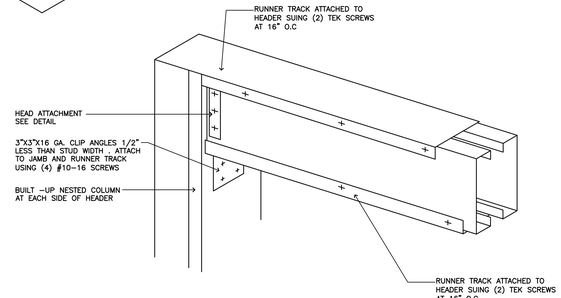
5 A-210 Vestibule Post Detail
SCALE: 3"=1'-0"



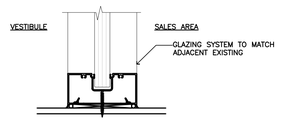
8 A-210 Nested Stud Column & Beam Details
SCALE: 3"=1'-0"



7A A-210 Floor Transition Detail
SCALE: 3"=1'-0"



7 A-210 Nested Stud Details
SCALE: 3"=1'-0"



6 A-210 Base Detail
SCALE: 3"=1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

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07-23-25	BD COMMENTS	3

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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

Vestibule Details

Drawing Number:

A-210

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

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Professional Seal:

STATE OF COLORADO
WILLIAM J COMMER
LICENSED ARCHITECT

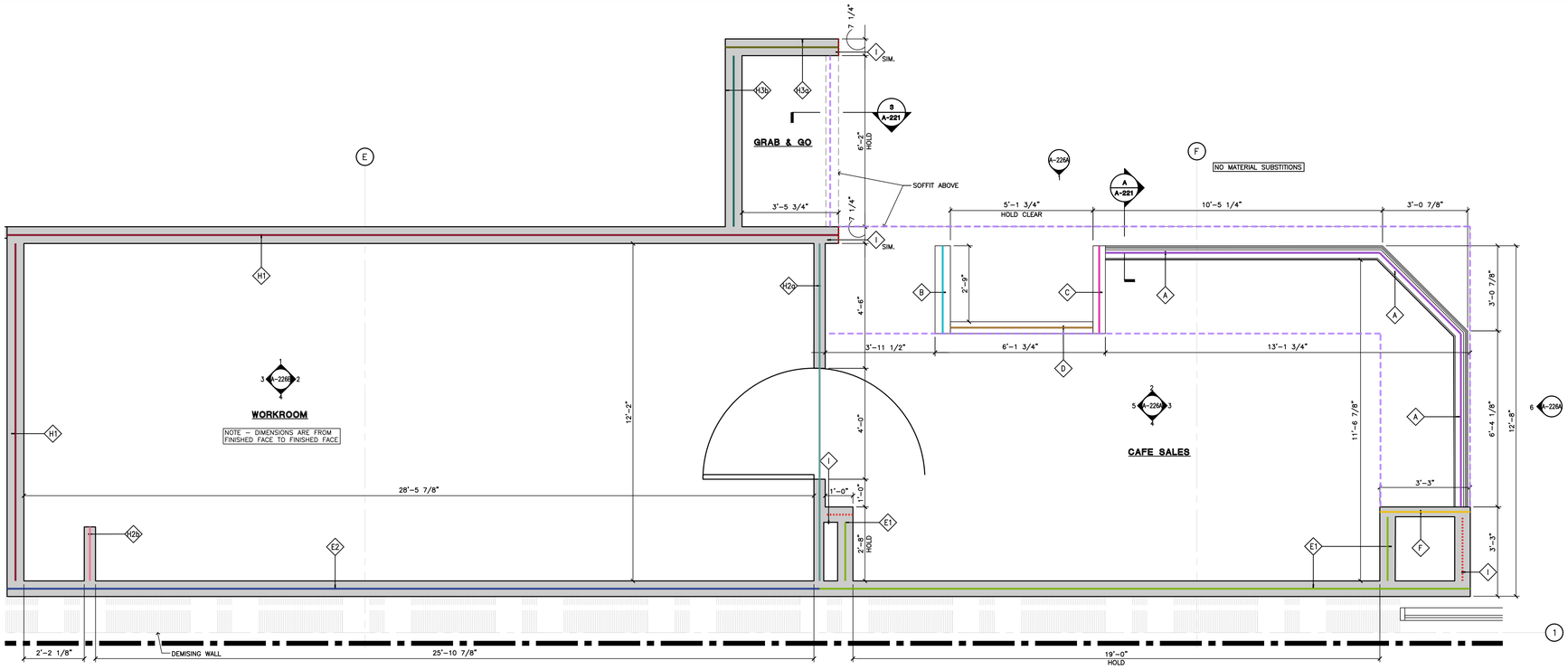
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

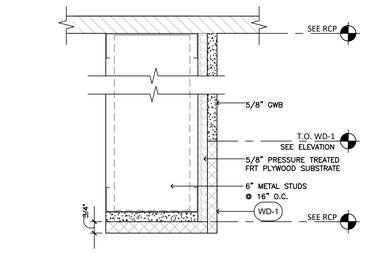
Cafe Render View

Drawing Number:

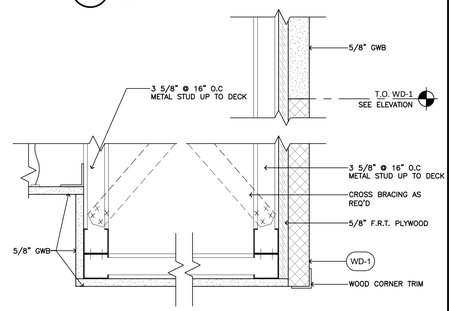
A-220



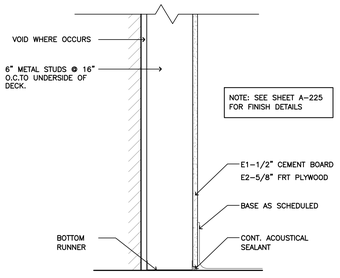
1 A-221 Enlarged Cafe & Workroom Plan
SCALE: 1/2" = 1'-0"



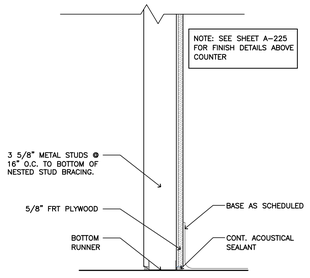
9 A-221 Soffit Section – Grab & Go
SCALE: 8" = 1'-0"



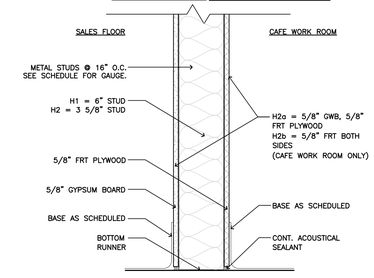
2 A-221 Soffit Section – Cafe Sales Area
SCALE: 8" = 1'-0"



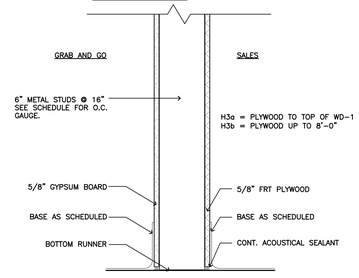
E A-221 Section Wall Typ. E1/E2
SCALE: 1-1/2"=1'-0"



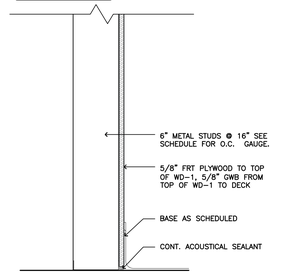
F A-221 Section Wall Typ. F
SCALE: 1-1/2"=1'-0"



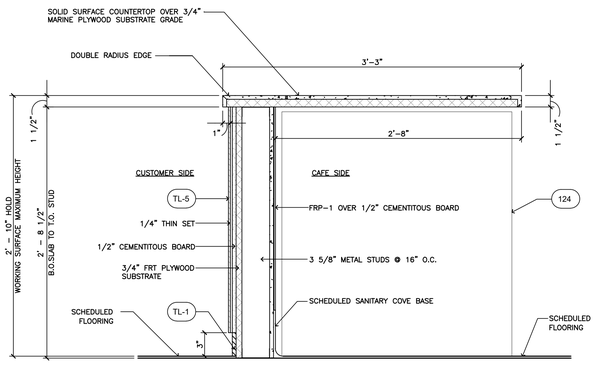
G A-221 Section Wall Typ. H1/H2
SCALE: 1-1/2"=1'-0"



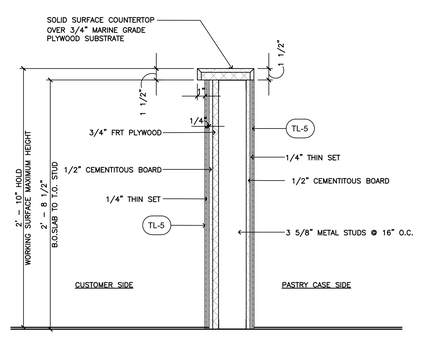
H A-221 Section Wall Typ. H3
SCALE: 1-1/2"=1'-0"



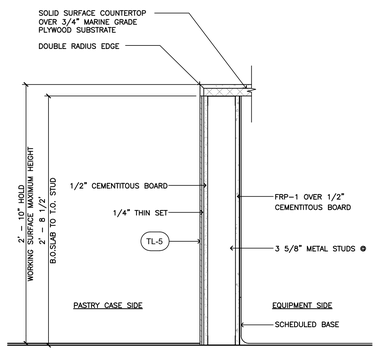
I A-221 Section Wall Typ. I
SCALE: 1-1/2"=1'-0"



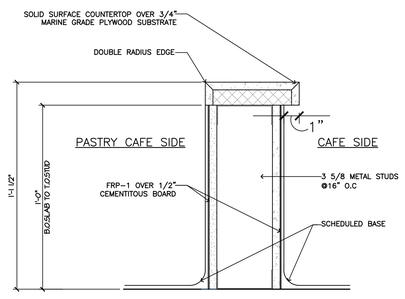
A Section Through Front Counter
A-221 SCALE: 1-1/2"=1'-0"



B A-221 Section Wall Typ.B
SCALE: 1-1/2"=1'-0"



C A-221 Section Wall Typ.C
SCALE: 1-1/2"=1'-0"



D A-221 Section Wall Typ.D
SCALE: 8" = 1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
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3173 CINEMA POINT,
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80922

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Professional Seal:

STATE OF COLORADO WILLIAM J COMMER 38353 LICENSED ARCHITECT

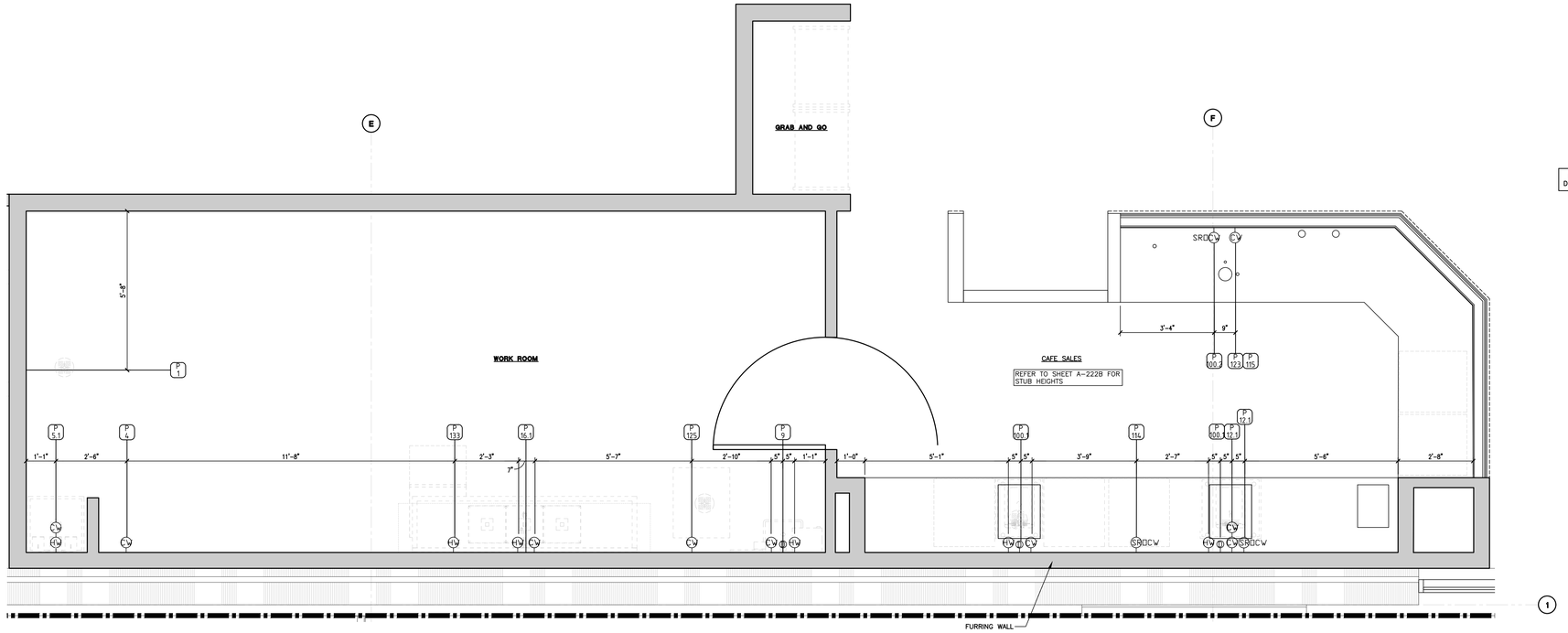
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

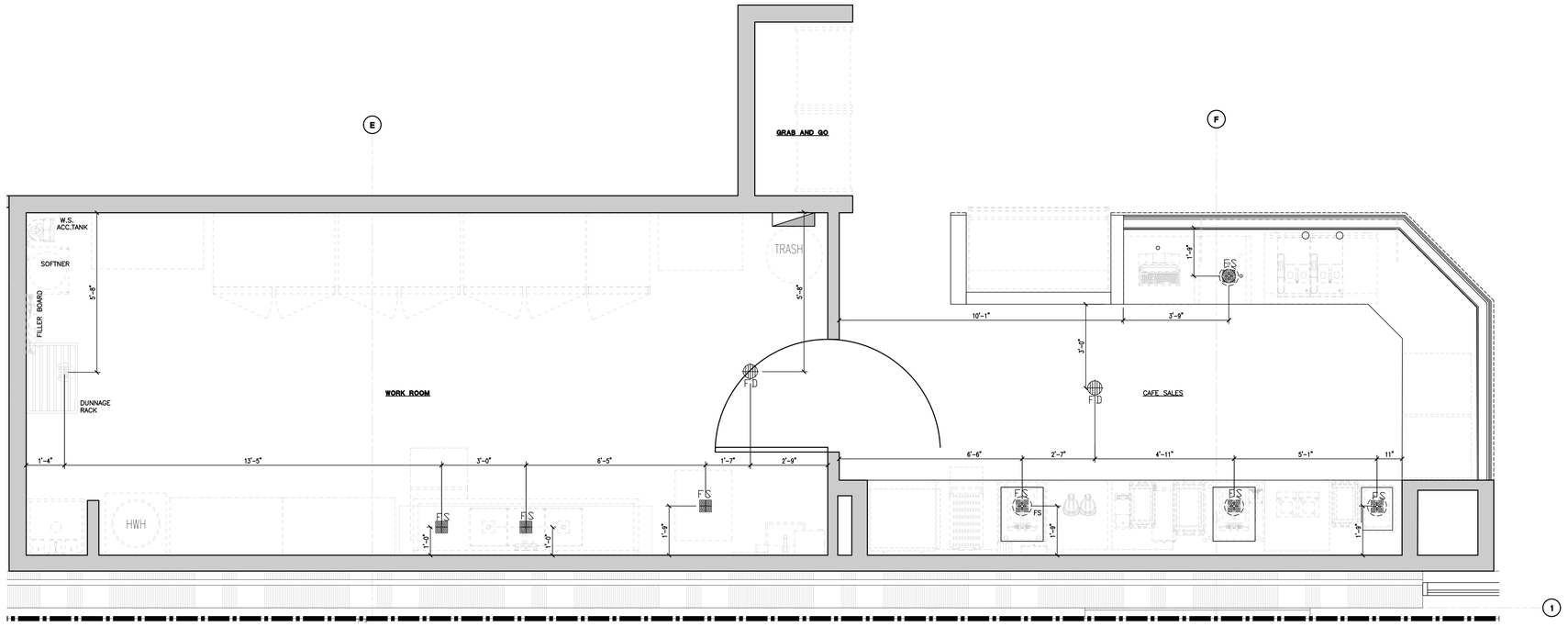
Cafe & Cafe Workroom
Enlarged Plan and Details

Drawing Number:

A-221



1 A-222A **Cafe & Cafe Workroom Plumbing Supply Plan**
SCALE: 1/2" = 1'-0" (COORDINATE WITH M.E.P. DWGS.)



2 A-222A **Cafe & Cafe Workroom Plumbing Sanitary Plan**
SCALE: 1/2" = 1'-0" (COORDINATE WITH M.E.P. DWGS.)

PLUMBING SYMBOLS

- HW — HOT WATER SUPPLY
- CW — COLD WATER SUPPLY
- D — DIRECT DRAIN
- FS — FLOOR SINK
- FD — FLOOR DRAIN
- U — UTILITY SLEEVE
- DRAIN LINE CONNECTION
- SUPPLY LINE CONNECTION

REFER TO SHEETS A-224 FOR EQUIPMENT SCHEDULE

COORDINATE WITH PLUMBING SHEETS

DO NOT SCALE THIS DRAWING. REFERENCE EQUIPMENT SPECIFICATIONS AND CUT SHEETS FOR ALL DIMENSIONS, UTILITIES, SERVICE ACCESS, REQUIRED CLEARANCES AND OTHER TECHNICAL INFORMATION

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
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80922

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Professional Seal:

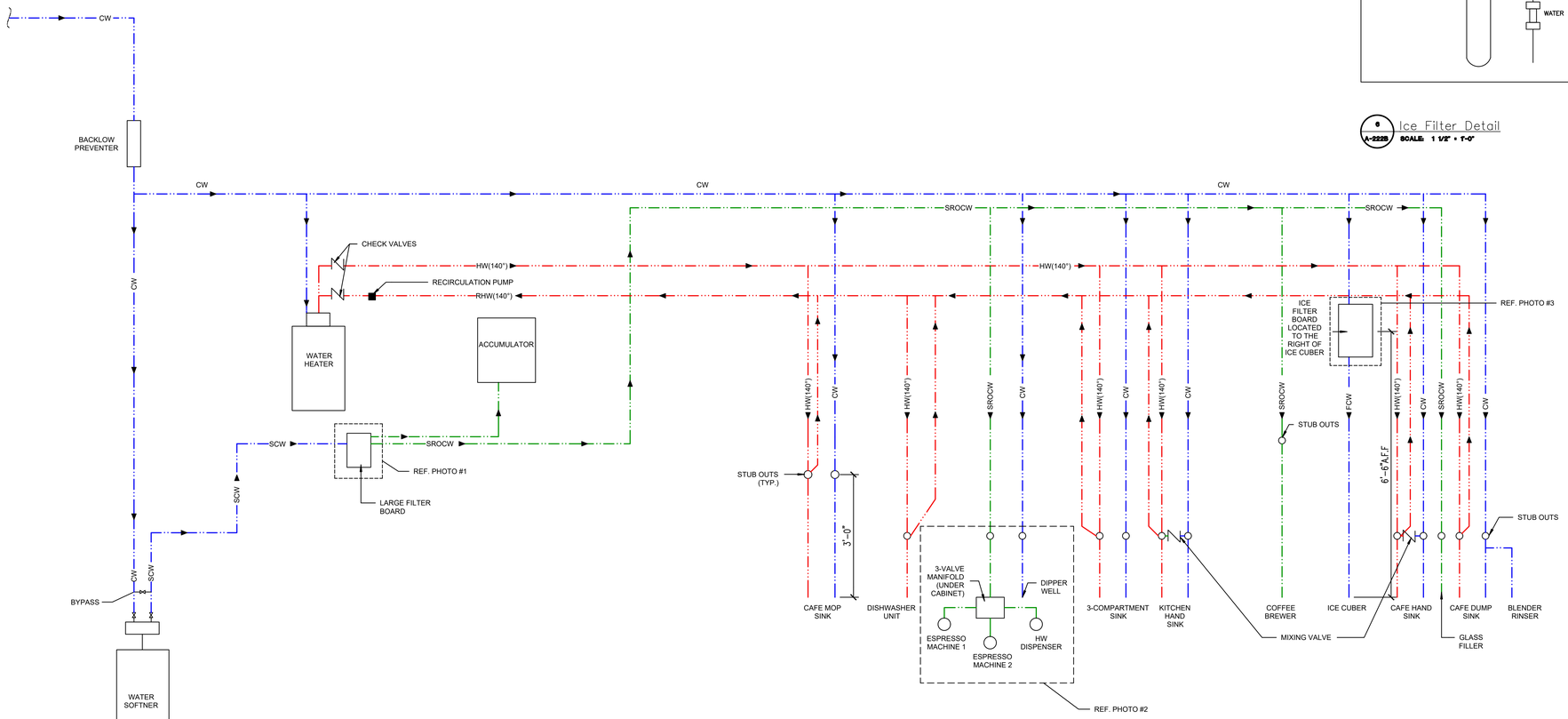
STATE OF COLORADO
WILLIAM J. COMMER
LICENSED ARCHITECT

LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:
Cafe & Cafe Workroom Floor Penetration & Plumbing Plan

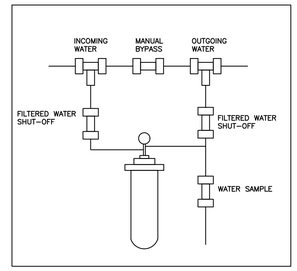
Drawing Number:

A-222A



1 A-222B — Cafe & Cafe Workroom Plumbing Riser Schematic

SCALE: 3/4" = 1'-0" (COORDINATE WITH M.E.P. DWGS.)



6 A-222B — Ice Filter Detail

SCALE: 1 1/2" = 1'-0"

RED = HW

BLUE = CW or SCW

GREEN = SROCW

REFER TO PLUMBING ISOMETRIC FOR SIZING

THIS SCHEMATIC IS DIAGRAMMATIC AND INTENDED FOR PIPE ROUTING REFERENCE ONLY. INSTALLATION OF EQUIPMENT SHALL BE PER MANUFACTURERS GUIDELINES AND RECOMMENDATIONS

FURNISH AND INSTALL AN APPROVED BACKFLOW PREVENTING DEVICE IN LINE WITH THE DOMESTIC WATER SUPPLY TO ALL APPLIANCES REQUIRING SAME (COFFEE MAKER, ETC.)

INCOMING WATER PRESSURE MUST BE BETWEEN 50 - 55 P.S.I. CONTRACTOR TO PROVIDE A BID ALTERNATE FOR BOOSTER PUMP DESIGNED FOR APPROXIMATE FLOW OF 60 GPM, AND APPROXIMATE PRESSURE BOOST OF 30 PSI TO SERVE THE CAFE. PUMP SHALL HAVE A VFD DRIVE FOR VARIABLE FLOW AND BID ALTERNATE SHOULD INCLUDE COST FOR ALL REQUIRED ACCESSORIES AND CONTROLS. THRUSH BRAND MODEL PB-VS--60-30 OR EQUAL. ENGINEER TO SIZE PUMP BASED ON FLOW TEST DATA IF PRESSURE IS DEEMED IN ADEQUATE.

2 A-222B — Reference Photo #1

SCALE: 3/4" = 1'-0"

WATER FILTRATION BOARD IN WORK ROOM

3 A-222B — Reference Photo #2

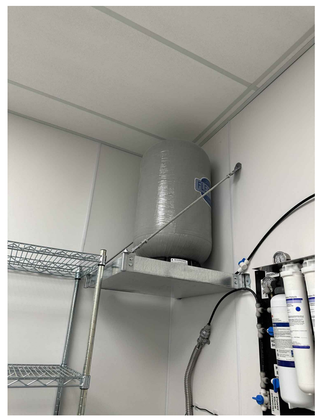
SCALE: NTS

DIPPING STATION IN CAFE

4 A-222B — Reference Photo #3

SCALE: NTS

HAND SINK AND ICE MACHINE FILTER IN WORKROOM



5 A-222B — Reference Photo #4

SCALE: NTS

ACCUMULATOR TANK IN WORKROOM

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

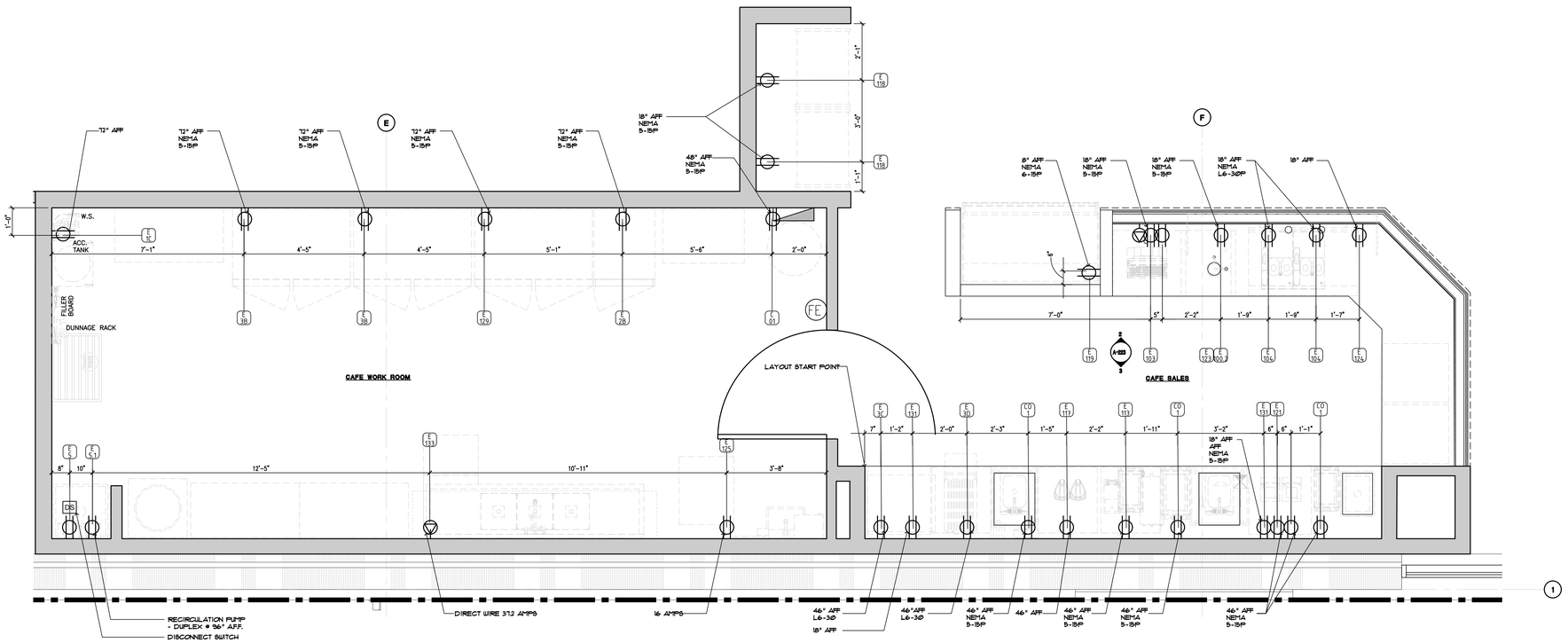
Professional Seal:

LIC.# : 305353
EXP. DATE : 10/31/25

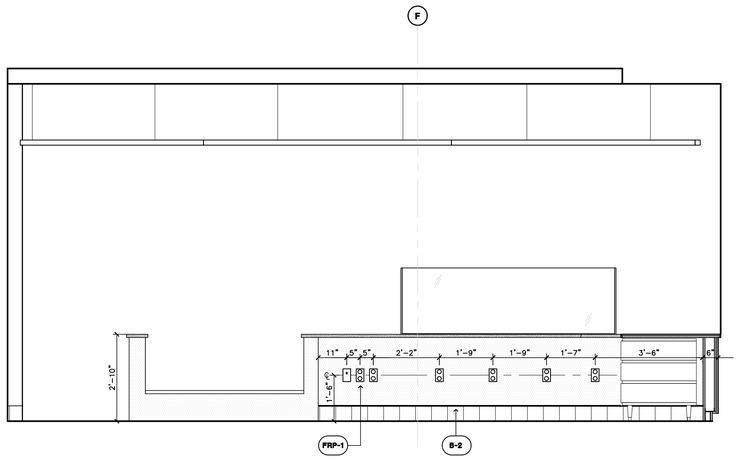
Drawing Description:
Cafe & Cafe Workroom Plumbing Schematic

Drawing Number:

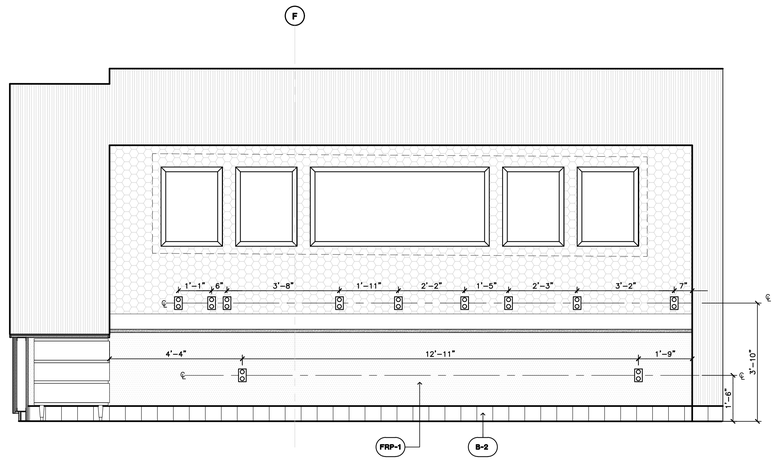
A-222B



1 A-223 Cafe & Cafe Workroom Electrical Plan
SCALE: 1/2" = 1'-0"



2 A223 Cafe Front Counter Elevation
SCALE: 1/2"-1'-0"



3 A223 Cafe Back Counter Elevation
SCALE: 1/2"-1'-0"

ELECTRICAL SYMBOLS

- WALL MOUNTED DUPLEX OUTLET
- FIXTURE MOUNTED DUPLEX OUTLET
- CEILING MOUNTED DUPLEX OUTLET
- WALL MOUNTED QUADPLEX OUTLET
- DEDICATED CIRCUIT OUTLET
- DATA OUTLET
- WALL MOUNTED JUNCTION BOX
- FLOOR MOUNTED JUNCTION BOX
- CEILING MOUNTED JUNCTION BOX
- UTILITY SLEEVE
- CONVENIENCE OUTLET
- DEDICATED OUTLET

REFER TO SHEETS A-224 FOR EQUIPMENT SCHEDULE

DO NOT SCALE THIS DRAWING. REFERENCE EQUIPMENT SPECIFICATIONS AND CUT SHEETS FOR ALL DIMENSIONS, UTILITIES, SERVICE ACCESS, REQUIRED CLEARANCES AND OTHER TECHNICAL INFORMATION

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
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80922

Project Number 33288
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Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
305353
LICENSED ARCHITECT

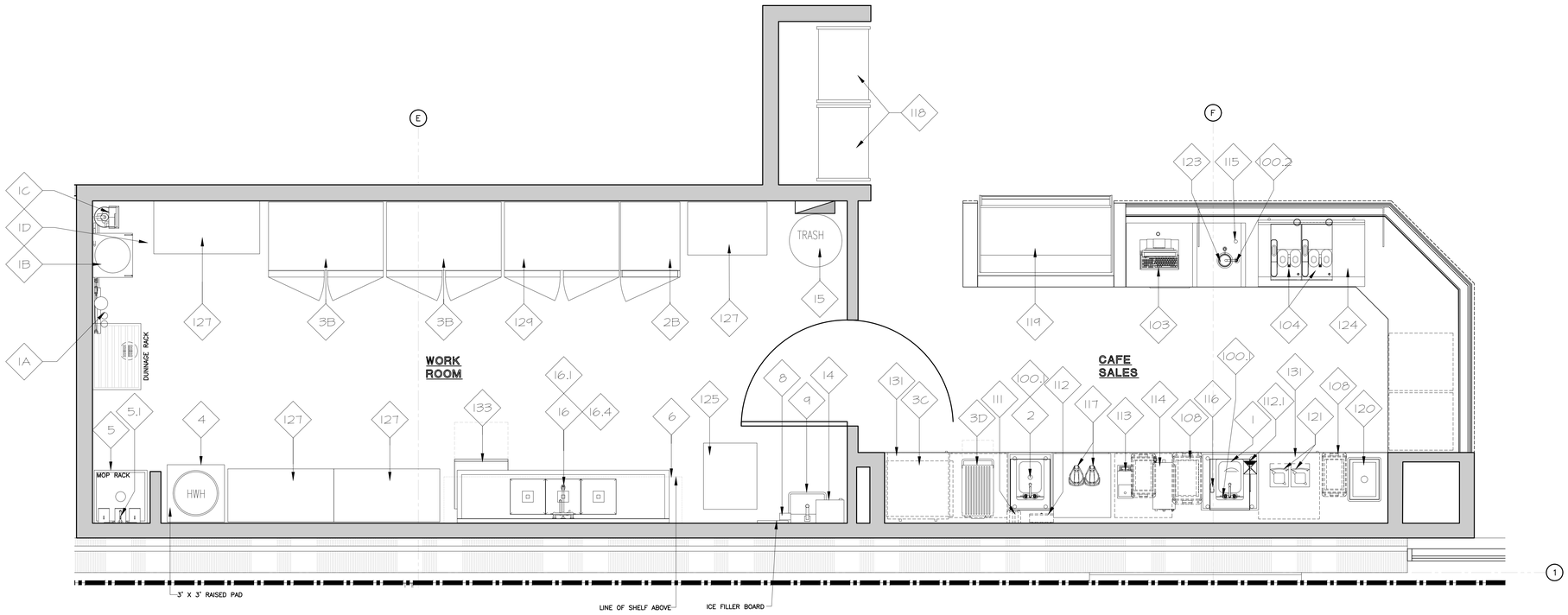
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

Cafe & Cafe Workroom Electrical Plan

Drawing Number:

A-223



1 A-224 Cafe & Cafe Workroom Equipment Plan
SCALE: 1/2" = 1'-0"

FOOD SERVICE EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY.
1	SINK BOWL, WELD-IN, DUMP SINK	N. WASSERSTROM & SONS	F4480234	1
2	SINK BOWL, WELD-IN, HAND SINK	N. WASSERSTROM & SONS	F4480234	1
3C	BLODGETT OVEN	BLODGETT	CTB SGL	1
3D	RAPID COOK OVEN	MERRYCHEF USA	E2S HIGH CLASSIC	1
100.1	PANTRY FAUCET	T&S BRASS	KL45-4000-WH	2
100.2	HOT WATER DISPENSER	IN SINK ERATOR	C1300	1
103	POS SYSTEM	-	-	1
104	ESPRESSO MACHINE	MELITTA	CT8	2
108	GARBAGE CAN	RUBBERMAID COMMERCIAL PRODUCTS	FG354060BLA	2
111	TOUCH-FREE SOAP DISPENSER	GOJO INDUSTRIES, INC.	PURELL CS6	1
112	SURFACE-MOUNTED PAPER TOWEL DISPENSER	BOBRICK	B-2621	1
112.1	BLENDER RINSER	BLENDTEC	JRE-610	1
113	COFFEE GRINDER	GRINDMASTER-UNIC-CRATHCO	890BS	1
114	COFFEE BREWER	BUNN	53100.0100	1
115	MANIFOLD	-		1
116	GLASS FILLER	T&S BRASS	B-1210	1
117	AIRPOT	SERVICE IDEAS	ECALS22SS	2
118	SELF-SERVICE REFRIGERATED CASE	STRUCTURAL CONCEPTS	B3424	2
119	PASTRY CASE, CURVED GLASS	FEDERAL INDUSTRIES	SSRPF-5052	1
120	DROP-IN ICE BIN	KROWNE	D278	1
121	BLENDER	VITAMIX	36019-ABAB	2
123	DIPPER WELL	T&S BRASS	B-2282-01-F05	1
124	DOUBLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-48-HC	1
131	SINGLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-27-HC	2

WORKROOM EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY
1A	R.O.WATER FILTRATION SYSTEM MANIFOLD BOARD	-	-	1
1B	R.O.WATER FILTRATION SYSTEM - ACCUMULATOR TANK	-	-	1
1C	R.O.WATER FILTRATION SYSTEM - WATER SOFTNER	-	-	1
1D	R.O.WATER FILTRATION SYSTEM - BRINE	-	-	1
2B	REFRIGERATOR, REACH-IN, SINGLE DOOR	TRUE MGF.	STG1R-1S-HC	1
3B	FREEZER, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2F-2S-HC	2
4	WATER HEATER	-	-	1
5	MOP SINK	FIAT	MSB2424	1
5.1	SERVICE FAUCET	CHICAGO FAUCET	897-RCF	1
6	WALL SHELF	SUPPLIED BY BN	-	1
8	SOAP DISPENSER	BN SUPPLIED PURELL	TFX	1
9	HAND SINK - WALL MOUNTED	HAND SINK PARTS & ACCESSORIES	DH-17-NO FAU	1
14	C-FOLD PAPER TOWEL DISPENSER	BN SUPPLIED C-FOLD	-	1
15	WASTE CONTAINER	-	-	1
16	THREE COMPARTMENT SINK	AMTEKCO INDUSTRIES LTD.	D724-03-74	1
16.1	PRE-RINSE FAUCET & ADD ON FAUCET			1
16.4	LEVER WASTE	T&S BRASS	B-3950	3
125	ICE MAKER, CUBE-STYLE	ITV ICE MAKER	SPIKA MS 500	1
127	WIRE SHELVING	1880 HOSPITALITY	FF2472C	4
129	REFRIGERATOR, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2DT-2S	1
133	DISHWASHER, UNDERCOUNTER	JACKSON WWS	DISHSTAR HT	1

CAFE EQUIPMENT POWER SCHEDULE

EQUIPMENT NO.	DESCRIPTION	QUANTITY	VOLTAGE AND PHASE	AMPS	HP	KVA	PANEL	CIRCUIT(S)	BREAKER SIZE	WIRE SIZE	CONDUIT	NEMA
129	REFRIGERATOR/FREEZER, REACH-IN, DOUBLE DOOR	1	115V/1Ø	11.6	1/2	1.334	K	19	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-20P
2B	REFRIGERATOR, REACH-IN, SINGLE DOOR	1	115V/1Ø	3.8	1/4	0.437	K	28	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
3B	FREEZER, REACH-IN, DOUBLE DOOR	2	115V/1Ø	9.4	1-1/4	1.081	K K	6 14	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
3C	BLODGETT OVEN	1	208V/3Ø	24.0	1/4	5.6	K	36,38,40	30A-3P	(3)#10 THWN & (1)#10 GND	3/4"	VERIFY WITH MANUFACTURER
3D	RAPID COOK OVEN	1	208V/1Ø	30.0	-	6.0	K	39,41	30A-2P	(2)#10 THWN & (1)#10 GND	3/4"	6-30P
133	DISHWASHER, UNDERCOUNTER	1	208V/1Ø	37.2	-	7.74	K	21,23	40A-2P	(2)#8 THWN & (1)#10 GND	3/4"	DIRECT
125	ICE MAKER, CUBE STYLE	1	115V/1Ø	16.0	-	1.289	K	1	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	VERIFY WITH MANUFACTURER
100.2	HOT WATER DISPENSER	1	115V/1Ø	11.3	-	1.300	K	22	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
103	POS SYSTEM	1	115V/1Ø	-	VERIFY	VERIFY	K K	35 18	20A-1P	(2)#12 THWN & (1)#12 GND & (1)#12 ISO. GND	3/4"	5-20P
104	ESPRESSO MACHINE	2	208V-220V/1Ø	30.0	-	6.240	K K	5,7 9,11	30A-2P	(2)#10 THWN & (1)#10 GND	3/4"	L6-30P
131	REFRIGERATOR, UNDERCOUNTER	2	115V/1Ø	2.0	1/6	0.24	K K	12 15	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
121	BLENDER	2	120V/1Ø	15.0	3.0	1.800	K K	13 20	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-20P
113	COFFEE GRINDER	1	120V/1Ø	8.0	1/2	0.920	K	37	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
114	COFFEE BREWER	1	208V/1Ø	14.0	-	2.9	K	8,10	20A-2P	(2)#12 THWN & (1)#12 GND	3/4"	VERIFY WITH MANUFACTURER
118	SELF-SERVICE REFRIGERATED CASE	2	120V/1Ø	15.6	-	1.198	K K	2 4	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-20P
119	DISPLAY CASE, REFRIGERATED, SELF SERVE	1	208V/1Ø	10.0	1/2	2.08	K	32,34	15A-2P	(2)#12 THWN & (1)#12 GND	3/4"	6-15P
124	DOUBLE UNDER COUNTER REFRIGERATOR	1	115V/1Ø	3.0	1/5	0.345	K	30	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
CO1	CONVENIENCE RECEPTACLE	-	120V/1Ø	1.5	-	0.180	SEE PLANS	SEE PLANS	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-20P

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
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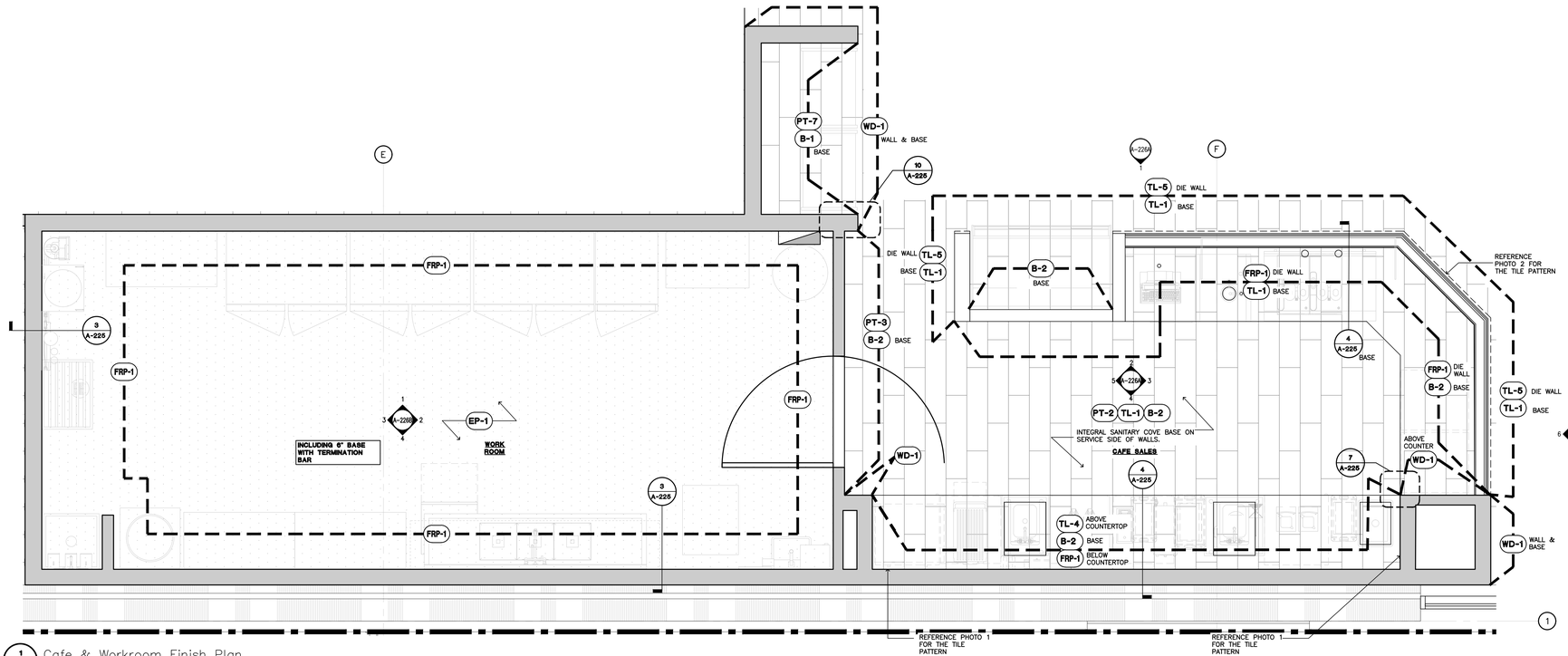
Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
COMMER ARCHITECT
LICENSED ARCHITECT

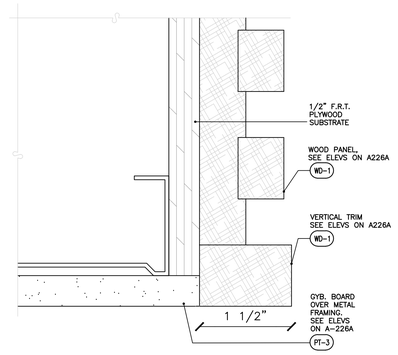
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:
Cafe & Cafe Workroom Equipment Plan and Schedule

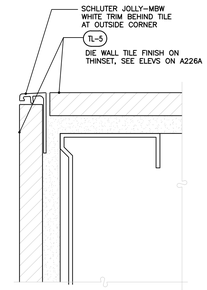
Drawing Number:
A-224



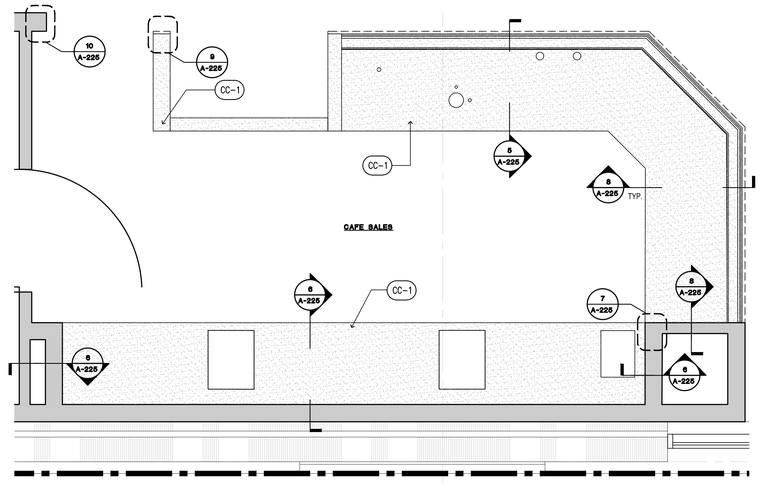
1 A-225 Cafe & Workroom Finish Plan
SCALE: 1/2" = 1'-0"



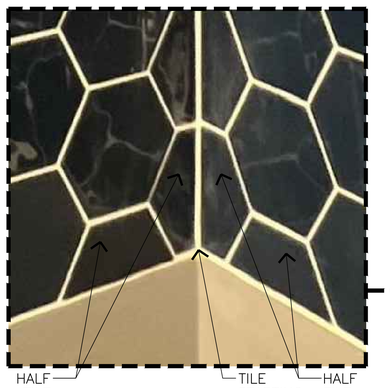
10 A-225 Detail – GWB & Wood Corner
SCALE: FULL SIZE



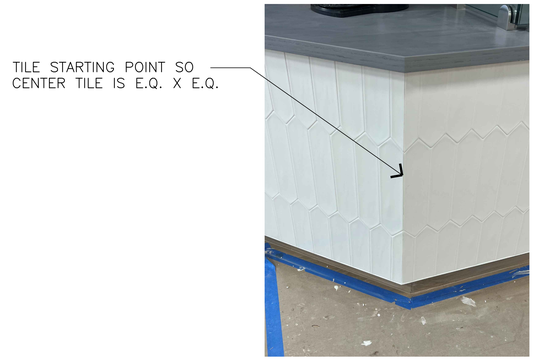
9 A-225 Detail – Tile Corner Edge
SCALE: FULL SIZE



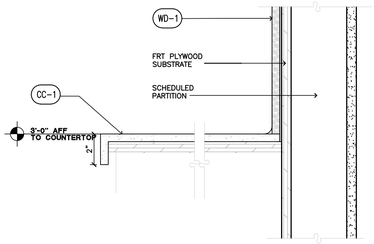
2 A-225 Cafe Countertop Plan
SCALE: 1/2" = 1'-0"



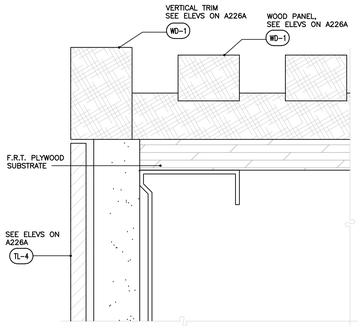
A A-225 Reference Photo 1
SCALE: N.T.S.



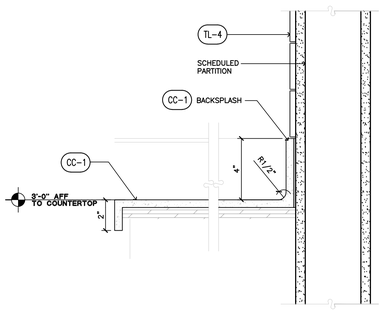
B A-225 Reference Photo 2
SCALE: N.T.S.



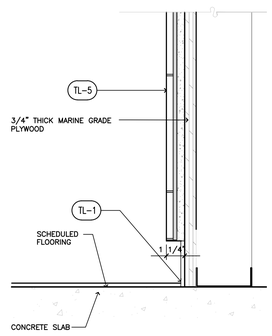
8 A-225 Back Countertop Detail
SCALE: 3"=1'-0"



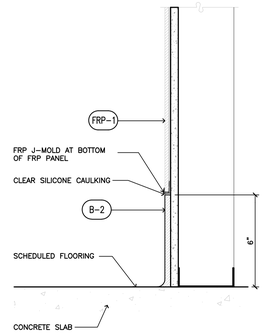
7 A-225 Detail – Tile & Wood Corner Edge
SCALE: FULL SIZE



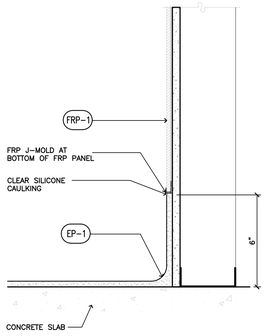
6 A-225 Back Countertop Detail
SCALE: 3"=1'-0"



5 A-225 Wall Base Detail at Die Wall
SCALE: 3"=1'-0"



4 A-225 Wall Base Detail at Cafe Sales
SCALE: 3"=1'-0"



3 A-225 Wall Base Detail at Workroom
SCALE: 3"=1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
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Professional Seal:

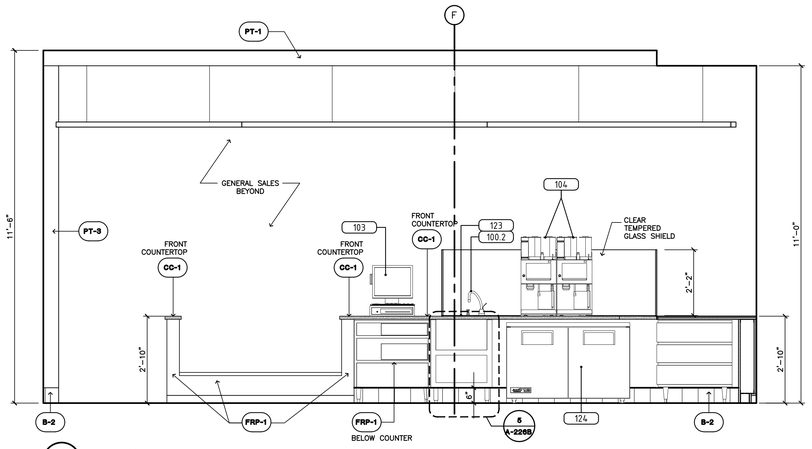
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

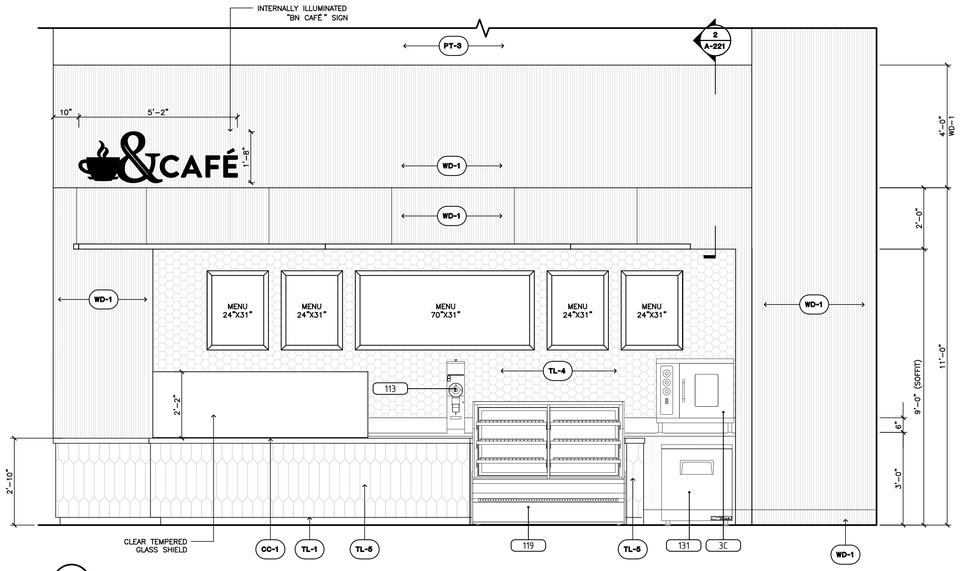
Cafe & Cafe Workroom Millwork Details

Drawing Number:

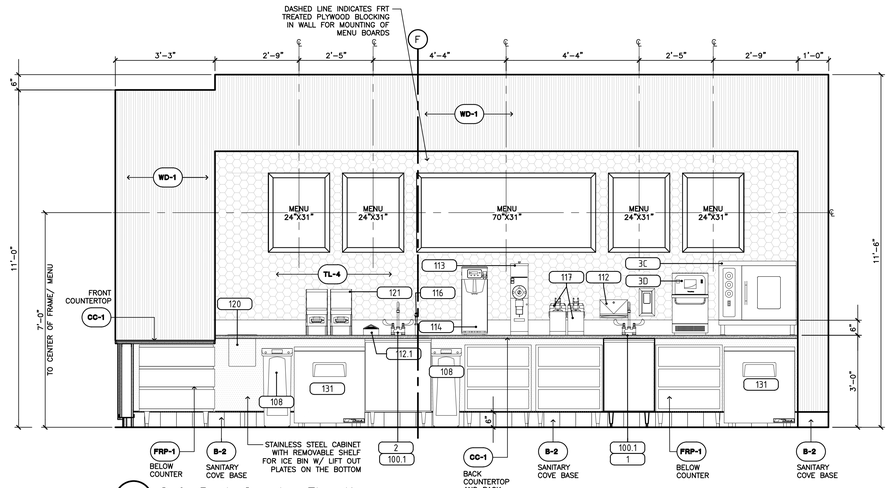
A-225



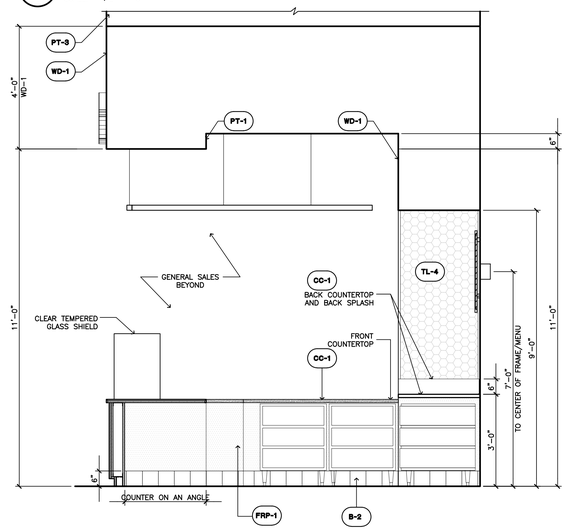
2 A-226A Cafe Counter Elevation
SCALE: 1/2"=1'-0"



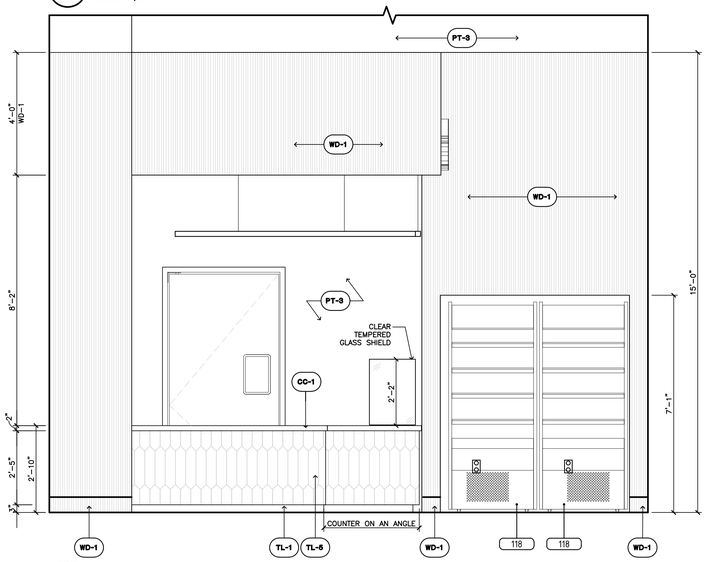
1 A226A Cafe Elevation
SCALE: 1/2"=1'-0"



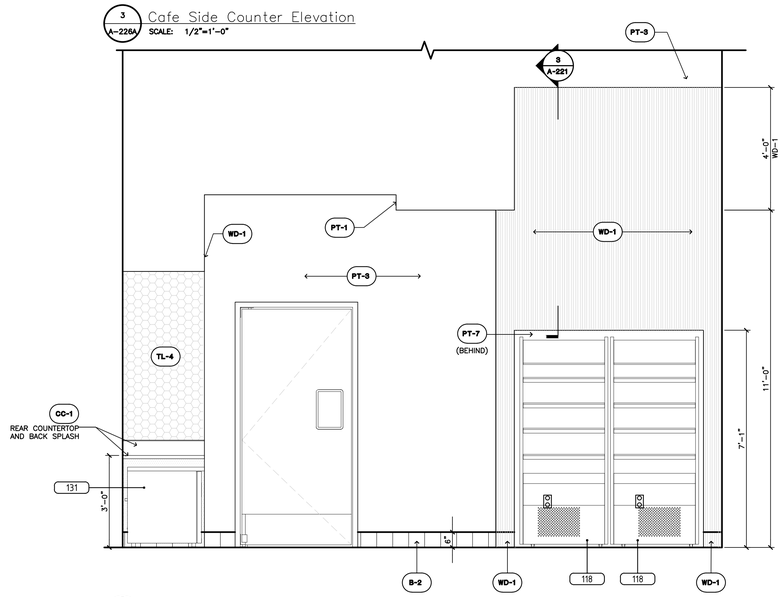
4 A226A Cafe Back Counter Elevation
SCALE: 1/2"=1'-0"



3 A-226A Cafe Side Counter Elevation
SCALE: 1/2"=1'-0"



6 A-226A Cafe Counter Elevation
SCALE: 1/2"=1'-0"



5 A-226A Cafe Service Line Elevation
SCALE: 1/2"=1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

These plans are an instrument of service and the property of the Architect. Infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

STATE OF COLORADO
WILLIAM J COMMER
305353
LICENSED ARCHITECT

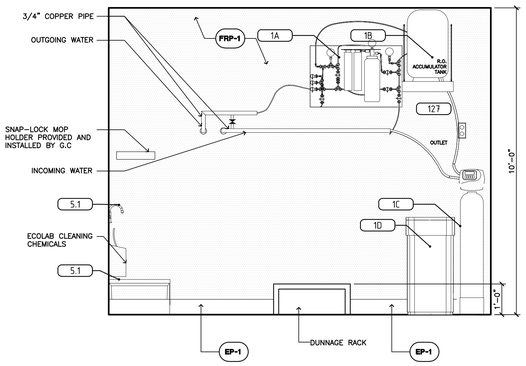
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

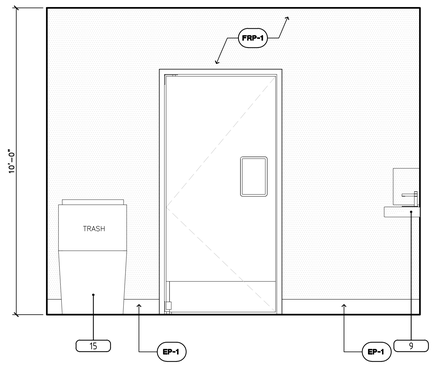
Cafe Elevations

Drawing Number:

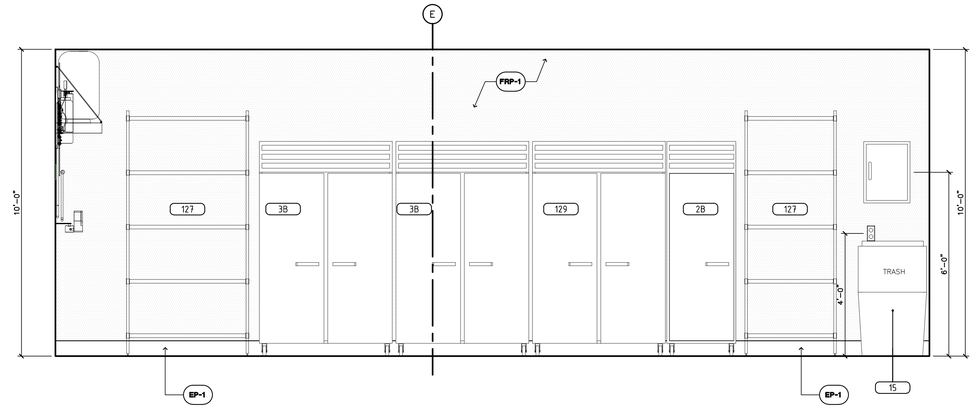
A-226A



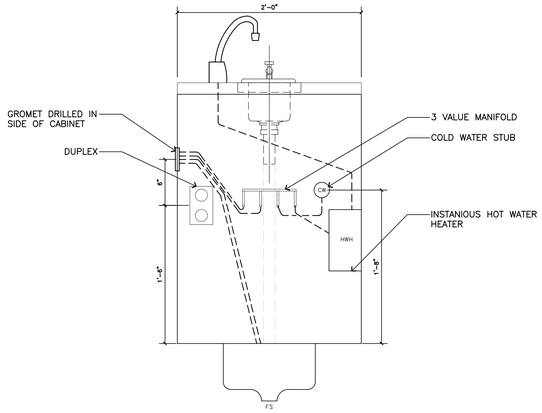
3 A-226B Workroom Elevation
SCALE: 1/2"=1'-0"



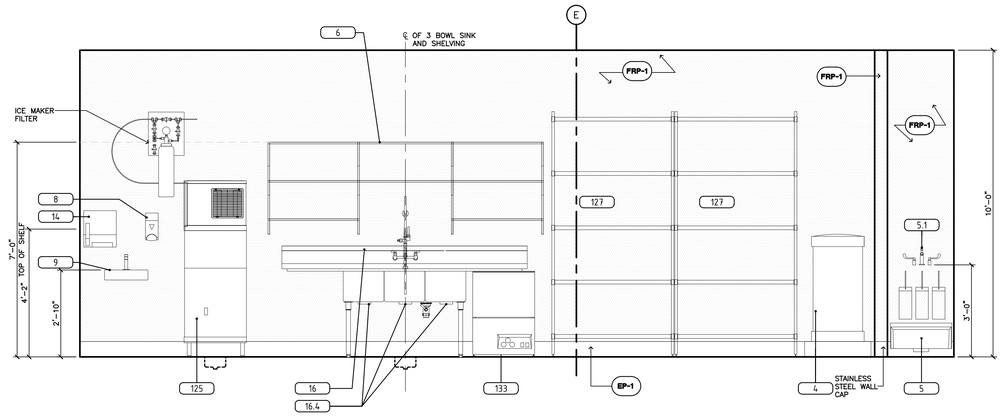
2 A-226B Workroom Elevation
SCALE: 1/2"=1'-0"



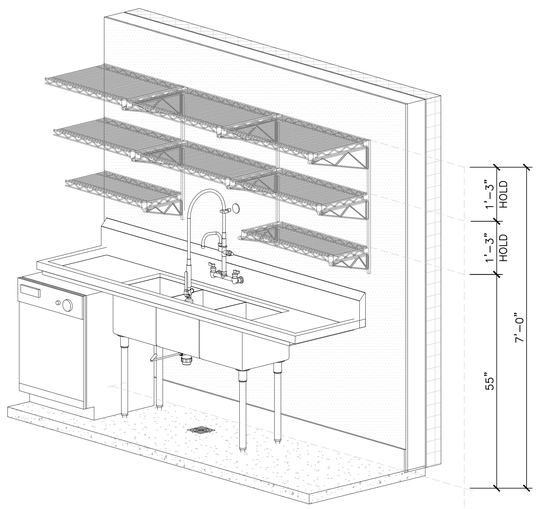
1 A-226B Workroom Elevation
SCALE: 1/2"=1'-0"



5 A-226B Dipperwell With Faucet Detail
SCALE: N.T.S.



4 A-226B Workroom Elevation
SCALE: 1/2"=1'-0"



7 A-226B AXONOMETRIC 3-BOWL SHELVING DETAIL
SCALE: N.T.S.
SIM.

6 A-226B Reference Photo #1
SCALE: N.T.S.
SIM.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

STATE OF COLORADO
WILLIAM J COMMER
LICENSED ARCHITECT

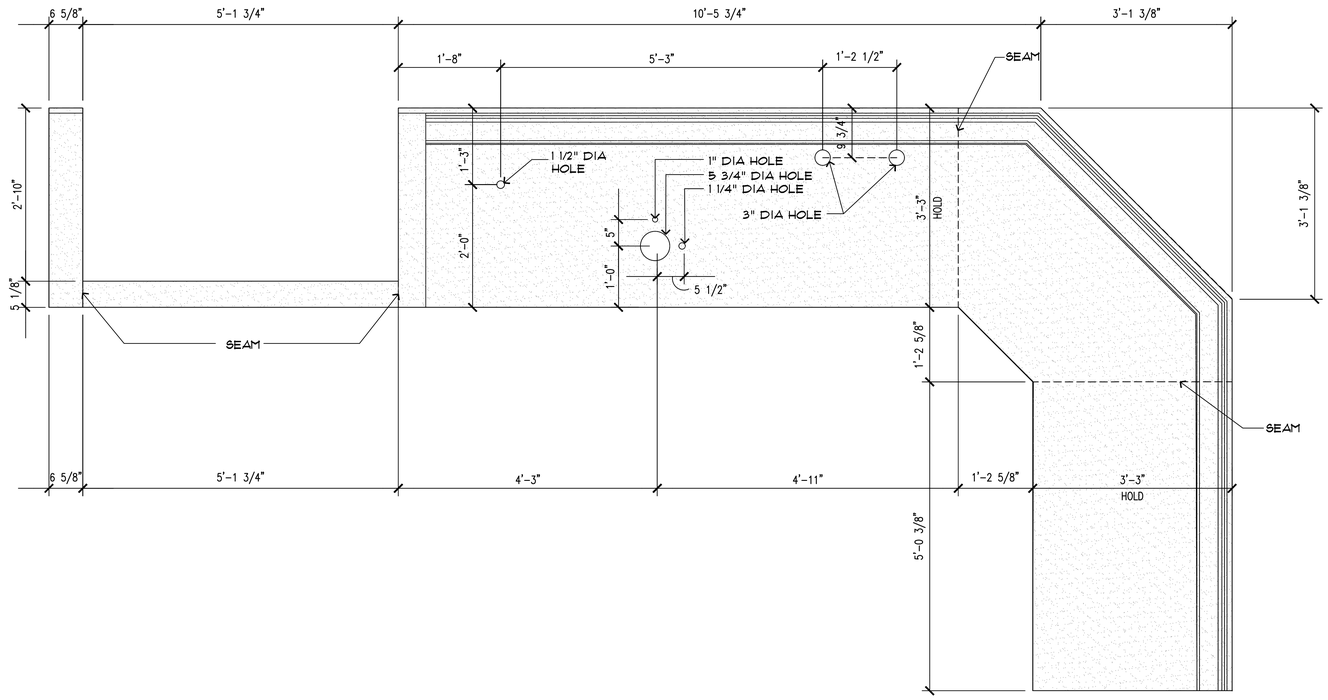
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

Cafe Workroom Elevations

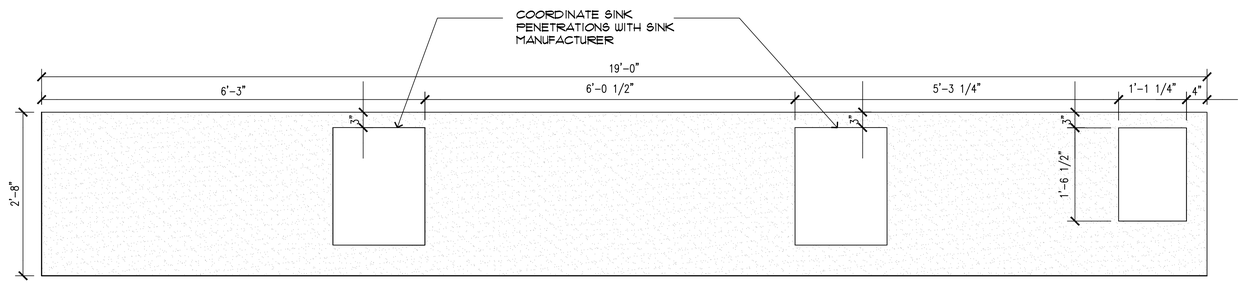
Drawing Number:

A-226B



1 A-227 Enlarged Cafe Sales Countertop Plan

SCALE: 1" = 1'-0"



2 A-227 Enlarged Cafe Sales Countertop Plan

SCALE: 1" = 1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
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80922

Project Number 33288
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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

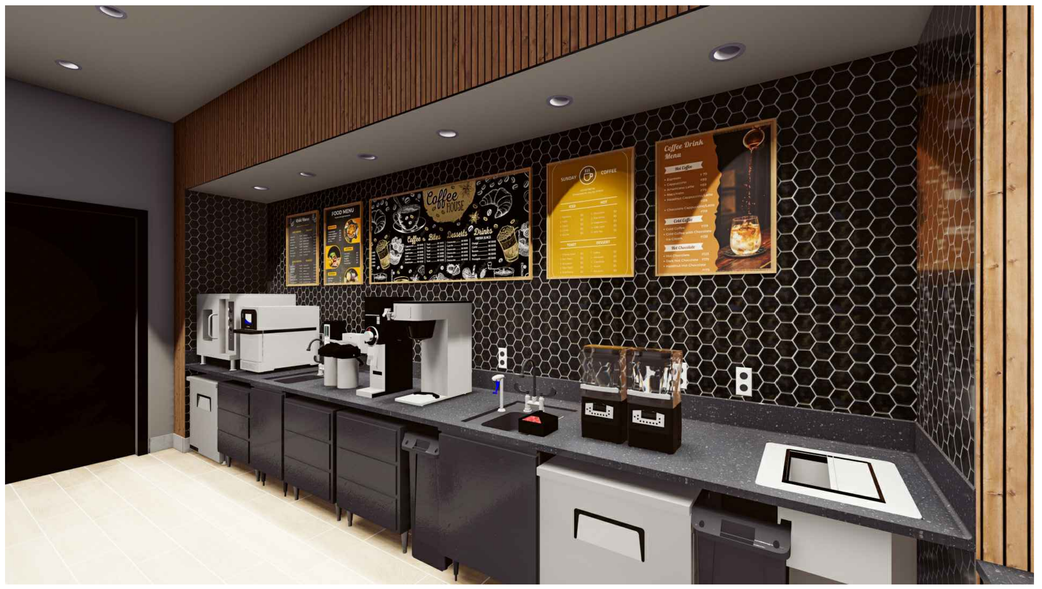
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

Enlarged Countertop
Penetration Plan

Drawing Number:

A-227



BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
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80922

Project Number 33288
Store Number 3566

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Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
LICENSED ARCHITECT

LIC.# : 305353
EXP. DATE : 10/31/25

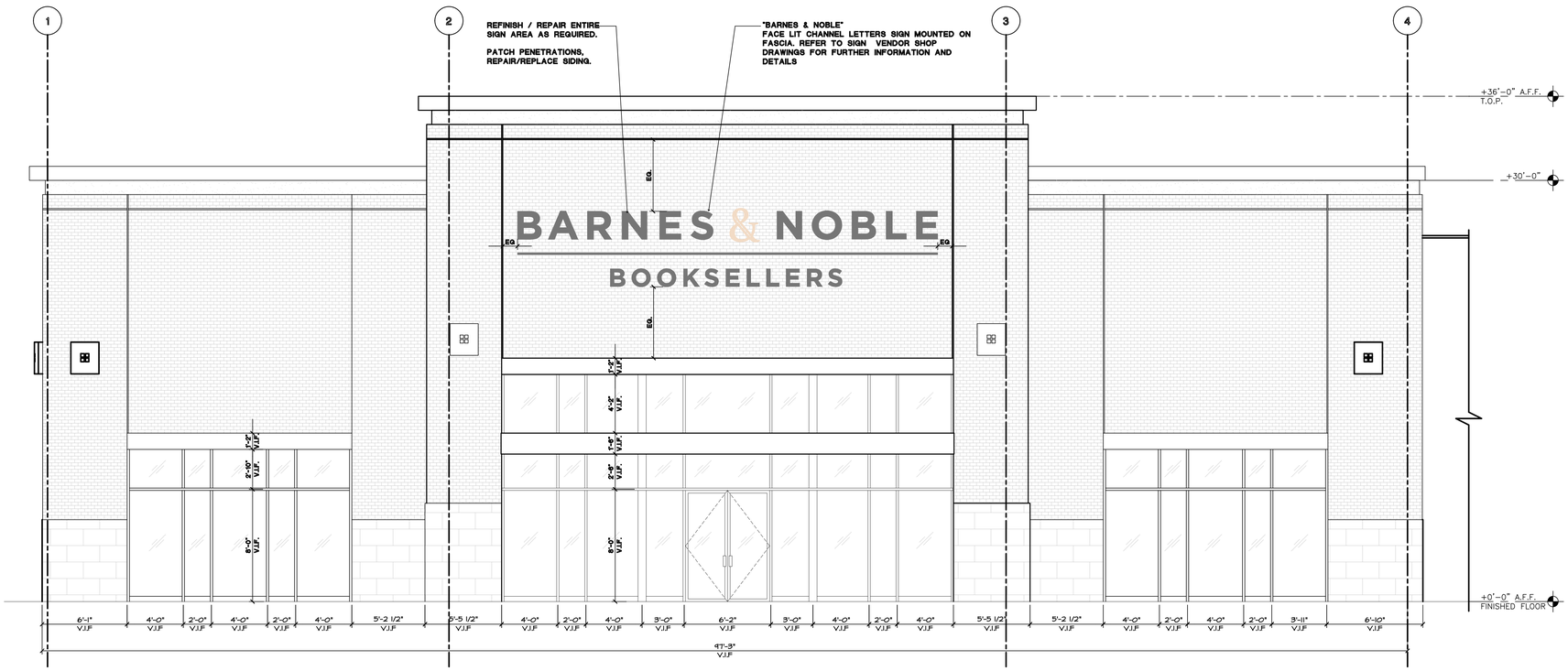
Drawing Description:

Cafe Renderings

Drawing Number:

A-228

NOTE: ALL SIGNAGE UNDER SEPARATE PERMIT.
EXTERIOR SCOPE LIMITED TO REPAIR/PATCHING EXISTING MATERIALS TO MATCH



1 / A-300 Front Elevation
SCALE: 1/4"=1'-0"

NOTE:

TENANT SIGNAGE PROVIDED BY B&N. PERMIT TO BE OBTAINED BY SIGN VENDOR. REFER TO VENDOR SHOP DRAWINGS FOR FURTHER INFORMATION AND DETAILS

THE FINAL LOCATION OF SIGN POWER NEEDS TO BE DETERMINED WITH B&N SIGN VENDOR

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

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Professional Seal:

STATE OF COLORADO
WILLIAM J COMMER
LICENSED ARCHITECT

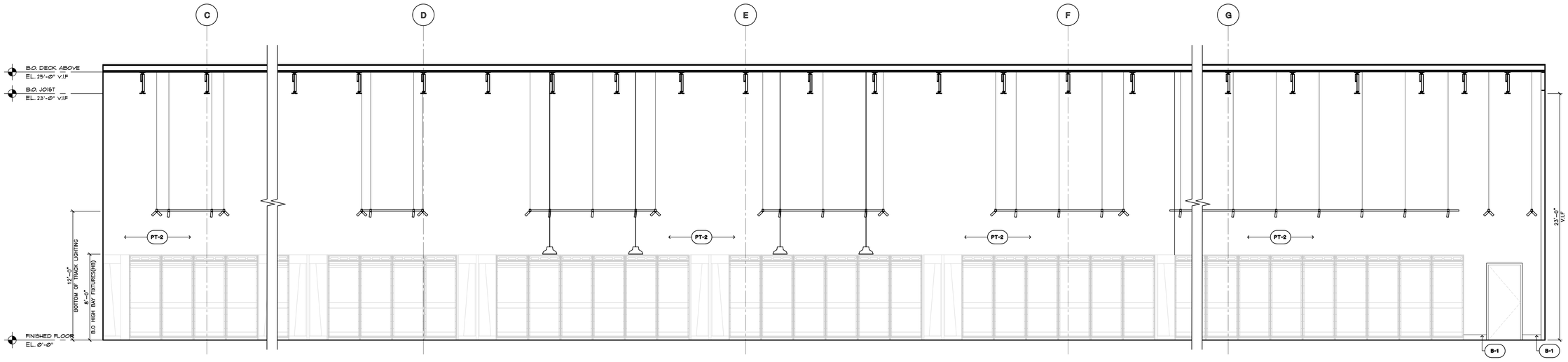
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

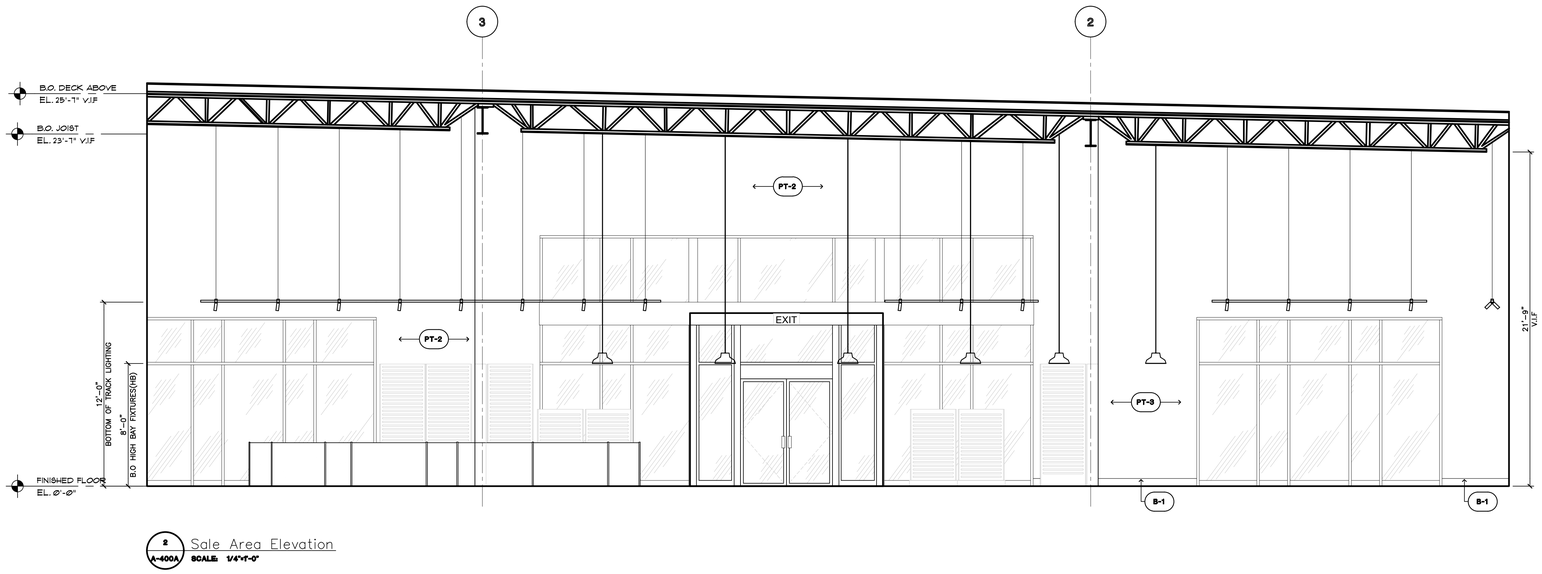
Front Exterior Elevation

Drawing Number:

A-300



1 A-400A Sale Area Elevation
SCALE 1/4"=1'-0"



2 A-400A Sale Area Elevation
SCALE 1/4"=1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

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Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
REGISTERED
LICENSED ARCHITECT

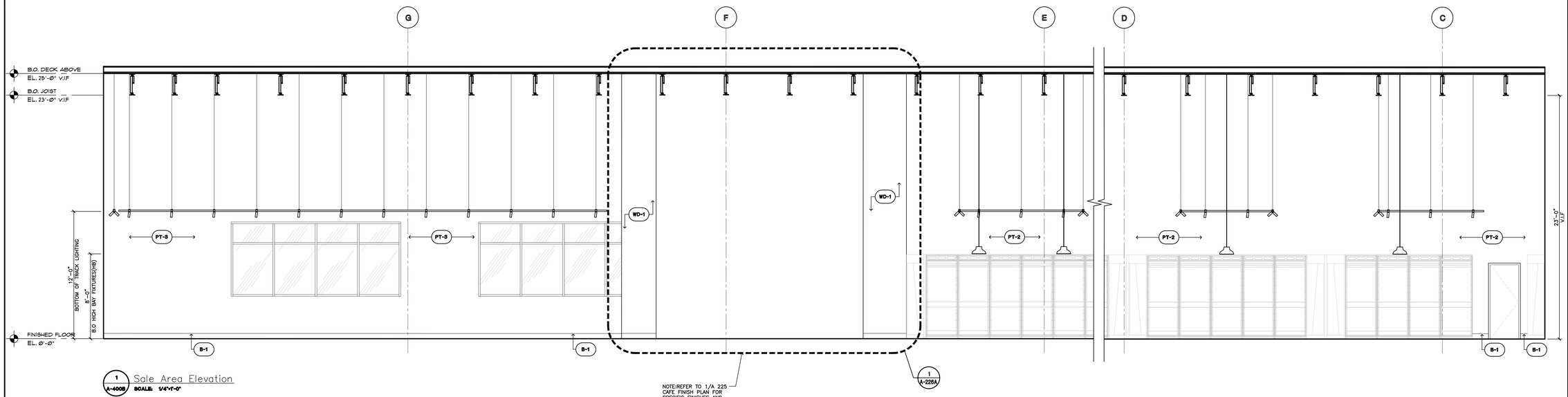
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

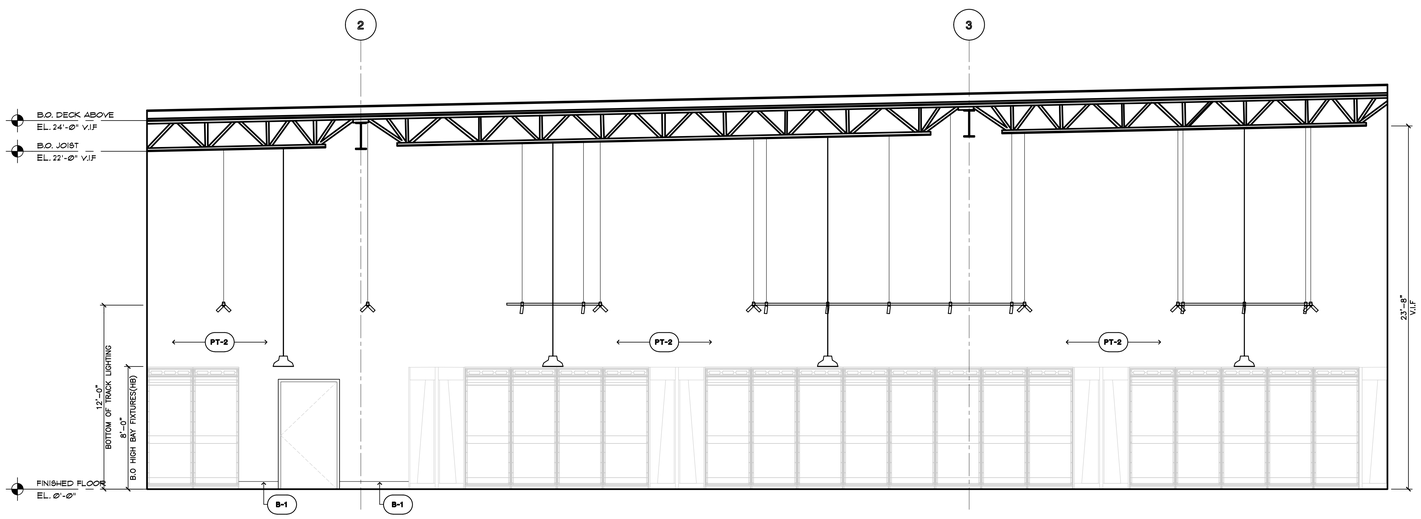
Sales Area Elevations

Drawing Number:

A-400A



1 A-400B Sale Area Elevation
SCALE: 1/4"=1'-0"



2 A-400B Sale Area Elevation
SCALE: 1/4"=1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
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WILLIAM J. COMMER
LICENSED ARCHITECT

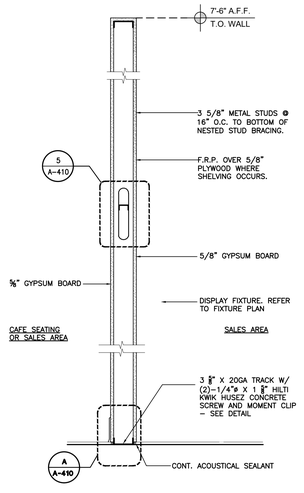
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

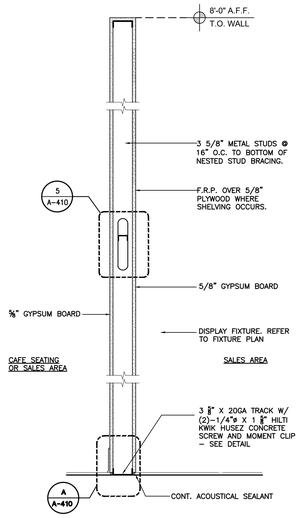
Sales Area Elevations

Drawing Number:

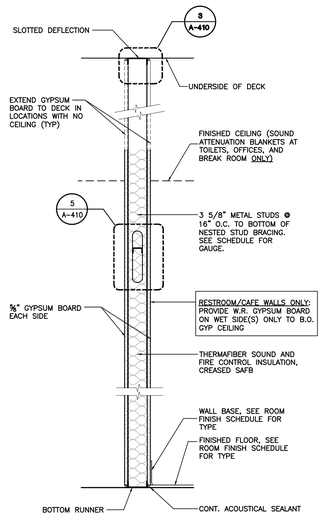
A-400B



4 Non-Rated Partial Height Partition
SCALE: 1"=1'-0"



3 Non-Rated Partial Height Partition
SCALE: 1"=1'-0"

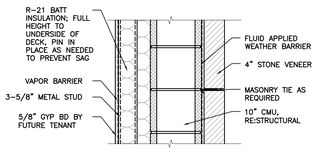


2 Non-Rated Full Height Partition
SCALE: 1"=1'-0"

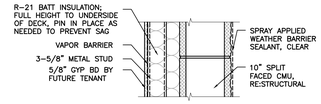
* PROVIDE ⅝" WATER RESISTANT GYP. BD AT JANITOR'S CLOSET, DRINKING FOUNTAIN, KITCHEN AND ALL TOILET LOCATIONS.

** SEE ELEVATIONS FOR ALL FINISHES. COLOR AND TEXTURE TBD. SEE SCHEDULE FOR MORE INFORMATION.

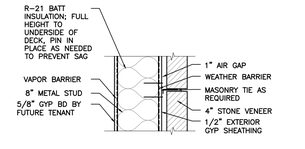
*** FINISH EXPOSED SIDE ONLY



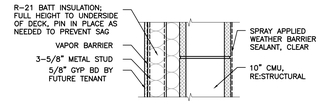
1C Exterior Wall-Stone at Stud
SCALE: 1"=1'-0"



1A Exterior-Split-Faced CMU
SCALE: 1"=1'-0"



1B Exterior Wall-Stone at CMU
SCALE: 1"=1'-0"



1 Exterior CMU
SCALE: 1"=1'-0"

METAL STUD AND BRACE SCHEDULE

METAL STUD SCHEDULE

SIZE	GA.	TYPE	16" O.C. MAX. UNSUPPORTED HEIGHT (1) (2) (3)	24" O.C.
2 1/2"	20	ST	" " 13'-10"	12'-0"
3 5/8"	20	ST	" " 17'-11"	15'-7"
	20	SJ	" " 18'-6"	16'-9"
	18	SJ	" " 19'-3"	
	16	SJ	" " 20'-0"	
	14	SJ	" " 22'-0"	
6"	20	ST	" " 26'-1"	22'-10"
	20	SJ	" " 26'-8"	23'-8"
	18	SJ	" " 28'-0"	
	16	SJ	" " 30'-0"	
	14	SJ	" " 32'-0"	

1. BASED ON INTERIOR NON-BEARING PARTITIONS WITH ONE LAYER OF GYPSUM BOARD EACH SIDE. USE GAUGE, TYPE AS SCHEDULED UNLESS NOTED OTHERWISE ON DRAWINGS.
2. RUNNER TRACK GAUGE TO MATCH STUD.
3. WALLS OVER 12'-0" HIGH TO BE BRACED W/ 1 1/2", 16 GA. COLD-ROLLED CHANNELS CLIP ANCHORED TO METAL STUDS.

METAL STUD BRACE SCHEDULE

METAL STUD SIZE	MAXIMUM UNBRACED HEIGHT
3 5/8" X 25 GA.	11'-3"
3 5/8" X 20 GA.	14'-0"
3 5/8" X 16 GA.	18'-0"
6" X 25 GA.	16'-0"
6" X 20 GA.	21'-0"
6" X 16 GA.	25'-0"

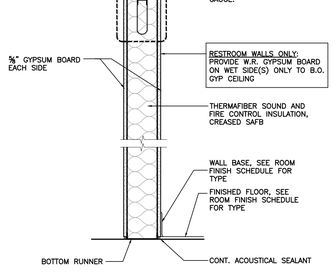
STUD NOTES

"e" - WEB STIFFENERS REQUIRED AT ENDS.

1. STUD GAUGE MAY VARY PER WALL TYPE DEPENDING ON THE ACTUAL HEIGHT AND SHEATHING TYPE REQUIRED.
2. WEB-HEIGHT TO THICKNESS RATIO EXCEEDS 200. WEB STIFFENERS ARE REQUIRED AT ALL SUPPORT POINTS AND CONCENTRATED LOADS.
3. 5 PSF (POUNDS PER SQUARE FOOT), 7.5 PSF AND 10 PSF LOADS HAVE NOT BEEN REDUCED FOR STRENGTH DEFLECTION CHECKS, FULL LATERAL LOADS ARE APPLIED.
4. LIMITING HEIGHTS ARE BASED ON PROPERLY FASTENED SHEATHING ON BOTH FLANGES OVER THE FULL HEIGHT OF THE STUD.
5. LIMITING HEIGHTS ARE BASED ON STEEL PROPERTIES ONLY (NON-COMPOSITE).
6. WEB CRIPPLING CHECK BASED ON 1 INCH END BEARING. WHERE LIMITING HEIGHTS ARE FOLLOWED BY "e", WEB STIFFENERS ARE REQUIRED.
7. ALLOWABLE MOMENT IS THE LESSER OF *Mo!* AND *Mod.* STUD DISTORTIONAL BUCKLING BASED ON AN ASSUMED Kϕ = 0.
8. REFER TO SPECIFICATIONS FOR DEFLECTION LIMIT.

PARTITION NOTES

1. ALL METAL STUDS TO CONFORM TO LOCAL BUILDING CODES.
2. ALL STUD WALLS AND PARTITIONS REQUIRE GYP. BOARD ON BOTH SIDES FOR THE FULL HEIGHT OF THE STUDS UNLESS SPECIFICALLY DETAILED OTHERWISE.
3. ALL SUSPENDED WALLS SHALL HAVE GYP. BOARD WHICH EXTENDS 6" ABOVE THE SUSPENDED CEILING UNLESS SHOWN AND/OR NOTED OTHERWISE BY THE ARCHITECT.
4. ALL STUD WALLS AND PARTITIONS SHALL HAVE CONTINUOUS LINES OF BRIDGING SPACED AT 4'-0" MAXIMUM ON CENTER. THE BRIDGING SHALL BE SECURELY FASTENED TO THE STUDS WITH EITHER SCREWS OR WELDS.
5. ALL STUDS SHALL BE "CEE" STUDS WITH FLANGE STIFFENERS.
6. THE MATERIALS AND DETAILS SHOWN ARE FOR TYP. INSTALLATIONS. WHERE THE STUD MANUFACTURER'S RECOMMENDATIONS OR LOCAL ORDINANCES ARE MORE RESTRICTIVE, THEY SHALL APPLY.
7. G.C. TO INSTALL GREENBOARD (OR EQUAL MOLD-RESISTANT GYP. BD) ON ALL TOILET ROOM, JANITORS CLOSET, OR SIMILAR TYPE OF WETROOM AREAS.
8. TYPICAL FASTENER:
   A. STEEL STUDS TO STEEL STUDS OR TRACKS:
      - 25/20 GAUGE - #8 - 18 X 1/2" TEKS WITH PHILIPS PAN HEAD - MIN. 2 PER CONNECTION
      - 18/16 GAUGE - #10 - 16 X 3/4" TEKS WITH PHILIPS PAN HEAD - MIN. 3 PER CONNECTION
   B. STEEL STUDS OR TRACKS TO WOOD PURLINS, GIRDERS AND BEAMS:
      - #14 - 10 X 1-1/2" H.W.H. TYPE "S" METAL-TO-WOOD TEKS AT 12" O.C., 2" FROM EACH END
   C. STEEL STUDS OR TRACKS TO STRUCTURAL STEEL (TUBE/WIDE FLANGE COL'S, BM'S, GIRDERS, ETC.):
      - 0.145" DIA. HILTI X-EDNI FASTENERS, 5/8" MIN, LENGTH AND MIN. OF 2 PER CONNECTION, OR 2 ROWS AT 16" O.C. FOR CONTINUOUS APPLICATIONS SUCH AS TRACKS, U.O.N.
   D. PLYWOOD TO STEEL STUDS:
      - #10 - 24 X 1-1/4" TEKS/3 (PLYMETALTEKS) WITH THIN WAFER HEAD - 24" O.C. FILED, 12" O.C. PERIMETER
   E. GYPSUM BOARD TO STEEL STUDS:
      - #7 X 1-1/4" HI-LOW TYPE "S" BUGLE HEAD SCREWS FOR 3/8" TO 5/8" GYPSUM BOARD TO 25 GAUGE OR 20 GAUGE STUDS
      - #6 X 1-1/4" TYPE S-12 BUGLE HEAD SCREWS FOR 3/8" TO 5/8" TO 18 GAUGE OR 16 GAUGE STUDS OR TRACKS
      - FIELD - 12" O.C. CEILINGS, 24" O.C. WALLS
      - BUTT JOINTS 12" O.C.
   F. STEEL STUDS OR TRACKS TO CONCRETE:
      - 0.145" DIA. HILTI X-DNI FASTENER, 1" MIN. LENGTH, 2 ROWS AT 16" O.C.



5 A-410 Non-Rated Full Height Partition
SCALE: 1"=1'-0"

BUILDING DEPARTMENT NOTE:

ALL PLYWOOD NEEDS TO BE FRT.

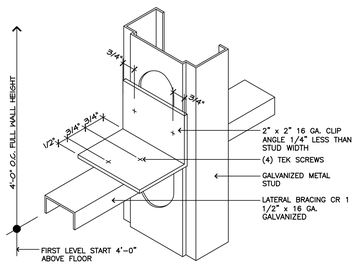
NOTE:

PROVIDE UL-LISTED METALLIC ELECTRICAL OUTLET BOXES IN FIRE-RESISTANCE-RATED WALL ASSEMBLIES.

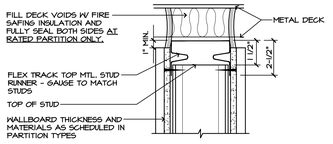
BOXES TO BE INSTALLED IN ACCORDANCE WITH UL DESIGN AND IBC REQUIREMENTS.

WHERE BOXES EXCEED SIZE / SPACING LIMITATIONS, APPLY INTUMESCENT PUTTY (E.G., 3M, HILTI, OR STI) PER MANUFACTURER'S UL-LISTED DETAIL TO MAINTAIN FIRE-RESISTANCE RATING.

THE LOW VOLTAGE WIRING AND ASSOCIATED CONDUITS MUST BE A UL-RATED ASSEMBLY.

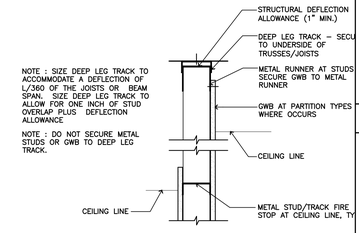


5 A-410 Lateral Bracing Detail at Partition
SCALE: N.T.S.



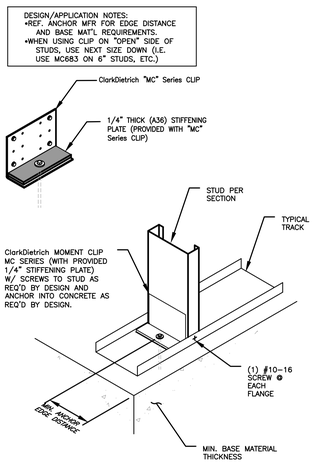
NOTES:
1. WALLBOARD SHALL BE FASTENED TO TOP HORIZONTAL RUNNER LEG BELOW BELLOWS.
2. FIRE TAPE ALL DRYWALL JOINTS ABOVE CEILINGS, SEAL AROUND ALL PENETRATIONS THRU OR INTO WALLS. FOR RATED PARTITIONS ONLY.
3. PROVIDE FIRE STOP CAULKING AT ALL U.L. RATED PARTITIONS.
4. PROVIDE ACOUSTICAL SEALANT AT ALL ACOUSTICAL PARTITIONS.

4 A-410 Slip track / Firestop Detail at Partition
SCALE: N.T.S.

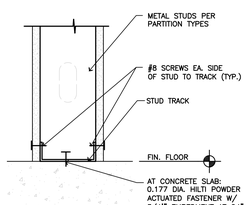


PROVIDE DIAGONAL BRACING AT 32" O.C. TO STRUCTURE ABOVE (TYP.)

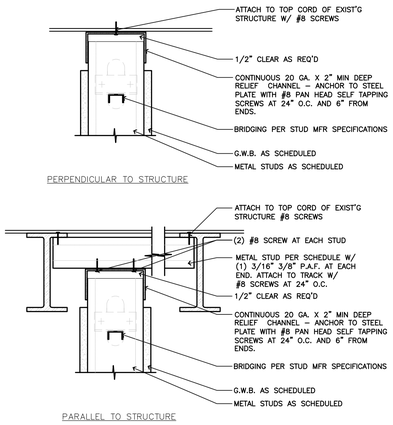
8 A-410 Top Track Detail
SCALE: N.T.S.



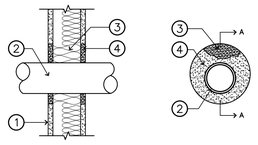
A Moment Clip Connection Detail
SCALE: 1"=1'-0"



9 A-410 Typ. Floor Track Detail
SCALE: 3" = 1'-0"



8 A-410 Typ. Connection Detail
SCALE: 3" = 1'-0"

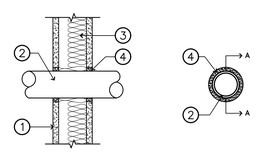


SECTION A-A

1. WALL ASSEMBLY - 1 HOUR FIRE RESISTANT GYPSUM PANEL WALL ASSEMBLY.
2. UL DESIGN NO. U465
   - NOMINAL 4"ø OR SMALLER SCHEDULE 10 OR HEAVIER STEEL PIPE. A MIN. 1/4" TO MAX. 1-1/4" ANNULAR SPACE BETWEEN PIPE AND PERIPHERY OF OPENING IS REQUIRED - 0 HOUR T RATING.
   - NOMINAL 4"ø OR SMALLER CAST OR DUCTILE IRON PIPE. A MIN. 1/4" TO MAX 1-1/4" ANNULAR SPACE BETWEEN PIPE AND PERIPHERY OF OPENING IS REQUIRED - 0 HOUR T RATING.
   - NOMINAL 4"ø OR SMALLER ELECTRICAL METALLIC TUBING OR STEEL CONDUIT. A MIN. 1/4" TO MAX 1-1/4" ANNULAR SPACE BETWEEN PIPE AND PERIPHERY OF OPENING IS REQUIRED. A 1/2"ø OR SMALLER ELECTRICAL METALLIC TUBING OR STEEL CONDUIT HAS 1 HOUR RATING; LARGER THAN 1/2"ø HAS 0 HOUR T RATING.
   - NOMINAL 3/4"ø OR SMALLER TYPE L OR HEAVIER COPPER TUBING. A MIN. 1/4" TO MAX 1-13/16" ANNULAR SPACE BETWEEN PIPE AND PERIPHERY OF OPENING IS REQUIRED - 0 HOUR T RATING.
   - NOMINAL 4"ø OR SMALLER TYPE L OR HEAVIER COPPER PIPE. A MIN. 3/8" TO MAX 1-1/4" ANNULAR SPACE BETWEEN PIPE AND PERIPHERY OF OPENING IS REQUIRED - 0 HOUR T RATING.
3. FORMING MATERIAL - MINIMUM 2-1/2" SAFING THERMAFIBER SAFING INSULATION WITH MINIMUM DENSITY OF 3.5 PCF - U.L. R-10905 LABEL.
4. FILL VOID OR CAVITY MATERIAL - MINIMUM 1/2" DEPTH OF FIRECODE COMPOUND INSTALLED ON EACH SIDE OF THE WALL.

Thru Penetration Firestopping Detail and Notes
U.L. Design No. WL-1039

7 A-410 Typ. Rated Thru Wall Penetration Detail
SCALE: N.T.S.



SECTION A-A

1. WALL ASSEMBLY - 2 HOUR FIRE RESISTANT GYPSUM PANEL WALL ASSEMBLY. DESIGN NO. U491
2. METALLIC PIPE - NOMINAL 3-1/2" ø SCHEDULE 10 OR HEAVIER STEEL PIPE. **PIPE SHOULD BE CENTERED WITHIN OPENING.** A MIN. 3/8" TO MAX 5/8" ANNULAR SPACE BETWEEN PIPE AND PERIPHERY OF OPENING IS REQUIRED.
3. FORMING MATERIAL - MINIMUM 1/2" SAFING THERMAFIBER SAFING INSULATION RECESSED FROM BOTH SURFACES TO ACCOMMODATE THE CAULK FILL MATERIAL.
4. FILL, VOID OR CAVITY MATERIAL - CAULK- MINIMUM 1/2" DEPTH OF FIRECODE COMPOUND INSTALLED ON EACH SIDE OF THE WALL.

Thru Penetration Firestopping Detail and Notes
U.L. Design No. WL-1064

6 A-410 Typ. Rated Thru Wall Penetration Detail
SCALE: N.T.S.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

STATE OF COLORADO WILLIAM J COMMER 305353 LICENSED ARCHITECT

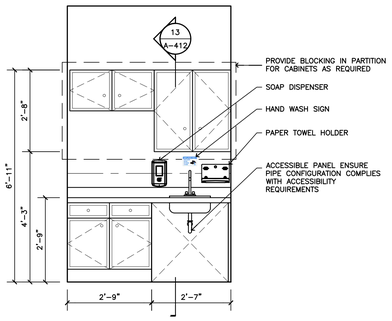
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

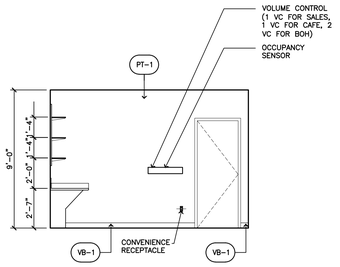
Partition Types

Drawing Number:

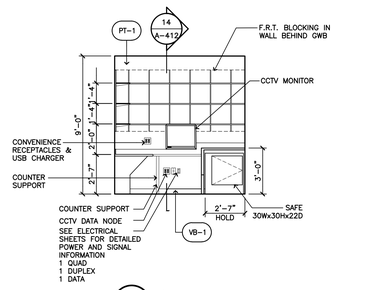
A-410



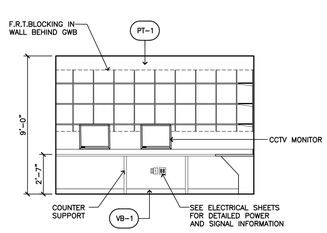
6 A-412 Typical Breakroom Kitchen Unit Elevation
SCALE: 1/2" = 1'-0"



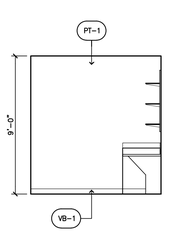
5 A-412 Office Elevation
SCALE: 1/4"=1'-0" (COORDINATE WITH M.E.P. DWGS.)



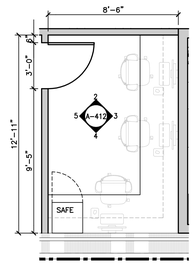
4 A-412 Office Elevation
SCALE: 1/4"=1'-0" (COORDINATE WITH M.E.P. DWGS.)



3 A-412 Office Elevation
SCALE: 1/4"=1'-0" (COORDINATE WITH M.E.P. DWGS.)



2 A-412 Office Elevation
SCALE: 1/4"=1'-0" (COORDINATE WITH M.E.P. DWGS.)



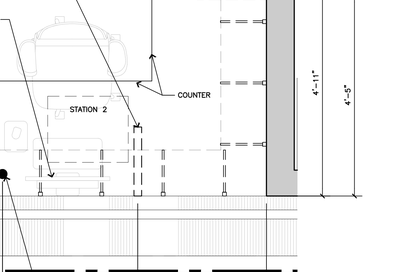
1 A-412 Office Plan
SCALE: 1/4"=1'-0"

11 A-412 Break Room Elevation
SCALE: 1/4"=1'-0"

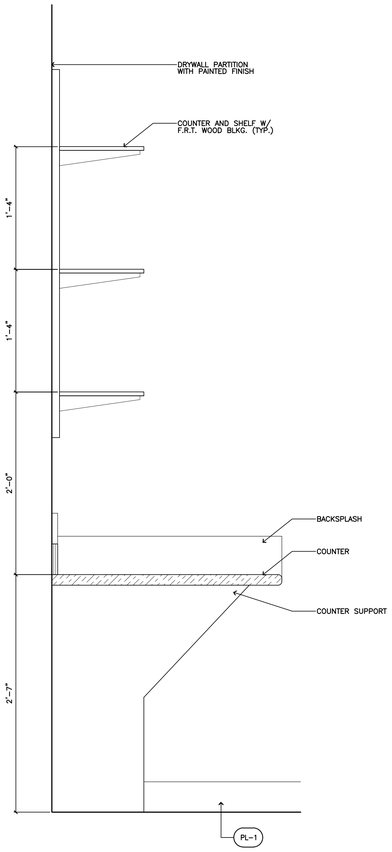
10 A-412 Break Room Elevation
SCALE: 1/4"=1'-0"

9 A-412 Break Room Elevation
SCALE: 1/4"=1'-0"

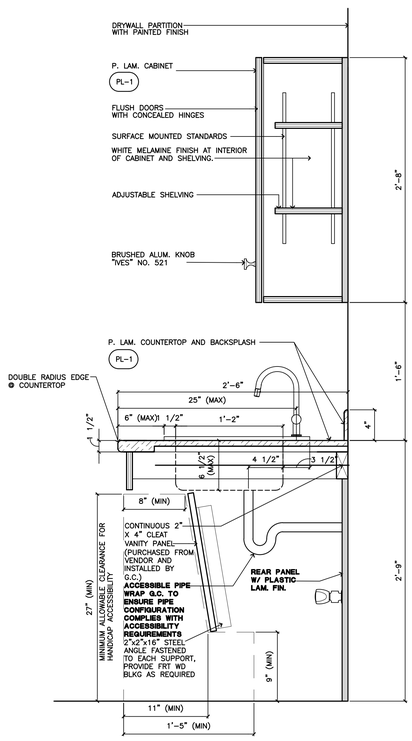
8 A-412 Break Room Elevation
SCALE: 1/4"=1'-0"



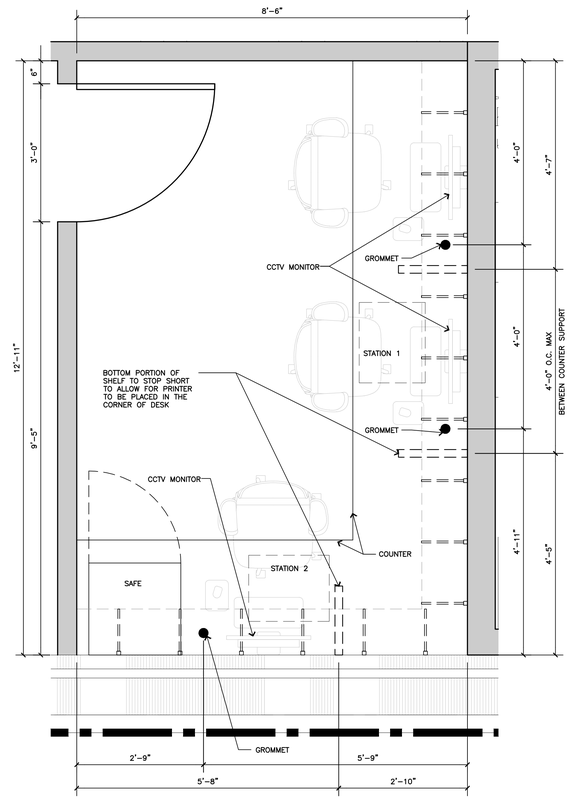
7 A-412 Break Room Plan
SCALE: 1/4"=1'-0"



14 A-412 Counter Section / Detail
SCALE: 1 1/2" = 1'-0"



13 A-412 Cabinet Section / Detail
SCALE: 1 1/2" = 1'-0"



12 A-412 Counter and Shelving Detail at Office
SCALE: 3/4" = 1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
305353
LICENSED ARCHITECT

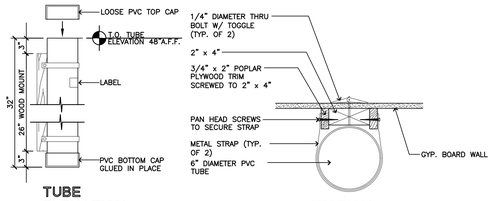
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

Enlarged Break Room and Office Plan, Elevations and Details

Drawing Number:

A-412



A TUBE ELEVATION

NOTE:
ENTIRE ASSEMBLY IS CONSTRUCTED BY GENERAL CONTRACTOR IN FIELD

G.C. MUST PROVIDE AND ATTACH A LABEL (WITH CLEAR TAPE) STATING:

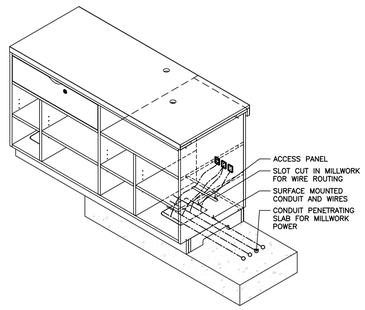
THIS TUBE CONTAINS STORE DRAWINGS AND IS NOT TO BE REMOVED WITHOUT THE CONSENT OF THE STORE MANAGER.

THANK YOU.

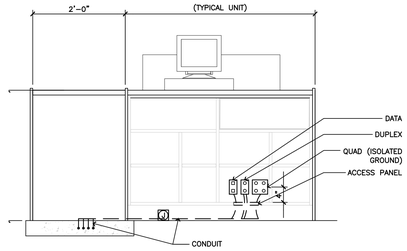
B SECTION THRU TUBE

NOTE:
UPON COMPLETION OF PROJECT, GENERAL CONTRACTOR MUST RUN A NEW BLUEPRINT SET FOR THE COMPLETED STORE DRAWINGS (ARCHITECTURAL, MECHANICAL & ELECTRICAL) AND PLACE BLUEPRINTS INTO THE TUBE HOLDER

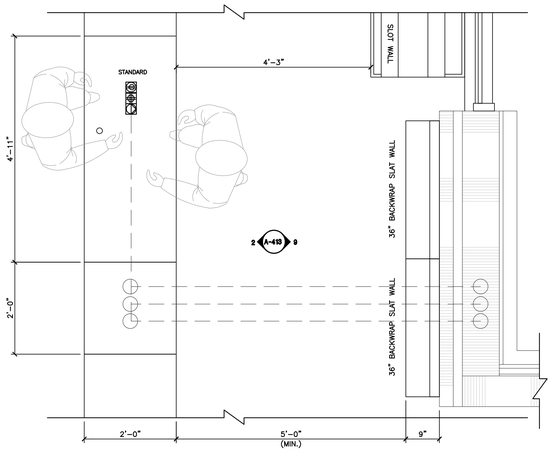
4 Reference Photo
A-413 SCALE: 3/4"=1'-0"



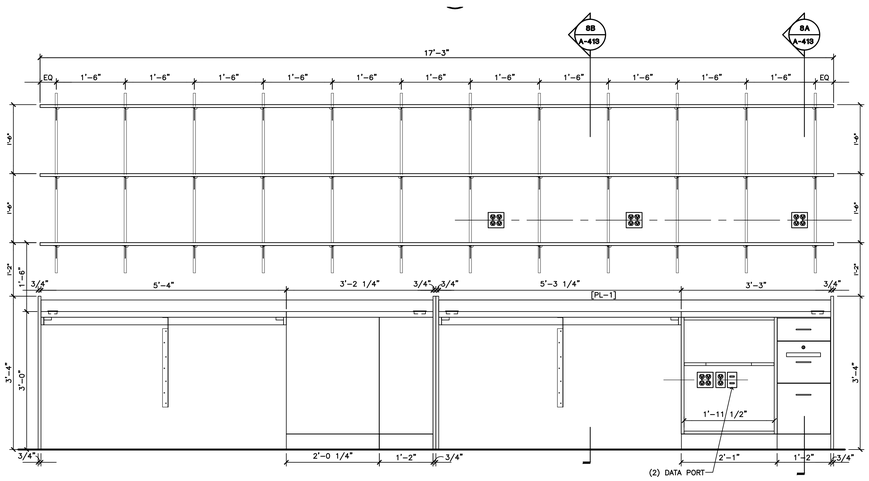
3 Isometric Detail
A-413 SCALE: 1"=1'-0"



2 Typical Cash Wrap Elevation
A-413 SCALE: 3/4"=1'-0" (COORDINATE WITH MEP DRAWINGS)

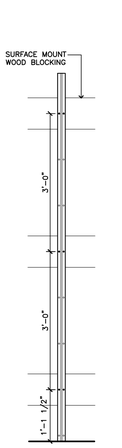


1 Typical Cash Wrap Plan Detail
A-413 SCALE: 3/4"=1'-0"

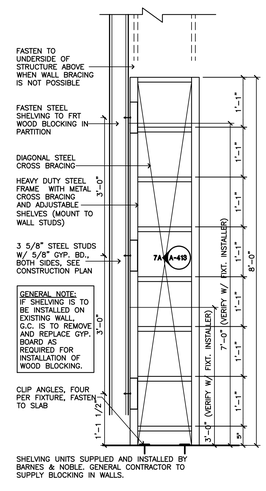


5 Plan Tube Holder
A-413 SCALE: N.T.S.

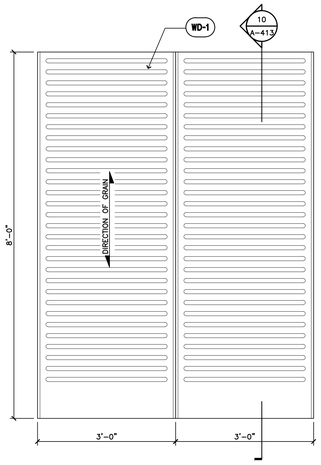
6 Typical Receiving Table Elevation
A-413 SCALE: 3/4" = 1'-0" (COORDINATE WITH MEP DRAWINGS)



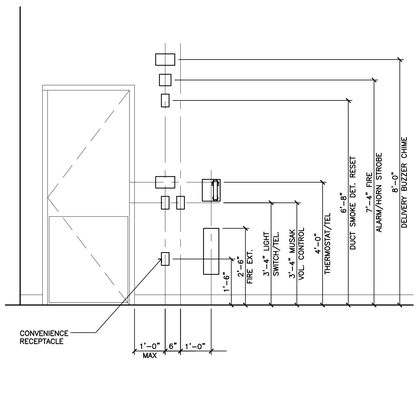
7A Shelving Post Detail
A-413 SCALE: 3/4"=1'-0"



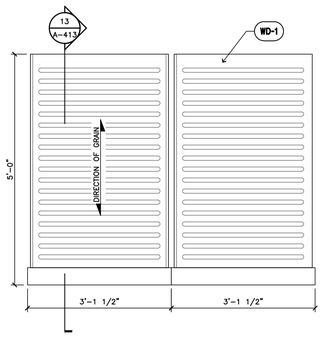
7 Shelving Detail at Stock Rm
A-413 SCALE: 3/4"=1'-0"



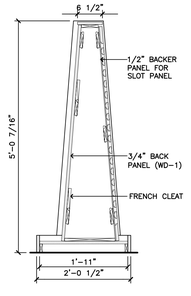
9 TYP. BW-Tall Slatwall Elevation
A-413 SCALE: 3/4"=1'-0"



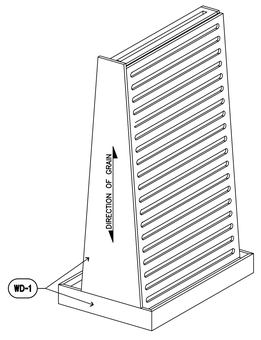
6 Typical Stock Room Elevation
A-413 SCALE: 1/2"=1'-0" (COORDINATE WITH MEP DRAWINGS)



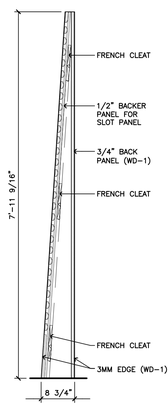
12 TYP. BW-Tall Slatwall Elevation
A-413 SCALE: 3/4"=1'-0"



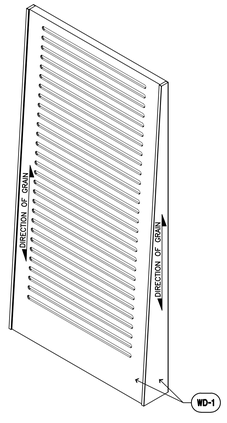
13 Typ. BW-Tall Slatwall Elev.
A-413 SCALE: 3/4"=1'-0"



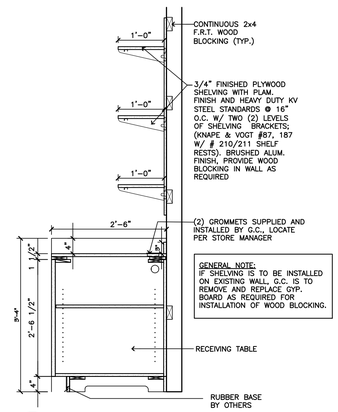
14 Typ. BW-Tall Slatwall 3D
A-413 SCALE: 3/4"=1'-0"



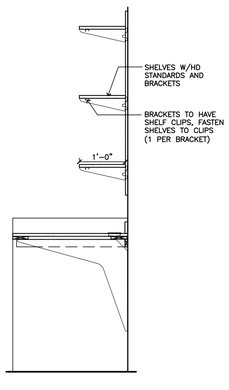
11 Typ. BW-Tall Slatwall Elev.
A-413 SCALE: 3/4"=1'-0"



10 Typ. BW-Tall Slatwall 3D
A-413 SCALE: 3/4"=1'-0"



8A Section at Receiving Table
A-413 SCALE: 3/4"=1'-0"



8B Section at Receiving Table
A-413 SCALE: 3/4"=1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
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Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
LICENSED ARCHITECT

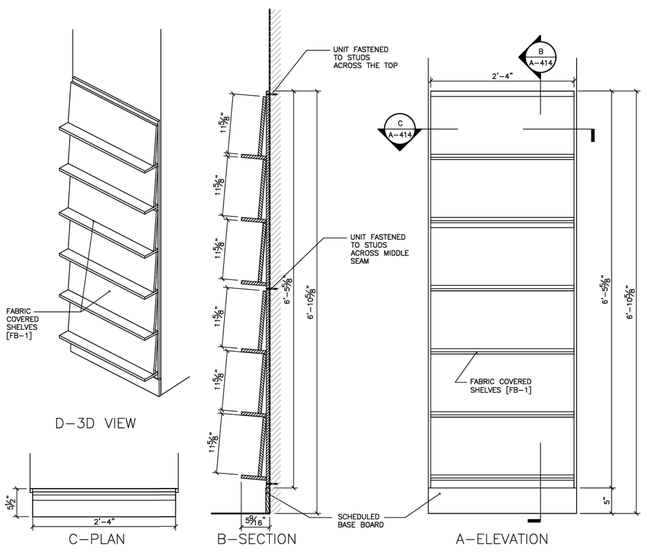
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

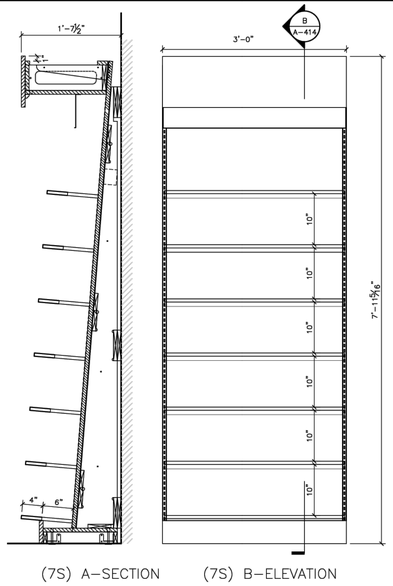
Cashwrap, Receiving Table & Miscellaneous Details

Drawing Number:

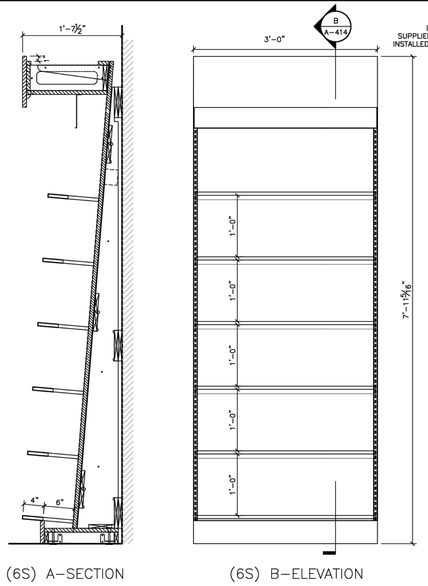
A-413



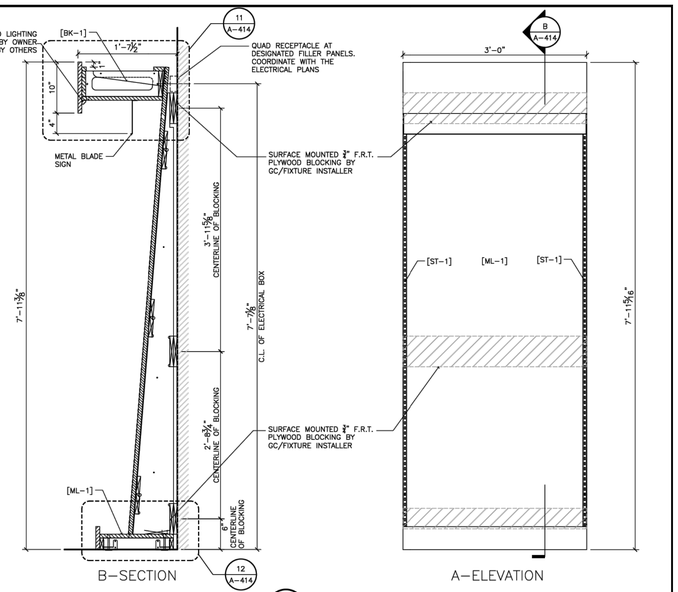
4 A-414 **Custom Column Fabric (6) Shelf**
SCALE: 1"=1'-0"



3 A-414 **W-3 W/ (7) Shelves @10" Spacing**
SCALE: 1"=1'-0"



2 A-414 **W-3 W/ (6) Shelves @12" Spacing**
SCALE: 1"=1'-0"



1 A-414 **W-3 Wall Fixture**
SCALE: 1"=1'-0"

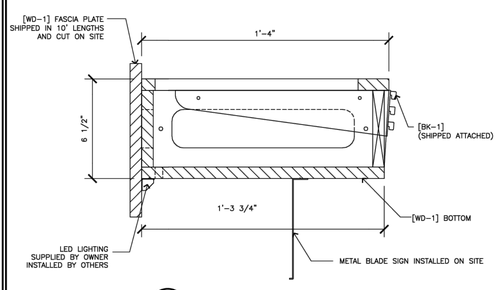
7 A-414 **W-3 Wall Fixture W/ Fabric Shelves**
SCALE: 1"=1'-0"

6 A-414 **W-3 Wall Fixture W/ Slotwall**
SCALE: 1"=1'-0"

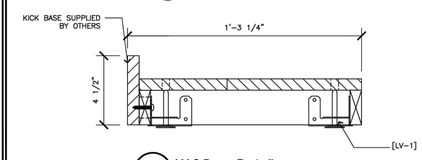
5 A-414 **W3FE-42**
SCALE: 1"=1'-0"

NOTE:

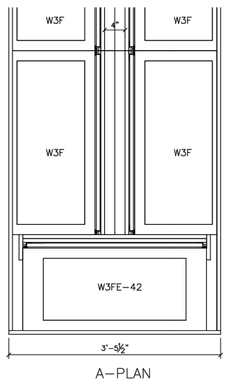
1. **DETAILS FOR MERCHANDISING ONLY. G.C. TO REFER TO MILLWORK DRAWINGS BY OTHERS FOR INSTALLATION GUIDELINES.**

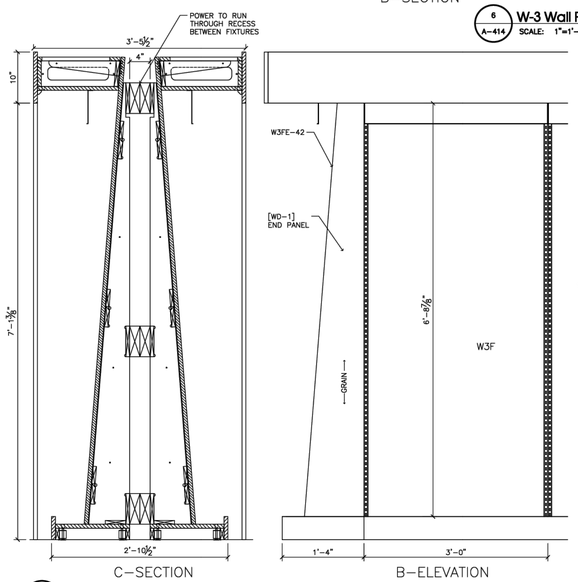


11 A-414 **W-3 Header**
SCALE: 3"=1'-0"

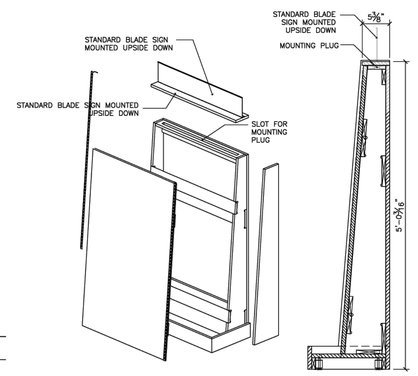


12 A-414 **W-3 Base Details**
SCALE: 3"=1'-0"

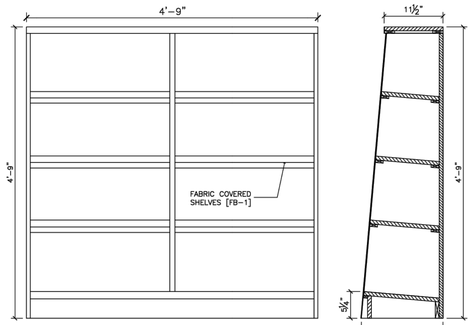




10 A-414 **W3F End Cap Configuration**
SCALE: 1"=1'-0"



9 A-414 **Short Perimeter**
SCALE: 1"=1'-0"



8 A-414 **Custom Fabric (6) Low Book Case**
SCALE: 1"=1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
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06-12-25	LL AND BD COMMENTS	
07-23-25	BD COMMENTS	

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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

STATE OF COLORADO
WILLIAM J. COMMER
LICENSED ARCHITECT

LIC.# : 305353
EXP. DATE : 10/31/25

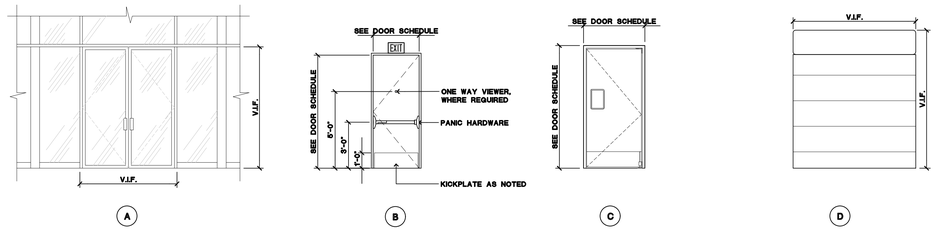
Drawing Description:

Millwork Details

Drawing Number:

A-414

DOOR TYPES

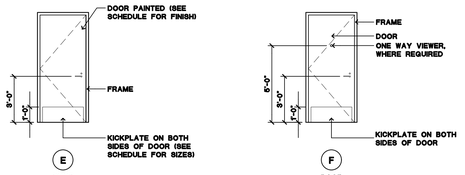


A — GLASS/ALUMN DOOR

B — EXISTING EMERGENCY EXIT DOOR

C

D



E — DOOR

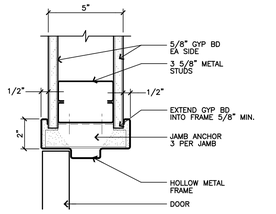
F — DOOR

NO PEEPHOLE ON BREAK ROOM DOOR (TYP.)

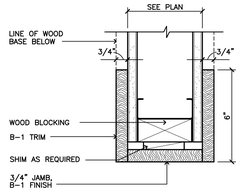
G — NOT USED

H — NOT USED

HEADER AND JAMB DETAILS



1 / A-500 — Typical Hollow Metal Frame
SCALE: 3" = 1'-0"



2 / A-500 — Cased Opening Detail
SCALE: 3" = 1'-0"

DOOR SCHEDULE

NO.	TYPE	LOCATION FROM	LOCATION TO	DOOR SIZE	DOOR THK.	DOOR MAT.	DOOR FIN.	FRAME JAMB/HEAD	FRAME MAT.	SADDLE	HARDWARE GROUP	FIRE RATING	REMARKS
1	A	VESTIBULE	EXTERIOR	EXISTING V.I.F.	-	EXIST.	PREFIN	EXIST.	EXIST.	-	-	-	EX. DOOR TO REMAIN REPAIR/REPLACE HARDWARE AS REQUIRED, PANIC HARDWARE EXISTING
2	A	SALES AREA	VESTIBULE	(2)3'-0" X 8'-0"	1 3/4"	ALUM/GLASS	PREFIN	-	ALUM	-	1A	-	TO HAVE PANIC HARDWARE
3	C	CAFE	CAFE WORK ROOM	4'-0" X 8'-0"	-	-	PREFIN(BLACK)	-	-	-	-	-	ELIASON ENG-1 DOOR WITH 9"X14" WINDOW AND FRAME
4	F	SALES AREA	CORRIDOR	3'-0" X 7'-0"	1 3/4"	H.M.	PT-2	1/A-500	H.M.	-	-	-	-
5	F	CORRIDOR	BREAK ROOM	3'-0" X 7'-0"	1 3/4"	H.M.	PT-1	1/A-500	H.M.	-	4	-	-
6	F	BREAK ROOM	MANAGER'S OFFICE	3'-0" X 7'-0"	1 3/4"	H.M.	PT-1	1/A-500	H.M.	-	4	-	-
7	F	CORRIDOR	JANITOR CLOSET	3'-0" X 7'-0"	1 3/4"	H.M.	PT-1	1/A-500	H.M.	-	3	-	-
8	E	CORRIDOR	MEN'S RESTROOM	3'-0" X 7'-0"	1 3/4"	H.M.	PT-3	1/A-500	H.M.	-	8	-	-
9	E	CORRIDOR	WOMEN'S RESTROOM	3'-0" X 7'-0"	1 3/4"	H.M.	PT-3	1/A-500	H.M.	-	8	-	-
10	B	CORRIDOR	EXTERIOR	EXISTING V.I.F.	-	EXIST.	PT-3	EXIST.	EXIST.	-	2	-	REFURBISH DOOR & HARDWARE AS NEEDED, PANIC HARDWARE EXISTING
11	F	SALES AREA	STOCK ROOM	4'-0" X 7'-0"	1 3/4"	H.M.	PT-1	1/A-500	H.M.	-	10	-	-
12	D	STOCK ROOM	EXTERIOR	EXISTING V.I.F.	-	EXIST.	PT-1	EXIST.	EXIST.	-	-	-	REFURBISH DOOR & HARDWARE AS NEEDED
13	B	STOCK ROOM	EXTERIOR	EXISTING V.I.F.	-	EXIST.	PT-1	EXIST.	EXIST.	-	-	-	REFURBISH DOOR & HARDWARE AS NEEDED, PANIC HARDWARE EXISTING
14	F	SALES AREA	STORAGE	3'-0" X 7'-0"	1 3/4"	H.M.	PT-1	1/A-500	H.M.	-	10	-	-

** NOTE:
GLAZING IN AN INDIVIDUAL FIXED OR OPERABLE PANEL THAT MEETS ALL OF THE FOLLOWING CONDITIONS SHALL BE CONSIDERED A HAZARDOUS LOCATION:
THE EXPOSED AREA OF AN INDIVIDUAL PANEL IS GREATER THAN 9 SQUARE FEET (0.84 M2).
THE BOTTOM EDGE OF THE GLAZING IS LESS THAN 18 INCHES (457 MM) ABOVE THE FLOOR.
THE TOP EDGE OF THE GLAZING IS GREATER THAN 36 INCHES (914 MM) ABOVE THE FLOOR.
ONE OR MORE WALKING SURFACE(S) ARE WITHIN 36 INCHES (914 MM), MEASURED HORIZONTALLY AND IN A STRAIGHT LINE, OF THE PLANE OF THE GLAZING.

ALL DOORS WHICH FACE THE GENERAL PUBLIC AT GENERAL SALES AREA TO BE FINISHED TO MATCH THE ADJACENT SURFACE.
ALL NEW & EXISTING LOCKS TO HAVE 7 – PIN BEST CYLINDERS, TYP.

DOOR NOTES

1. PROVIDE PAINT FINISH AT ALL H.M. DOORS TO MATCH ADJACENT WALL COLOR (U.O.N.).
2. PAINT HOLLOW METAL DOORS & FRAMES SEMI-GLOSS ENAMEL.
3. ALL LOCKS SHALL BE COORDINATED TO SUIT B&N LOCKING SYSTEMS - BEST 7 PIN INTERCHANGABLE
4. RATED DOORS SHALL BE A TIGHT FITTING SMOKE AND DRAFT CONTROL ASSEMBLY.
5. ALL REQUIRED SIGNAGE SHALL CONFORM WITH LOCAL and ADA ACCESSIBILITY GUIDELINES, INCLUDING BUT NOT LIMITED TO PROPORTION, COLOR, CONTRAST, RELIEF AND GRADE 2 BRAILLE REQUIREMENTS.
6. ALL DOORS SHALL BE PROVIDED WITH LEVER OR PUSH-PULL TYPE HARDWARE, NOT REQUIRING GRASPING OR TWISTING, NOR MORE THAN ONE OPERATION.
7. ALL CORES TO BE BEST, CONSTRUCTION CORE TYPE (U.O.N.) - 7 PIN AND CONTROL (KEY AVAILABLE AT TURN OVER)
8. EXIT DOORS (INCL. BUT NOT LIMITED TO MALL SECURITY GRILLES) SHALL BE OPERABLE FROM THE INSIDE WITHOUT THE USE OF A KEY OR ANY SPECIAL KNOWLEDGE OR EFFORT. EXCEPTION FOR "B" OCCUPANCIES: MAY HAVE A SIGN ON OR ADJACENT TO THE DOOR STATING "THIS DOOR TO REMAIN UNLOCKED DURING BUSINESS HOURS". THE SIGN SHALL BE IN LETTERS NOT LESS THAN 1" HIGH ON A CONTRASTING BACKGROUND. THE LOCKING DEVICE MUST BE A TYPE THAT WILL BE READILY DISTINGUISHED AS LOCKED. (U.F.C. 12.106(c))
9. THE MINIMUM CLEAR AISLE WIDTH SHALL BE 44" WHEN MERCHANDISE OR OTHER SIMILAR OBSTRUCTIONS ARE PLACED ON BOTH SIDES OF THE AISLE AND A MINIMUM OF 36" WHEN PLACED ON ONE SIDE OF THE AISLE.
10. MAXIMUM EFFORT TO OPERATE AN INTERIOR DOOR IS 5LBS. 8.5LBS FOR EXTERIOR DOORS AND 15LBS FOR FIRE EXIT DOORS.
11. ALL HARDWARE MANUFACTURERS ARE TO BE AS SPECIFIED, NO SUBSTITUTIONS ALLOWED.
12. ALL THRESHOLDS NOT TO EXCEED 0.5 INCH, RAISED THRESHOLDS (AND FLOOR CHANGES) GREATER THEN 0.25 INCH AT DOORWAYS SHALL BE BEVELED WITH A SLOPE NOT GREATER THAN ONE UNIT VERTICAL IN TWO UNITS HORIZONTAL.

HARDWARE GROUPS

NOTE: DOOR HARDWARE AND SCHEDULE TO BE VERIFIED AND APPROVED BY TENANT PRIOR TO INITIAL PURCHASE ORDER AND CONSTRUCTION

NO.	FROM TO	QTY	DESCRIPTION	MANUFACTURER	ITEM NO.	FINISH
1	ENTRY DOOR	1	THRESHOLD			ALUMINUM
		2	CLOSER		7300	
		1	CYLINDER	BEST (7-PIN CORED)		ALUMINUM
		2	PULL HANDLE			
		4	PUSHBAR			
		1	HINGE	KAWNEER		
1A	VESTIBULE	1	HINGE	KAWNEER	TOP OFFSET PIVOT	MATCH STOREFRONT
		2	HINGE	KAWNEER	STANDARD INTERMEDIATE PIVOT	
		1	HINGE	KAWNEER	BOTTOM OFFSET PIVOT	
		2	CLOSER	DORMA	7300	
		2	PULL HANDLE	BID	MATCH EX.	MATCH EX.
		4	PUSHBAR (TWO BARS PER DOOR - UPPER AND LOWER BAR)			
		1	BOTTOM RAIL WEATHER STRIPPING	KAWNEER	38-560	
		1	THRESHOLD	ZERO	4", 872A X W/ (2) 288A 6", 874A X W/ (2) 288A	ALUMINUM
2	EMERGENCY EXIT DOOR «EXISTING - VERIFY HARDWARE IN FIELD, PROVIDE IF NOT EXISTING	3	HINGES	STANLEY	FBB191NRP-5x4.5	US32D
		1	CLOSER	LCN	4110-NON-CUSTOMER SIDE OF DOOR	US26D
		1	PANIC ALARM	VON DURPIN	33EO	SP-28
		1	THRESHOLD	ZERO	573 A W/288A	ALUMINUM
		1	PERIMETER SEAL	NATIONAL GUARD PRODUCTS	160 V-WITH VINYL SEAL	ALUMINUM
		1	EXIT ALARM	DETEX	EA500	
		1	CORE	BEST 7-PIN	CONSTRUCTION CORE	
		1	THRESHOLD	HAGER	626S	AL
		1	WEATHERSTRIP	HAGER	891SV	AL
3	JANITOR CLOSET		HINGES	HAGER	BB1279 4-1/2 X 4-1/2	26D
		-	CLOSER	LCN	4030 H	AL
		1	CLASSROOM LOCKSET (W/ CONST. CORE)	SCHLAGE	ND70HD RHO	26D
		1	PROTECTION PLATE	HAGER	190S 12 X 34	32D
		1	FLOOR STOP	HAGER	242F	26D
		3	SILENCERS	HAGER	307D	GR
4	MANAGER'S OFFICE, BREAK ROOM, «EXISTING - VERIFY HARDWARE IN FIELD, PROVIDE IF NOT EXISTING	3	HINGES	HAGER	BB1279 4-1/2 X 4-1/2	26D
		1	PUSHBUTTON LOCK (W/CONST. CORE)	ALARM CLOCK	DL 2700 IC	26D
		1	CLOSER	LCN	4030 H	26D
		1	PROTECTION PLATE	HAGER	190S 12" X 34"	32D
		1	FLOOR STOP	HAGER	242F	26D
		3	SILENCERS	HAGER	307D	GR
		1	DOOR VIEWER	HAGER	1756	26D
5	STOCK OUT SWING DOUBLE DOORS «EXISTING - VERIFY HARDWARE IN FIELD, PROVIDE IF NOT EXISTING	2	CONTINUOUS HINGES	HAGER	780-112 HD	AL
		2	CLOSER	LCN	4111 H CUSH (110 DEG. OPEN)	AL
		1	EXIT DEVICE	VON DUPRIN	2260 X 1909 STRIKE	AL
		1	ALARM	DETEX	EAX500	AL
		1	CYLINDER	SCHLAGE	BO-122	26D
		2	PROTECTION PLATE	HAGER	190S 12" X 34" (INT. ONLY)	32D
		1	COORDINATOR	HAGER	297D	BLK
		2	MOUNTING BRACKET	HAGER	297M	BLK
		2	FLUSHBOLT	HAGER	282D	26D
		1	ASTRAGAL	HAGER	8378 X 96"	AL
		1	DOOR VIEWER	HAGER	1756	26D
		1	THRESHOLD	HAGER	626S	AL
		1	WEATHERSTRIP	HAGER	891SV	AL
		1	DOOR PULL	HAGER	34G (INT. OF ACTIVE LEAF)	26D
		1	RAIN DRIP GAURDS NOP SWC		16A	AL
6	ELECTRIC ROOM	3	HINGES	HANGAR	BB1279 4-1/2 X 4-1/2	26D
		1	STORE ROOM LOCKSET(w/const. core)	SCHLAGE	ND 10 RHO	26D
		1	CLOSER	LCN	4030 or 4031	MTLPC
		1	PROTECTION PLATE	HAGER	190S 12" X 34"	32D
		1	FLOOR STOP	HAGER	242F	26D
		3	SILENCERS	HAGER	307D	GR
7	NOT USED					
8	RESTROOMS (STORE WITH CAFE)	3	HINGES	HANGAR	BB1279 4-1/2 X 4-1/2	26D
		1	CLOSER	LCN	4030 or 4031	MTLPC
		1	CLASSROOM SET	SCHLAGE	ND 70HD RHO (BEST 7-PIN CYLINDER)	26D
		1	DEAD BOLT	SCHLAGE	B560B 626 OS-OCC(BEST 7-PIN CYLINDER)	26D
		1	PROTECTION PLATE	HAGER	190S 12 X 34	32D
		1	FLOOR STOP	HAGER	242F	26D
		3	SILENCERS	HAGER	307D	GR
		1	FLOOR STOP	HAGER	242F	26D
6A	RESTROOMS (STORE WITHOUT CAFE)	3	HINGES	HANGAR	BB1279 4-1/2 X 4-1/2	26D
		1	CLOSER	LCN	4030 or 4031	MTLPC
		1	TRILOGY LEVER	ALARM LOCK	DL 2700 IC (BEST 7-PIN CYLINDER)	26D
		1	DEAD BOLT	SCHLAGE	B560B 626 OS-OCC(BEST 7-PIN CYLINDER)	26D
		1	PROTECTION PLATE	HAGER	190S 12 X 34	32D
		1	FLOOR STOP	HAGER	242F	26D
		3	SILENCERS	HAGER	307D	GR
		1	FLOOR STOP	HAGER	242F	26D
10	WORK ROOM, STOCK ROOM, STORAGE	3	HINGES	HAGER	BB1279 4-1/2 X 4-1/2	26D
		1	CLOSER (Door swing in)	LCN	4030 H CUSH (mount on J.C. side)	AL
		1	PUSHBUTTON LOCK (w/ const. core)	ALARM CLOCK	DL 2700 IC	26D
		2	PROTECTION PLATE	HAGER	190S 12"x 35"	32D
		1	FLOOR STOP	HAGER	242F	26D
		3	SILENCERS	HAGER	307D	GR
11	SALES CLOSET	3	HINGES	HAGER	BB1279 4-1/2 X 4-1/2	26D
		1	CLOSER (Door swing in)	LCN	4030 H CUSH (mount on J.C. side)	AL
		1	PUSHBUTTON LOCK (w/ const. core)	ALARM CLOCK	DL 2700 IC	26D
		2	PROTECTION PLATE	HAGER	190S 12"x 34"	32D
		1	FLOOR STOP	HAGER	242F	26D
		3	SILENCERS	HAGER	307D	GR

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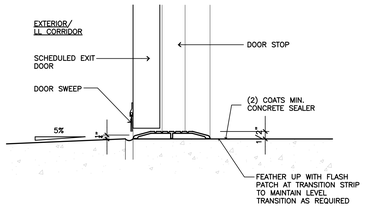
Professional Seal:

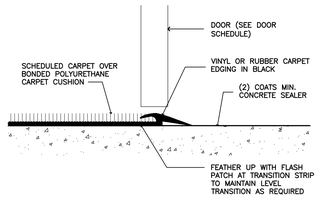
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:
Door Types, Schedules, Notes, and Details

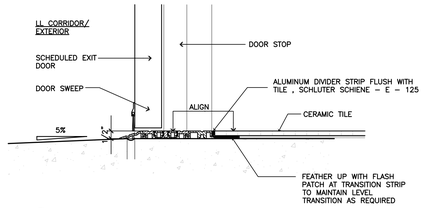
Drawing Number:
A-500



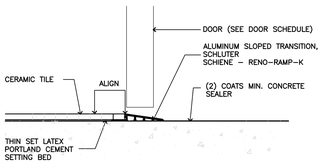
1 A-510 Exterior Conc. to Sealed Conc.
SCALE: 3"=1'-0"



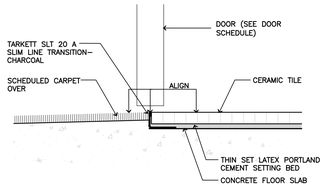
2 A-510 Carpet to Sealed Concrete
SCALE: 3"=1'-0"



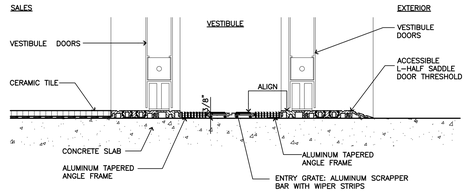
3 A-510 Sealed Conc. to Ceramic Tile
SCALE: 3"=1'-0"



6 A-510 Tile to Exterior
SCALE: 3"=1'-0"



5 A-510 Carpet to Ceramic Tile
SCALE: 3"=1'-0"



4 A-510 Exterior to Walk Off Mat
SCALE: 3"=1'-0"

7 A-510 Ceramic Tile to Epoxy Flooring
SCALE: 3"=1'-0"

8 A-510 Ceramic Tile to VCT
SCALE: 3"=1'-0"

9 A-510 Carpet to VCT
SCALE: 3"=1'-0"

ROOM FINISH SCHEDULE

NO.	ROOM	FLOORING – FINISH							BASE – FINISH							WALLS – FINISH								CEILING – FINISH				REMARKS
		CARPET	ENTRY MAT	VINYL COMPOSITE TILE	CERAMIC/PORCELAIN TILE	SEALED CONCRETE FLOOR	EPOXY FLOORING	EXISTING FLOORING	VINYL BASE	WOOD BASE	CERAMIC COVE TILE BASE	SCHLUTER COVE SHAPED BASE	EPOXY SYSTEM BASE	STAINLESS STEEL BASE	EXISTING BASE	PAINTED GYP. BD.	WALL COVERING	WOOD VENEER	CERAMIC/PORCELAIN TILE	SLATWALL	FIBER REINFORCED PANEL	FIRE TAPED, GYP. BD. – NO FINISH	WAINSCOT	EXISTING	ACOUSTIC TILE	PAINTED GYP. BD.	PAINTED STRUCTURE	OPEN TO STRUCTURE	EXISTING CEILING	
100	VESTIBULE		●							●																●			PAINT GWB CEILING PT–4
101	SALES AREA	●			●					●						●										●			PAINT WALLS PT–2 FROM 7'–6" TO UNDERSIDE OF DECK, BASE B–1
102	CAFE SEATING AREA				●					●						●										●	●		PAINT WALLS PT–3 TO UNDERSIDE OF DECK, BASE B–1
103	CAFE				●						●					●		●	●		●						●		FRP BELOW COUNTERTOP & TL–5 ABOVE COUNTERTOP, BASE B–2
104	CAFE WORK ROOM						●						●			●					●				●				PAINT WALLS FRP–1, BASE EP–1
105	MANAGER'S OFFICE	●							●							●									●				PAINT WALLS PT–1, BASE VB–1
106	BREAK ROOM			●					●							●									●				PAINT WALLS PT–1, BASE VB–1
107	JANITOR CLOSET			●					●												●					●	●		PAINT WALLS PT–1, BASE VB–1
108	MEN'S RESTROOM				●						●					●				●						●			SEE A–200
109	WOMEN'S RESTROOM				●						●					●				●						●			SEE A–200
110	CORRIDOR				●					●						●				●					●				PAINT WALLS PT–3, BASE B–1
111	STOCK ROOM					●			●							●											●	●	PAINT WALLS PT–1, BASE VB–1
112	STORAGE					●			●							●											●	●	PAINT WALLS PT–1, BASE VB–1

FINISH NOTES:

1. FOR LOCATION AND TYPE OF BASE IN SALES AREA SEE FINISH PLAN.

2. ALL INTERIOR FINISHES, SPECIFIED WALL COVERINGS AND CEILING MATERIALS, SHALL HAVE A FLAME SPREAD RATING NOT TO EXCEED 75 AND SMOKE-DEVELOPED INDEX NOT TO EXCEED 450.

3. ALL CARPETING SHALL HAVE A CLASS 'A' CLASSIFICATION WITH A FLAME SPREAD RATING NOT TO EXCEED 25 AND A SMOKE-DEVELOPED INDEX NOT TO EXCEED 450.

4. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR PROPER PREPARATION OF ALL NEW AND EXISTING SURFACES IN A BATHFACTORY MANNER TO RECEIVE NEW FINISHES. THIS INCLUDES THE DEMOLITION AND REMOVAL OF NECESSARY ITEMS. TOUCH-UP AND/OR REFINISH OF SURFACES DAMAGED BY SUBSEQUENT WORK SHALL BE THE RESPONSIBILITY OF THE GENERAL CONTRACTOR. ALL WORK SHALL BE PERFORMED IN CONFORMANCE WITH THE MANUFACTURERS RECOMMENDED INSTALLATION METHODS. THE GENERAL CONTRACTOR SHALL PREPARE THE EXISTING FLOOR PRIOR TO THE APPLICATION OF FINISH FLOORING. THE EXISTING CONCRETE SLAB MUST BE SMOOTH AND LEVEL WITH A TOLERANCE OF ONE-EIGHT INCH PER FOOT. LATEX CEMENT PATCHING COMPOUND SHALL BE UTILIZED ON ASPHALT BASED COMPOUNDS.

5. ALL NON-CARPETED MEANS OF EGRESS WALKING SURFACES (IE – CERAMIC TILE, WOOD FLOOR, ETC.) SHALL HAVE A SLIP-RESISTANT SURFACE (0.6 MIN. COEFFICIENT OF FRICTION OR AS PER CODE) AND BE SECURELY ATTACHED.

6. EPOXY GROUT TILE SHOULD BE CLEANED IMMEDIATELY AFTER INSTALLATION, FOLLOW MANUFACTURERS SPECIFICATIONS FOR CORRECT INSTALLATION. IF GROUT IS LEFT ON TILE MORE THAN 48 HOURS CONTRACTOR IS RESPONSIBLE FOR CLEANING IT USING KERACLEAN-S BY MAPEI.

7. G.C. TO PROVIDE GROUT SAMPLE FOR APPROVAL AT ALL TILE LOCATIONS, ARCHITECT TO APPROVE PRIOR TO COMPLETE INSTALLATION. GROUT SPECIFIED DOES NOT REQUIRE SEALING.

8. ALL JOINTS IN GYPSUM BOARD WALLS SHALL BE FINISHED WITH PAPER TAPE 3" WIDE AND THREE COATS OF VINYL, DRY OR PREMIXED JOINT COMPOUND. ALL OUTSIDE CORNERS SHALL BE FINISHED WITH METAL CORNER BEADS, TAPED AND SPACKLED. ALL AREAS TO BE PAINTED SHALL BE SANDED SMOOTH, DUST PRIOR TO APPLICATION OF FIRST COAT OF PAINT. WIPE SANDED SURFACES WITH DAMP CLOTH IN ORDER TO LAY FLAT ANY NAP WHICH MAY HAVE FORMED IN SANDING

SUBMIT FINISH SAMPLES TO ARCHITECT FOR APPROVAL PRIOR TO COMMENCEMENT OF WORK. G.C. TO COORDINATE TIME LINE FOR ALL APPROVALS AND PRODUCT LEAD TIMES. NO EXCEPTIONS.

GENERAL FINISH NOTES:

1. DUE TO THE LONG LEAD TIMES OF TILE, THE G.C. IS ADVISED TO ORDER MATERIALS AS EARLY AS POSSIBLE. THERE WILL BE NO SUBSTITUTIONS ALLOWED.

FLAME/SPREAD AND SMOKE DEVELOPMENT INFORMATION PER CBC TABLE 803.9, THE INTERIOR WALL AND CEILING FINISH REQUIREMENTS FOR GROUP M ARE:

B – FOR EXIT RAMPS AND PASSAGEWAYS
C – FOR CORRIDORS, EXIT ENCLOSURES, RAMPS AND ROOMS

WALL BASE (WOOD LAMINATE) BY BN

FLAME SPREAD < 40 (CLASS B)
SMOKE DEVELOPMENT < 150 (CLASS A)

WALL BASE (VINYL) BY BN

FLAME SPREAD ≤ 25 (CLASS A)
SMOKE DEVELOPMENT ≤ 450 (CLASS A)

PAINTS (ACRYLIC COATING)

FLAME SPREAD ≤ 25 (CLASS A)
SMOKE DEVELOPMENT ≤ 450 (CLASS A)

ACOUSTIC CEILING TILE

FLAME SPREAD ≤ 25 (CLASS A)
SMOKE DEVELOPMENT ≤ 50 (CLASS A)

FIBERGLASS REINFORCED PANEL (FRP)

FLAME SPREAD ≤ 25 (CLASS A)
SMOKE DEVELOPMENT ≤ 450 (CLASS A)

COUNTERS TOP (CORIAN) BY BN

FLAME SPREAD ≤ 25 (CLASS A)
SMOKE DEVELOPMENT ≤ 450 (CLASS A)

NOTE: ALL FLOOR TRANSITIONS ARE FURNISHED BY BARNES & NOBLE AND INSTALLED BY GENERAL CONTRACTOR.

FINISH LEGEND

BASE

FINISH	DESCRIPTION	LOCATION	MANUFACTURER	NOTES
B–1	6" HIGH 3/4" THICK WOOD BASE STAINED TO MATCH FIXTURE	SALES FLOOR, CORRIDOR	–	PROVIDED BY BN
B–2	6X6 FLAT TOP COVE A3601 CERAMIC TILE	RESTROOMS, CAFE	AMERICAN OLEAN	–
VB–1	VINYL BASE # P198 IVORY (USE COVE BASE ON VT–1)	BACK AREAS/OFFICE	ROPPE	STANDARD 5/8" TOE BY BN

FLOOR

TL–1	SALES FLOOR FIELD TILE AFC BN WPT 914 SIZE: 9 X 36 X 3/8". GROUT: MAPEI #05 CHAMOIS, ⅛" THICK OR LESS	SALES FLOOR, CAFE	ASSOCIATED FLOORING	SEE A–131 PROVIDED BY ASSOCIATED
TL–3	AFC BN FLAV 12"x 24" TITANIUM.	RESTROOM, CORRIDOR	ASSOCIATED FLOORING	ADD GROUT GT–1 SILVER
VT–1	VINYL COMPOSITE TILE #51811 ANTIQUE WHITE	BREAK AREA , J.C.	ARMSTRONG	SEE A–131
CP–1	SAMPLE NO. 008–1003318–001–02 FACE YARN – NYLON FINISH	SALES FLOOR, MANAGER'S OFFICE	ASSOCIATED FLOORING	SEE A–131
SC–1	CONCRETE SEALER – PORCH & DECK – SATIN LATEX – PEWTER GRAY – TWO COATS (FILL / CAULK ALL C.J.'S PRIOR TO SEALING)	STOCK ROOM, STORAGE	PITTSBURGH PAINTS	SEE A–131
CON–1	POLISHED CONCRETE + CLEAR CONCRETE PENETRATING SEALER, MATTE (NO GLOSS OR SURFACE FILM)	SALES FLOOR		SEE A–131
CP–2	FIELD CARPET	MANAGER'S OFFICE	ASSOCIATED FLOORING	PROVIDED BY ASSOCIATED
FM–1	GRATE MAT – GRAY	VESTIBULE	MATSINC	SEE A–131
EP–1	POURED EPOXY FLOORING W/ 6" HIGH COVE BASE, POLY–CRETE MOB W/ ARMOR TOP TOPCOAT, PEWTER COLOR – 1/4" THICK – MATTE	WORKROOM	STONHARD OR EQUAL	SEE A–131

WALL

FINISH	DESCRIPTION	LOCATION	MANUFACTURER	NOTES
PT–1	DESIGNER WHITE FINISH: EGGSHELL	BACK AREAS AND OFFICE, CAFE	BENJAMIN MOORE	SEE PAINT FINISH NOTES BELOW
PT–2	2100–70 VICTORIAN LACE FINISH: EGGSHELL	SALES FLOOR WALLS (LIGHT CONCEPT)	BENJAMIN MOORE	SEE PAINT FINISH NOTES BELOW
PT–3	HC–172 REVERE PEWTER FINISH = EGGSHELL	CAFE, RESTROOM, CORRIDOR	BENJAMIN MOORE	SEE PAINT FINISH NOTES BELOW
PT–4	WHITE, FINISH = FLAT	ALL CEILINGS	BENJAMIN MOORE	SEE PAINT FINISH NOTES BELOW
PT–5	COROTECH ACRYLIC EPOXY V450: WHITE	UNDERSIDE OF CAFE BACK SOFFIT ONLY	BENJAMIN MOORE	WASHABLE EPOXY FINISH TO MATCH PT–2
PT–6	2121–70 CHANTILLY LACE, FINISH = SEMI–GLOSS	UNDERSIDE OF ROOF DECK, STRUCTURE AND DUCTWORK	BENJAMIN MOORE	SALES FLOOR OPEN DECK CEILING (LIGHT CONCEPT STORE)
PT–7	2132–10 BLACK, FINISH = SEMI–GLOSS	DRINK COOLER ALCOVE	BENJAMIN MOORE	–
PT–8	CHALKBOARD PAINT, FINISH = EGGSHELL & BLACK	SALES SIDE OF THE TONY WALL AT CASHWRAP	BENJAMIN MOORE	SEE PAINT FINISH NOTES BELOW
PT–9	KENDALL CHARCOAL (HC–166)	EXTERIOR SIGNBAND	BENJAMIN MOORE	–
TL–2	SUBWAY TILE 3"X6"– WITH ⅛" GROUT JOINTS – FINISH: WHITE D100 (1) SEM GLOSS GROUT TO MATCH TILE COLOR.	RESTROOM WAINSCOT	ASSOCIATED FLOORING	SEE A200 FOR LOCATIONS
TL–4	GLAZED CERAMIC TILE – AMERICAN OLEAN – PLAYSCAPES – 4 1/4" X 4 7/8" HEXAGON MIDNIGHT BLUE	CAFE BACKSPLASH	ASSOCIATED FLOORING	SEE A225 & A226A FOR LOCATIONS
TL–5	CERAMIC TILE – Roca Flow White 3"x12" Picket	CAFE DIE WALLS	ASSOCIATED FLOORING	SEE A225 & A226A FOR LOCATIONS
WD–1	WOOD TILE – NATURAL FINISH	CAFE WALLS	–	SUPPLIED BY B&N
FRP–1	FIRE REINFORCED PANELS CLASS A WHITE SMOOTH FINISH	WORK ROOM/ JAN CL., CAFE	KEMLITE	–

NOTE:
APPLY ALL PAINT FINISH AS FOLLOWS:
• SOFFITS – EGGSHELL
• WALLS – EGGSHELL.
• CAFE WORK ROOM / TOILET / J.C. – SEMI–GLOSS.
• MTL'S / DR'S & FRM'S / MURALS – SEMI–GLOSS
• **ALL EXPOSED STRUCTURE, ROOF, DECK, HVAC DUCTS, ELECTRICAL CONDUITS, ETC. IN EXPOSED CEILING AREAS TO BE SEMI–GLOSS ENAMEL

SOLID SURFACE / PLASTIC LAMINATE / METALS

FINISH	DESCRIPTION	LOCATION	MANUFACTURER	NOTES
CC–1	SOLID SURFACE "MILKY WAY"	CAFE COUNTERTOP	LIVINGSTONE	PROVIDED BY CAFE MILLWORK VENDOR
PL–1	DESIGNER WHITE MATTE FINISH D354–60	OFFICE MILLWORK	WILSONART	SEE A–412

SEALANT

CLEAR (MATTE FINISH	WOOD BASE TO CERAMIC TILE FLOOR
WHITE (MATTE FINISH)	COUNTERTOPS / CABINETS TO PAINTED SURFACES FRP TO SANITARY COVE BASE / PAINTED SURFACE

CEILING

FINISH	DESCRIPTION	LOCATION	MANUFACTURER	NOTES
ACT–1	CLEAN ROOM VL: 868, 24 X 48, SQUARE LAY–IN 15/16	WORK ROOM	ARMSTRONG	–
ACT–2	OPTIMA 24 X 48 SQUARE LAY–IN FINE TEXTURE WITH PRELUDE XL 15/16" SUSPENSION SYSTEM	STORAGE	ARMSTRONG	–
ACT–3	OPTIMA 24 X 24 SQUARE LAY–IN FINE TEXTURE WITH PRELUDE XL 15/16" SUSPENSION SYSTEM	BREAK ROOM, MANAGER'S OFFICE, CORRIDOR	ARMSTRONG	–

GROUTS

GT–1	CHAMOIS 05	–	MAPEI– KERAPOXY CQ	USE WITH TL–1
GT–2	AVALANCHE 38	–	MAPEI– KERAPOXY CQ	USE WITH TL–2,TL–4 & TL5
GT–3	SILVER 27	–	MAPEI– KERAPOXY CQ	USE WITH TL–3

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

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04–25–25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

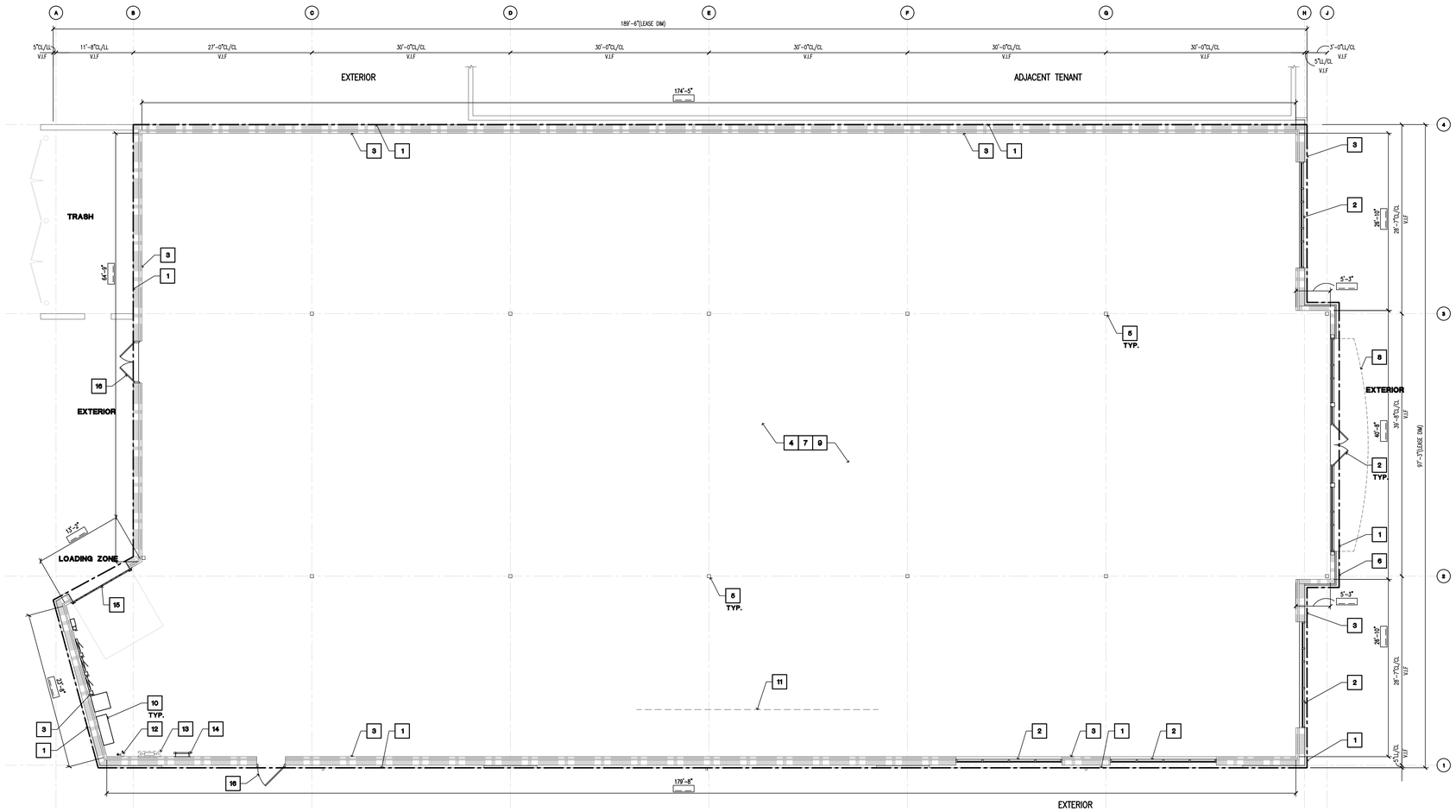
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

Finish Schedule and
Floor Transition Details

Drawing Number:

A-510



1 Demolition Plan
DM-101 SCALE: 1/8" = 1'-0"

LEASE CONFIRMATION PLAN NOTES

1. BEFORE CONSTRUCTION OF THE TENANT FIT-UP, CONFIRM AND NOTE INDICATED DIMENSIONS OF DEMISING PARTITIONS AND COLUMN LOCATIONS.
2. FORWARD COMPLETED ' LEASE CONFIRMATION PLAN' TO THE PROJECT REPRESENTATIVE PRIOR TO LAYOUT OF PARTITIONS
3. --- FIELD DIMENSION BOX
4. ———————— LEASELINE
5. UPON INITIAL SITE VISIT, VERIFY ALL EXISTING CONDITIONS (I.E., OVERALL DIMENSIONS, CONDITION OF SUB FLOOR, LOCATION OF SERVICES, STRUCTURE, CLEARANCES, ETC.) AND REPORT ANY DISCREPANCIES, OMISSIONS, OR OBSTRUCTIONS THAT WILL AFFECT THE LAYOUT OR DESIGN OF THE SPACE IMMEDIATELY TO THE PROJECT REPRESENTATIVE.
6. LOCATE ALL VERTICAL PIPES, CONDUIT, SHAFTS, LEADERS, COLUMNS AND DRAINS UNCOVERED DURING DEMOLITION

DEMOLITION LEGEND

- EXISTING DEMISING WALL TO REMAIN (V.I.F.)
- EXISTING PARTITIONS AND DOOR TO REMAIN
- EXISTING PARTITIONS, DOORS, FRAMES, FIXTURES, SOFFITS, FINISHES ETC. TO BE REMOVED

DEMOLITION GENERAL NOTES

1. COORDINATE WITH M.E.P. DRAWINGS IF EXISTING EQUIPMENT IS TO REMAIN.
2. PROCEED WITH DEMOLITION IN ACCORDANCE WITH ALL APPLICABLE CODES AND REGULATIONS.
3. GENERAL CONTRACTOR SHALL VERIFY FIELD CONDITIONS AND NOTIFY THE ARCHITECT OF ALL DISCREPANCIES BEFORE PROCEEDING WITH WORK.
4. DEMOLISH IN AN ORDERLY AND CAREFUL MANNER AS REQUIRED TO ACCOMMODATED NEW WORK, PREVENT MOVEMENT OR SETTLEMENT OF STRUCTURE(S). PROVIDE BRACING OR SHORING AS REQUIRED. G.C. WILL ASSUME LIABILITY FOR SUCH MOVEMENT, SETTLEMENT, DAMAGE OR INJURY. PHASE DEMOLITION IN ACCORDANCE WITH CONSTRUCTION SCHEDULE.
5. EXISTING SPRINKLER HEAD GRID, MAIN AND BRANCH PIPES OF ADJACENT USES TO REMAIN FUNCTIONAL DURING THE DURATION OF THE DEMOLITION WORK. COORDINATE ANY ALTERATIONS OF SPRINKLER SYSTEM WITH SPRINKLER CONTRACTOR.
6. CEASE OPERATIONS AND NOTIFY THE OWNER IMMEDIATELY IF SAFETY OF STRUCTURE APPEARS TO BE ENDANGERED. TAKE ALL NECESSARY PRECAUTIONS TO PROPERLY SUPPORT THE STRUCTURE. DO NOT RESUME OPERATIONS UNTIL SAFETY IS RESTORED.
7. G.C.TO COORDINATE THE REMOVAL, DISCONNECTION AND/OR CAPPING OF ANY UTILITIES WITH LANDLORD'S FIELD REPRESENTATIVE. ALL ABANDONED ELECTRICAL/TELEPHONE CONDUIT, PLUMBING LINES, DRAIN LINES, AND MECHANICAL DUCTWORK AS THE RESULT OF DEMOLITION SHALL BE REMOVED BACK TO THE LANDLORD'S ORIGINAL POINT. G.C. IS TO PLACE MARKERS TO INDICATE LOCATION OF DISCONNECTED SERVICES.
8. HVAC SUBCONTRACTOR TO INSPECT EXISTING CONDITIONS PRIOR TO DEMOLITION AND DETERMINE WHETHER ANY COMPRESSOR REFRIGERANT LINES ARE PRESENT THAT MUST BE PROPERLY DRAINED (TO E.P.A. STANDARDS) PRIOR TO DEMOLITION.
9. PLACE MARKERS TO INDICATE LOCATIONS OF DISCONNECTED SERVICES. IDENTIFY SERVICE LINES AND CAPPING LOCATIONS ON PROJECT RECORD DOCUMENTS.
10. EXCEPT WHERE NOTED OTHERWISE, MAINTAIN POSSESSION OF MATERIALS BEING DEMOLISHED. IMMEDIATELY REMOVE FROM SITE.
11. BURNING OF MATERIALS ON SITE IS NOT PERMITTED.
12. REPAIR ALL DEMOLITION PERFORMED IN EXCESS OF THAT REQUIRED AT NO COST TO THE OWNER.
13. PROVIDE TEMPORARY SERVICES AS NECESSARY FOR CONTINUOUS OPERATION FOR ALL LIFE/SAFETY DEVICES.
14. REMOVE DEMOLISHED MATERIALS, TOOLS AND EQUIPMENT FROM SITE UPON COMPLETION OF WORK. LEAVE SITE IN A CONDITION ACCEPTABLE TO THE ARCHITECT.
15. GENERAL CONTRACTOR TO COORDINATE ALL WORK BELOW SLAB WITH LANDLORD AND SHALL PATCH & REPAIR ANY/ALL DAMAGE THAT MAY OCCUR DURING DEMOLITION WORK.
16. WHERE THE EXISTING CONCRETE SLAB IS TO BE PATCHED AND REPAIRED, THE GENERAL CONTRACTOR SHALL MAINTAIN THE EXISTING FIRE RATING OF THE EXISTING SLAB. PATCH WORK TO BE STRUCTURALLY SOUND AND CAPABLE OF SUPPORTING LIVE LOADS FOR MERCANTILE OCCUPANCIES AS PER LOCAL CODES AND JURISDICTION.
17. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO VERIFY THE EXTENT OF DEMOLITION WITH THE OWNER AND THE LANDLORD AND BRING ANY DISCREPANCIES TO BOTH PARTIES ATTENTION PRIOR TO FINALIZING THE BID AND COMMENCEMENT OF WORK.
18. G.C. TO PATCH AND REPAIR ALL DEMISING WALLS TO REMAIN, AS REQUIRED.
19. G.C. IS RESPONSIBLE FOR PATCHING AND REPAIRING EXISTING FLOOR SLAB AND DEMISING WALLS AFTER COMPLETION OF ALL DEMOLITION AND PRIOR TO THE COMMENCEMENT OF NEW CONSTRUCTION. G.C. IS TO PATCH, REPAIR AND LEVEL EXISTING FLOOR SLAB AND INFORM OWNER OF ALL SEVERE IRREGULARITIES IN EXISTING SLAB.
20. G.C. TO CUT ALL NECESSARY OPENINGS. ALL PROTRUSIONS, MARKS, CRACKS, BROKEN AND/OR OTHER EVIDENCE OF DAMAGED OR UNSOUND STRUCTURAL CONDITIONS SHALL BE REMOVED, REPLACED, REPAIRED, OR RESTORED AS DIRECTED BY GOVERNING AUTHORITIES.
21. ALL CLEANUP OF CONSTRUCTION ON OPERATING SIDE OF SPACE MUST BE HANDLED DAILY.
22. G.C. TO PROVIDE ALL LIFE SAFETY SYSTEMS INCLUDING, BUT NOT LIMITED TOO TEMPORARY LIGHTING, BARRICADES, GUARD RAILS AND VENTILATION SYSTEMS AS REQUIRED BY LOCAL, STATE AND FEDERAL JURISDICTIONS.

DEMOLITION KEY NOTES

1. LEASE LINE. VERIFY EXACT LOCATION IN FIELD
2. STOREFRONT SYSTEM AND ENTRY DOORS TO BE PROTECTED DURING ALL PHASES OF CONSTRUCTION, CLEAN, PATCH AND REPAIR AS REQUIRED
3. EXTERIOR WALL PATCH AND REPAIR AS REQUIRED TO MAINTAIN EXISTING FIRE RATING.
4. LIGHT FIXTURES AND EMERGENCY LIGHTING TO BE REMOVED. SEE REFLECTED CEILING PLAN AND POWER PLAN FOR FURTHER INFORMATION.
5. COLUMN, TO BE PROTECTED DURING CONSTRUCTION. REPAIR EXISTING FIRE PROOFING AS NEEDED. VERIFY EXACT LOCATION AND SIZE IN THE FIELD.
6. EXTERIOR FACADE SYSTEM TO REMAIN. PROTECT DURING CONSTRUCTION AND REPAIR AS REQUIRED.
7. REFER TO MECHANICAL DRAWINGS FOR MECHANICAL SCOPE. COORDINATE WITH LANDLORD OPERATIONS ON SITE PRIOR TO WORK.
8. CANOPY TO REMAIN
9. PREPARE SURFACES FOR NEW INSTALLATIONS.
10. ELECTRICAL EQUIPMENT. SEE POWER PLAN FOR FURTHER INFORMATION
11. TRENCH FLOOR FOR SERVICES-SEE MEP DRAWINGS
12. FIRE RISER TO REMAIN.
13. DOMESTIC WATER TO REMAIN.
14. ROOF ACCESS HATCH AND LADDER TO REMAIN.
15. OVERHEAD ROLL UP DOOR TO REMAIN.
16. EXTERIOR DOOR TO REMAIN

BARNES&NOBLE

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BUILDING VALUE SINCE 1994

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Professional Seal:

STATE OF COLORADO
WILLIAM J COMMER
LICENSED ARCHITECT

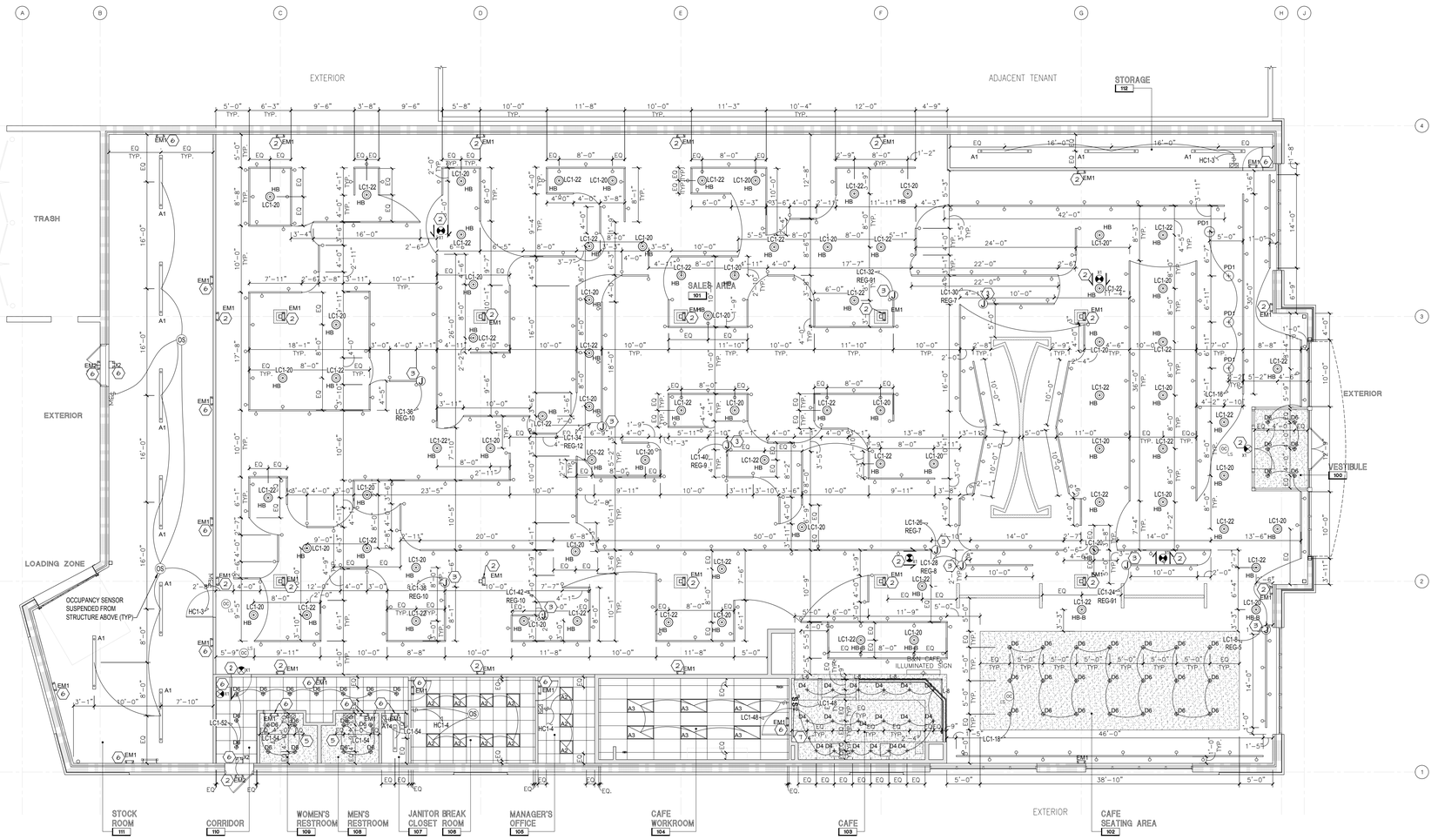
LIC.# : 305353
EXP. DATE : 10/31/25

Drawing Description:

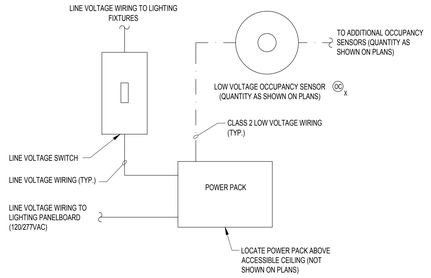
Demolition Plan

Drawing Number:

DM-101



NOTE:
A FINAL COMMISSIONING REPORT SHALL BE DELIVERED TO THE BUILDING OWNER PER SECTION C408.2.5 OF THE 2021 IECC.



TYPICAL LOW VOLTAGE OCCUPANCY SENSOR WIRING DIAGRAM (NON-LIGHTSTAT)

N.T.S.

NOTES:

1. SENSOR SWITCH IS SHOWN AS THE BASIS OF DESIGN. REFER TO SPECIFICATIONS FOR ADDITIONAL INFORMATION AND OTHER ACCEPTABLE MANUFACTURERS. IF ANY OF THE OTHER ACCEPTABLE MANUFACTURERS ARE USED, NOTE THAT THEIR EXACT REQUIREMENTS, COMPONENTS AND WIRING WILL LIKELY DIFFER AND THE CONTRACTOR IS RESPONSIBLE TO PROVIDE A COMPLETE, OPERATIONAL SYSTEM EQUIVALENT TO THAT SHOWN.
2. WIRING DIAGRAMS ARE DIAGRAMMATIC AND SHOWN FOR REFERENCE ONLY. REFER TO MANUFACTURER'S WIRING DIAGRAMS FOR ALL REQUIRED COMPONENTS AND WIRING.
3. INSTALL LOW VOLTAGE WIRING IN 3/4" CONDUIT (MIN.).
4. REFER TO MANUFACTURER'S LATEST LITERATURE FOR REQUIRED WIRING SIZES. INCREASE AS REQUIRED FOR VOLTAGE DROP.
5. MOUNT POWER PACK IN DEEP JUNCTION BOX IF REQUIRED PER LOCAL AHJ.
6. LOW VOLTAGE WIRING IS NOT SHOWN ON PLANS AND NOT ALL LINE VOLTAGE WIRING IS SHOWN ON PLANS. PROVIDE ALL REQUIRED LINE AND LOW VOLTAGE WIRING FOR A COMPLETE, FUNCTIONAL SYSTEM.
7. SOME AREAS HAVE OCCUPANCY SENSOR CONTROL ONLY (NO LINE VOLTAGE SWITCHES) AS SHOWN ON PLANS.

CONTRACTOR TO FURNISH AND INSTALL CURRENT LIMITERS TO LIMIT THE WATTAGE AVAILABLE FOR THE TRACK. CONTRACTOR TO UTILIZE CON-TECH LIGHTING LA-23-R (TITLE 24 COMPLIANT-CALIF. ONLY) OR LA-23-RN (NON-TITLE 24 COMPLIANT) CURRENT LIMITING DEVICE OR EQUIVALENT.

CON-TECH CURRENT LIMITING DEVICES		
REG-05	60W	0.5A
REG-1	120W	1A
REG-2	210W	1.75A
REG-21	240W	2A
REG-3	300W	2.5A
REG-4	360W	3A
REG-5	480W	4A
REG-6	600W	5A
REG-7	720W	6A
REG-8	840W	7A
REG-9	900W	7.5A
REG-91	960W	8A
REG-10	1,200W	10A
REG-12	1,440W	12A
REG-14	1,680W	14A

LIGHTING KEY NOTES (#)

1. NOT USED.
2. CONNECT EXIT SIGNS AND EM/NL IN SALES AREA TO UNSWITCHED CIRCUIT LC1-5. ALL FIXTURES ON THIS CIRCUIT ARE EMERGENCY LIGHTS OR EXIT SIGNS.
3. FURNISH AND INSTALL IN-LINE CURRENT LIMITER.
4. NOT USED.
5. CONTRACTOR TO TIE TOILET EXHAUST FANS TO WALL MOUNTED OCCUPANCY SENSOR.
6. LIGHTING NOTE: NEC SECTION 700.12 (F) ALL UNIT EQUIPMENT (EMERGENCY LIGHTING) SHALL BE WIRED AHEAD OF SWITCH OR CONTROL OF NORMAL LIGHTING CIRCUIT SERVING THAT AREA. THE BRANCH CIRCUIT FEEDING THE UNIT EQUIPMENT SHALL BE THE SAME BRANCH CIRCUIT AS THAT SERVING THE NORMAL LIGHTING IN THE AREA AND CONNECTED AHEAD OF ANY LOCAL SWITCHES. THE BRANCH CIRCUIT THAT FEEDS UNIT EQUIPMENT SHALL BE CLEARLY IDENTIFIED AT THE DISTRIBUTION PANEL IN A SEPARATE AND UNINTERRUPTED AREA SUPPLIED BY A MINIMUM OF THREE NORMAL LIGHTING CIRCUITS. A SEPARATE BRANCH CIRCUIT FOR UNIT EQUIPMENT SHALL BE FROM THE SAME PANELBOARD AS THAT OF THE NORMAL LIGHTING CIRCUITS AND PROVIDED WITH A LOCK-ON FEATURE.
7. LOCAL SWITCHES FOR CAFÉ FIXTURES. VERIFY FINAL LOCATIONS AND MOUNTING HEIGHTS WITH OWNER PRIOR TO ROUGH-IN.

LIGHTING GENERAL NOTES

A. REFER TO SHEET E-201 FOR SYMBOL LEGEND, LIGHTING FIXTURE SCHEDULES AND NOTES.

B. ALL LIGHTING FIXTURES IN SALES AND STOCK AREA S TO BE CONTROLLED BY THE LIGHTSTAT LIGHTING CONTROL SYSTEM. SEE DETAILS ON SHEET E-203.

C. CONNECT SWITCHES AND BRANCH CIRCUITS TO OUTLETS, EQUIPMENT AND LIGHTING FIXTURES AS INDICATED ON PLANS AND PANELBOARD SCHEDULES WITH #12 MINIMUM CONDUCTORS IN CONCEALED RACEWAYS. CONDUCTOR SIZES AND QUANTITIES OTHER THAN #12 AWG SHALL BE AS NOTED. ALL FINAL CONNECTIONS TO TRANSFORMERS, MOTORIZED AND VIBRATING EQUIPMENT SHALL BE WITH FLEXIBLE METAL CONDUIT.

D. CEILING MOUNTED OCCUPANCY SENSOR SHALL CONTROL LIGHTING IN ROOMS IDENTIFIED EXCEPT UNSWITCHED EMERGENCY LIGHTING. COORDINATE WITH ALL OTHER TRADES BEFORE FINAL PLACEMENT. SET TIME DELAY FOR ALL CEILING SENSORS AT 15 MINUTES.

E. LIGHTING CIRCUITS UP TO 100'-0" IN LENGTH SHALL UTILIZE #12 CONDUCTORS MINIMUM. LIGHTING CIRCUITS FROM 101'-0" TO 150'-0" IN LENGTH SHALL UTILIZE #10 CONDUCTORS MINIMUM. LIGHTING CIRCUITS FROM 151'-0" TO 200'-0" IN LENGTH SHALL UTILIZE #8 CONDUCTORS MINIMUM. LIGHTING CIRCUITS ABOVE 251'-0" IN LENGTH SHALL UTILIZE #6 CONDUCTORS MINIMUM.

F. ALL CONDUIT IN EXPOSED AREAS SHALL BE PAINTED. CONDUIT'S PAINT TO MATCH THE SAME COLOR OF DECK ABOVE. ALL CAMERA LOCATIONS TO BE PROVIDED WITH CONDUIT (PENDANDTS).

G. ALL SHADED FIXTURES INCLUDING EXIT SIGNS SHALL BE USED EMERGENCY LIGHTING PURPOSES (WITH BUILT-IN BATTERY).

H. ✕ INDICATES AN ELECTRICAL BREAK IN TRACK. (TYPICAL) AN ELECTRICAL BREAK IN THE TRACK INDICATES AN LIFEED TRACK ACCESSORY, AN ELECTRICAL BREAK AT THE END OF THE TRACK INDICATES A LIVE END TRACK ACCESSORY.

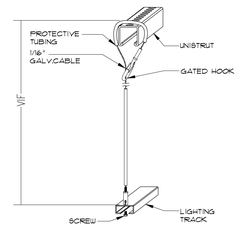
I. LOCATE EXIT SIGNS AS REQUIRED BY LOCAL CODE.

GENERAL NOTES:

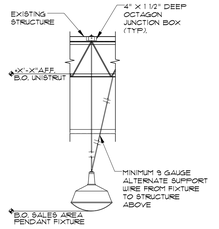
1. VERIFY ALL EXISTING CONDITIONS AND ALL NEW PROPOSED WORK, I.E. DUCTWORK, ELECTRICAL CONDUIT, SPRINKLER LINES, ETC. CONTACT PM IF THERE ARE ANY CONFLICTS WITH PROPOSED DESIGN.

2. ALL UNISTRUT, J-BOXES, THREADED ROD, ETC ABOVE TO BE PAINTED

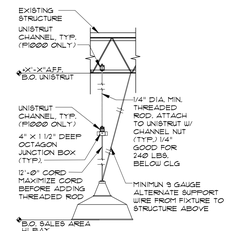
3. MOUNTING HEIGHT OF UNISTRUT AND LENGTH OF THREADED ROD TO BE COORDINATED IN FIELD AROUND EXISTING STRUCTURE CONDITIONS AND HVAC DUCTWORK. MAINTAIN HEIGHTS SHOWN FOR LIGHTING FIXTURES ON PLAN



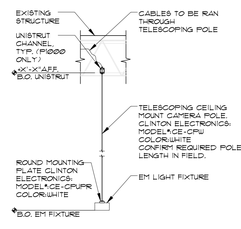
A DETAIL AT TRACK LIGHT FIXTURE



C DETAIL AT CASH WRAP PENDANT LIGHT FIXTURE



B DETAIL AT SALES AREA HI-BAY FIXTURE



D DETAIL AT EM LIGHT FIXTURE

BARNES&NOBLE

PROJECT DESIGNER:
WJCA
BUILDING VALUE SINCE 1994

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CO. LICENSE NUMBER: 42050
EXPIRES: 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	0
06-12-25	LL AND BD COMMENTS	1
07-23-25	BD COMMENTS	2

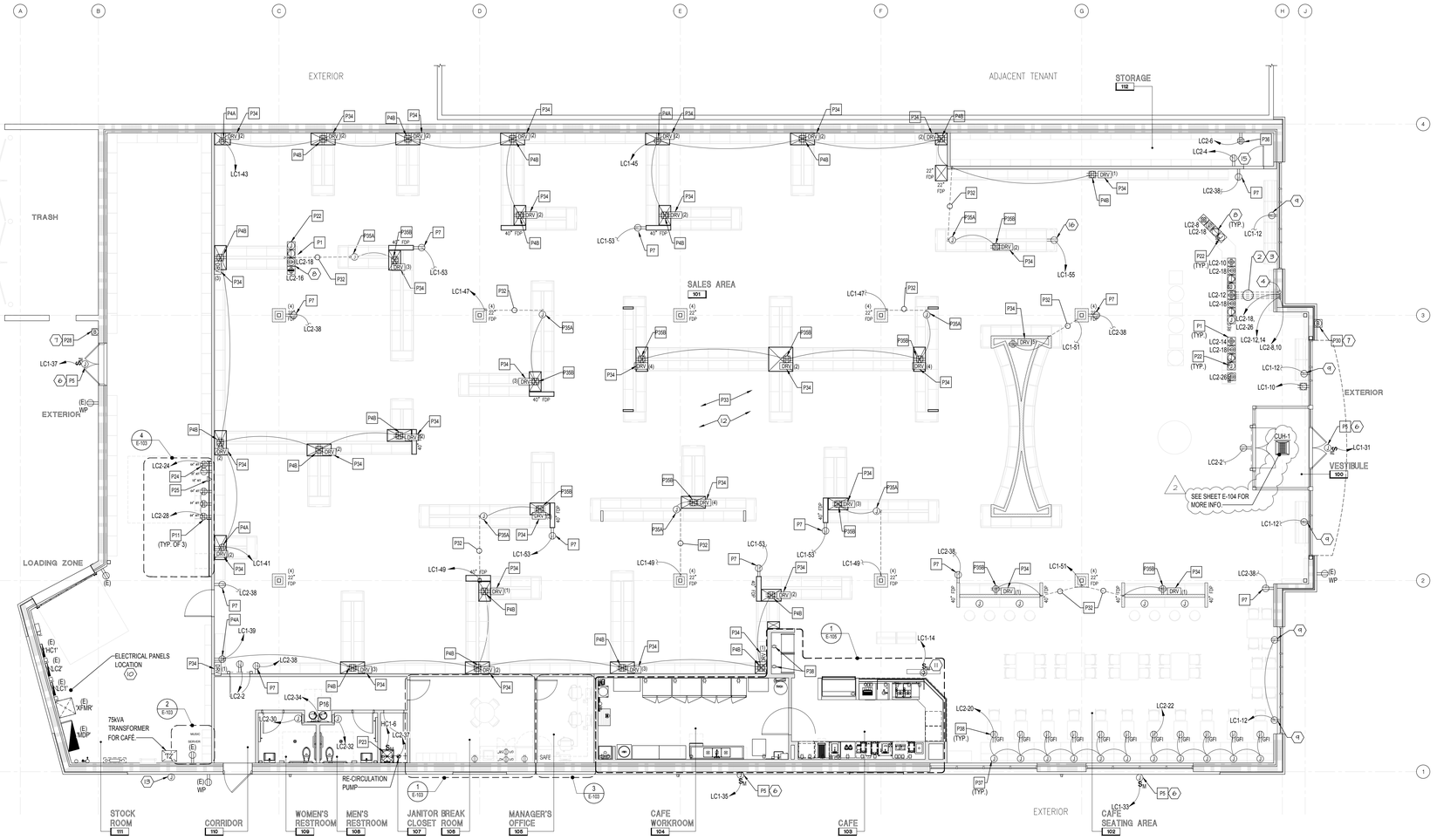
These plans are an instrument of service and the property of the Architect, infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:
LIGHTING PLAN

Drawing Number:
E-101



POWER PLAN

NOTE:
A FINAL COMMISSIONING REPORT SHALL BE DELIVERED TO THE BUILDING OWNER PER SECTION C408.2.5 OF THE 2021 IECC.

GENERAL NOTES

1. COORDINATE FIXTURE GANGBOX LOCATIONS WITH FIXTURE VENDOR PRIOR TO INSTALLATION.
2. COORDINATE FIXTURE JUNCTION BOX LOCATIONS AND REQUIREMENTS WITH FIXTURE VENDOR.
3. PROVIDE CHASE IN FLOOR SLAB FOR ELECTRICAL CONDUITS REQUIRED. COORDINATE WITH ARCHITECTURAL AND ELECTRICAL DRAWINGS. PATCH FLOOR AS REQUIRED FOR FINISHED FLOOR INSTALLATION.
4. NOT USED.
5. GENERAL CONTRACTOR SHALL COORDINATE ALL HOLES/GROMMET LOCATIONS IN COUNTERS WITH MILLWORK AND EQUIPMENT SUPPLIER. HOLES PREDRILLED BY MILLWORK VENDOR. GROMMETS FURNISHED AS WELL.
6. REFER TO A-101 FOR DIMENSIONAL INFORMATION.
7. GENERAL CONTRACTOR SHALL PROVIDE AND INSTALL ALL CONDUIT & PULL STRINGS AS SPECIFIED.
8. B&N VENDOR TO PROVIDE AND INSTALL ALL CATEGORY 6 CABLE, RJ45 TERMINATIONS, HUBS, TRANSFORMERS, AND SURGE PROTECTORS AS REQUIRED.
9. GENERAL CONTRACTOR SHALL PROVIDE JUNCTION BOXES AND CONDUIT STUBS (ABOVE CEILING) AS REQUIRED FOR HVAC THERMOSTATS, REMOTE T-STAT SENSORS, MUSIC VOLUME CONTROLS, FIRE ALARM DEVICES, AND SECURITY SYSTEM DEVICES. COORDINATE WITH MECHANICAL DOCUMENTS, BARNES & NOBLE VENDORS, AND LANDLORD (IF APPLICABLE) FOR QUANTITY AND LOCATIONS.

   DUCT SMOKE DETECTOR RESET/TEST STATIONS SHALL BE LOCATED AS FOLLOWS:

   LOCATE IN MANAGER'S OFFICE OR LOCATE ON REAR FACE OF COLUMN NEAREST TO ASSOCIATED HVAC UNIT.

OUTLET AND DEVICE SCHEDULE

NOTE: GENERAL CONTRACTOR SHALL PROVIDE "KROY" TYPED LABELS INDICATING PANEL AND CIRCUIT NUMBER FOR ALL OUTLETS AND JUNCTION BOXES. ADDITIONALLY, THOSE OUTLETS DESIGNATED FOR USE BY THE MUSIC SYSTEM SHALL BE LABELED WITH RDN/PANEL/CIRCUIT AND AN ADHESIVE RED DOT SHALL BE APPLIED AT BOTH THE OUTLET AND PANEL BREAKER.

PROVIDE ELECTRICAL DEVICES AND COVER PLATES IN THE FOLLOWING COLORS PER LOCATION:

ISOLATED GROUND RECEPTACLES SHALL BE IDENTIFIED BY AN ORANGE TRIANGLE LOCATED ON THE FACE OF THE RECEPTACLE, PER NEC SECTION 410-56(c).

A. GENERAL SALES AREA
   DEVICES - WHITE
   COVER PLATES - WHITE

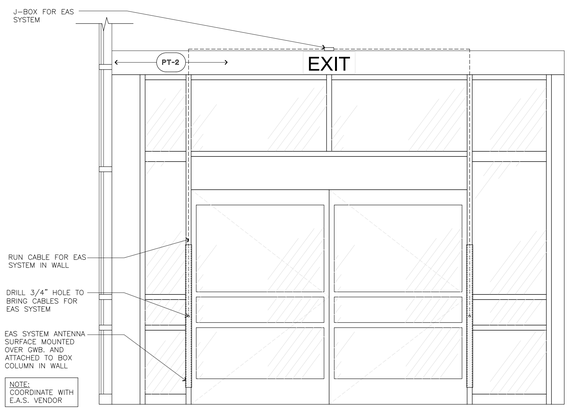
B. GENERAL SALES AREA AND CAFE SEATING AREA
   DEVICES - WHITE
   COVER PLATES - WHITE

C. JUNIOR'S SALES AREA AND MUSIC SALES AREA.
   DEVICES - WHITE
   COVER PLATES - WHITE

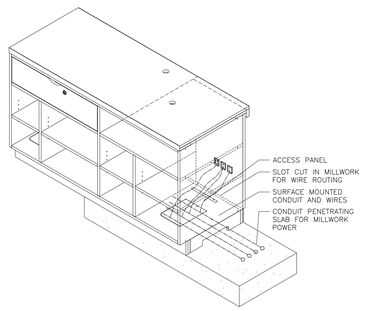
D. TOILETS
   DEVICES - WHITE
   COVER PLATES - WHITE

E. BREAK ROOM, OFFICES, STORAGE (BACK OF HOUSE), AND PAINTED STOREFRONT HEADS.
   DEVICES - WHITE
   COVER PLATES - WHITE

F. GENERAL SALES AREA (WITHIN ACT-1 FINISH) & JANITOR'S CLOSET
   DEVICES - WHITE
   COVER PLATES - WHITE



TYPICAL EAS SYSTEM ELEVATION DETAIL



CASHWRAP ISOMETRIC DETAIL

POWER PLAN KEY NOTES P#

P1	CASHWRAP: RECESSED MOUNTED POWER DEVICES INSTALLED ON FLOOR OR WALL. BELOW FIXTURE ACCESS PANEL FOR WIRE ROUTING. ALL CONDUIT FOR TELEPHONE, DATA SYSTEM CABLE AND EMPTY CONDUIT FOR SPARE SHALL RUN IN FLOOR CHASE TO NEAREST WALL. RUN UP WALL AND OVERHEAD TO TELEPHONE BOARD. VERIFY REQUIREMENTS IN FIELD AND WITH SYSTEM SUPPLIER. METHOD OF CORE DRILLING, TRENCHING, ETC. SHALL BE PER BUILDING REQUIREMENTS. PROVIDE A 109" WHIP TO EXTEND BELOW FIXTURE TOE-KICK. ALL WIRING TO BE RUN IN CONDUIT FOR THE ENTIRE RUN BACK TO PANEL.
P2	VC AND KEY SHUT OFF SWITCH FOR MUSIC SYSTEM AT CASHWRAP BY BN I.T. VENDOR.
P3	NOT USED.
P4A P4B	FIXTURE SHELF LIGHTING TO BE ASSEMBLED IN TWO (2) PHASES: • PHASE # 1 (PRIOR TO MILLWORK CABINET INSTALLATION). PROVIDE A WALL MOUNTED QUAD OUTLET "P4A" AT EACH DESIGNED FILLER PANEL LOCATION. QUAD OUTLET "P4A" TO BE MOUNTED AT 91-1/8" AFF, HORIZONTALLY MOUNTED, TO CENTER OF QUAD OUTLET. • PHASE # 2 (AFTER MILLWORK CABINET INSTALLATION). PROVIDE SURFACE MOUNTED QUAD OUTLET "P4B" ON TOP OF EACH FILLER PANEL AS SHOWN. DISTRIBUTE POWER THROUGH METAL CONDUIT FROM THE WALL MOUNTED QUAD OUTLET "P4A", IN A DAISY CHAIN ARRANGEMENT, TO EACH OF THE OTHER SURFACE MOUNTED QUAD OUTLETS "P4B". METAL CONDUIT TO BE ROUTED ON THE TOP SURFACE OF THE FILLER PANELS AND CABINETS. QUAD OUTLETS "P4A" AND "P4B" TO BECOME THE SOURCE OF POWER OF ALL REQUIRED DRIVERS FOR SHELF LIGHTING. POWER SHOULD BE EXTENDED AND DISTRIBUTED IN METAL CONDUIT TO ALL SHELF LIGHTING WITHIN MILLWORK CABINETS . REFER TO POWER KEY NOTE # P34 FOR MORE INFORMATION.
P5	BUILDING SIGNS: JUNCTION BOXES WITH 20 AMP CIRCUITS (120V/60HZ) FOR BARNES & NOBLE SIGN (MOUNT WITHIN CANOPY BULKHEAD -OR- BEHIND WALL ABOVE FINISHED CEILING). VERIFY EXACT QUANTITY & LOCATION WITH SHOP DRAWINGS PROVIDED BY B&N SIGN VENDOR PRIOR TO INSTALLATION. VERIFY ALL SIGN LOCATIONS WITH SIGN VENDOR DRAWINGS. ALL WIRING, CONNECTIONS, AND TRANSFORMERS SHALL BE CONCEALED ON THE INTERIOR PORTION OF THE TENANT STOREFRONT AREA.
P6	NOT USED.
P7	UTILITY OUTLET: AT 18" AFF (TYPICAL)
P11	PDT CHARGERS: INSTALL THREE (3) QUAD RECEPTACLES AT 64" A.F.F. ABOVE FIRST SHELF OF THE RECEIVING TABLE SHELVING SPACED AS SHOWN ON A-413, DETAIL 7A.
P16	DRINKING FOUNTAIN: JUNCTION BOX AT 18" AFF (CORRIDOR).
P20	NOT USED.
P22	36" WHIP AND J-BOX - SEE CASHWRAP ISOMETRIC DETAIL ON THIS SHEET E-102
P23	NEW ELECTRIC WATER HEATER: (JANITORS CLOSET) PROVIDE 30 AMP DISCONNECT SWITCH
P24	TELEPHONE SYSTEM: OUTLET AT 48" A.F.F. IN STOCK ROOM. GENERAL CONTRACTOR SHALL PROVIDE CONDUIT AS REQUIRED FROM LL TERMINATION POINT TO PANEL AS SHOWN. VERIFY DISTANCE IN FIELD.
P25	TELEPHONE: AT 48" AFF WITHIN BREAK ROOM (SEE DWG A-412) AND STOCK ROOM (SEE DWG A-413) (SEE DWG A-222).
P28	DELIVERY BUZZER ACTUATOR: AT EXTERIOR STOCK ROOM DOORS. PROVIDE J-BOX WITH WEATHER PROOF SPRING LOADED COVER.
P30	EMPLOYEE BUZZER ACTUATOR: AT EXTERIOR ENTRY DOORS MOUNTED AT 45" AFF. PROVIDE RECESSED J-BOX WITH WEATHER PROOF, SPRING LOADED COVER. VERIFY EXACT LOCATION (HIDDEN FROM PLAIN VIEW) WITH B&N DEVELOPMENT MANAGER / ARCHITECT.
P31	NOT USED.
P32	ROUTE CONDUIT AND WIRE CONCEALED TO WALL. ROUTE CONDUITS BELOW FLOOR. TRENCH & PATCH FLOOR AS REQUIRED. ROUTE TO NEAREST WALL OR COLUMN AND THEN UP TO ACCESSIBLE CEILING SPACE FOR CONTINUATION.
P33	20 NEW CONDUITS WITH SQUARE J-BOX AT BOTTOM WITH PULL STRING FROM DECK TO 12'-0" A.F.F. COORDINATE LOCATION WITH BN VENDOR.
P34	PHASE # 2 (AFTER MILLWORK CABINET INSTALLATION). EACH MILLWORK CABINET IS 3'-0" WIDE AND RUNS HORIZONTAL SHELF LIGHTING FROM SIDE TO SIDE. EVERY 18'-0" OF SHELF LIGHTING (OR EVERY 6 MILLWORK CABINETS), REQUIRE A 100-WATT DRIVER FOR POWER. CONTRACTOR TO VERIFY EXACT ELECTRICAL REQUIREMENTS FOR SHELF LIGHTING WITH VENDOR OF MILLWORK CABINETS PRIOR TO BID. DRIVERS TO BE PROVIDED BY BARNES & NOBLE. DRIVERS TO BE INSTALLED BY BARNES & NOBLE'S MILLWORKERS.
P35A P35B	FIXTURE SHELF LIGHTING TO BE ASSEMBLED IN TWO (2) PHASES: • PHASE # 1 (PRIOR TO MILLWORK CABINET INSTALLATION). PROVIDE A FLOOR MOUNTED J-BOX "P35A" AT EACH DESIGNED FILLER PANEL LOCATION WITH TRENCHING AS SHOWN. • PHASE # 2 (AFTER MILLWORK CABINET INSTALLATION). PROVIDE SURFACE MOUNTED QUAD OUTLET "P35B" ON TOP OF EACH FILLER PANEL AS SHOWN. DISTRIBUTE POWER THROUGH METAL CONDUIT FROM THE FLOOR MOUNTED J-BOX "P35A" TO QUAD OUTLET "P35B". QUAD OUTLET "P35B" TO BECOME THE SOURCE OF POWER OF ALL REQUIRED DRIVERS FOR SHELF LIGHTING. POWER SHOULD BE EXTENDED AND DISTRIBUTED IN METAL CONDUIT TO ALL SHELF LIGHTING WITHIN MILLWORK CABINETS . REFER TO POWER KEY NOTE # P34 FOR MORE INFORMATION.
P36	QUAD RECEPTACLE FOR BACK-UP SERVER.
P37	J-BOX WITH PULL WIRE TO FEED GFI RECEPTACLE IN CAFE SEATING AREA.
P38	COVER PLATES AND DEVICES IN CAFE BENCH SEATING AND CAFE BAR COUNTERS ARE BLACK/BLACK

CODED NOTES

1. NOT USED.
2. PROVIDE POWER CONDUIT(S) WITHIN FLOOR, BETWEEN FLOOR BOX(S) AND/OR NEAREST WALL OR COLUMN. METHOD OF CORE DRILLING, TRENCHING, ETC. SHALL BE PER BUILDING REQUIREMENTS. CONTRACTOR SHALL SIZE CONDUIT PER NEC.
3. PROVIDE DATA/TELECOMMUNICATION CONDUIT WITH PULL WIRE WITHIN FLOOR, BETWEEN FLOOR BOX(S) AND/OR NEAREST WALL OR COLUMN. METHOD OF CORE DRILLING, TRENCHING, ETC. SHALL BE PER BUILDING REQUIREMENTS. CONTRACTOR SHALL SIZE CONDUIT PER NEC.
4. PROVIDE 1-1/2" CONDUITS WITH PULL WIRE BACK TO TELEPHONE BOARD FOR DATA.
5. NOT USED.
6. BARNES & NOBLE SIGN WITH A TOTAL OF (1) CIRCUIT ((2)#10-1"C). VERIFY WIRING ARRANGEMENT WITH SIGN SUPPLIER AND PROVIDE TRANSFORMERS AS REQUIRED. PROVIDE LOCAL DISCONNECT MEANS AS PER NEC REQUIREMENTS (TYPICAL ALL SIGNS). VERIFY MOUNTING HEIGHT AND LOCATION WITH SIGN VENDOR APPROVED DRAWINGS.
7. IF NOT EXISTING, BUZZER (PUSH BUTTON) SHALL BE EDWARDS #600 ROUND LOW VOLTAGE PUSH BUTTON MOUNTED 45" ABOVE FLOOR.
8. FURNISH AND INSTALL RECESSED MOUNTED MULTIPLE GANG (MOUNTED ON FLOOR) BOXES, STEEL CITY, HUBBELL, OR LEW (OR APPROVED EQUAL). FLOOR BOXES SHALL HAVE SEPARATE SECTIONS FOR DATA AND TELEPHONE OUTLET SEPARATE SECTION WITH HUBBELL 1GS362 RECEPTACLE ON SEPARATE CIRCUIT FOR CASH-REGISTER CONNECTION AND ONE SECTION WITH JUNCTION BOX FOR NORMAL POWER. RUN SEPARATE CIRCUIT AND GROUND WIRE IN SEPARATE CONDUIT FOR EACH CASH REGISTER CIRCUIT. SEPARATE HOMERUN REQUIRED AND CONDUIT GROUND NOT PERMITTED FOR CASH REGISTER CIRCUITS. THE GROUND WIRE ON CASH REGISTER CIRCUITS SHALL BE #12 AWG SOLID GREEN ISOLATED. RUN CONTINUOUS FROM ISOLATED GROUND RECEPTACLES THROUGH LOCAL PANEL(S) MUST BE TERMINATED IN SPACES, DISTRIBUTION PANELBOARD OR AT THE DERIVED SYSTEM TRANSFORMER. GROUNDING TO LOCAL PANEL(S) NOT PERMITTED. TAPED SPLIT-BOLTS SOLDERLESS CONNECTORS SHALL BE USED TO ENSURE CONTINUOUS ISOLATED GROUND. (WIRENUTS NOT PERMITTED). VERIFY EXACT LOCATIONS AND CONDITIONS WITH CASHWRAP INSTALLER AT SITE.
9. PROVIDE SHOW WINDOW RECEPTACLE. E.C. SHALL INSTALL RECEPTACLE WITHIN 18" OF THE TOP OF THE SHOW WINDOW FOR EACH 12' OR MAJOR FRACTION THEREOF OF SHOW WINDOW AREA MEASURED HORIZONTALLY AT ITS MAXIMUM WIDTH.
10. NON-SALES ELECTRICAL EQUIPMENT IS EXISTING TO REMAIN. MAINTAIN EXISTING CIRCUITING & CONTROLS.
11. PROVIDE JUNCTION BOX IN CEILING FOR ILLUMINATED SIGN. COORDINATE WITH ARCHITECTURAL SHEETS FOR MORE INFORMATION.
12. ALL FIRE ALARM EQUIPMENT IS EXISTING TO REMAIN. IF DEVICES ARE REQUIRED TO BE REMOVED FOR DEMOLITION - KEEP/RE-INSTALL IN SAME LOCATION.
13. JUNCTION BOX MOUNTED ON EXTERIOR OF BUILDING TO RUN INSIDE TO CCTV MONITOR LOCATION.
14. NOT USED.
15. KEY PAD RECEPTACLE AT 80" AFF.
16. RECEPTACLE FOR NOOK.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
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NEW JERSEY LICENSED PROFESSIONAL ENGINEER
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CO. LICENSE NUMBER: 42050
EXPIRES: 10-31-25

3173 CINEMA POINT,
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80922

Project Number 33288
Store Number 3566

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07-23-25	BD COMMENTS	3

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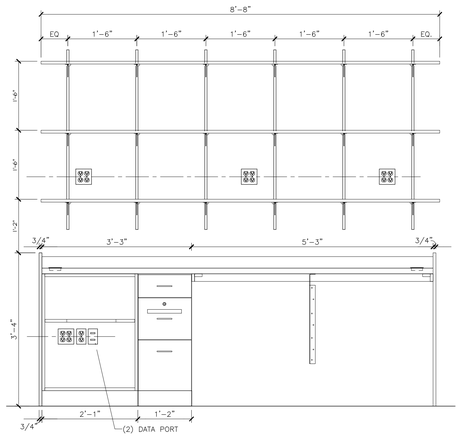
Professional Seal:

Drawing Description:
POWER PLAN

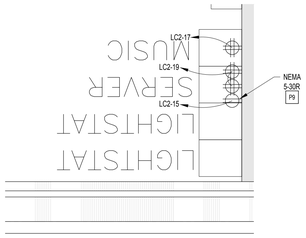
Drawing Number:
E-102



4
E-103

TYPICAL RECEIVING TABLE ELEVATION

3/4" = 1' - 0"



NOTE:
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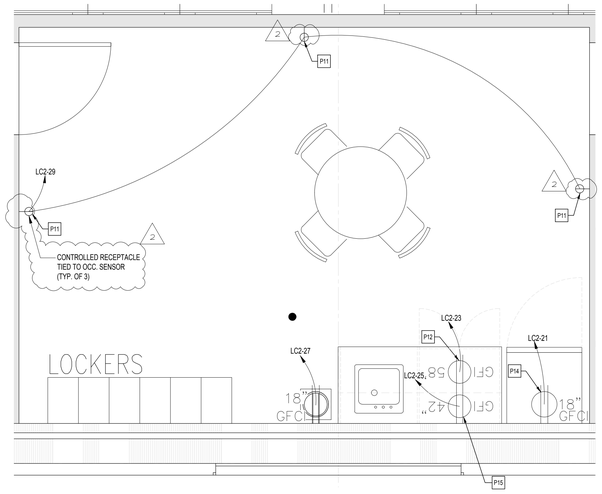
REFERENCE ARCHITECTURAL ELEVATION SHEETS FOR ADDITIONAL INFORMATION.

CODED NOTES (#)

1. FURNISH AND INSTALL RECESSED MOUNTED MULTIPLE GANG (MOUNTED ON FLOOR OR WALL) BOXES, STEEL CITY, HUBBELL OR LEW (OR APPROVED EQUAL). FLOOR AND WALL BOXES SHALL HAVE SEPARATE SECTION FOR TELEPHONE OUTLET, SEPARATE SECTION WITH HUBBELL 105362 RECEPTACLE ON SEPARATE CIRCUIT, ONE SECTION WITH (2) HUBBELL 5362 (QUAD RECEPTACLE) FOR NORMAL POWER, ONE SECTION WITH 6"x6" JUNCTION BOX FOR NETWORK, ONE SECTION WITH 4"x4" JUNCTION BOX FOR A SPARE. SEPARATE HOMERUN REQUIRED AND CONDUIT GROUND NOT PERMITTED ON CLEAN POWER CIRCUITS. THE GROUND WIRE ON CLEAN POWER CIRCUITS SHALL BE #12 AWG SOLID GREEN ISOLATED, RUN CONTINUOUS FROM ISOLATED GROUND RECEPTACLES THROUGH LOCAL PANEL(S) MUST BE TERMINATED IN SPACES, DISTRIBUTION PANELBOARD OR AT THE DERIVED SYSTEM TRANSFORMER. GROUNDING TO LOCAL PANEL(S) NOT PERMITTED. TAPE SPLIT-BOLTS SOLDERLESS CONNECTORS SHALL BE USED TO ENSURE CONTINUOUS ISOLATED GROUND. (WIRENUTS NOT PERMITTED). VERIFY EXACT LOCATIONS AND CONDITIONS AT SITE.
2. NOT USED.
3. NOT USED.

POWER PLAN KEY NOTES

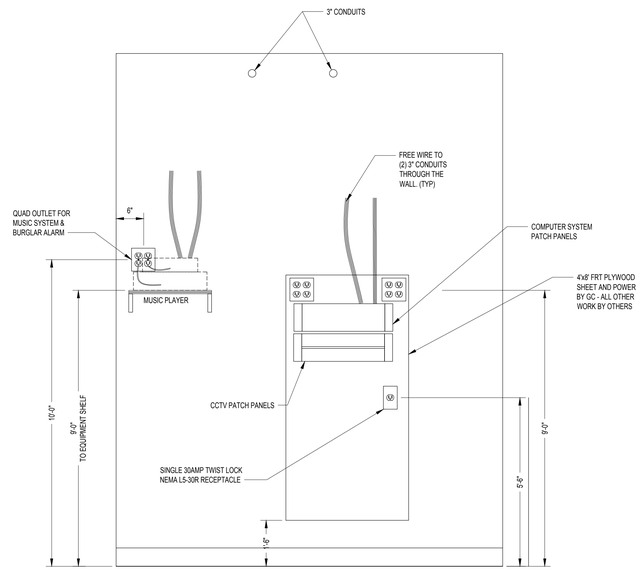
P9	POS SERVER RACK; DEDICATED 30AMP CIRCUIT WITH TWIST LOCK NEMA L5-30R RECEPTACLE
P11	UTILITY OUTLET; AT 18" AFF (TYPICAL)
P12	MICROWAVE; RECESSED/CLOCK GFI OUTLET (BREAK ROOM). SEE DWG A-412.
P14	REFRIGERATOR; GFI DUPLEX RECEPTACLE (BREAK ROOM)
P15	GENERAL CONVENIENCE; GFI OUTLET ABOVE COUNTER (BREAK ROOM). SEE DWG. A-412.
P17	NOT USED.
P18	NOT USED.
P19	NOT USED.
P23	NOT USED.
P24	NOT USED.
P25	NOT USED.
P27	NOT USED.
P29	NOT USED.
P37	NOT USED.
P39	NOT USED.
P40	NOT USED.
P41	NOT USED.



1
E-103

ENLARGED BREAK ROOM POWER PLAN

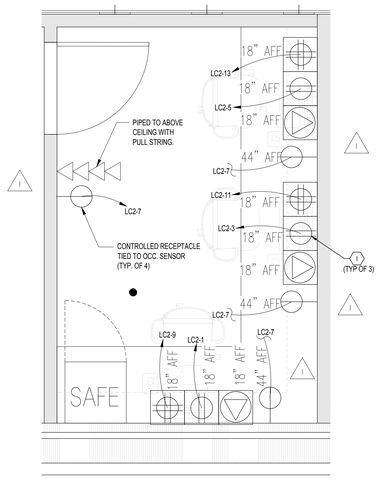
1/2" = 1' - 0"



2
E-103

SERVER ROOM DETAIL

1/2" = 1' - 0"



3
E-103

ENLARGED ELECTRICAL POWER PLAN

1/2" = 1' - 0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
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07-23-25	BD COMMENTS	3

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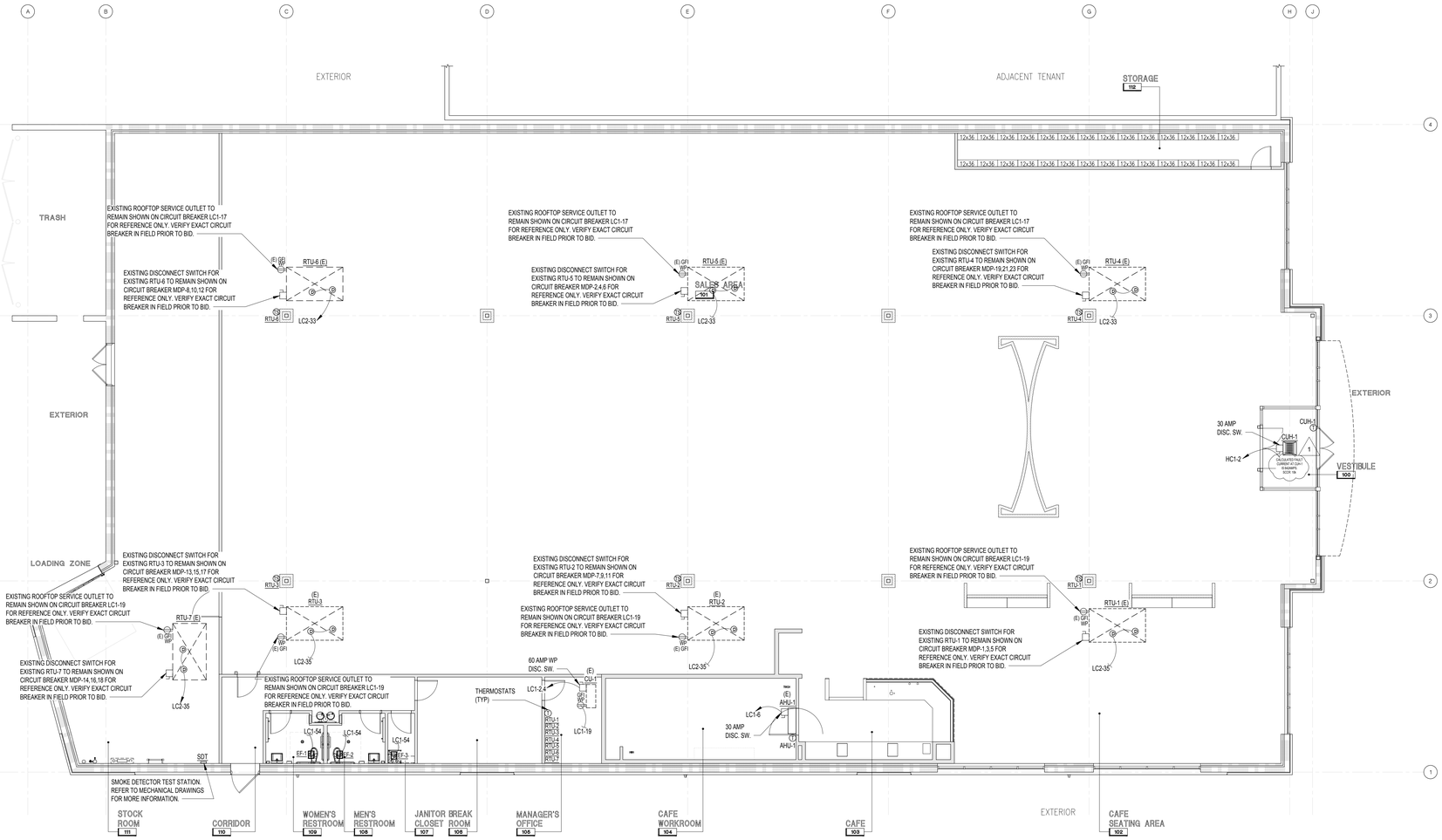
General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:
ENLARGED
POWER PLANS

Drawing Number:
E-103



ELECTRICAL HVAC PLAN

1/8" = 1' - 0"

EXAMPLE HVAC STAT SET UP

1

NOTE:
A FINAL COMMISSIONING REPORT SHALL BE DELIVERED TO THE BUILDING OWNER PER SECTION C408.2.5 OF THE 2021 IECC.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J. COMMER
763 SUSQUEHANNA AVE
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NEW JERSEY LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES: 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
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Professional Seal:

Drawing Description:
ELECTRICAL
HVAC PLAN

Drawing Number:
E-104

GENERAL NOTE:

VERIFY FINAL LOCATIONS WITH KITCHEN DRAWINGS PRIOR TO ROUGH-IN.

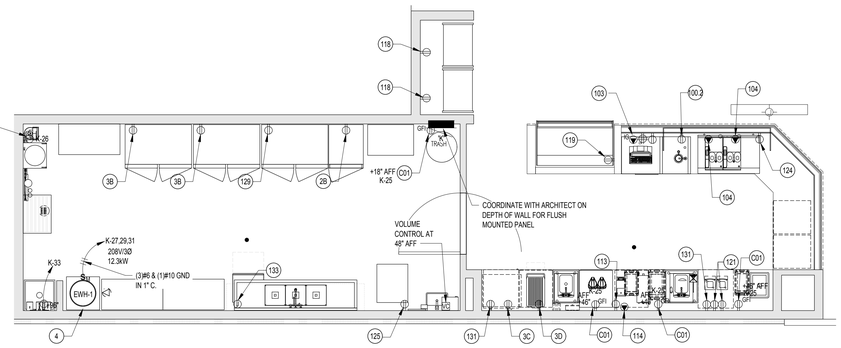
GENERAL NOTE:

VERIFY ALL KITCHEN EQUIPMENT, SPECIFICATIONS, REQUIREMENTS, ETC. WITH KITCHEN VENDOR DRAWINGS & OWNER PRIOR TO BID.

ELECTRICAL KITCHEN SYMBOL LEGEND

SYMBOL	DESCRIPTION
	WALL/FIXTURE MOUNTED DUPLEX OUTLET
	SPECIAL OUTLET. SEE CAFE EQUIPMENT POWER SCHEDULE FOR PLUG TYPE.
J	WALL MOUNTED JUNCTION BOX

MOUNT RECEPTACLE +64-72" AFF FOR WATER SOFTENER. COORDINATE WITH MANUFACTURE CUT SHEETS FOR FINAL MOUNTING HEIGHT, LOCATION, AND POWER REQUIREMENTS.



1 E-105 ENLARGED KITCHEN POWER PLAN

1/4" = 1' - 0"

GENERAL NOTES:

1. ALL DEVICES IN CAFE SHALL BE GFI.
2. FOR LOCATION OF ELECTRICAL DEVICES, REFER TO ARCHITECTURAL EQUIPMENT/CABINET PLANS (A-227 & A-227a). ALL DEVICES TO ALIGN WITH DESIGNATED EQUIPMENT AND CABINET LOCATIONS AS SHOWN. NO EXCEPTIONS SHALL BE TAKEN.
3. GENERAL CONTRACTOR SHALL COORDINATE ELECTRICAL INSTALLATION WITH PLUMBING FOR INSTALLATION OF CARLON. SCHEDULE 40 CONDUIT CHASE FOR FILTER TUBING AS REQUIRED.
4. SERVICE COUNTER MILLWORK NOTE: EACH ELECTRICAL DEVICE WITHIN REAR COUNTER OF SERVICE AREA SHALL BE STUBBED-OUT WITH M.C. CONDUIT. GENERAL CONTRACTOR SHALL PROVIDE AND COORDINATE FINAL RECESSED INSTALLATION OF JUNCTION BOXES WITH MILLWORK INSTALLATION.

KITCHEN ELECTRICAL NOTES

1. ALL COVERPLATES AND DISCONNECT SWITCHES IN KITCHEN AREA SHALL BE STAINLESS STEEL.
2. ALL ELECTRICAL WORK FOR FOOD SERVICE EQUIPMENT SHALL BE COMPLETELY INTERWIRED BY ELECTRICAL CONTRACTOR. FINAL CONNECTIONS TO EQUIPMENT JUNCTION BOX OR PULL BOX, AND ALL ELECTRICAL WORK FROM PANEL BOARDS, TO BE BY THE ELECTRICAL CONTRACTOR.
3. THE ELECTRICAL CONTRACTOR SHALL PROVIDE ALL ROUGH-IN AND FINAL CONNECTION TO THE FOOD SERVICE EQUIPMENT. ALL WORK TO BE IN COMPLIANCE WITH ALL NATIONAL, STATE AND LOCAL CODES APPLICABLE.
4. VERIFY OUTLET RATING AND CONFIGURATION WITH EQUIPMENT SUPPLIER PRIOR TO ROUGH-IN.
5. VERIFY EXACT LOCATION AND MOUNTING HEIGHTS OF ALL OUTLETS WITH EQUIPMENT SUPPLIER PRIOR TO ROUGH-IN.
6. ELECTRICAL CONTRACTOR TO FURNISH AND INSTALL ALL PLUGS AND CORDS REQUIRED. ALL CORDS SHALL BE NEMA RATED AND UL APPROVED FOR MANUFACTURER AND EQUIPMENT.
7. ELECTRICAL CONTRACTOR TO FURNISH AND INSTALL ALL JUNCTION BOXES, PVC OR METAL CONDUIT, CONVENIENCE OUTLETS WITH COVERS, SWITCHES CONNECTORS, CONTROLS, AND OTHER ACCESSORIES THAT ARE NOT AN INTEGRAL PART OF THE FOOD SERVICE EQUIPMENT AS REQUIRED TO MAKE FINAL CONNECTIONS TO THE FOOD SERVICE EQUIPMENT FOR A COMPLETE AND FUNCTIONAL OPERATION MEETING ALL APPLICABLE NATIONAL, STATE, AND LOCAL CODES AND ORDINANCES.
8. ELECTRICAL CONTRACTOR TO FURNISH AND INSTALL ALL DISCONNECTS OR CIRCUIT BREAKERS AS REQUIRED BY CODES FOR EACH CONNECTION. COORDINATE LOCATION WITH THE KITCHEN EQUIPMENT CONTRACTOR.
9. ELECTRICAL CONTRACTOR TO FURNISH AND INSTALL WALL SWITCH FOR FACTORY INSTALLED LIGHTING FIXTURES IN EXHAUST VENTILATOR HOODS PER APPLICABLE STATE AND LOCAL CODES APPLICABLE. ELECTRICAL CONTRACTOR SHALL PROVIDE WIRE AND CONNECTION TO EACH LIGHT FIXTURE. THE ELECTRICAL CONTRACTOR SHALL FULLY CONCEAL ALL WIRING BETWEEN POWER SOURCE, WALL SWITCH, AND JUNCTION BOX ON HOOD. THE ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING ANY INNER WIRING OF LIGHT FIXTURES BETWEEN VENTILATOR HOOD SECTIONS AS REQUIRED. ALL WIRING WITHIN HOOD AND POWER SOURCE TO BE IN STRICT ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS, NFPA #96 AND ALL OTHER APPLICABLE CODES.
10. IN ACCORDANCE WITH NFPA #96 AND MANUFACTURER'S RECOMMENDATIONS, THE ELECTRICAL CONTRACTOR SHALL FURNISH AND INSTALL A PUSH BUTTON STATION WITH PILOT LIGHT FOR VENTILATOR FAN MOTOR(S). THE ELECTRICAL CONTRACTOR IS TO BE RESPONSIBLE FOR, AND TO PROVIDE ALL REQUIRED WIRING FROM POWER SUPPLY THROUGH FAN SWITCH TO FAN MOTOR(S) AND PROVIDE MAGNETIC STARTERS AND FULLY INTERWIRE SYSTEM WITH ALL POWER INTERRUPTION DEVICES(S) BUILT INTO HOOD AND FIRE PROTECTION SYSTEM AS REQUIRED BY NFPA #96, NATIONAL, STATE AND/OR LOCAL CODES APPLICABLE. ELECTRICAL CONTRACTOR TO PROVIDE LOCK-OUT DEVICES ON CONTROL BOXES FOR EXHAUST HOOD FANS, SYSTEM AND FIRE PROTECTION SYSTEM.
11. ELECTRICAL CONTRACTOR TO PROVIDE, INSTALL AND FULLY WIRE SHUNT-TRIP BREAKERS FOR SHUT DOWN OF FUEL AND POWER TO COOKING EQUIPMENT AS REQUIRED BY NFPA #96 AND ALL OTHER NATIONAL, STATE, OR LOCAL CODES APPLICABLE. THE HOLDING COILS FOR SHUNT-TRIP BREAKERS SHALL BE WIRED TO A 120 VOLT/SINGLE PHASE CONTROL CIRCUIT BY THE ELECTRICAL CONTRACTOR AND EXTENDED THROUGH A CONTACTOR AND MAINTAINED BY A PRESSURE SWITCH LOCATED AT THE MOUNTING BRACKET OF THE CHEMICAL CYLINDER FOR HOOD PROTECTION. THE ELECTRICAL CONTRACTOR SHALL ALSO PROVIDE, INSTALL, AND FULLY INTERWIRE WITH POWER SHUTDOWN RELAY SWITCH, AND ADDITIONAL RELAY OR SWITCHES REQUIRED TO INTERFACE FIRE PROTECTION SYSTEM WITH FAN VENTILATOR MOTORS AND BUILDING ALARM SYSTEMS AS REQUIRED BY NFPA #96, NATIONAL, STATE, AND LOCAL CODES APPLICABLE. COORDINATE WITH FIRE SUPPRESSION CONTRACTOR FOR LOCATION OF FIRE SUPPRESSION SYSTEM, AND GAS SHUT-OFF VALVE AS PART OF THE COMPLETE SYSTEM AS APPLICABLE.
12. AT THE REMOTE FIRE CABLE PULL ELECTRICAL TRADES TO PROVIDE EMPTY JUNCTION BOX AT 54" AFF AND CONDUIT CONCEALED IN WALL TO 6" ABOVE FINISHED CEILING. COORDINATE EXACT REQUIREMENTS WITH FOOD SERVICE EQUIPMENT TRADE AND FIRE SUPPRESSION CONTRACTOR.
13. ELECTRICAL CONTRACTOR SHALL INTERWIRE DISPOSER CONTROL SWITCH AND TO TIME DELAY/RELAY, MAGNETIC STARTER, DISPOSER MOTOR, AND SOLENOID VALVE WITH WATER TIGHT CONDUIT AS REQUIRED PER LOCAL CODES.
14. ELECTRICAL CONTRACTOR SHALL INTERWIRE THROUGH TIME CLOCK FOR LOW TEMPERATURE COMPRESSOR AND WALK-IN COMPARTMENT BLOWER COIL FAN MOTORS AND DEFROST ELEMENT POWER SOURCE AS PART OF MAIN POWER SOURCE. PROVIDE ALL WIRING AND CONDUIT WITH DISCONNECT. (VERIFY REQUIREMENTS WITH EQUIPMENT SUPPLIER)
15. ELECTRICAL CONTRACTOR SHALL PROVIDE FINAL CONNECTION TO A JUNCTION BOX MOUNTED ON TOP OF A PREFABRICATED REFRIGERATOR AND/OR FREEZER WALL AT APPROXIMATELY 8'-6" AFF. INTERWIRE THE LIGHT ADJACENT TO THE DOOR WITH THE FACTORY MOUNTED LIGHT SWITCH. ALL CONDUIT SHALL BE RUN EXPOSED ON TOP OF WALK-IN. NO EXPOSED CONDUIT WILL BE ALLOWED ON INSIDE OF WALK-IN. ELECTRICAL SERVICE REQUIRED FOR WALK-IN SHALL BE AS SHOWN FOR LIGHTS, DOOR AND DOOR FRAME HEATER, THRESHOLD PLATE HEATERS (WHERE SPECIFIED), HEATED PRESSURE RELIEF PORT (ON FREEZERS) AND ALARM SYSTEMS (WHERE SPECIFIED). (VERIFY REQUIREMENTS WITH EQUIPMENT SUPPLIER)
16. ELECTRICAL CONTRACTOR TO PROVIDE THE REQUIRED POWER SUPPLY AND FINAL CONNECTIONS TO THE TERMINAL BLOCK AT THE CONDENSING UNIT AND TO FULLY INTERWIRE TO ANY ADDITIONAL COMPONENTS, INCLUDE THE PROPER SIZE DISCONNECTS OR CIRCUIT BREAKERS. ALL WIRING FOR LOW AND MEDIUM TEMPERATURE CONDENSING UNIT TO BE ROUTED THROUGH DEFROST TIME CLOCK AND THEN WIRED TO EVAPORATOR COIL FOR PROPER POWER SUPPLY WITH THE REQUIRED QUANTITY OF WIRES. THE EVAPORATOR COIL DEFROST HEATER AND FAN MOTOR VOLTAGES AND LOADS ARE AS NOTED ON PLAN. VERIFY LOCATION OF COMPRESSORS AND COORDINATE WITH REFRIGERATION CONTRACTOR FOR FINAL CONNECTIONS. (VERIFY REQUIREMENTS WITH EQUIPMENT SUPPLIER)
17. ALL 50AMP OR LESS - SINGLE PHASE AND 100AMP OR LESS - THREE PHASE, RECEPTACLES IN KITCHEN AREAS SHALL BE GFCI PROTECTION PER N.E.C. 210-8(B).

FOOD SERVICE EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY.
1	SINK BOWL, WELD-IN, DUMP SINK	N. WASSERSTROM & SONS	F4480234	1
2	SINK BOWL, WELD-IN, HAND SINK	N. WASSERSTROM & SONS	F4480234	1
3C	BLODGETT OVEN	BLODGETT	CTB SGL	1
3D	RAPID COOK OVEN	MERRYCHEF USA	E2S HIGH CLASSIC	1
100.1	PANTRY FAUCET	T&S BRASS	KL45-4000-WH	2
100.2	HOT WATER DISPENSER	IN SINK ERATOR	C1300	1
103	POS SYSTEM	-	-	1
104	ESPRESSO MACHINE	MELITTA	CT8	2
108	GARBAGE CAN	RUBBERMAID COMMERCIAL PRODUCTS	FG354060BLA	2
111	TOUCH-FREE SOAP DISPENSER	GOJO INDUSTRIES, INC.	PURELL CS6	1
112	SURFACE-MOUNTED PAPER TOWEL DISPENSER	BOBRICK	B-2621	1
112.1	BLENDER RINSER	BLENDTEC	JRE-610	1
113	COFFEE GRINDER	GRINDMASTER-UNIC-CRATHCO	890BS	1
114	COFFEE BREWER	BUNN	53100.0100	1
115	MANIFOLD	-		1
116	GLASS FILLER	T&S BRASS	B-1210	1
117	AIRPOT	SERVICE IDEAS	ECALS22SS	2
118	SELF-SERVICE REFRIGERATED CASE	STRUCTURAL CONCEPTS	B3424	2
119	PASTRY CASE, CURVED GLASS	FEDERAL INDUSTRIES	SSRPF-5052	1
120	DROP-IN ICE BIN	KROWNE	D278	1
121	BLENDER	VITAMIX	36019-ABAB	2
123	DIPPER WELL	T&S BRASS	B-2282-01-F05	1
124	DOUBLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-48-HC	1
131	SINGLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-27-HC	2

WORKROOM EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY
1A	R.O.WATER FILTRATION SYSTEM MANIFOLD BOARD	-	-	1
1B	R.O.WATER FILTRATION SYSTEM - ACCUMULATOR TANK	-	-	1
1C	R.O.WATER FILTRATION SYSTEM - WATER SOFTNER	-	-	1
1D	R.O.WATER FILTRATION SYSTEM - BRINE	-	-	1
2B	REFRIGERATOR, REACH-IN, SINGLE DOOR	TRUE MGF.	STG1R-1S-HC	1
3B	FREEZER, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2F-2S-HC	2
4	WATER HEATER	-	-	1
5	MOP SINK	FIAT	MSB2424	1
5.1	SERVICE FAUCET	CHICAGO FAUCET	897-RCF	1
6	WALL SHELF	SUPPLIED BY BN	-	1
8	SOAP DISPENSER	BN SUPPLIED PURELL	TFX	1
9	HAND SINK - WALL MOUNTED	HAND SINK PARTS & ACCESSORIES	DH-17-NO FAU	1
14	C-FOLD PAPER TOWEL DISPENSER	BN SUPPLIED C-FOLD	-	1
15	WASTE CONTAINER	-	-	1
16	THREE COMPARTMENT SINK	AMTEKCO INDUSTRIES LTD.	D724-03-74	1
16.1	PRE-RINSE FAUCET & ADD ON FAUCET			1
16.4	LEVER WASTE	T&S BRASS	B-3950	3
125	ICE MAKER, CUBE-STYLE	ITV ICE MAKER	SPIKA MS 500	1
127	WIRE SHELVING	1880 HOSPITALITY	FF2472C	4
129	REFRIGERATOR, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2DT-2S	1
133	DISHWASHER, UNDERCOUNTER	JACKSON WWS	DISHSTAR HT	1

CAFE EQUIPMENT POWER SCHEDULE

EQUIPMENT NO.	DESCRIPTION	QUANTITY	VOLTAGE AND PHASE	AMPS	HP	KVA	PANEL	CIRCUIT(S)	BREAKER SIZE	WIRE SIZE	CONDUIT	NEMA
129	REFRIGERATOR/FREEZER, REACH-IN, DOUBLE DOOR	1	115V/1Ø	11.6	1/2	1.334	K	19	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-20P
2B	REFRIGERATOR, REACH-IN, SINGLE DOOR	1	115V/1Ø	3.8	1/4	0.437	K	28	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
3B	FREEZER, REACH-IN, DOUBLE DOOR	2	115V/1Ø	9.4	1-1/4	1.081	K K	6 14	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
3C	BLODGETT OVEN	1	208V/3Ø	24.0	1/4	5.6	K	36,38,40	30A-3P	(3)#10 THWN & (1)#10 GND	3/4"	VERIFY WITH MANUFACTURER
3D	RAPID COOK OVEN	1	208V/1Ø	30.0	-	6.0	K	39,41	30A-2P	(2)#10 THWN & (1)#10 GND	3/4"	6-30P
133	DISHWASHER, UNDERCOUNTER	1	208V/1Ø	37.2	-	7.74	K	21,23	40A-2P	(2)#8 THWN & (1)#10 GND	3/4"	DIRECT
125	ICE MAKER, CUBE STYLE	1	115V/1Ø	16.0	-	1.289	K	1	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	VERIFY WITH MANUFACTURER
100.2	HOT WATER DISPENSER	1	115V/1Ø	11.3	-	1.300	K	22	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
103	POS SYSTEM	1	115V/1Ø	-	VERIFY	VERIFY	K K	35 18	20A-1P	(2)#12 THWN & (1)#12 GND & (1)#12 ISO. GND	3/4"	5-20P
104	ESPRESSO MACHINE	2	208V-220V/1Ø	30.0	-	6.240	K K	5,7 9,11	30A-2P	(2)#10 THWN & (1)#10 GND	3/4"	L6-30P
131	REFRIGERATOR, UNDERCOUNTER	2	115V/1Ø	2.0	1/6	0.24	K K	12 15	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
121	BLENDER	2	120V/1Ø	15.0	3.0	1.800	K K	13 20	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-20P
113	COFFEE GRINDER	1	120V/1Ø	8.0	1/2	0.920	K	37	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
114	COFFEE BREWER	1	208V/1Ø	14.0	-	2.9	K	8,10	20A-2P	(2)#12 THWN & (1)#12 GND	3/4"	VERIFY WITH MANUFACTURER
118	SELF-SERVICE REFRIGERATED CASE	2	120V/1Ø	15.6	-	1.198	K K	2 4	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-20P
119	DISPLAY CASE, REFRIGERATED, SELF SERVE	1	208V/1Ø	10.0	1/2	2.08	K	32,34	15A-2P	(2)#12 THWN & (1)#12 GND	3/4"	6-15P
124	DOUBLE UNDER COUNTER REFRIGERATOR	1	115V/1Ø	3.0	1/5	0.345	K	30	15A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-15P
CO1	CONVENIENCE RECEPTACLE	-	120V/1Ø	1.5	-	0.180	SEE PLANS	SEE PLANS	20A-1P	(2)#12 THWN & (1)#12 GND	3/4"	5-20P

REFERENCE NOTES:

1. OWNER/VENDOR PROVIDED ITEM. VERIFY ALL UTILITY REQUIREMENTS AND LOCATIONS PRIOR TO ROUGH-IN.
2. POS SYSTEM. PROVIDE ISOLATED, DEDICATED, GROUNDED CIRCUIT. PROVIDE 3/4" CONDUIT WITH PULL WIRE. GENERAL CONTRACTOR TO COORDINATE WITH SYSTEM INSTALLER.
3. THE ELECTRICAL CONTRACTOR SHALL BE REQUIRED TO FURNISH AND INSTALL A CODE-COMPLIANT DISCONNECTING MEANS FOR EACH PIECE OF EQUIPMENT IDENTIFIED IN THE ELECTRICAL CONNECTION SCHEDULE AS REQUIRING A DIRECT CONNECTION OR PROVIDE A LOCK-OFF TYPE BREAKER AS SHOWN IN THE PANEL SCHEDULE ON SHEET E202. ALL DISCONNECTS SHALL BE NEMA 4X. DISCONNECTS SHALL BE FUSED WHERE REQUIRED BY N.E.C. (TYPICAL).

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

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763 SUSQUEHANNA AVE
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NEW JERSEY LICENSED PROFESSIONAL ENGINEER

KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES: 09-30-25

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Professional Seal:

Drawing Description:

ENLARGED
KITCHEN
POWER PLANS

Drawing Number:

E-105

LIGHT FIXTURE SCHEDULE (LIGHT CONCEPT)

SYMBOL	TAG	LOAD (VAE)	DESCRIPTION	AREA
	HB	15	HIGH BAY HALOGEN LIGHTINGS NUVO SF76-283, L16A19N1530K FINISH : WHITE	SUSPENDED IN SALES FLOOR AT 8'-0" A.F.F.
	HB-B	15	HIGH BAY HALOGEN LIGHTINGS NUVO SF76-283, L16A19N1530K FINISH : BLACK	SUSPENDED ABOVE CAFE SEATING AREA AT 8'-0" A.F.F.
	LW-01	22	NEW TRACK HEADS MANUFACTURER: ATTAIN LED TRACK HEAD HP MODEL: AT-25-35-95-24-WH-J	TRACK MOUNTED IN SALES AREA
	LWT-01 TRACK	-	NEW PENDANT MOUNTED MANUFACTURER: CONTECH SINGLE CIRCUIT LINE VOLTAGE TRACK VARIOUS LENGTHS- WHITE UNISTRUT AND HANGING HARDWARE	SUSPENDED IN SALES FLOOR AT 12'-0" A.F.F.
	A1	68	BOH LIGHTING MANUFACTURER: LITHONIA LIGHTING MODEL: TZL1N LED L96 10000LM FST MVOLT 30K 80CRI PLR E7W WH	SUSPENDED AT 12'-0" A.F.F. IN STOCK ROOM
A2	A2	-	OFFICE AND BREAK ROOM LIGHTING MANUFACTURER: DT BY TCP MODEL:DTF2UZD3835K	LAY-IN CEILING
A3	A3	46	ATTAIN 2 FT. X 4 FT. INTEGRATED LED 4000 LUMENS 4000K 120V COMMERCIAL GRADE RECESSED TROFFER W/ PRISMATIC LENSES	LAY-IN CEILING
	A14	34	MANUFACTURER: LITHONIA LIGHTING MODELS: ZL1N LED STRIPLIGHT L48 5000LM FST MVOLT 30K 80CRI PLR E7W WH	SURFACE MOUNTED TO CEILING OR SUSPENDED AT 12FT A.F.F.
X1	X1	3.8	THERMOPLASTIC LED EXIT SIGN MANUFACTURER: BEST LIGHTING MODEL: EZXTEU-2RW-EM	SUSPENDED OR WALL MOUNTED AT 8'-0" A.F.F. MIN. B.O. SIGN
X2	X2	5	LED EXIT SIGN AND EMERGENCY LIGHT, THERMOPLASTIC COMBO MANUFACTURER: BEST LIGHTING MODEL: LEDCXTEU-2-R-W-RC	SUSPENDED OR WALL MOUNTED AT 8'-0" A.F.F. MIN. B.O. FIXTURE
EM1	EM1	5	EMERGENCY LIGHT MANUFACTURER: ASTRALITE MODEL: EU-5-LED	SUSPENDED OR WALL MOUNTED AT 10'-0" A.F.F. MIN. B.O. FIXTURE (SALES AREA) OR AS HIGH AS POSSIBLE (NON-SALES)
EM2	EM2	2	EXTERIOR EMERGENCY LIGHT MANUFACTURER: BEST LIGHTING MODEL: LED HIGH LUMEN THERMOPLASTIC HPEM	WALL MOUNTED AT 8'-0" A.F.F. MIN. B.O. FIXTURE
	D4	12	4" DIAMETER LED DOWNLIGHT MANUFACTURER: ATTAIN LED MODEL: AT-DL4-12-35-WH	RECESSED IN CAFE CEILING
	D6	32	6" DIAMETER LED DOWNLIGHT MANUFACTURER: MAXI LUME MODEL: H6-LED-2000L-DIM10-120-MD-35K-90/ L607 SHZ WH	RECESSED IN RESTROOM CEILING
	PD1	60	22" DIAMETER LED PENDANT MANUFACTURER: WAYFAIR MODEL: METAIRIE 1 LIGHT SINGLE DOME PENDANT	SUSPENDED ABOVE CASHWRAP AT 7'-6" AFF FROM J-BOX AT DECK
	L-8 L-6 L-4 L-2	80 60 40 20	LED TUBE PENDANT ADJUSTABLE MANUFACTURER: CORONET LED MODEL: 8ft: CRD3-8-30-HIGH-UNV-DB-BLK -AC-SD 6ft: CRD3-6-30-HIGH-UNV-DB-BLK -AC-SD 4ft: CRD3-4-30-HIGH-UNV-DB-BLK -AC-SD 2ft: CRD3-2-30-HIGH-UNV-DB-BLK -AC-SD	SUSPENDED ABOVE COUNTER AT 9'-0" AFF
OC LS	OCCUPANCY SENSOR (LIGHTSTAT)			DECK MOUNTED J-BOX WITH 3/4" CONDUIT DROP 10'-6" AFF

LIGHTING FIXTURE SCHEDULE NOTES:

1. PROVIDE WITH INTEGRAL EMERGENCY BATTERY PACK. EMERGENCY BATTERY PACKS SHALL PROVIDE A MINIMUM OF NINETY (90) MINUTES OF CODE REQUIRED EMERGENCY LIGHTING. EACH BATTERY PACK PROVIDED SHALL PRODUCE THE MAXIMUM LUMEN OUTPUT AVAILABLE WITH THE MAXIMUM NUMBER OF LAMPS. BODINE OR APPROVED EQUAL. CIRCUIT BATTERY PACK AHEAD OF ALL LIGHTING CONTROLS.
2. REFER TO ARCHITECTURAL REFLECTED CEILING PLAN AND/OR ROOM FINISH SCHEDULE TO DETERMINE PROPER TYPE OF LIGHT FIXTURE REQUIRED FOR THE CEILING CONSTRUCTION PRIOR TO ORDERING THE FIXTURES & PROVIDE FIXTURES THAT ARE COMPATIBLE WITH THE CEILING SYSTEM.
3. NOT USED
4. NOT USED
5. NOT USED
8. VERIFY ALL INFORMATION MARKED AS "TBD" WITH BARNES & NOBLE PROJECT MANAGER AND THE ARCHITECT PRIOR TO PURCHASE.
9. VERIFY ALL MOUNTING HEIGHTS AND METHODS WITH ARCHITECT PRIOR TO INSTALLATION / ROUGH-IN.

LOW VOLTAGE SCHEDULE

SYMBOL	TAG	DESCRIPTION	MOUNTING HEIGHT
S1	SPEAKER	-	LAY -IN CEILING
S2	SPEAKER	PROVIDE J-BOX MOUNTED TO DECK	MATCH HEIGHT BOTTOM OF LIGHT GRID
CC P	CCTV	PROVIDE SURFACE MOUNTED J-BOX WITH 3/4" CONDUIT AND SWIVEL MOUNTED J-BOX	MATCH HEIGHT BOTTOM OF LIGHT GRID
CC C	CCTV	PROVIDE 3/4" CONDUIT THROUGH CEILING TO ABOVE VESTIBULE	CEILING MOUNTED
CC W	CCTV	PROVIDE 3/4" CONDUIT SLEEVE THRU WALL 12"-15" ABOVE CENTRE OF DOOR FRAME	WALL MOUNTED
M	MOTION DETECTOR	DECK MOUNTED J-BOX WITH 3/4" CONDUIT DROP	10'-6" AFF
KP	KEYPAD	PROVIDE 3/4" CONDUIT FLUSH AND CENTERED 48" AFF TO ABOVE VESTIBULE WITH PULL STRING. SEE ELECTRICAL DRAWINGS FOR OUTLET	48" AFF
CBA	CELL BOOST ANTENA	DUPLEX OUTLET MOUNTED IN THE CEILING. SEE ELECTRICAL DRAWINGS FOR OUTLET.	MATCH HEIGHT BOTTOM OF LIGHT GRID
AP	ACCESS POINT	PROVIDE SURFACE MOUNTED J-BOX WITH 3/4" CONDUIT AND SWIVEL MOUNTED J-BOX	MATCH HEIGHT BOTTOM OF LIGHT GRID
RECT	ANTENNA	PROVIDE SURFACE MOUNTED J-BOX WITH 3/4" CONDUIT AND SWIVEL MOUNTED J-BOX, FOR BN INTERNAL PHONE SYSTEM, 2-TOTAL, 1-FRONT, 1-BACK SALES ONLY	MATCH HEIGHT BOTTOM OF LIGHT GRID
EAS-WM	SECURITY PADDLE	PROVIDE 3/4" CONDUIT FLUSH AND CENTERED ON WOOD FRAME FROM 48" AFF TO ABOVE CEILING IN CORRIDOR. SEE ELECTRICAL DRAWINGS FOR OUTLET	WALL MOUNTED 48" AFF
EAS-FM	SECURITY PADDLE	PROVIDE 1" CONDUIT FROM SLAB TO ABOVE VESTIBULE CEILING. BOTH OUTLETS ON SAME CIRCUIT. SEE ELECTRICAL DRAWINGS FOR OUTLET	FLOOR MOUNTED 24" FROM VESTIBULE TO CENTRELINE
GB	-	-	WALL MOUNTED

LOW VOLTAGE GENERAL NOTES

1. GENERAL CONTRACTOR SHALL COORDINATE FINAL LOCATIONS FOR LOW VOLTAGE DEVICES WITH INDIVIDUAL LV VENDORS.
2. CAMERAS SHOULD NOT BE PLACED WHERE NOT OBSTRUCTED BY AWNING, STOREFRONT ETC.

NOTE: AS PER LANDLORD SPECIFICATIONS ALL MECHANICAL, PLUMBING, AND ELECTRICAL EQUIPMENTS BE PROPERLY SUPPORTED FROM STEEL STRUCTURE, NOT ROOF DECK.

GENERAL NOTES

1. REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATION OF LIGHT FIXTURES. CONTRACTORS TO COORDINATE LOCATIONS OF LIGHTING, SPEAKERS, AIR DIFFUSERS, GRILLES, SPRINKLER HEADS & THE LIKE, WITH REFLECTED CEILING LAY-OUTS AS REQUIRED & DIRECTED BY THE ARCHITECT.
2. ALL DEVICES, EQUIPMENT, FIXTURES, & THE LIKE, MUST BE GROUNDED BY USE OF A PROPERLY SIZED GROUNDING CONDUCTOR. MECHANICAL/ ELECTRICAL BONDS OF THE METALLIC RACEWAY SYSTEM SHALL ALSO BE MAINTAINED.
3. REFER TO MECHANICAL, PLUMBING, AND FIRE PROTECTION PLANS FOR EXACT LOCATION OF MECHANICAL AND PLUMBING EQUIPMENT. COORDINATE LOCATION OF DISCONNECT SWITCH ASSOCIATED WITH EACH PIECE OF EQUIPMENT WITH RESPECTIVE CONTRACTOR AND INSTALL IN ACCORDANCE WITH THE NEC.
4. REFER TO DIVISION 15 (21, 22 & 23) SPECIFICATIONS, HVAC, PLUMBING AND FIRE PROTECTION PLANS FOR ADDITIONAL ELECTRICAL WORK REQUIREMENTS & COORDINATION.
5. ALL RECEPTACLES SHOWN BACK-TO-BACK IN WALLS SHALL BE SEPARATED HORIZONTALLY BY 8" MINIMUM.
6. WHERE OPEN WIRING METHODS FOR LOW VOLTAGE SYSTEMS ARE PERMITTED BY THE CONTRACT DOCUMENTS AND LOCAL AUTHORITY, THE CONDUCTOR INSULATION MUST BE PLENUM RATED.
7. BRANCH CIRCUIT CONDUCTOR SIZES (& CONDUITS) SHALL BE INCREASED FROM THOSE INDICATED ON THE PLANS TO PREVENT EXCESSIVE VOLTAGE DROP. BRANCH CIRCUITS SHALL BE INSTALLED WITH WIRES OF SUFFICIENT SIZE SO THAT VOLTAGE DROP BETWEEN THE PANEL & THE LOADS DO NOT EXCEED A LIMIT OF 3%.
8. REGARDLESS OF THE TEMPERATURE RATING OF THE CONDUCTOR INSULATION, ALL CONDUCTOR AMPACITY RATINGS FOR THIS PROJECT SHALL BE DETERMINED FROM THE 75°C CONDUCTOR TEMPERATURE RATINGS INDICATED IN THE NEC TABLES. WHERE EQUIPMENT OR DEVICES ARE PROVIDED WITH TERMINALS/LUGS RATED FOR 60°C, THE AMPACITY RATING OF THE 75°C CONDUCTOR SHALL BE LIMITED TO ITS ASSOCIATED 60°C RATING AS INDICATED IN THE NEC TABLES. THE ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE TO INCREASE THE CONDUCTORS AND CONDUIT SIZE AS REQUIRED.
9. ALL 120V AND 277V BRANCH CIRCUITS SHALL BE PROVIDED WITH SEPARATE NEUTRAL CONDUCTORS. SHARED NEUTRALS WILL NOT BE PERMITTED FOR MULTI-CIRCUIT INSTALLATIONS. WHERE MULTIPLE CIRCUITS ARE RUN IN A COMMON RACEWAY, THE AMPACITY OF THE CONDUCTORS SHALL BE PROPERLY DERATED & CONDUIT SHALL BE SIZED PER CODE. UNDER NO CIRCUMSTANCES SHALL MORE THAN SIX (6) CURRENT CARRYING CONDUCTORS BE RUN IN A SINGLE CONDUIT. REFERENCE NEC ARTICLE AND TABLE 310.15(B)(2)(a).
10. ALL CONDUITS SHALL CONTAIN A GROUND CONDUCTOR SIZED PER NEC TABLE #250.122. IN ADDITION, WHERE AN ISOLATED, INSULATED GROUND IS REQUIRED, A SEPARATE GROUND CONDUCTOR WITH GREEN INSULATION SHALL BE RUN FROM THE PANEL GROUND BUS TO THE ISOLATED GROUND CONNECTION OF THE DEVICE. IN NO CASE SHALL THE SYSTEM GROUND (CONDUCTOR & ASSOCIATED OUTLET BOXES, CONDUIT & BUILDING STEEL) BE ALLOWED TO CONTACT THE ISOLATED GROUND (CONDUCTOR & DEVICE). WHERE CIRCUIT CONDUCTORS ARE INCREASED IN SIZE FOR VOLTAGE DROP, THE GROUND CONDUCTOR SIZE SHALL BE INCREASED PROPORTIONATELY (ACCORDING TO CIRCULAR MIL AREA) FROM THE SIZE REQUIRED BY NEC TABLE #250.122.
11. ELECTRICAL INSTALLATION REQUIREMENTS FOR ALL HVAC, PLUMBING, FIRE PROTECTION, SPECIAL SYSTEMS AND OWNER EQUIPMENT BEING FURNISHED BY OTHERS SHALL BE REVIEWED AND COORDINATED WITH OTHER TRADES PRIOR TO ROUGH-IN. OBTAIN EQUIPMENT SHOP DRAWINGS FROM INSTALLER/SUPPLIER/CONTRACTOR/OWNER FURNISHING EQUIPMENT, AS REQUIRED, FOR REVIEW AND COORDINATION. CONTACT ARCHITECT/ENGINEER WITH ANY DISCREPANCIES FOUND BETWEEN CONSTRUCTION DRAWINGS AND EQUIPMENT BEING FURNISHED PRIOR TO ROUGH-IN.
12. THE ELECTRICAL CONTRACTOR SHALL FURNISH ALL ACCESS PANELS, AS REQUIRED FOR SERVICING AND TESTING, FOR EQUIPMENT AND/OR DEVICES FURNISHED UNDER HIS CONTRACT. THE GENERAL CONTRACTOR SHALL INSTALL ACCESS PANELS. THE ELECTRICAL CONTRACTOR SHALL COORDINATE THE SIZE AND LOCATION OF EACH ACCESS PANEL WITH THE ARCHITECT AND GENERAL CONTRACTOR PRIOR TO ROUGH-IN.
13. ELECTRICAL CONTRACTOR SHALL INCLUDE IN HIS BID ALL CUTTING, TRENCHING AND PATCHING ASSOCIATED WITH THE ELECTRICAL INSTALLATION.
14. ALL PENETRATIONS THROUGH FIRE RATED WALLS ASSOCIATED WITH THE ELECTRICAL INSTALLATION SHALL BE SLEEVED AND FIRE-STOPPED USING A UL APPROVED METHOD. UL APPROVED METHOD SHALL MEET OR EXCEED FIRE RATING OF STRUCTURE BEING PENETRATED. REFERENCE ARCHITECTURAL PLANS FOR FIRE RATED STRUCTURES.
15. NO CONDUIT, BOXES, WIRING, OR CABLES SHALL BE INSTALLED WITHIN 1 1/2" OF THE LOWEST POINT OF THE UNDERSIDE OF THE ROOF DECKING, NOR SHALL THEY BE INSTALLED CONCEALED WITHIN METAL-CORRUGATED ROOF DECKING. FOR EXISTING INSTALLATIONS, THE ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE TO REPLACE AND/OR REWORK EXISTING CONDUIT, BOXES, WIRING, AND CABLING THAT IS NOT IN COMPLIANCE WITH THIS REQUIREMENT.
16. ALL ELECTRICAL EQUIPMENT AND DEVICES FOR THIS PROJECT MUST BE UL LISTED. DEVICES, EQUIPMENT, SYSTEMS SHALL BE INSTALLED PER N.E.C. REQUIREMENTS AND MANUFACTURER'S INSTRUCTIONS.
17. ALL CONDUIT AND CABLING SHALL BE PROPERLY SUPPORTED AS REQUIRED BY THE NATIONAL ELECTRICAL CODE. FOR EXISTING INSTALLATIONS, THE ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE TO REPLACE AND/OR REWORK EXISTING CONDUIT AND/OR CABLING THAT IS NOT IN COMPLIANCE WITH THIS REQUIREMENT.
18. CONTRACTOR SHALL FIELD VERIFY SLAB ON GRADE FLOOR CONSTRUCTION TYPE PRIOR TO CUTTING. UNDER NO CIRCUMSTANCES SHALL THE CONTRACTOR CUT A STRUCTURAL FLOOR SLAB THICKER THAN FOUR (4") INCHES WITHOUT PRIOR WRITTEN APPROVAL FROM ENGINEER OF RECORD. NOTIFY ENGINEER OF RECORD OF ANY SLAB THICKNESS GREATER THAN FOUR (4") INCHES PRIOR TO PROCEEDING WITH ANY SAW CUTTING.

ELECTRICAL ABBREVIATIONS

A	AMPS
AC	AIR CONDITIONING UNIT
AFC	ABOVE FINISH COUNTER
AFF	ABOVE FINISH FLOOR
AFG	ABOVE FINISH GRADE
AHU	AIR HANDLING UNIT
AIC	ASYMMETRICAL INTERRUPTING CURRENT
ARCH.	ARCHITECTURAL
ATS	AUTOMATIC TRANSFER SWITCH
AWG	AMERICAN WIRE GAGE
BKR	BREAKER
BLDG.	BUILDING
C	CONDUIT
CATV	CABLE TELEVISION
CCTV	CLOSED CIRCUIT TELEVISION
CH	CHILLER
CONTR	CONTRACTOR
CT	COOLING TOWER
CU	COPPER
CUH	CABINET UNIT HEATER
DE	DUAL ELEMENT
DN	DOWN
DS	DISCONNECT SWITCH
DWG.	DRAWING
(E) or EXIST.	EXISTING
EBB	ELECTRIC BASEBOARD
E.C.	ELECTRICAL CONTRACTOR
EF	EXHAUST FAN
EH	ELECTRIC HEATER
ELEC.	ELECTRICAL
EM	EMERGENCY
EMT	ELECTRICAL METALLIC TUBING
EQ.	EQUAL
ETR	EXISTING TO REMAIN
EUH	ELECTRIC UNIT HEATER
EWC	ELECTRIC WATER COOLER
EWH	ELECTRIC WATER HEATER
F	FUSE
FA	FIRE ALARM
FACP	FIRE ALARM CONTROL PANEL
FC	FAN COIL UNIT
FLUOR	FLUORESCENT
FPB	FAN POWER BOX (VAV)
F.P.C.	FIRE PROTECTION CONTRACTOR
FS	FLOW SWITCH
FT.	FOOT/FEET
G.C.	GENERAL CONTRACTOR
GFI	GROUND FAULT INTERRUPTING PROTECTION
GND	GROUND
HID	HIGH INTENSITY DISCHARGE
HOA	HAND-OFF-AUTOMATIC
HP	HORSEPOWER
HPS	HIGH PRESSURE SODIUM
HVAC	HEATING, VENTILATION, AIR CONDITIONING
IG	ISOLATED GROUND
INCAND.	INCANDESCENT
JB	JUNCTION BOX
KCMIL	ONE THOUSAND CIRCULAR MILS
K.E.C.	KITCHEN EQUIPMENT CONTRACTOR
KVA	KILOVOLT AMPERE
KW	KILOWATT
LTG	LIGHTING
MATV	MASTER ANTENNA TV
MAU	MAKE-UP AIR UNIT
MAX	MAXIMUM
MCB	MAIN CIRCUIT BREAKER
MCC	MOTOR CONTROL CENTER
M.C.	MECHANICAL CONTRACTOR
MECH.	MECHANICAL
MFR	MANUFACTURER
MH	METAL HALIDE
MIN	MINIMUM
MLO	MAIN LUGS ONLY
MOD	MOTOR OPERATED DAMPER
MSB	MAIN SWITCHBOARD
MTD	MOUNTED
NEC	NATIONAL ELECTRIC CODE
NF	NON FUSED
NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
NIC	NOT IN CONTRACT
NL	NIGHTLIGHT
NTS	NOT TO SCALE
Ø or PH	PHASE
P	POLE
PB	PULL BOX
P.C.	PLUMBING CONTRACTOR
PNL	PANEL
PRE	POWER ROOF EXHAUSTER
PVC	POLYVINYL CHLORIDE
RTU	ROOF TOP UNIT
SPKR	SPEAKER
SPST	SINGLE POLE SINGLE THROW
TIE	MULTIPLE OUTLETS WIRED ON SAME BRANCH CICRUIT
TS	TAMPER SWITCH
TTB	TELEPHONE TERMINAL BOARD
TV	TELEVISION
TYP.	TYPICAL
UH	GAS FIRED UNIT HEATER
UL	UNDERWRITER'S LABORATORY
UNO	UNLESS NOTED OTHERWISE
UV	UNIT VENTILATOR
V	VOLTS
W	WATTS
WP	WEATHER-PROOF
WR	WEATHER-RESISTANT TYPE DEVICE (NEMA 3R RATED)
XFMR	TRANSFORMER

ELECTRICAL SYMBOL LEGEND

SYMBOL	DESCRIPTION
	CONDUIT WITH WIRING RUN CONCEALED IN OR ABOVE CEILING OR WALL, OR RUN EXPOSED IN UNFINISHED AREAS. CROSS HATCHING INDICATES NUMBER OF CONDUCTORS (#12 AWG - MINIMUM). PROVIDE A CODE-SIZED GROUND WIRE IN ALL CONDUITS IN ADDITION TO THE CONDUCTORS SHOWN. SHADED DOT INDICATES CODE-SIZED ISOLATED GROUND WIRE IN CONDUIT.
	CONDUIT WITH WIRING RUN CONCEALED BELOW FLOOR. CROSS HATCHING INDICATES NUMBER OF CONDUCTORS (#12 AWG - MINIMUM). PROVIDE A CODE-SIZED GROUND WIRE IN ALL CONDUITS IN ADDITION TO THE CONDUCTORS SHOWN. SHADED DOT INDICATES CODE-SIZED ISOLATED GROUND WIRE IN CONDUIT.
	PANELBOARD MOUNTED 6'-6" TO TOP. SEE PANEL SCHEDULES & ELECTRICAL ONE-LINE DIAGRAM.
	DISCONNECT SWITCH - TYPE & RATING AS SHOWN ON PLANS
S_A	20A - 120V/277V SINGLE POLE TOGGLE SWITCH MOUNTED 48" AFF TO TOP OF BOX, UNLESS NOTED OTHERWISE. IF APPLICABLE, LOWER CASE SUBSCRIPT "x" - KEYS SWITCH TO FIXTURES BEING CONTROLLED (TYPICAL OF ALL SWITCH SYMBOLS)
S_{OC}	LINE VOLTAGE PASSIVE INFRARED WALL SWITCH OCCUPANCY SENSOR MOUNTED 48"AFF TO TOP OF BOX, UNLESS NOTED OTHERWISE (OC2 = DUAL RELAY). MANUFACTURER SHALL BE WATTSTOPPER, HUBBELL, SENSOR SWITCH, COOPER CONTROLS, OR LUTRON)
$(OS)_{TYPE}$	LOW VOLTAGE CEILING MOUNTED OCCUPANCY SENSOR. LOCATE AS DIRECTED PER MANUFACTURER AND PROVIDE STANDARD VS. EXTENDED COVERAGE RANGE AS DIRECTED BY MANUFACTURER. REFER TO TYPICAL LOW VOLTAGE OCCUPANCY SENSOR WIRING DIAGRAM ON SHEET E101. (1 = DUAL TECHNOLOGY 360° COVERAGE PATTERN, 2 = DUAL TECHNOLOGY CORNER MOUNTED WITH WIDE VIEW COVERAGE PATTERN, 3 = PIR NARROW HALLWAY COVERAGE PATTERN). MANUFACTURER SHALL BE WATTSTOPPER, HUBBELL, SENSOR SWITCH, COOPER CONTROLS, OR LUTRON.)
	20A - 125V GROUNDING TYPE SIMPLEX RECEPTACLE MOUNTED 18" AFF TO TOP OF BOX, UNLESS NOTED OTHERWISE
	20A - 125V GROUNDING TYPE DUPLEX RECEPTACLE MOUNTED 18" AFF, UNLESS NOTED OTHERWISE
	20A - 125V GROUNDING TYPE QUADRAPLEX RECEPTACLE MOUNTED 18" AFF, UNLESS NOTED OTHERWISE
IG	20A - 125V ISOLATED GROUND TYPE DUPLEX RECEPTACLE MOUNTED 18" AFF, UNLESS NOTED OTHERWISE
GFI WP	20A - 125V GROUND FAULT INTERRUPTING TYPE DUPLEX RECEPTACLE, WEATHER RESISTANT LISTED WITH DIE-CAST ALUMINUM "WHILE IN-USE COVER" AND MOUNTED 18" AFF TO TOP OF BOX, UNLESS NOTED OTHERWISE
GFI	20A - 125V GROUND FAULT INTERRUPTING TYPE DUPLEX RECEPTACLE MOUNTED 18" AFF TO TOP OF BOX, UNLESS NOTED OTHERWISE
USB	20A - 125V GROUNDING TYPE DUPLEX RECEPTACLE WITH (2) USB PORTS MOUNTED 18" AFF TO TOP OF BOX, UNLESS NOTED OTHERWISE
USB	20A - 125V GROUNDING TYPE DUPLEX RECEPTACLE WITH (4) USB PORTS MOUNTED 18" AFF, UNLESS NOTED OTHERWISE
	FURNISH AND INSTALL A COMPLETE FLOOR BOX SYSTEM WHICH INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING: SINGLE GANG, CAST IRON, FULLY ADJUSTABLE FLOOR BOX RECESSED IN CONCRETE FLOOR SLAB WITH (1) 20A DUPLEX RECEPTACLE. FLUSH ALUMINUM FINISHED CARPET FLANGE AND COVER. FOR TILE FLOOR INSTALLATION, PROVIDE APPROPRIATE FLANGE SO THAT TOP COVER OF FLOOR BOX IS FLUSH WITH TILE. COORDINATE EXACT LOCATION WITH ARCHITECT PRIOR TO INSTALLATION. FLOOR BOX IS CAPABLE OF ACCEPTING (4) 1"CONDUITS. FURNISH AND INSTALL SYSTEM PER MANUFACTURER'S INSTRUCTIONS. WIREMOLD #880CS1-1, #818TCAL, #828R-TCAL
T	FURNISH AND INSTALL A COMPLETE FLOOR BOX SYSTEM WHICH INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING: 2-GANG, CAST IRON, FULLY ADJUSTABLE MULTI-SERVICE FLOOR BOX RECESSED IN CONCRETE FLOOR SLAB WITH (1)20A DUPLEX RECEPTACLE, (1) 6-PORT TELE/DATA MOUNTING BEZEL, FLUSH ALUMINUM FINISHED CARPET FLANGE AND COVERS, AND VOLTAGE DIVIDER. THE FURNISHED FLOOR BOX SHALL BE SIZED TO ACCEPT THE QUANTITY AND SIZE OF CONDUITS REQUIRED FOR THE TELE/DATA CABLING. FOR TILE FLOOR INSTALLATION, PROVIDE APPROPRIATE FLANGE SO THAT TOP COVER OF FLOOR BOX IS FLUSH WITH TILE. COORDINATE EXACT LOCATION WITH ARCHITECT PRIOR TO INSTALLATION. COORDINATE BRACKET AND TELE/DATA DEVICE REQUIREMENTS WITH TECHNOLOGY INSTALLER PRIOR TO ORDERING MATERIALS. FLOOR BOX IS CAPABLE OF ACCEPTING (6) 1"CONDUITS. FURNISH AND INSTALL SYSTEM PER MANUFACTURER'S INSTRUCTIONS. WIREMOLD #880CS2-1, #828TCAL, #828R-TCAL, #828COMTCAL
	STANDARD GANG BOX. REFER TO TERMINAL LIST BELOW FOR INDIVIDUAL DEVICES: DATA, QUAD, DUPLEX
B	DOOR BELL ASSEMBLY. MOUNTING, TYPE, RATINGS AS INDICATED ON PLANS. MAKE CONNECTIONS PER MANUFACTURER'S INSTRUCTIONS.
CH	CHIME ASSEMBLY. MOUNTING, TYPE, RATINGS AS INDICATED ON PLANS. MAKE CONNECTIONS PER MANUFACTURER'S INSTRUCTIONS.
B	PUSH-BUTTON ASSEMBLY MOUNTED AT 48" AFF TO TOP OF BOX. TYPE AND RATINGS PER LOAD BEING SERVED. MAKE CONNECTIONS TO EQUIPMENT PER MANUFACTURER'S INSTRUCTIONS.
J	JUNCTION BOX - SIZE AS REQUIRED BY NEC
▼	TELEPHONE OUTLET MOUNTED 18" AFF UNLESS NOTED OTHERWISE. PROVIDE 1"C WITH PULLWIRE TO ACCESSIBLE CEILING SPACE. PROVIDE CONDUIT BUSHING.
▽	DATA OUTLET MOUNTED 18" AFF, UNLESS NOTED OTHERWISE, PROVIDE 1"C WITH PULLWIRE TO ACCESSIBLE CEILING SPACE. PROVIDE CONDUIT BUSHING.
▼▽	COMBINATION TELEPHONE/DATA OUTLET MOUNTED 18" AFF, UNLESS NOTED OTHERWISE. 1"C WITH PULLWIRE TO ACCESSIBLE CEILING SPACE. PROVIDE CONDUIT BUSHING.
	PLUG MOLD MNTD. ABOVE COUNTERTOP.
DRV (#)	DRIVER FOR SHELF LIGHTING. (#) QUANTITY OF UNITS.

NOTE: NOT ALL SYMBOLS MAY APPLY TO PROJECT.

BARNES&NOBLE

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BUILDING VALUE SINCE 1994

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3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

These plans are an instrument of service and the property of the Architect. Infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:
ELECTRICAL SCHEDULES & DETAILS

Drawing Number:
E-201

ELECTRICAL PANEL SCHEDULE

PANELBOARD	MDP	VOLTAGE	277 / 480 V	PHASE	3	WIRE	4																
PANEL TYPE	(EXIST)	MAINS	1,000	BUS RATING	1,000	AIC RATING	(EXIST)																
NEMA TYPE ENCLOSURE	NEMA 1	MOUNTING	SURFACE	OPTIONS		NOTE	(BY L.L.)																

CKT. NO.			DESCRIPTION	POLE	WIRE SIZE	BKR. SIZE	TOTAL WATTS	PHASE	TOTAL WATTS	BKR. SIZE	WIRE SIZE	POLE	DESCRIPTION			CKT. NO.
1							6640	A	6640							2
3		HACR	RTU-1	3	10	30	6640	B	6640	30	10	3	RTU-5	HACR		4
5							6640	C	6640							6
7							6640	A	6640							8
9		HACR	RTU-2	3	10	30	6640	B	6640	30	10	3	RTU-6	HACR		10
11							6640	C	6640							12
13							6640	A	6640							14
15		HACR	RTU-3	3	10	30	6640	B	6640	30	10	3	RTU-7	HACR		16
17							6640	C	6640							18
19							6640	A					SPACE			20
21		HACR	RTU-4	3	10	30	6640	B					SPACE			22
23							6640	C					SPACE			24
25							23,473	A					SPACE			26
27			150kVA TRANSFORMER (PANELS LC1 & LC2)	3	2/0	175	23,595	B					SPACE			28
29							23,027	C					SPACE			30
31								A	22,298							32
33			SPARE	3		125		B	21,685	100	1	3	75kVA TRANSFORMER (PANEL 'K')		(N)	34
35								C	22,755							36
37					(2 SETS)		5,000	A								38
39			HC1	3	250	400	1,155	B		100		3	SPARE			40
41							4,000	C								42
43			SPACE					A								44
45			SPACE					B		200		3	SPARE			46
47			SPACE					C								48
49			SPACE					A					SPACE			50
51			SPACE					B					SPACE			52
53			SPACE					C					SPACE			54
55			SPACE					A					SPACE			56
57			SPACE					B					SPACE			58
59			SPACE					C					SPACE			60

ALL PHASES TO BE BALANCED TO WITHIN 7%

A= 97,251 WATTS
B= 92,915 WATTS
C= 96,262 WATTS

LO CIRCUITS WITH HANDLE LOCK-OFF DEVICE
HACR HEATING. A/C & REFRIGERATION

NOTE: ALL CIRCUITS ARE EXISTING UNLESS NOTED OTHERWISE (N) NEW

TOTAL CONNECTED LOAD	286428	WATTS	345	AMPS	(N) NEW CIRCUIT/LOAD
TOTAL DEMAND LOAD	271264	WATTS	326	AMPS	

ELECTRICAL PANEL SCHEDULE

PANELBOARD	LC1	VOLTAGE	120 / 208 V	PHASE	3	WIRE	4																
PANEL TYPE	(EXIST)	MAINS	200	BUS RATING	200	AIC RATING	(EXIST)																
NEMA TYPE ENCLOSURE	NEMA 1	MOUNTING	SURFACE	OPTIONS		NOTE	(BY L.L.)																

CKT. NO.			DESCRIPTION	POLE	WIRE SIZE	BKR. SIZE	TOTAL WATTS	PHASE	TOTAL WATTS	BKR. SIZE	WIRE SIZE	POLE	DESCRIPTION			CKT. NO.
1			SPARE	1		20		A	2704	40	8	2	CU-1	HACR	(E)	2
3	(E)		LOADING DOCK DOOR RECEPT	1	12	20	600	B	2704							4
5		LO	EM LIGHTS/EXIT SIGNS	1	12	20	500	C	120	20	12	1	AHU-1	HACR	(E)	6
7	(E)		EXTERIOR LIGHTS NW	1	12	20	930	A	500	20	12	1	S/W TRACK/VESTIBULE LTS	LCP		8
9	(E)		NORTH RECEPTACLE	1	12	20	360	B	500	20	12	1	STOREFRONT CAFÉ SIGN	LCP		10
11	(E)		EAST RECEPTACLES	1	12	20	180	C	900	20	12	1	S/W OUTLETS	LCP		12
13	(E)		SOUTH RECEPTACLES	1	12	20	180	A	500	20	12	1	CAFÉ SIGN	LCP		14
15			SPARE	2		20		B	240	20	12	1	SALES PENDANT LIGHTS	LCP		16
17	(E/R)		ROOFTOP OUTLETS	3	12	20	540	C	672	20	12	1	CAFÉ SITTING AREA LTS	LCP		18
19	(E/R)		ROOFTOP OUTLETS	1	12	20	900	A	570	20	12	1	SALES HIGH BAY LIGHTS	LCP		20
21							1667	B	600	20	12	1	SALES HIGH BAY LIGHTS	LCP		22
23	(E)		EUH-1 (ELECTRIC HEATER)	3	12	20	1666	C	960	20	12	1	SALES TRACK LIGHTING	LCP		24
25							1667	A	720	20	12	1	SALES TRACK LIGHTING	LCP		26
27	(E)		POLE LIGHTS	1	12	20	440	B	840	20	12	1	SALES TRACK LIGHTING	LCP		28
29	(E)		EAST BLDG EXTERIOR LIGHTS	1	12	20	240	C	720	20	12	1	SALES TRACK LIGHTING	LCP		30
31	(E/R)	LCP	SIGN (ELEC. ACCESS PANEL)	1	12	20	1200	A	960	20	12	1	SALES TRACK LIGHTING	LCP		32
33		LCP	SIGN	1	12	20	1200	B	1440	20	12	1	SALES TRACK LIGHTING	LCP		34
35		LCP	SIGN	1	12	20	1200	C	1200	20	12	1	SALES TRACK LIGHTING	LCP		36
37		LCP	REAR SIGN	1	12	20	1200	A	1200	20	12	1	SALES TRACK LIGHTING	LCP		38
39		LCP	PERIMETER SHELF REC	1	12	20	1300	B	900	20	12	1	SALES TRACK LIGHTING	LCP		40
41		LCP	PERIMETER SHELF REC	1	12	20	1400	C	1200	20	12	1	SALES TRACK LIGHTING	LCP		42
43		LCP	PERIMETER SHELF REC	1	12	20	1000	A		20		1	SPACE			44
45		LCP	PERIMETER SHELF REC	1	12	20	1100	B	200	20	12	1	LCP			46
47		LCP	FLOOR MILLWORK REC	1	12	20	1300	C	797	20	12	1	CAFÉ/WORKRM LIGHTS			48
49		LCP	FLOOR MILLWORK REC	1	12	20	1000	A		20		1	SPARE			50
51		LCP	FLOOR MILLWORK REC	1	12	20	700	B	304	20	12	1	CORRIDOR LIGHTS	LCP		52
53		LCP	FIXTURE UTILITY OUTLETS	1	12	20	900	C	542	20	12	1	JAN. CL/RR LTS/EF-1,2,3			54
55			NOOK OUTLET	1	12	20	180	A					SPACE			56
57			SPACE					B					SPACE			58
59			SPACE					C					SPACE			60
61			SPACE					A					SPACE			62
63			SPACE					B					SPACE			64
65			SPACE					C					SPACE			66
67			SPACE					A					SPACE			68
69			SPACE					B					SPACE			70
71			SPACE					C					SPACE			72
73			SPACE					A					SPACE			74
75			SPACE					B					SPACE			76
77			SPACE					C					SPACE			78
79			SPACE					A					SPACE			80
81			SPACE					B					SPACE			82
83			SPACE					C					SPACE			84

ALL PHASES TO BE BALANCED TO WITHIN 7%

A= 15,411 WATTS
B= 15,095 WATTS
C= 15,037 WATTS

LO CIRCUITS WITH HANDLE LOCK-OFF DEVICE
LCP CIRCUITS ON LIGHTING CONTROL PANEL
HACR HEATING. A/C & REFRIGERATION

TOTAL CONNECTED LOAD	45543	WATTS	126	AMPS	(E) EXISTING CIRCUIT/LOAD
TOTAL DEMAND LOAD	50567	WATTS	140	AMPS	(E/R) EXISTING/REVISED CIRCUIT/LOAD

ELECTRICAL PANEL SCHEDULE

PANELBOARD	LC2	VOLTAGE	120 / 208 V	PHASE	3	WIRE	4																
PANEL TYPE	(EXIST)	MAINS	200	BUS RATING	200	AIC RATING	(EXIST)																
NEMA TYPE ENCLOSURE	NEMA 1	MOUNTING	SURFACE	OPTIONS		NOTE	(BY L.L.)																

CKT. NO.			DESCRIPTION	POLE	WIRE SIZE	BKR. SIZE	TOTAL WATTS	PHASE	TOTAL WATTS	BKR. SIZE	WIRE SIZE	POLE	DESCRIPTION			CKT. NO.
1		IG	OFFICE RECEPTACLE	1	12	20	180	A	1000	20	12	1	E.A.S SYSTEM	LO		2
3		IG	OFFICE RECEPTACLE	1	12	20	180	B	180	20	12	1	KEYPAD OUTLET			4
5		IG	OFFICE RECEPTACLE	1	12	20	180	C	360	20	12	1	BACK-UP SERVER REC	IG	LO	6
7			OFFICE RECEPTACLES	1	12	20	720	A	360	20	12	1	CASHWRAP RECEPT	IG		8
9		IG	OFFICE QUAD RECEPT.	1	12	20	360	B	360	20	12	1	CASHWRAP RECEPT	IG		10
11		IG	OFFICE QUAD RECEPT.	1	12	20	360	C	360	20	12	1	CASHWRAP RECEPT	IG		12
13		IG	OFFICE QUAD RECEPT.	1	12	20	360	A	360	20	12	1	CASHWRAP RECEPT	IG		14
15		LO	IT RACK	1	10	30	2000	B	360	20	12	1	CASHWRAP/INFO RECEPT	IG		16
17			MUSIC SYSTEM	1	12	20	360	C	900	20	12	1	CASHWRAP OUTLETS			18
19			PATCH PANEL RECEPTS	1	12	20	720	A	1080	20	12	1	CAFÉ SITTING OUTLETS			20
21			BREAKRM REFRIGERATOR	1	12	20	1000	B	900	20	12	1	CAFÉ SITTING OUTLETS			22
23			BREAKRM MICROWAVE	1	12	20	1500	C	540	20	12	1	RECEIVING TABLE			24
25			BREAKRM COUNTER REC	1	12	20	180	A	300	20	12	1	INFO. RECEPTACLE			26
27			WATER COOLER	1	12	20	500	B	1080	20	12	1	PDT CHARGERS			28
29			BREAKRM OUTLETS	1	12	20	540	C	1450	20	12	1	HAND DRYER			30
31			SPARE	1		20		A	1450	20	12	1	HAND DRYER			32
33		LO	DUCT DETECTORS	1	12	20	1080	B	500	20	12	1	DRINKING FOUNTAIN			34
35		LO	DUCT DETECTORS	1	12	20	1440	C		20		1	SPARE			36
37			RE-CIRCULATION PUMP	1	12	20	92	A	1260	20	12	1	UTILITY CONV. OUTLETS			38
39			SPACE					B					SPACE			40
41			SPACE					C					SPACE			42
43			SPACE					A					SPACE			44
45			SPACE					B					SPACE			46
47			SPACE					C					SPACE			48
49			SPACE					A					SPACE			50
51			SPACE					B					SPACE			52
53			SPACE					C					SPACE			54
55			SPACE					A					SPACE			56
57			SPACE					B					SPACE			58
59			SPACE					C					SPACE			60
61			SPACE					A					SPACE			62
63			SPACE					B					SPACE			64
65			SPACE					C					SPACE			66
67			SPACE					A					SPACE			68
69			SPACE					B					SPACE			70
71			SPACE					C					SPACE			72
73			SPACE					A					SPACE			74
75			SPACE					B					SPACE			76
77			SPACE					C					SPACE			78
79			SPACE					A					SPACE			80
81			SPACE					B					SPACE			82
83			SPACE					C					SPACE			84

ALL PHASES TO BE BALANCED TO WITHIN 7%

A= 8,062 WATTS
B= 8,500 WATTS
C= 7,990 WATTS

LO CIRCUITS WITH HANDLE LOCK-OFF DEVICE
IG CIRCUITS WITH ISOLATED GROUND

TOTAL CONNECTED LOAD	24552	WATTS	68	AMPS
TOTAL DEMAND LOAD	18726	WATTS	52	AMPS

ELECTRICAL PANEL SCHEDULE

PANELBOARD	HC1	VOLTAGE	277 / 480 V	PHASE	3	WIRE	4																
PANEL TYPE	(EXIST)	MAINS	MLO	BUS RATING	400	AIC RATING	(EXIST)																
NEMA TYPE ENCLOSURE	NEMA 1	MOUNTING	SURFACE	OPTIONS		NOTE	(BY L.L.)																

CKT. NO.			DESCRIPTION	POLE	WIRE SIZE	BKR. SIZE	TOTAL WATTS	PHASE	TOTAL WATTS	BKR. SIZE	WIRE SIZE	POLE	DESCRIPTION			CKT. NO.
1			SPARE	1		20		A	5000	25	10	1	CUH-1			2
3			STOCKRM/STORAGE LTG	1	12	20	727	B	428	20	12	1	OFFICE/BREAKRM LTG			4
5			SPARE	1		20		C	4000	20	12	1	(EWH-2) WATER HEATER	LO		6
7			SPARE	1		20		A					SPACE			8
9			SPARE	1		20		B					SPACE			10
11			SPARE	1		20		C					SPACE			12
13			SPARE	1		20		A					SPACE			14
15			SPARE	1		20		B					SPACE			16
17			SPARE	1		20		C					SPACE			18
19			SPARE	1		20		A					SPACE			20
21			SPACE					B					SPACE			22
23			SPACE					C					SPACE			24
25			SPACE					A					SPACE			26
27			SPACE					B					SPACE			28
29			SPACE					C					SPACE			30
31			SPACE					A					SPACE			32
33			SPACE					B					SPACE			34
35			SPACE					C					SPACE			36
37			SPACE					A					SPACE			38
39			SPACE					B					SPACE			40
41			SPACE					C					SPACE			42

ALL PHASES TO BE BALANCED TO WITHIN 7%

A= 5,000 WATTS
B= 1,155 WATTS
C= 4,000 WATTS

TOTAL CONNECTED LOAD	10155	WATTS	13	AMPS	
TOTAL DEMAND LOAD	11444	WATTS	14	AMPS	

ELECTRICAL PANEL SCHEDULE

PANELBOARD	K	VOLTAGE	120 / 208 V	PHASE	3	WIRE	4																
PANEL TYPE	NQ	MAINS	200	BUS RATING	225	AIC RATING	22,000																
NEMA TYPE ENCLOSURE	NEMA 1	MOUNTING	SURFACE	OPTIONS		NOTE																	

CKT. NO.		EQUIP #	DESCRIPTION	POLE	WIRE SIZE	BKR. SIZE	TOTAL WATTS	PHASE	TOTAL WATTS	BKR. SIZE	WIRE SIZE	POLE	DESCRIPTION	EQUIP #		CKT. NO.
1		125	ICE MAKER, CUBE STYLE	1	12	20	1289	A	1198	20	12	1	SELF-SERVICE REF. CASE	118	GFCI	2
3			SPARE	1		20		B	1198	20	12	1	SELF-SERVICE REF. CASE	118	GFCI	4
5	GFCI	104	ESPRESSO MACHINE	2	10	30	3120	C	1081	15	12	1	REACH-IN FREEZER	3B	GFCI	6
7							3120	A	1456	20	12	2	COFFEE BREWER	114		8
9	GFCI	104	ESPRESSO MACHINE	2	10	30	3120	B	1456							10
11							3120	C	240	15	12	1	REFRIGERATOR U/C	131	GFCI	12
13	GFCI	121	BLENDER	1	12	20	1800	A	1081	15	12	1	REACH-IN FREEZER	3B	GFCI	14
15	GFCI	131	REFRIGERATOR U/C	1	12	15	240	B		20		1	SPARE			16
17			SPARE	1		20		C	600	20	12	1	CAFÉ POS CASHWRAP	103		18
19		129	REACH-IN FRIDGE/FREEZER	1	12	20	1334	A	1800	20	12	1	BLENDER	121	GFCI	20
21		133	DISHWASHER, U/C	2	8	40	3869	B	1300	15	12	1	HOT WATER DISPENSER	100.2	GFCI	22
23							3869	C		20		1	SPARE			24
25			KITCHEN CONVEN. REC.	1	12	20	720	A	1000	20	12	1	WATER SOFTENER			26
27							4100	B	437	15	12	1	REACH-IN REFRIGERATOR	2B		28
29	LO	4	WATER HEATER (EWH-1)	3	6	50	4100	C	346	15	12	1	U/C REFRIGERATOR	124		30
31							4100	A	1040	15	12	2	PASTRY CASE, REFRIG.	119		32
33			RE-CIRCULATING PUMP	1	12	20	125	B	1040							34
35	IG	103	CAFÉ POS CASHWRAP	1	12	20	400	C	2880							36
37	GFCI	113	COFFEE GRINDER	1	12	15	920	A	1440	30	10	3	BLODGETT OVEN	3C		38
39		3D	RAPID COOK OVEN	2	10	30	3000	B	1800							40
41							3000	C		20		1	SPARE			42

ALL PHASES TO BE BALANCED TO WITHIN 7%

A= 22,298 WATTS
B= 21,688 WATTS
C= 22,755 WATTS

LO CIRCUITS WITH HANDLE LOCK-OFF DEVICE
IG CIRCUITS WITH ISOLATED GROUND
GFCI GROUND FAULT CIRCUIT INTERRUPTER

TOTAL CONNECTED LOAD	66738	WATTS	186	AMPS	
TOTAL DEMAND LOAD	69813	WATTS	194	AMPS	

INFORMATION ABOUT EXISTING CONDITIONS WAS TAKEN FROM LANDLORD DRAWINGS SUBMITTED BY YOW ARCHITECTS PC, ON 04/02/2025, REVISION 1. VERIFY EXISTING CONDITIONS PRIOR TO BID.

ELECTRICAL LOAD SUMMARY

DESCRIPTION	NEC CONNECTED kW	VOLT	PHASE	NEC DEMAND FACTOR	NEC DEMAND kW
LIGHTING- 120V	6.0	120	1	1.25	7.5
LIGHTING - 277V	1.2	277	1	1.25	1.4
INTERIOR SIGN	1.0	120	1	1.25	1.3
TRACK LIGHTS	10.6	120	1	TRACK LENGTH=	10.6
RECEPTACLES	36.1	120	1	>10kW=10+[0.5*(kW-10)]	23.1
STOREFRONT SIGN	4.8	120	1	1.25	6.0
S/W OUTLETS	0.9	120	1	1.25	1.1
HAND DRYER	2.9	120	1	1.00	2.9
CAFÉ EQUIPMENT (208V,3PH)	6.1	208	3	1.00	6.1
WATER HEATER	16.3	277-208	1	1.25	20.4
CAFÉ EQUIPMENT (208V, 1PH)	31.2	208	1	1.00	31.2
(EXIST) RTU-1,2,3,4,5,6,7	139.4	480	3	1.00	139.4
CUH-1	5.0	277	1	1.00	5.0
SPLIT SYSTEM	5.5	208	1	1.00	5.5
(EXIST) EUH-1	5.0	208	3	1.00	5.0
ADD'L TO MEET NEC 220.12 ****	18.15	120	1	1.00	18.1
CAFÉ EQUIPMENT	14.3	120	1	1.00	14.3
TOTALS	304.6				299.1

NOTES:
* USE GREATER VALUE OF THE TWO CATEGORIES.
** 125% OF THE LARGEST MOTOR OR COMPRESSOR IN SYSTEM APPLIED ONLY ON ONE UNIT.
*** N.E.C. ARTICLE 220-12 REQUIREMENT (200 VA PER FOOT OF SHOW WINDOW)
**** MINUS ACTUAL SHOW WINDOW LIGHTING kVA.
***** N.E.C. 2014 ARTICLE 220.43(B) EXCEPTION (CURRENT LIMITERS)

N.E.C. DEMAND kVA x 1,000 / SYSTEM VOLTAGE x 1.732 = MINIMUM FEEDER AMPERAGE

299.1 x 1000 = 299,057
480 x 1.732 = 831

359.7 AMPS USE (EXIST) 1000 AMP SERVICE

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J. COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

Kenneth A. Fuller, P.E.
153 East Main St.
Rockaway, NJ 07866
Tel: 973-784-4258
kfullerpe@msn.com

COLORADO LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES: 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

These plans are an instrument of service and the property of the Architect. Infringements will be prosecuted.

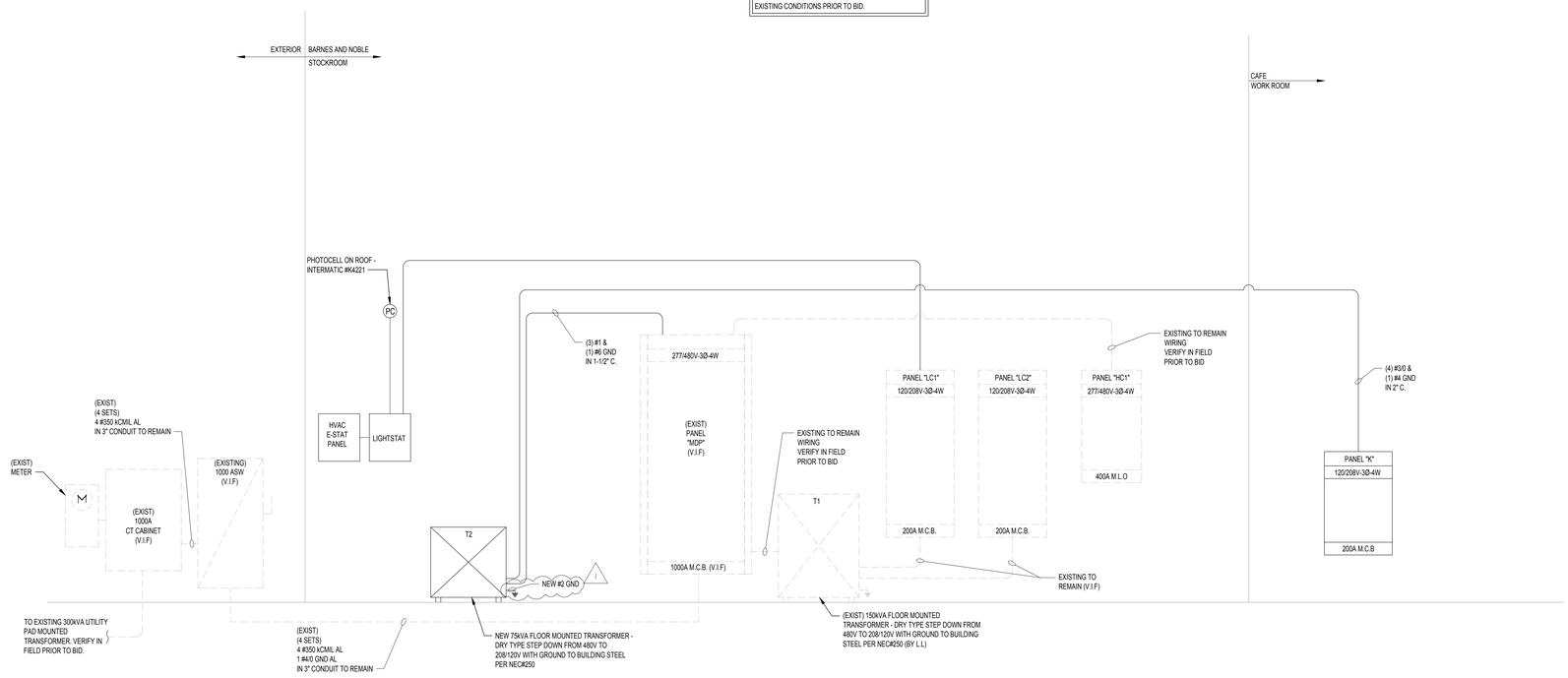
General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:
ELECTRICAL
PANEL SCHEDULES

Drawing Number:
E-202.1

INFORMATION ABOUT EXISTING CONDITIONS WAS TAKEN FROM LANDLORD DRAWINGS SUBMITTED BY YOW ARCHITECTS PC, ON 04/02/2025, REVISION 1. VERIFY EXISTING CONDITIONS PRIOR TO BID.



ELECTRICAL ONE-LINE DIAGRAM

N.T.S. 277/480V-3Ø-4W

INDICATES EXISTING EQUIPMENT/CONDUIT/WIRE, UNLESS NOTED OTHERWISE

INDICATES EQUIPMENT/CONDUIT/WIRE INSTALLED UNDER THIS CONTRACT

XFORMER TO PANEL K = 2,570 AMPS

TRANSFORMER EATON		
TRANSFORMER	75	kVA
VOLTAGE E	480	VOLTS
TRANSFORMER FLA	90	AMPS
POWER FACTOR	100	%
Z	3.51	%
ISCA	2,570	AMPS

XFORMER TO PANEL K	SERVICE	
Fault Current	2,570	Amps
Secondary Voltage	208	Volts
Length of Conductor	102	Feet
Short Circuit Current RMS Symmetrical	2,570	Isca
Number of Conductors per Phase	1	Conductor
Phase Conductor Size	#3/0	MCM
Phase conductor Constant	12,844	
Voltage Line to Line	208	Volts
'f' Factor	0.170	
Neutral Conductor Size	#3/0	MCM
Neutral conductor Constant	12,844	
Voltage Line to Neutral	120	Volts
'f' Factor	0.442	
Multiplier m Line to Line	0.855	
Multiplier m Line to Neutral	0.694	
Fault Current at Main Terminals of Disc. Line to Line	2,197	Amps
Fault Current at Main Terminals of Disc. Line to Neutral	1,782	Amps

1

NOTE:
A FINAL COMMISSIONING REPORT SHALL BE DELIVERED TO THE BUILDING OWNER PER SECTION C408.2.5 OF THE 2021 IECC.

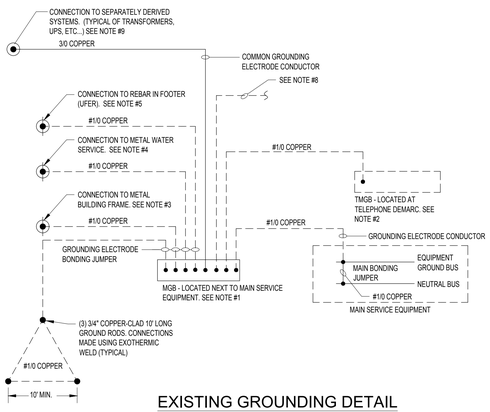
1

REFERENCED DETAIL NOTES:

1. MGB (MAIN GROUND BAR) SHALL BE ERICO #EGBA14412 - (INCLUDES MOUNTING BRACKET AND INSULATORS; HOLE PATTERN AS REQUIRED FOR INSTALLATION. INCLUDING SPARE HOLES FOR FUTURE USE.) OR EQUAL BY HARGER. USE COMPRESSION CONNECTORS (LISTED FOR GROUNDING) TO CONNECT COPPER WIRE TO MGB.
2. TMGB (TELECOMMUNICATION MAIN GROUND BAR) SHALL BE ERICO #TMGBA12L15PT (INCLUDES MOUNTING BRACKET AND INSULATORS) OR EQUAL BY HARGER. USE COMPRESSION CONNECTORS (LISTED FOR GROUNDING) TO CONNECT COPPER WIRE TO TMGB.
3. METAL BUILDING FRAME MUST BE USED AS A GROUNDING ELECTRODE WHERE IT IS EFFECTIVELY GROUNDED PER NEC SECTION 250.52. PROVIDE #3/0 BONDING JUMPERS AS REQUIRED TO PROVIDE CONTINUOUS CONDUCTIVITY WHERE BUILDING FRAME IS SEPARATED INTO ELECTRICALLY ISOLATED SECTIONS AS MAY OCCUR FROM EXPANSION JOINTS, BUILDING ADDITIONS, BUILDING ELEVATION CHANGES, AND THE LIKE.
4. WHERE A WATER METER, FILTERING EQUIPMENT, OR NON-METAL PIPING IS IN THIS METAL WATER PIPING SYSTEM, IT MUST BE BONDED AROUND TO MAINTAIN CONTINUITY EVEN IF THE WATER METER OR FILTER IS REMOVED PER NEC SECTION 250.52.
5. THE REBAR MUST NOT BE LESS THAN 20 FEET IN LENGTH AND 1/2" IN DIAMETER OR 20 FEET OR MORE OF BARE COPPER CONDUCTOR NOT SMALLER THAN NO. 4 IN THE BOTTOM OF THE FOOTER PER NEC SECTION 250.52.
6. IF INSTALLED IN OR ATTACHED TO A BUILDING OR STRUCTURE, METAL PIPING SYSTEM(S), INCLUDING GAS PIPING, THAT IS LIKELY TO BECOME ENERGIZED SHALL BE BONDED PER NEC 250.104 (B). THE BONDING CONDUCTOR(S) OR JUMPER(S) SHALL BE SIZED IN ACCORDANCE WITH TABLE 250.102(C)(1) OR 250.122. AND EQUIPMENT GROUNDING CONDUCTORS SHALL BE SIZED IN ACCORDANCE WITH TABLE 250.102(C)(1) OR 250.122 USING THE RATING OF THE CIRCUIT THAT IS LIKELY TO ENERGIZE THE PIPING SYSTEM(S). THE POINTS OF ATTACHMENT OF THE BONDING JUMPER(S) SHALL BE ACCESSIBLE
7. THE METAL FRAME OF THE BUILDING SHALL BE USED AS A COMMON GROUNDING ELECTRODE CONDUCTOR. IF BUILDING DOES NOT HAVE A METAL FRAME, A #3/0 COMMON GROUNDING ELECTRODE CONDUCTOR SHALL BE RUN FROM THE MAIN GROUND BAR FOR ALL SEPARATELY DERIVED SYSTEMS. MULTIPLE SEPARATELY DERIVED SYSTEMS MAY BE CONNECTED TO THE GROUNDING ELECTRODE SYSTEM BY INDIVIDUAL CONDUCTORS OR A COMMON CONDUCTOR WITH TAPS. THE GROUNDING ELECTRODE TAP CONDUCTOR FOR EACH SYSTEM SHALL BE CONNECTED TO THE COMMON GROUNDING ELECTRODE CONDUCTOR WITH THE CONNECTION IN AN ACCESSIBLE LOCATION.
8. GROUNDING CONDUCTORS TO ADDITIONAL EQUIPMENT AS REQUIRED BY NEC 250 AND SPECIFICATIONS.

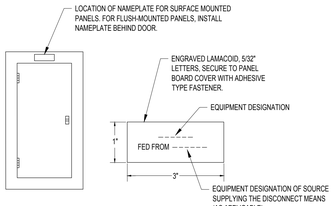
GENERAL GROUNDING DETAIL NOTES:

1. ALL GROUNDING WORK AND MATERIALS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH REQUIREMENTS IN NEC 250.
2. ALL GROUNDING CONDUCTORS SHALL BE COPPER. PROVIDE INSULATED CONDUCTORS FOR INTERIOR INSTALLATIONS AND BARE COPPER FOR EXTERIOR INSTALLATIONS.
3. ALL UNDERGROUND CONNECTIONS TO ELECTRODES AND CONDUCTORS SHALL BE MADE BY EXOTHERMIC WELDING.
4. CONDUCTOR CONNECTIONS TO MAIN GROUND BAR SHALL BE MADE WITH COMPRESSION CONNECTORS LISTED FOR GROUNDING.
5. GROUNDING ELECTRODE AND BONDING JUMPERS SHALL BE INSTALLED SO THAT THEY ARE NOT ANY LONGER THAN NECESSARY AND ANY UNNECESSARY BENDS AND LOOPS ARE AVOIDED.
6. REFERENCE ONE-LINE DIAGRAM FOR CONDUCTOR SIZE OF THE SUPPLY SIDE BONDING JUMPER AND GROUNDING ELECTRODE TAP CONDUCTOR SERVING SEPARATELY DERIVED SYSTEM(S).
7. REFERENCE ONE-LINE DIAGRAM FOR CONDUCTOR SIZE OF THE EQUIPMENT GROUNDING CONDUCTOR SERVING ELECTRICAL DISTRIBUTION EQUIPMENT.
8. INSTALL SLEEVES, SLEEVE SEALS, FIRE STOPPING, RACEWAYS TO PROTECT AGAINST PHYSICAL DAMAGE AND ENVIRONMENTAL CONDITIONS WHERE REQUIRED. FURNISH AND INSTALL BONDING BUSHINGS WHERE FERROUS METALLIC RACEWAYS ARE UTILIZED.



EXISTING GROUNDING DETAIL

N.T.S.



NOTE:
NAMEPLATES ARE REQUIRED ON ALL SWITCHBOARDS, DISTRIBUTION PANELS, PANELBOARDS, ENCLOSURES AND ELECTRICAL CABINETS. ACCESS DOORS/PANELS FOR CONCEALED ELECTRICAL EQUIPMENT, EMERGENCY SYSTEMS/BOXES, MOTOR CONTROL CENTERS, ENCLOSED SWITCHES/CIRCUIT BREAKERS/CONTROLLERS, POWER-TRANSFER DEVICES, PUSH-BUTTONS, CONTACTORS, LIGHTING CONTROL SYSTEMS, INVERTERS, GENERATORS, UPS, MONITORING EQUIPMENT, STARTERS, DISCONNECT SWITCHES, METER SOCKETS, RELAYS, TRANSFORMERS, AND JUNCTION BOXES GREATER THAN 4 11/16" SQUARE. ALSO PROVIDE NAMEPLATES ON BRANCH SWITCHES/BREAKERS OF SWITCHBOARDS AND DISTRIBUTION PANELS.

STANDARD COLORS:
1. NORMAL POWER - WHITE BACKGROUND, BLACK LETTERS
2. EMERGENCY POWER - RED BACKGROUND, WHITE LETTERING
3. IN ADDITION TO THE FUNCTION NAMEPLATE, PROVIDE NAMEPLATES IDENTIFYING ALL "MAIN SERVICE NAMEPLATE DISCONNECTS" - RED BACKGROUND - WHITE LETTERING

NAMEPLATE DETAIL

N.T.S.

ONE-LINE DIAGRAM NOTES

1. THE ELECTRICAL CONTRACTOR SHALL VERIFY ALL POWER COMPANY REQUIREMENTS PRIOR TO BIDDING & INCLUDE THE COST OF ALL ASSOCIATED LABOR, MATERIALS, & CHARGES IN HIS BID.
2. THE ELECTRICAL CONTRACTOR SHALL VERIFY THE AVAILABLE FAULT CURRENT WITH THE UTILITY COMPANY PRIOR TO BIDDING AND PROVIDE EQUIPMENT RATED ACCORDINGLY. SUBMIT FAULT CURRENT CALCULATIONS WITH SHOP DRAWING SUBMITTAL.
3. ALL BUSSING SHALL BE COPPER.
4. PROVIDE FULL LENGTH VERTICAL BUSSING IN ALL SWITCHBOARDS, DISTRIBUTION PANELS, & PANELBOARDS.
5. PROVIDE FULL SIZE HORIZONTAL BUSSING IN ALL SWITCHBOARDS.
6. ALL CIRCUIT BREAKERS SHALL BE BOLT-ON TYPE.
7. ALL WALL-MOUNTED EQUIPMENT SHALL BE MOUNTED ON 3/4" FIRE RATED BACKBOARD.
8. ALL FLOOR-MOUNTED EQUIPMENT SHALL BE MOUNTED ON 4" HIGH CONCRETE HOUSEKEEPING PAD.
9. PROVIDE NAMEPLATES PER NAMEPLATE DETAIL.
10. COORDINATE SPACE WITH ALL OTHER TRADES TO MAINTAIN ALL CODE-REQUIRED CLEARANCES.
11. REFER TO ELECTRICAL SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.
12. THE ELECTRICAL CONTRACTOR SHALL PROVIDE FINAL SETTINGS REQUIRED FOR ALL ADJUSTABLE ELECTRONIC TYPE CIRCUIT BREAKERS WITH LONGTIME, SHORT TIME, GF, INSTANTANEOUS, ETC. THE ELECTRICAL CONTRACTOR SHALL PROVIDE FINAL COORDINATION STUDY.
13. THE ELECTRICAL CONTRACTOR SHALL COMPLY WITH NEC ARTICLE 110.16 FOR LABELING OF PANELS FOR ARC FLASH HAZARD WARNING AS WELL AS FOLLOWING REQUIRED SAFETY PRECAUTIONS WHEN SERVICING OR MAINTAINING ELECTRICAL EQUIPMENT.
14. HVAC CIRCUIT BREAKERS TO BE "HACR" TYPE WHERE REQUIRED BY EQUIPMENT NAMEPLATE PER N.E.C
15. ELECTRICAL CONTRACTOR SHALL BALANCE PANELS AND ELECTRICAL EQUIPMENT TO ±10% BETWEEN PHASES: A/B, B/C, A/C REGARDLESS OF CIRCUITING INDICATED.
16. PROPER CLEARANCE MUST BE MAINTAINED ABOUT ELECTRICAL EQUIPMENT PER N.E.C. FIELD VERIFY EXACT MOUNTING SPACE AVAILABLE IN ELECTRICAL ROOM / AREA PRIOR TO INSTALLATION OF ELECTRICAL EQUIPMENT.
17. ELECTRICAL CONTRACTOR SHALL MAKE FINAL ELECTRICAL CONNECTIONS FOR A COMPLETE ELECTRICAL DISTRIBUTION SYSTEM.
18. EXTERIOR MOUNTED ELECTRICAL EQUIPMENT SHALL BE NEMA 3R RATED AND BE FURNISHED WITH HEATERS, THERMOSTAT AND DISCONNECTING MEANS INTEGRAL TO EQUIPMENT.
19. DRY TYPE TRANSFORMERS SHALL BE GROUNDED TO THE BUILDING GROUNDING ELECTRODE SYSTEM PER NEC.
20. GROUNDING ELECTRODE SYSTEM CONDUCTORS SHALL BE COPPER.
21. CONDUCTORS SHALL BE COPPER, UNLESS NOTED OTHERWISE.
22. FEEDER ROUTING IS DIAGRAMMATIC ONLY. ACTUAL ROUTING OF FEEDERS (OVERHEAD OR UNDERGROUND) IS THE RESPONSIBILITY OF THE ELECTRICAL CONTRACTOR.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J. COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

Kenneth A. Fuller, P.E.
153 East Main St.
Rockaway, NJ 07666
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COLORADO LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES: 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

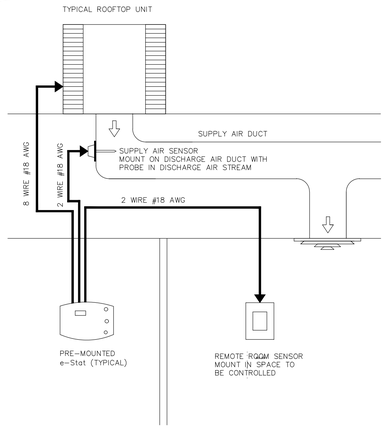
These plans are an instrument of service and the property of the Architect, infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

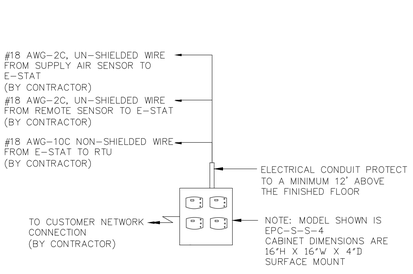
Professional Seal:

Drawing Description:
ELECTRICAL
ONE-LINE DIAGRAM

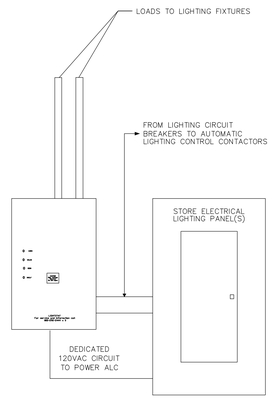
Drawing Number:
E-202



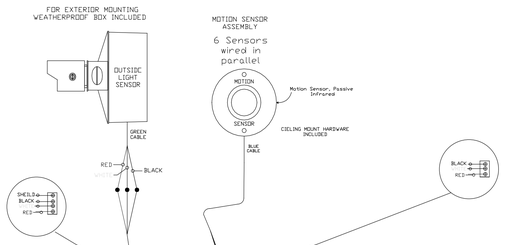
TYPICAL E-STAT WIRING SCHEMATIC

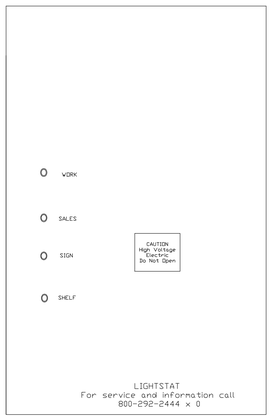


TYPICAL E-STAT PRE MOUNTED ON CABINET

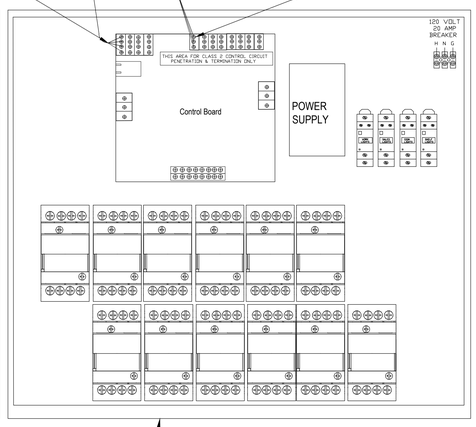


LCC CONTROL RISER DIAGRAM



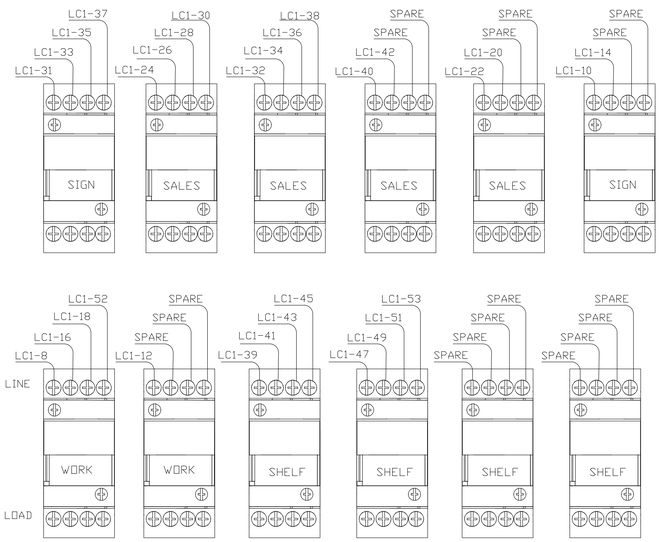


DOOR EXTERIOR



INTERIOR

32"H X 24"W X 6"D



AUTOMATIC LIGHTING CONTROL CONTACTOR TERMINATIONS

General Notes

No.	Revision/Issue	Date

Firm Name and Address

LIGHTSTAT INC.
22 W. WEST HILL ROAD
BARKHAMSTED, CT. 06063
800-292-2444

Project Name and Address

BARNES & NOBLE

Project	Sheet
Date 10-2-24	
Scale	

DRAWINGS FOR INFORMATION ONLY - BN VENDOR TO VERIFY ALL INFORMATION PRIOR TO WORK

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J. COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

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NEW JERSEY LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:
ELECTRICAL
DETAILS

Drawing Number:
E-203

ELECTRICAL SPECIFICATIONS

Section 260010 - General Provisions

A. General
1. Requirements specified in Division 1, instructions to bidders, supplemental general conditions, special conditions, addenda, alternates, contract and proposal, along with Division 26, 27, 28 and all its sections, comprise the contract documents for the electrical contract, along with these specifications as though they were one, and anything implied by the specifications shall be interpreted as also implied by the drawings and vice versa. Provide necessary items for a complete installation of all electrically operated equipment listed in the specifications or shown on the contract drawings.
2. The architectural, structural, mechanical, plumbing and equipment drawings and specifications are incorporated into, and become a part of this division. This contractor shall examine all such drawings and specifications and become thoroughly familiar with the provisions contained therein. The submission of his bid shall indicate such knowledge.
3. Electrical drawings are diagrammatic. They are intended to show the approximate locations of equipment and conduit. Dimensions given on the plans, in figures, shall take precedence over scaled dimensions and shall be verified in the field. The electrical contractor shall layout all equipment rooms to make sure the equipment, as purchased, fits in the room or space shown. Exact location of all equipment shall be verified in the field and routing of conduits shall suit field conditions.
4. Until the time of installation, the architect reserves the right to make minor changes in the location of conduit and equipment without additional cost to the contract.
5. The electrical drawings and specifications are intended to supplement each other. Material and labor necessary to the project shall be furnished and installed even though not specifically mentioned in both. Labor and/or materials neither shown nor specified, but obviously necessary for the completion and proper functioning of the system, shall be furnished and installed by the electrical contractor.
6. Arrange all equipment substantially as shown on the drawings. Make deviations only where necessary to avoid interference. Check all equipment sizes against available space prior to shipment to avoid interference.
7. Examine the work of other trades insofar as their work comes in contact with or is covered by this work in no case attach to, or finish against any defective work or install work in a manner which will prevent proper installation of the work of other trades.
8. Electrical contractor shall verify with other trades all electrical characteristics of equipment requiring electrical connections, contractor shall verify same and shall notify engineer of any discrepancies prior to start of work. Electrical contractor shall provide disconnecting means and overload protection for all equipment, unless furnished integral with equipment package.
9. It is the intent of these drawings that this be a complete electrical job, any errors or omissions shall be brought to the attention of the engineer prior to bidding the job.
10. Should any of the general notes, specifications, details or instructions on plans conflict, the strictest provision shall govern.
11. The existing conditions of these documents are based upon existing drawings prepared by Cortland Morgan Architect dated 12/13/2011 and may not reflect current installations or as-built conditions. Prior to initiating material procurement and construction, it is the contractor's responsibility to verify existing conditions are consistent with the contract documents. This may require removal of existing finishes and possible selective demolition to verify as-built conditions.
12. Do NOT scale drawings.
13. The contractor shall make provisions for the delivery and safe storage of his/her materials and equipment in coordination with the work of other trades. Materials and equipment shall be delivered at such stages of the work as will expedite the work as a whole and shall be marked and stored in such a way as to be easily checked and inspected. The arrival and placing of large equipment items shall be scheduled early enough to permit entry and setting when there is no restriction or problem due to size and weight. Protection of all finishes during delivery is the responsibility of the contractor.

B. Visit to the Site
1. This contractor shall visit the site of the work and familiarize himself with all conditions affecting his work. The submission of his proposal shall indicate such knowledge. No additional payment shall be made on claims that arise from a lack of knowledge of the existing conditions

C. Code and Permits
1. Installation shall be in full accordance with all codes, rules and regulations of municipal, city, county, state and public utilities and all other authorities having jurisdiction over the premises.
2. Comply with any specification requirements that are in excess but not in conflict with code requirements.
3. The contractor shall secure and pay for all permits, plan reviews and certificates of inspection in connection with his work, required by the foregoing authorities. Before final payment of the contract is allowed, all certificates shall be delivered to the architect in duplicate.
4. Electrical material and equipment shall bear the UL label except where UL does not label such types of material and equipment.

D. Shop Drawings Submittals
1. The electrical contractor shall submit product data and shop drawings. Each submittal shall be identified using the respective specification numbering system and titles. Each submittal shall clearly identify which products and options are applicable. The submittals shall be submitted through the architect to the engineer and then, if necessary, resubmitted for final approval. Submittals shall be submitted for the following items:
   - D.a. Wiring devices
   - D.b. Switchboards, Panelboards, transformers and safety switches including fault current study based on equipment being supplied.
   - D.c. Lighting control system and devices
   - D.d. Lighting fixtures
   - D.e. Fire alarm system
2. Electronic Submittals: Prepare submittals as PDF package, incorporating complete information into each PDF file. Name PDF file with submittal number.
3. Each submittal shall be provided with a cover identifying the following:
   - D.a. Name of the job
   - D.b. Location of the job, address, city and state.
   - D.c. Name and address of the company issuing the submittal.
   - D.d. Date of the submittal
4. All submitted product data and shop drawings (manufacturers' equipment descriptive sheets or vendors' prepared drawings) shall have the general contractor's or subcontractor's "stamp of approval" indicating that the item submitted is as called for on the plans and specifications, is approved by the general contractor or subcontractor, the date of approval and initialed by the person approving the submittal and the name of the company submitting said equipment for approval.
5. Any submittal not as specified shall be returned without review for corrections and re-submittal.
6. Every effort shall be made, in checking the shop drawings, to detect and correct all errors, omissions and inaccuracies. Failure to do this will not relieve the electrical contractor of the responsibility for the proper and complete installation in accordance with the contract documents.

E. As-built Drawings
1. Submit three paper-copy set(s) of marked-up record prints to the architect. Contractor shall use red ink for all contractor mark-ups on record prints.
2. Submit PDF electronic files of scanned record prints. Scanned record prints shall be in color.
3. Print and scan each drawing, whether or not changes and additional information were recorded.

F. Standards and Substitutions
1. Wherever the words "approved by", "approved equal", "as directed" or similar phrases are used in the following specifications, they shall be understood to refer to the owner as the approving agency. The name or make of any equipment or materials named in this specifications (whether or not the words "or approved equal" are used) shall be known as the "standard".
2. These specifications establish quality standard of materials and equipment to be provided. Specific items are identified by manufacturer, trade name or catalog designation. This contractor shall submit his base bid price based upon standard specified equipment described herein and as detailed on drawings and associated contract documents. These specifications are not to be considered proprietary. The contractor may submit information on materials and manufacturers (other than those listed) for review by the architect and engineer no later than ten (10) days before bids are submitted. Manufacturers of products accepted by the architect and engineer will be listed in an addendum to the specifications as an acceptable substitution equipment accepted as detailed below and shall be shown as a separate add or deduct price to be factored into the base bid price by the architect and owner if accepted.
3. Should the contractor propose to furnish materials and equipment other than those specified or approved by addendum, submit a written request for substitutions to the architect at the bid opening. The request shall be an alternate to the original bid; be accompanied with complete descriptive (manufacturer, brand name, catalog number, etc.) and technical data for all items. Failure by this contractor to submit the requisite documentation detailed above shall be understood by the architect and engineer to indicate that substitute equipment will not be presented by the contractor for consideration. Such substitutions will not be considered after the bid opening date and delay of project will not be permitted for further inspection and evaluation after this date.
4. Where such substitutions alter the design or space requirements indicated on the drawings, include all items of cost for the revised design and construction including cost of all allied trades involved.
5. Acceptance or rejection of the proposed substitutions shall be subject to approval of the architect and engineer. If requested, the contractor shall submit (at his cost) inspection samples of both the specified and proposed substitute items.
6. In all cases where substitutions are permitted, the contractor shall bear any extra cost of evaluating the quality of the material and equipment to be provided.

G. Testing and Placing in Service
1. Any material or equipment failing a test shall be repaired or replaced at the contractor's expense.
2. Tests shall include the following:
   - G.a. Measure the load on each phase of the main service and each phase of every feeder under full load conditions.
   - G.b. Measure the no-load and full-load voltages (phase to phase, phase to neutral and phase to ground for each phase of each service, of each separately derived system, and at each panelboard or transformer).
   - G.c. Measure the ground resistance of the main service grounding electrode and the ground resistance of each separately derived system's grounding electrode.
   - G.d. Make insulation resistance tests on all dry type transformers and motors.
3. Provide performance testing as required per N.E.C. or local authority having jurisdiction.

H. Interferences
1. Before the installation of any item begins, the electrical contractor shall carefully ascertain that it does not interfere with clearances for the erection of finish beams, columns, pilasters, walls or other structural or architectural members as shown on the architectural drawings. If any work is installed and the architectural design cannot be followed, this contractor shall, at his own expense, make changes in his work as directed by the architect to permit the completion of the architectural work in accordance with drawings and specifications.
2. It shall be the duty of this contractor to report any interferences between his work and that of any of the other contractors as soon as they are discovered. The architect shall determine which equipment will be relocated, regardless of which was installed first. His decision will be final.

I. Quality Assurance
1. All products shall be new and of the type and quality specified. Where materials, equipment, apparatus or other products are specified by manufacturer, brand name, type of catalog number, such designation shall establish the standards of the desired quality and style. It is the intent of these specifications to establish a standard of quality of materials and equipment installed.

J. Special Inspections
1. Special Inspection (as applicable) is to be provided in addition to inspections conducted by the department of building safety and shall not be construed to relieve the owner or his/her authorized agent from requesting periodic and called inspections required by the building code. Special Inspection shall be paid by the owner.
2. Special Inspector shall meet the qualifications as stated in the applicable building code and shall perform the duties and responsibilities as outlined in the applicable building code.
3. The electrical contractor shall provide access to areas requiring testing or inspections, and provide requested documentation (if required by the Special Inspector).

Section 260050 - Basic Electrical Materials and Methods

A. Nameplates
1. General: furnish and mount on each panelboard, switchboard (including branch devices), large junction box, safety switch, starter, remote control, push button station, and all similar controls, a nameplate descriptive of the equipment or equipment controlled.
2. Provide black and white nameplates constructed from laminated phenolic with a white center core. Letters shall be engraved in the phenolic to form white letters 3/8" high. Fasten the nameplates with an adhesive type fastener.

B. Mounting Accessories
1. This contractor shall furnish and install all angle iron, channel iron, rods, supports, hangers, concrete or plywood required to install, mount and support any electrical equipment or device called for on the plans.
2. Supporting material shall be complete with hangers, connectors, bolts, clamps and necessary accessories for complete installation. Supporting material shall be galvanized, painted or otherwise suitably finished. Products by Binkley, Steel City, or Raco will be acceptable.
3. All surface-mounted equipment on block walls shall be mounted on 3/4" plywood backboard. All floor-mounted equipment shall be installed on a 4" high concrete housekeeping pad.

C. Execution
1. The electrical work for construction proposed shall conform to all federal (OSHA), state, all specific safety requirements and the requirements of the current edition of the NEC.
2. Review the HVAC and plumbing specifications for electrical requirements and include the same in the contract cost.
3. Equipment connections, starters, disconnect switches, control transformers and pushbutton stations for the equipment furnished by the owner or under a separate contract shall be installed and connected under this division, as indicated on the contract drawings.
4. All cutting, patching, excavating, backfilling and concrete work related to this contract will be the responsibility of the electrical contractor. This contractor shall assume the responsibility of providing the sleeves, chases and openings necessary for the electrical installation and for their repair in an acceptable manner, as determined by the architect. All holes shall be core-drilled. Provide fire stopping materials, UL Listed for application, in all openings created through fire-rated walls, floors or ceilings. Contractor shall field verify slab on grade floor construction type prior to cutting. Under no circumstances shall the contractor cut a structural floor slab thicker than four (4") inches without prior written approval from Engineer of Record. Notify Engineer of Record of any slab thickness greater than four (4") inches prior to proceeding with any saw cutting.
5. This contractor shall be responsible for providing all required access panels necessary for his work, coordinate with architect prior to installation.

D. Materials and Workmanship
1. All work shall be installed in a practical and workmanlike manner, by mechanics skilled in the several trades necessary.
2. All materials shall be new and free from defects and shall be the best of their several kinds unless specified or indicated on the drawings to the contrary.
3. During each phase and at the completion of the construction, this contractor shall remove all debris and excess materials caused by his work. He shall leave the area of operation broom clean.
4. All electrical equipment shall bear the underwriters laboratories label or ETL label.
5. This contractor shall guarantee his workmanship and material (lamps excepted) for a period of one year from the date of building opening and leave his work in perfect order at the completion. Should defects develop within the guarantee period, the contractor shall, upon notice of the same, remedy the defects and have all damages to other work or furnishings caused by the repairs corrected at his expense to the condition before such damage.

E. Scope of Work
1. The electrical contractor shall provide all labor, material, storage, unpacking and placement; to include but not be limited to, the following items:
   - E.a. Demolition
   - E.b. Emergency lighting and power.
   - E.c. Complete electrical distribution system including, but not limited to, switchboards, distribution and appliance panelboards, transformers, safety switches and feeders.
   - E.d. Complete branch circuit wiring system.
   - E.e. Complete power wiring for all air conditioning equipment, plumbing equipment, heating equipment, ventilating and exhaust equipment.
   - E.f. Complete lighting fixture installation, including all lamps.
   - E.g. Complete communications conduit system including but not limited to, back boxes, plates, j-hooks, cable trays, etc., as specified on the drawings and as required by the local service provider and/or owner.
   - E.h. Temporary electrical power and lighting, as required for construction.
   - E.i. Testing of all cables and circuit wiring after installation.
   - E.j. Exit light system.
   - E.k. Wiring devices, floor boxes, multi-outlet assemblies.
   - E.l. Lighting control system and devices
   - E.m. Grounding and Bonding of the electrical system.
   - E.n. Outdoor lighting and controls.
   - E.o. Fire alarm system.
   - E.p. Communication service
   - E.q. Electric service.

F. Temporary Service
1. The electrical contractor shall furnish, install and remove as required all temporary power and temporary lighting in all areas and individual rooms when needed by the individual trades in the performance of their work. This contractor shall provide a minimum of twenty (20) footcandles of illumination for temporary lighting. Any additional lighting required by individual trades shall be provided by the individual trades including power for the lighting. The electrical work for construction purposes shall conform to all federal (OSHA), state, specific safety requirements, as well as the requirements of the national electric code and national electrical safety code. The electrical contractor shall obtain and pay for all required applications, permits and inspections pertaining to this work. This cost shall be included in the contractor's price.
2. New light fixtures shall not be used for temporary lighting.

G. Electric Service
1. Provide trenching and backfill to the power company specifications.
2. Provide conduit for primary service where required by the power company.
3. Concrete encase conduits where required by the power company and where indicated on the plans.
4. Provide metering to power company specifications.
5. Make provisions for the pad-mount transformer as required by the power company including the transformer pad and grounding.
6. Pay the cost of all power company charges connected with permanent electric service to the building.
7. Coordinate all work with the power company and perform any work necessary to assure a complete, working installation. The entire service installation shall be in complete conformance with the power company's requirements.
8. Verify the exact routing of the primary and secondary services, and all service requirements, with the power company prior to bidding.

Section 260519 - Wiring and Cable

A. Color code conductors (except control and instrumentation conductors) as follows:

	208/120 System	480/277 System
Phase A	Black	Brown
Phase B	Red	Orange
Phase C	Blue	Yellow
Neutral	White	Grey
Ground	Green	Green

1. #12 and #10 conductors shall have continuous insulation color, as listed above.
2. Color code conductors larger than above, which do not have continuous insulation color by application of at least two laps of colored tape on each conductor at all points of access including junction boxes. Color tape shall be the equal of 3M products Scotch #35.
3. Conductors shall be soft annealed copper insulated for 600 volts unless specifically indicated otherwise. Aluminum conductors are not allowed on this project.

B. Insulation type shall be type THWN for wire sizes #8 AWG and larger and THHN or THWN for #10 AWG and smaller. THHN shall not be used in wet or damp locations.

C. Flexible cord shall be heavy duty type so with an equipment ground conductor in addition to the current carrying conductors.

D. Provide #12 conductors, unless otherwise indicated.
1. Control conductors shall be #14 minimum for NEC class I and #16 for NEC class II.

E. Conductors #8 AWG and larger shall be stranded.

F. Conductors #10 AWG and smaller shall be solid.

G. Install wiring in conduit.

H. Connect #10 and smaller wires with constant pressure expandable spring type connectors, "Scotchlok" by 3M or B-Cap by Buchanan.

I. Connect #8 and larger wires with compression connectors or splices as manufactured by Burndy or T&B.

J. Insulate splicing connectors to at least 200% of the wire insulation. Use pre-stretched tubing connector insulators, 3M PST for #2 and larger conductors.

K. Pull conductors using recognized methods and equipment leaving at least 6" wire at all junction boxes for connections.
1. Clean out each conduit system before pulling wire.

L. Form and tie all wiring in panelboards.

M. There shall be no wirenut joints or splices made inside switchboards/panelboards.

N. Branch circuit wire sizes (and conduits) shall be increased from those indicated on the plans to prevent excessive voltage drop. Branch circuits shall be installed with wires of sufficient size so that voltage drop between the panel and the loads does not exceed limit of 3%.

O. Regardless of the temperature rating of the conductor insulation, all conductor ampacity rating for this project shall be determined from the 75°C conductor temperature ratings indicated in the NEC tables. Where equipment or devices are provided with terminals/lugs rated for 60°C, the ampacity rating of the 75°C conductor shall be limited to its associated 60°C rating as indicated in the NEC tables. The electrical contractor shall be responsible to increase the conductors and conduit size as required.

P. Circuits may be multi-plexed in conduit provided wire is properly derated and conduit sized per code. Under no circumstances shall more than six (6) current carrying conductors be run in a single conduit.

Section 260526 - Grounding and Bonding

A. Ground all equipment per N.E.C.

B. Ground each outside lighting pole separately with one ground rod and a #6 ground wire.

C. Ground all dry type transformers as per drawings and NEC #450-10.

Section 260526 (cont.)

D. All conduits shall contain a code-sized ground wire size per N.E.C. in addition to the conductors shown on the plans. Where circuit conductors are increased in size for any reason (i.e. voltage drop, derating, etc.), the ground wire size shall be increased proportionately (according to circular mil area).

E. Where an isolated, insulated ground is required a separate isolated green ground shall be run from the panel isolated ground bus to the isolated ground connection of the device served. In no case shall the system ground (green wire and associated outlet boxes, conduit and building steel) be allowed to contact the isolate ground (green wire with white stripe).

Section 260533 - Raceways and Boxes

A. Raceways
1. All wire shall be run in accordance with code in corrosion resistant, rigid, threaded, metal conduit or electrical metallic tubing (E.M.T.) unless otherwise specifically stated herein.
   - A.a. Conduit in exterior walls, below floor slab, or underground shall be rigid, threaded, galvanized, heavy wall type.
   - A.b. Carlon PVC type 40 heavy wall conduit with ground wire may be used below floor slab or underground in lieu of rigid, threaded, galvanized conduit. PVC 40 conduit shall not be run in or above floor slab. PVC conduit shall terminate below floor slab with rigid, threaded metal conduit adapter. Conduit above slab shall be metal.
   - A.c. Conduit run exposed to the weather shall be heavy wall, metal threaded type.
2. Conduit size shall be 3/4" minimum.
3. Conduit shall be securely fastened in place.
4. All conduit shall be concealed in walls, floors and ceilings wherever possible. Exposed conduit in finished areas will not be permitted. Exposed conduit will be permitted in the unfinished areas with the specific approval of the architect.
5. Use flex conduit for the connection to recessed or semi-recessed lighting fixtures (6' length maximum). Use liquid tight metal conduit for all connections to motors and other equipment subject to vibration and in areas subject to moisture.
6. Use watertight joints with buried and concrete encased conduit. All buried conduits outside of buildings shall have a minimum of 24" of cover. Metal conduits buried in earth shall be painted (two coats) with heavy asphaltum paint.
7. Support runs of conduit as detailed in the appropriate table of the national electrical code (NEC).
8. Install exposed runs of conduit and conduit above lay-in ceilings parallel or perpendicular to the walls, structural members of intersections of vertical planes and ceilings. Provide right angle turns using fittings or symmetrical bends. Support conduits within 1° of all changes in direction.
9. If a conduit is suspended, it shall be supported on trapeze hangers which use "all-thread" rods from the structural steel. The use of ceiling support wire or similar material will not be accepted.
10. Install empty conduit for future use as indicated on the drawings. Conduit shall be complete with jetline or pull rope, junction/outlet boxes, tie rings and appropriate cover plates.
11. Provide pitch pockets where conduits penetrate the roof.
12. Thread lubrication/sealant is required on outdoor and underground threaded metal joints.
13. Install fire seal fittings where conduits penetrate concrete floor slabs or masonry walls required to be fire rated.
14. Horizontal portion of conduit exposed on the roof and feeding equipment shall not be more than 5'-0" unless the written approval from architect or engineer is obtained.

B. Pull and Junction Boxes
1. Install pull and junction boxes where shown on the drawings, and where required for changes in direction, at junction points, and to facilitate wire pulling. Furnish box sizes in accordance with NEC unless larger boxes are indicated.
2. Provide steel boxes and removable covers of code gauge, hot rolled sheet steel, hot dipped galvanized inside and outside, for above ground work. Furnish weatherproof boxes when installed above ground outside.
3. Provide cast iron boxes, hot dipped galvanized inside and outside where shown on the drawings. Furnish removable covers with gaskets and stainless steel, brass or bronze screws.
4. Provide concrete boxes for underground work unless otherwise indicated on the drawings. Furnish steel frames and covers with the cover attached to the frame with hexagon head, brass or bronze cap screws, 3/8" in diameter. Provide a rubber gasket for sealing between the cover and the frame. Paint the cover with two coats of heavy asphaltum.

C. Outlet Boxes
1. Use sheet steel boxes, zinc coated or cadmium plated, for concealed interior work.
2. Use cast boxes, zinc-cadmium finish malleable iron, for exposed interior work, and for exposed or concealed work in wet, damp or exterior locations. Cast boxes shall be series FD by Crouse Hinds or Appleton.
3. Wall box sizes (minimum) shall be 4" square X 2-1/2" deep where wall construction permits. Where wall construction dictates, the depth may be reduced to 2-1/8" or 1-1/2" under special conditions.
4. Fixture outlets in ceilings (minimum) shall be 4" octagonal X 1-1/2" deep (4-11/16" octagonal X 2-1/2" deep where required to accommodate larger conduit or larger number of wires).
5. Ganged boxes shall be one piece (minimum), 2-1/8" deep.
6. Provide cast iron, concrete-tite floor boxes with adjustable covers set flush and level with the finished floor, with outlets as indicated on the drawings. Provide Hubbell #B-2400, 4200, or 4300 series boxes with leveling screws. Flush type covers and openings to serve outlets used. Furnish flush caps for closing off box when not in use.
7. Flush mount boxes in all finished walls, install the plaster rings in drywalled plastered walls and raised covers as required in walls with other finishes so that the cover plates fit tightly against boxes or rings, 3/16" maximum gaps are allowed for noncombustible walls.
8. Adjust location of outlets in masonry or tile construction to occur in the nearest joint to the height specified. Heights shall meet A.D.A. requirements.
9. Support boxes to maintain proper alignment and rigidity.
10. Clean boxes of all foreign matter prior to the installation or wiring of devices.
11. Mounting heights on the drawings are to the centerline of the box unless otherwise noted.

Section 260923 - Lighting Control Devices

A. Sensor Layout: Utilizing project-specific floor plans, manufacturer shall produce a CAD layout of their recommended locations for all occupancy sensors and daylight sensors. Indicate where additional sensors are recommended or where any sensors can be eliminated. Contractor shall use this layout for rough-in of sensor locations.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
1. Acuity Controls
2. Hubbell Control Solutions
3. Wattstopper
4. Lutron
5. Greengate
6. Douglas Lighting Controls
7. Crestron
8. Steinel Professional
9. Touché Lighting Control
10. B.E.G. Controls

C. Daylight Harvesting Controls
1. Sequence of Operation: A daylight harvesting lighting control system shall be furnished and installed complete in rooms indicated on plans. The control sequence shall be as follows:
   - a. General Mode: Occupants shall have the ability to turn on/off and dim all lighting fixtures in the room to a desired light level via the wall dimmer switch(es). Refer to plans for the quantity of zones required.
   - b. Occupancy Detection (where indicated): The occupancy sensor shall turn off all lighting fixtures in the room if occupancy is not detected within 20 minutes. Sensor shall function as vacancy (manual-on).
   - c. Daylight Harvesting: The daylight sensor shall measure lighting levels within the space and automatically dim lighting fixtures according to their daylight zones. Daylight zones shall be dimmed independently to maintain a consistent lighting level across the space. Note: Some rooms may have fixtures that are not on daylight zones. Confirm lighting level setpoints with the Architect prior to programming.
   - d. Emergency Mode (rooms with lighting fixtures on an emergency circuit): Upon loss of normal power, any lighting fixtures on emergency circuits shall be forced on to full light output. All such rooms shall either have UL924 power packs as part of the lighting controls or UL924 devices furnished and installed as required.
2. Dimming Control Module
   - a. Provide an open loop continuous dimming daylight harvesting control module with three individually adjustable zones of control.
   - b. Module shall have pushbutton programming and automated setup.
   - c. LCD display shall provide "real-time" light-level readings.
   - d. Compatible with 2-wire 0-10 volt dimming ballasts, 50 ballasts per channel.
   - e. Module shall be capable of integrating with occupancy sensors and manual override control stations.
   - f. DIN rail mounting.
   - g. Shall have standard 2 year warranty and shall be UL listed.
3. On/Off Control Module
   - a. Provide an open loop on/off daylight harvesting stepped control module with three individually adjustable zones of control.
   - b. Module shall have pushbutton programming and automated setup.
   - c. LCD display shall provide "real-time" light-level readings.
   - d. Module shall be capable of integrating with occupancy sensors and manual override control stations.
   - e. DIN rail mounting.
   - f. Shall have standard 2 year warranty and shall be UL listed.
4. Power Pack (with 3 Relays)
   - a. Provide a 120/277VAC to +24VDC power pack to power the dimming control module with three individually controlled relays for ON/OFF control.
   - b. Built-in overload protection.
   - c. Quick connect.
   - d. DIN rail mounting.
   - e. Shall have standard 2 year warranty and shall be UL listed.
5. Power Pack (with 1 Relay)
   - a. Provide a 120/277VAC to +24VDC power pack with one relay for ON/OFF control.
   - b. Built-in overload protection.
   - c. Quick to install connector.
   - d. Plenum rated for mounting inside or outside a junction box or inside a fluorescent ballast cavity.
   - e. Shall have standard 5 year warranty and shall be UL listed.
6. Relay
   - a. Provide a 10 amp single-pole double-throw relay with 10-30 VAC/DC/120 VAC coil or 10-30 VAC/DC/208-277 VAC coil.
   - b. Normally open and normally closed isolated contacts.
   - c. LED status indicator.
   - d. Plenum rated housing.
   - e. Shall have standard 1 year warranty and shall be UL listed.
7. UL924 Relay
   - a. Provide a 10 amp single-pole double-throw relay with 10-30 VAC/DC/120 VAC coil or 10-30 VAC/DC/208-277 VAC coil.
   - b. Normally open and normally closed isolated contacts.
   - c. LED status indicator.
   - d. Plenum rated housing.
   - e. UL924 listed for emergency use.
   - f. Shall have standard 1 year warranty and shall be UL listed.

Section 260923 (cont.)

8. Photosensor
   - a. Provide an open loop photosensor that continuously measures footcandle levels using a photodiode element to provide input to the dimming control module.
   - b. Foot-candle range of 3 to 6,000 fc.
   - c. Sensor shall be capable of mounting both horizontally and vertically.
   - d. User selectable foot-candle range setting via jumper beneath front cover.
   - e. Shall have standard 2 year warranty and shall be UL listed.
9. Occupancy Sensor
   - a. Provide a digital dual-technology ultrasonic and passive infrared occupancy sensor to turn lighting ON/OFF based on occupancy via the dimming control module.
   - b. Automatic self-adaptive technology with no manual adjustment required.
   - c. Non-volatile memory for sensor settings.
   - d. 1,600 square-foot coverage area.
   - e. Auxiliary relay for building automation system integration.
   - f. Shall have standard 5 year warranty and shall be UL listed.
10. 4-Button Wall Dimmer Switch
    - a. Provide a low voltage four button wall switch for manual override control of the dimming control module.
    - b. Four buttons provide ON/OFF, dim up, dim down and automatic controls with LED indicators.
    - c. Mounts in standard single-gang box.
    - d. Contractor shall provide decorator style wall plate (not included).
    - e. Buttons shall have engraved labels.
    - f. Standard finish as selected by architect.
    - g. Shall have standard 2 year warranty and shall be UL listed.
11. 1-Button Wall Switch
    - a. Provide a low voltage one button latching wall switch for manual override control of lighting fixtures.
    - b. Single button provides ON/OFF control with LED indicator.
    - c. Mounts in standard single-gang box.
    - d. Contractor shall provide decorator style wall plate (not included).
    - e. Button shall have engraved label.
    - f. Standard finish as selected by architect.
    - g. Shall have standard 2 year warranty and shall be UL listed.

D. Occupancy Sensors, Line Voltage, Wall Switch Type
1. Shall use passive infrared motion detection.
2. Shall be compatible with incandescent, magnetic or electronic low voltage, and magnetic or electronic fluorescent, as well as motor loads.
3. Switch shall be microprocessor controlled.
4. Shall be capable of detecting occupancy with true, 180º field of view.
5. Shall utilize zero crossing circuitry, which increases relay life, protects from the effects of inrush current, and increases sensor longevity.
6. Wall switch shall have integral shutters that narrow the field of view from 180º.
7. Shall feature pushbutton for manual on and off, which times out based upon occupancy detection.
8. An LED shall indicate occupancy status.
9. Internal timer shall be factory set at 10 minutes, shall be push-button programmable from 30 seconds to 20 minutes and shall reset every time occupancy is re-detected. Requires no field calibration or sensitivity adjustments.
10. Manual range, photocell, and time settings shall be user-configurable.
11. Switch shall be rated at 120/277V in one unit.
12. Unit shall fit in a standard box and use a standard wallplate, which is gangable.
13. Wall switch shall not protrude more than .4 inches from box.
14. Shall be a Decora style unit with a matching wallplate available.
15. Shall have standard 5 year warranty and shall be UL listed.
16. Two-pole devices shall provide switching for 2 separate banks from a single unit.

E. Occupancy Sensors, Low Voltage, Ceiling Mount
1. Shall incorporate dual-technology passive infrared and ultrasonic motion detection.
2. Shall mount on ceiling.
3. Shall have 360° coverage with at least a 28 ft coverage pattern (when mounted at 9 ft) in all directions for walking motions.
4. Shall automatically adapt to changing room conditions–including background PIR levels and continuous airflow not less than 6 feet from sensor.
5. Shall incorporate a real-time motion indicator LED, which is visible from the front of unit.
6. Shall have mask inserts for PIR rejection to prevent false tripping.
7. Internal timer shall be factory set at 10 minutes, shall be push-button programmable from 30 seconds to 20 minutes and shall reset every time occupancy is re-detected. Requires no field calibration or sensitivity adjustments.
8. Shall be included with a low voltage relay for tie-in to building automation system.
9. Shall have standard 5 year warranty and shall be UL listed.

F. Power Pack
1. Transforms 120 or 277V to class 2, 15 to 24V DC, to power remote sensors.
2. Shall be compatible with incandescent, magnetic or electronic low voltage, and magnetic or electronic fluorescent, as well as motor loads.
3. Ratings: 20A incandescent, 20A fluorescent, 120 or 277V.
4. Shall be plenum rated. Mount in deep junction box where required per local AHJ.
5. Shall have elongated mounting nipple which can be mounted either directly through a ½" knockout in a junction box or to be located inside an adjacent box for specific local code requirements, contractor to verify.
6. Shall be capable of powering up to 14 sensors.
7. Shall have self-contained relays with relay circuit protection.
8. Provide 2-pole version for rooms with two levels of lighting control including inboard/outboard switching.
9. Shall have standard 5 year warranty and shall be UL listed.

G. Wall Timer Switches
1. The timer shall be an electronic interval timer with a manually operated toggle switch.
2. The timer switch shall be capable of 3-way operation.
3. Switch contacts shall break the current at the end of a preset time which is user adjustable from 1 minute to 18 hours. Timer adjustment shall be hidden after wallplate is installed.
4. Flicker feature shall be provided to provide a flick warn of the load controlled two minutes before the end of the timed cycle and again one minute before the end of the timed cycle.
5. Unit shall fit into a standard 2-1/2 inch deep wall box, single or multi-gang installation, and accept a standard toggle switchplate.
6. Unit shall be capable of switching fluorescent lights with electronic or electromagnetic ballasts, incandescent lights, or motor loads. Unit shall accept input of 24, 120, 208-240, or 277 volts AC at 50/60 Hz.
7. Time switch shall have 5 year warranty and shall be UL listed.

H. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust field-assembled components and equipment installation, including connections, and assist in field testing. Report results in writing. Remove and replace lighting control devices where test results indicate that they do not comply with specified requirements.

I. Adjusting
1. Final occupancy sensor locations shall be determined in accordance with manufacturer's recommendations and locations adjusted as required prior to rough-in. Refer to sensor layout submittal provided by manufacturer. All sensors shall have non-adjustable factory calibrated sensitivity for maximum performance. A factory-authorized service representative shall be engaged to meet on-site with the contractor to determine proper device locations prior to rough-in.
2. Once lighting fixtures and occupancy sensors have been installed, a factory-authorized service representative shall be engaged to set-up and program occupancy sensors and photosensors. Contractor and service representative shall meet on-site with the Owner to determine appropriate set points and programming.
   - a. All occupancy sensors shall be field adjusted/aimed to effectively detect motion and eliminate nuisance tripping.
   - b. Time Delay settings for occupancy sensors shall be factory set at 10 minutes, and shall not be field adjusted unless specifically instructed by Architect. This delay selection is based on lamp life vs. energy savings and sensor performance.
3. Once all occupancy sensors have been set-up, adjusted and programmed, contractor shall meet again with factory-authorized service representative and Owner to test operation of systems. Service representative shall be engaged to make adjustments to sensors, set points and programming as necessary for proper operation.

J. Occupancy Adjustments: When requested within 12 months of date of Substantial Completion, a factory-authorized service representative shall be engaged to provide on-site assistance in adjusting sensors to suit actual occupied conditions. Provide up to two visits to site outside normal occupancy hours for this purpose.

K. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain lighting controls. Refer to Division 1 Section "Closeout Procedures." Provide a minimum of four (4) hours of Owner training.

Section 262726 - Wiring Devices

A. Wiring device color shall be selected by architect, unless otherwise indicated.

B. Provide totally enclosed, 20 ampere, 120/277 volt, quiet A/C general use snap switches.

C. Switches shall be specification grade as manufactured by Hubbell, P&S, or Leviton.

D. Provide NEMA configuration 5-20R Duplex 125 volt grounding type receptacles rated for 20 amperes unless otherwise indicated on the drawings.

E. Receptacles shall be specification grade as manufactured by Hubbell, P&S or Leviton.

F. Receptacles requiring amperages, voltages or configurations different from the duplex convenience receptacles above shall be as indicated on the drawings.

G. Provide other receptacles of a quality, material and workmanship equal to that specified for duplex convenience receptacles.

H. Provide cover or device plates for outlet boxes as follows unless otherwise noted:
1. Finished areas: Thermoplastic - color to match device.
2. Unfinished areas: Zinc coated sheet metal, aluminum, or cast metal as appropriate for the type of box.
3. Exterior areas: Copper free aluminum with gray, powder epoxy finish, gasket, weatherproof, Crouse-Hinds "WLRD" for duplex receptacles and WLRS for single receptacles or equal.
4. Telephone, communication, and signal outlet plates, shall match those used for receptacles and switches. All outlet and/or junction boxes shall be complete with a cover plate by this contractor.
5. Where devices are ganged, they shall be installed under a common cover plate.

I. Locate the switches approximately 4'-0" above the finished floor elevation or nearest block course (within A.D.A. requirements), unless otherwise indicated. The long dimension of the switches shall be vertical.

J. Locate receptacles approximately 1'-6" above the finished floor elevation or nearest block course (within A.D.A. requirements), unless noted otherwise. The long dimension of receptacles shall be vertical.

Section 262816 - Safety Switches

A. Safety switches shall be the enclosed heavy-duty type (type HD) with quick-make, quick-break mechanism and external pad lockable operating handle.

B. Safety switches shall be rated for 240 or 600 volts as applicable. They shall be horsepower rated when used in motor circuits.

C. Safety switches shall be fusible or non-fusible 2, 3, or 4 pole as indicated on the drawings.

D. Safety switches shall be single throw unless otherwise indicated on the drawings.

E. Enclosures shall be NEMA 1 indoors and NEMA 3R outdoors unless otherwise indicated on the drawings.

F. Manufacturer shall be Square D, Siemens, G.E., or Cutler-Hammer. All safety switches shall be by one manufacturer.

G. Mount the safety switches securely between 3' X 6' levels above the floor unless otherwise indicated on the drawings.

H. Switches on block walls shall be mounted on a 3/4" plywood backboard, where located indoors.

Section 265119 - Lighting Fixtures

A. LED lighting fixtures:
1. Recessed Fixtures: Comply with NEMA LE 4.
2. Bulb shape complying with ANSI C79.1.
3. Lamp base complying with ANSI C81.61.
4. CRI of minimum 80.
5. CCT of 3500K, unless noted otherwise on the plans or fixture schedule.
6. Rated lamp life of 50,000 hours, minimum at 70 percent lumen maintenance.
7. Lamps dimmable from 100 percent to 10 percent of maximum light output, unless noted otherwise on the plans or fixture schedule.
8. Integral driver. Driver power factor shall be 40 percent or greater. Harmonic distortion shall be less than 10% THD. Drivers shall be equipped with automatic thermal protection and 20 KA surge protection with end of life LED indicator.
9. Nominal Operating Voltage: as indicated on plans and schedules.
10. Efficiency minimum of 80 lumens per watt.
11. Each LED luminaire type shall be binned within a three-step MacAdam Ellipse to ensure color consistency among luminaires.
12. Fixtures shall comply with UL 1598 and UL 8750.

B. Linear fluorescent lighting fixtures:
1. Lamps for new light fixtures shall be T8, 3500K, minimum 80 CRI of

ELECTRICAL SPECIFICATIONS (cont.)

Section 284621 - Alarm System

A. Fire Alarm System (addressable analog type)

1. This contractor shall submit fire alarm system drawings and specification to the local fire department for their approval before installation of any fire alarm components or wiring.
2. Equipment: all devices, combinations of devices, appliances and equipment shall be listed for the purpose for which they are used and shall be installed in compliance with applicable codes and standards.
3. Provide a complete, supervised, power-limited, fire alarm system. All equipment herein specified is that of Simplex Time Recorder Co. Equipment supplied by listed acceptable alternate manufacturers shall meet or exceed the quality of the Simplex equipment specified.
4. The fire alarm system shall be an electrically supervised system, which shall monitor the integrity of circuit conductors and power supplies.
5. The installation organization shall be a company specializing in the installation of fire alarm systems. This organization shall have a minimum of five years experience with installation of such systems.
6. The contractor shall provide and maintain on the site an up-to-date record set of approved shop drawings.
7. Record drawings shall include location of end-of-line device locations.
8. Upon completion of the work, and final acceptance by the local authority, the contractor shall submit record drawings to the owner.
9. A remote monitoring facility output circuit, selectable for interface to remote station reverse polarity, local energy master box, shunt master box, digital alarm communicator, or radio transmitter monitoring systems, shall be provided. The equipment supplier must contact the local authorities prior to bidding and supply all required equipment in the base bid.
10. Control panel shall be simplex 4010-9101 with Simplex Type 4010-9810 Internal D.A.C.T. The power supply shall be adequate to serve control panel modules, relays and alarm indicating appliances. Include a secondary emergency power supply with capacity for operating system for a period of time as specified per NFPA, the local authorities having jurisdiction and or Owner/Building Standards. The strictest provision shall govern.
11. LCD annunciator panel shall be Simplex #4606-9101 with Simplex #4603-9111 brushed aluminum trim plate.
12. Addressable pull stations shall be semi-flush, action push/pull type, simplex #4099-9003.
13. Smoke detectors shall be analog photoelectric type, Simplex #4098-9710.
14. Analog heat detectors shall be 135 degrees F fixed temperature with rate-of-rise type, Simplex #4098-9733 with addressable base, Simplex #4098-9788.
15. Duct-mounted smoke detectors shall have a base with auxiliary relay (Simplex Type 4098-9756), analog photoelectric detector (Simplex Type 4098-9714), and sampling tubes (length as required).
16. Duct detector remote key reset/test station with alarm LED shall be Simplex Type 2098-9806.
17. Semi-flush audio/visual unit shall be Simplex Type 4903-9301 Series with A.D.A. complying 15/75 Candela strobe and horn. All strobe lights shall be synchronized.
18. Visual only unit shall be Simplex Type 4904-9307 with A.D.A complying 15/75 Candela strobe. All strobe lights shall be synchronized.
19. Waterflow and tamper switches shall be furnished and installed by the sprinkler contractor. The electrical contractor shall connect each device to the fire alarm system using an individual addressable module (IAM) Simplex #4090-9001.
20. Magnetic door holders shall be electrically operated and magnetically hold smoke doors in an open position. Provide a flush, surface or floor mounted as required:
    - B.a. Flush mounting - Simplex Type: 2088-9579
    - B.b. Surface mounting - Simplex Type: 2088-9582
    - B.c. Floor mounting - Simplex Type: 2088-9573
21. Provide and install wiring per manufacturers' specifications. All wiring shall be in conduit (3/4" minimum).
22. The completed fire alarm system shall be fully tested in accordance with NFPA-72H, and local fire department requirements, by the installer, in the presence of the owner's representative and the local fire marshal. Upon completion of a successful test, the installer shall so certify, in writing, to the owner and general contractor.
23. Include on-site services of a certified technician to provide technical installation support for panel start up, program editing, troubleshooting of the fire alarm system control panel, and assistance to the installer for one complete final system checkout in accordance with the field quality control section of these specifications.
24. Acceptable manufacturers/suppliers shall be Simplex, Notifier (as supplied by a certified Nesco affiliate) or Siemens.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:

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80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

These plans are an instrument of service and the property of the Architect. Infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:

ELECTRICAL
SPECIFICATIONS

Drawing Number:

E-302

GENERAL NOTES

DESIGN CRITERIA

1. ASHRAE Design Requirements
   - A. Heating/Cooling Loads
     - 1) Heating
       - a) Outside air temperatures: -1 degrees F DB (ASHRAE 99.6%).
     - 2) Cooling
       - a) Outside air temperatures: 90 degrees F DB / 58 degrees F WB (ASHRAE 99%).
   - B. Building Design
     - 1) ASHRAE and Energy Code guidelines.
     - 2) Indoor relative humidity at 50 percent RH for cooling loads only.
       - a) Humidity is not being controlled or maintained in the heating or cooling equipment modes.
3. Design Ventilation
   - A. Use ASHRAE 62 Standard or 2021 International Mechanical Code.
   - B. See Ventilation Schedule on drawings.
   - C. See floor plans for rooms that need to be in a positive or negative relation to other rooms shown by the difference in supply and return or exhaust air quantities.

GENERAL

1. The term General Contractor (G.C.) as used in these documents refers to the Contractor / Construction Manager in responsible charge of the project in terms of coordination, scheduling, subcontractor coordination, etc. this term refers to, but is not limited to, General Contractor, Construction Manager, Design Build Contractor, Prime Contractor, etc. The term is referencing the entity that coordinates the work of other trades.
2. These drawings are diagrammatic and indicate the general extent of the work. The contractor shall be responsible for the coordination and proper installation of all mechanical systems. The contractor shall provide all necessary offsets and fitting which may be required due to space constraints or other conditions.
3. Existing building HVAC, Plumbing and Fire Protection systems shown on these drawings which are to be removed or modified where taken from the original drawings dated 2010 and may not show current installations or conditions. Each contractor shall field verify all existing systems.
4. The mechanical systems or its modifications are designed to be a complete operating system and stable after the building or its modifications are fully completed. It is solely the contractor's responsibility to determine construction, installation, and programming procedures and sequences to have a complete and working system and to insure the safety of the construction personnel, public, building and its component parts, and adjacent buildings and properties. This includes the addition of whatever temporary or permanent bracing, etc. that may be necessary to brace new or existing construction, walls, and framing to remain so that the structure is braced for construction loads, etc. and that no horizontal or vertical settlement or any damage occurs to the adjacent new or permanent supports and bracing that are installed. Design of these supports shall be provided by the contractor. Provide all materials, labor, equipment, and accessories required to furnish and install the systems identified in specifications and drawings.
5. It is the contractor's responsibility to enforce all applicable safety codes and regulations during all phases of construction.
6. Construction loads shall not exceed structural design live loads. The contractor shall be responsible for all design required to support construction equipment used in constructing this project. Verify and coordinate with structural drawings.
7. The contractor shall perform all construction for the project in a manner and sequence that are based on accepted industry standards that recognize the interaction of the components that comprise the systems, without causing distress, unanticipated movements or irregular load paths as a result of the construction means and methods employed.
8. The contractor shall provide all miscellaneous supporting steel, etc. for the proper installation of all mechanical systems.
9. Before fabrication and/or installing any work, contractor shall see that it does not interfere with clearance required for finish on beams, columns, plasters, walls, or other structural or architectural members, as shown on architectural drawings. If any work is so installed and it later develops that architectural design cannot be followed, contractor shall, at his own expense, make such changes in his work as architect may direct to permit completion of architectural work in accordance with plans and specifications.
10. All piping shall be protected as required by the applicable Mechanical, Plumbing, Fire Protection and Building Codes: " General Regulations" and other Code Chapters.
11. Pipes passing through or under walls shall be protected from breakage. Pipes passing through studs, joist, rafters or similar members less than 1 1/2" from the nearest edge of the members shall be protected by steel shield plates.
12. Piping shall be installed to prevent strains and stresses that exceed the structural strength of the pipe. Where necessary, provisions shall be made to protect piping from the damage resulting from pipe expansion and contraction and structural/soil settlement. Expansion joint fittings shall be used where necessary to provide for expansion and contraction of the pipes. Sleeved openings shall be sized appropriately to accommodate pipe movement and structural/soil settlement. Expansion joint fittings shall be of the typical material suitable for use with the type of piping in which fittings are installed. At a minimum install rubber mechanical joint couplings or CSA-certified expansion joints on all vertical piping at every other floor of the building and rigidly support the stack pipe on alternating floors to direct any movement into the appropriate expansion compensator. Design of these expansion fittings shall be provided by the contractor. Any analysis which requires additional support or expansion detailing shall be shared with the mechanical design professional and any stresses or point loads created by the engineered system shall be shared with the structural designer for review.
13. Install additional offsets on piping or ductwork where required to obtain maximum headroom or to avoid conflict with other work without additional cost to owner.
14. Report any interferences between work under this division and that of any other contractors to architect as soon as they are discovered. Architect will determine which equipment shall be relocated, regardless of which was first installed, and his decision shall be final.
15. The contractor shall coordinate floor, wall, and roof penetrations, louver sizes, etc. with general trades.
16. Principal openings on these drawings through the framing are shown on the structural drawings. The mechanical contractor shall examine the structural and mechanical drawings for the required openings and shall verify size and location of all openings with the general contractor. General contractor shall provide all openings required through the framing by the mechanical, electrical, plumbing, or other trades, whether or not shown on the structural drawings. Any deviation from the openings shown on the structural drawings shall be brought to the engineer's attention for review.
17. All mechanical and electrical work: Ductwork, plumbing, piping, wiring, lighting, etc. and all architectural items that need to be removed during the modification of or reinforcing of, existing structure shall be replaced in kind by the respective contractor. The contractors shall keep all existing systems in operation during the construction phase of the project.
18. All contractors are required to examine the drawings and specifications carefully, visit the site and fully inform themselves as to all existing conditions and limitations, prior to agreeing to perform the work. Failure to visit the site and familiarize themselves with the existing conditions and limitations will in no way relieve the contractor from furnishing any materials or performing any work in accordance with drawings and specification without additional cost to the owner to have a complete and working system.
19. Details labeled "Typical Details" or "Typical" on drawings apply to situations occurring on the whole project that are the same or similar to those specifically detailed. Such details apply whether or not details are referenced at each location on drawings. Notify engineer for clarifications regarding applicability of "Typical Details".
20. Work and coordinate these drawings with architectural, civil, structural, mechanical, plumbing, fire protection, electrical, and technology drawings.
21. Do not scale drawings.
22. Any discrepancies between mechanical and architectural drawings shall be brought to the attention of the architect and mechanical engineer.
23. Should any of the general notes conflict with any details or instructions on plans, or in the specifications, the strictest provision shall govern.
24. Shop drawings and submittals
    - A. Shop drawings and submittals shall be checked and coordinated with other materials and contracts by the general, mechanical and electrical contractors and shop drawings and submittals shall bear the prime contractor's review stamp with the checker's initials before being submitted to the architect for approval.
    - B. When the contractor has been authorized to use the architect and engineer's drawings as construction coordination drawings, the contractor must remove all title blocks, professional seals and any other references to the architect and engineer from those drawings. The contractors name and title shall be placed on the drawings.
    - C. Where voltage, amp draw, dimensions and elevations of existing construction could affect the new construction, it is the contractor's responsibility to make field verifications and measurements in time for their incorporation into the shop drawings.
25. Refer to architectural and electrical reflected ceiling plans for exact location of light fixtures. Contractors to coordinate locations of lighting, speakers, air diffusers, grilles, sprinkler heads and the like, with reflected ceiling lay-outs as required and directed by the architect.
26. Ductwork or piping shall not be located over the top of any electrical panels or equipment.
27. Contractor shall include in his bid all cutting, trenching, and patching associated with the installation of this projects work.
28. Cutting, Patching and Drilling
    - A. All cutting and patching of the building construction required for this work shall be by this contractor unless shown on architectural drawings and confirmed as to size and location prior to new construction. Cutting shall be in a neat and workmanlike manner.
    - B. Neatly saw cut all rectangular openings, set sleeve through opening, and finish patch or provide trim flange around opening.
    - C. Neatly saw cut floors and patch floor to match existing, including floor covering.
    - D. Contractor shall field verify slab-on-grade or supported floor construction type prior to cutting. Under no circumstances shall this contractor cut a floor thicker than 4 inches, a structural floor slab, whether on grade or supported, without prior written approval from the architect. If floor slab indicated to be cut on mechanical plans is found to be structural in nature, do not cut. Contact architect immediately for further directions.
    - E. Core drill and sleeve all round openings.
    - F. Do not cut any structural components without architect's written approval, including, but not limited to roof joists, columns, floor joists, beams, girders, structural floor slabs, rebar, etc.
    - G. Patch, and finish to match adjacent areas that have been cut, damaged or modified as a result of the installation of the mechanical systems. Fire-stop all penetrations of fire rated construction in a code approved manner.
    - H. All contractors shall confirm with owner, prior to bid, times available for noise producing work such as cutting and core drilling of floors, walls, etc. as well as times for work which requires access into adjoining tenant spaces. Include any premium time in bid.
    - I. Exact location of roof top air conditioning units shall be approved by the structural engineer. Mechanical contractor shall furnish and install all supplemental support steel for equipment and roof penetrations after approval of structural engineer.
    - J. The mechanical contractor shall coordinate work with the general contractor prior to construction. The mechanical contractor shall provide information regarding openings in walls, floors, etc., concrete equipment pads and foundations to the general contractor. If the mechanical contractor fails to comply with this request, or if incorrect information is given, the necessary cutting and patching will be performed by the general contractor, the mechanical contractor's expense.
    - K. All openings required for this branch of work shall be accomplished in time to be incorporated in, and be compatible with the construction program; otherwise this contractor shall be responsible and pay for all changes made necessary for his failure to do so. Pipe holes in floors and walls shall be core drilled if not sleeved during construction.
29. Refer to mechanical, plumbing, fire protection, and electrical plans for location of mechanical, plumbing, and electrical equipment. Coordinate location of disconnect switch associated with each piece of mechanical and plumbing equipment with electrical contractor.
30. Installation requirements for all HVAC, plumbing, and fire protection systems shall be reviewed and coordinated with all other trades involved prior to rough-in. Give equipment shop drawings from installer/supplier/contractor equipment, as required, for review and coordination to all other trades involved. Contact architect/engineer with any discrepancies found between construction drawings and equipment being furnished prior to rough-in.
31. The contractor shall furnish all access panels or doors in hard ceilings and walls with a size as required for servicing and testing, for equipment, valves and/or devices furnished under this contract. The general contractor shall install access panels. The contractor shall coordinate the size and location of each access panel with the architect and general contractor prior to rough-in.
32. Firestopping
    - A. All penetrations through fire rated walls associated with the installation shall be sleeved and fire-stopped using a UL approved method. UL approved method shall meet or exceed fire rating of structure being penetrated. Reference architectural plans for fire rated structures. If shown, reference architectural, mechanical and electrical drawings for penetration details.
    - B. All openings through fire rated walls, floors, and/or roofs for ductwork, piping, conduit, etc., shall be fire sealed with a calcium sallicate, silicone "RTV" foam, "3M" fire rated sealants, Hilti Firestop Systems, or approved equal to maintain the intended fire rating and associated UL ratings as recommended by the architect and/or sealant manufacturer.
    - C. All fire stopping sealants shall be thixotropic so as not so slump or sag and shall be trowelable. Fire stopping sealants shall be intumescent and shall be free of asbestos, halogens, and volatile solvents.
    - D. Fire stopping materials shall be classified in the Underwriters Laboratories (UL) fire resistance directory or listed in the Warnock Hersey International Directory.
33. All equipment and devices for this project must be UL listed. Devices, equipment, systems shall be installed per National Electrical Code requirements and manufacturer's instructions.
34. All conduit and cabling shall be properly supported as required by the National Electrical Code. For existing installations, the contractor shall be responsible to replace and/or rework existing conduit and/or cabling that is not in compliance with this requirement.
35. All materials and work in the ceiling return air plenum shall be approved for plenum rated application in accordance to the current building code. Where open wiring methods for low voltage systems is permitted by the contract documents and local authority, the conductor insulation must be plenum rated.
36. All hot water heating supply and return branch run-out piping shall be 3/4 inches unless otherwise noted on drawing.
37. Shop Areas and Material Storage
    - A. No plumbing or mechanical trade is permitted to use as shop working area, any concrete slab that is to receive metallic waterproofing, asphalt tile. plastic tile, etc., except by express permission of the architect.
    - B. The contractor shall make provisions for the delivery and safe storage of his materials and equipment in coordination with the work of others. Materials and equipment shall be delivered at such stages of the work as will expedite the work as a whole and shall be marked and stored in such a way as to be easily checked and inspected. The arrival and placing of large equipment items shall be scheduled early enough to permit entry and setting when there is no restriction or problem due to size and weight.
38. Temporary Heat
    - A. The HVAC contractor under this division shall set up temporary heat and other services as may be required and/or requested by the general contractor. See "General Conditions" and "Special Conditions." This contractor shall pay expenses resulting from temporary heat and services used solely by him.

DEMOLITION

1. The architectural drawings are to be used only as a guideline for demolition. The contractor must visit the site prior to bidding to verify all work required for a complete job and include the cost of such work in his bid.
2. The mechanical drawings are intended to show only the general existing building construction within the area of demolition. The drawings do not show all systems, quantities, sizes, obstructions, etc., and are not intended to be used by the contractor to define the complete scope of demolition. The contractor must field verify the actual building and systems conditions to define all elements within the scope of demolition.
3. Examine areas and conditions under which demolition work must be performed. This contractor shall coordinate his work with other trades performing demolition work and/or demolition work performed by the owner. In every instance of demolition and/or remodeling, the contractor shall figure a complete job as none other shall be accepted.
4. The extent of work shown or not shown shall include removal and legally dispose off site, all the items and systems being removed.
5. Where temperature controls are indicated for demolition, retain the services of a temperature control contractor to perform such demolition.
6. This contractor shall retain on the premises in neatly stacked piles where instructed for selection by the owner, all material, wire, fixtures and/or equipment which are specified to be removed or replaced. All such items, not selected for salvage by the owner, shall become the property of this contractor and shall be removed from the premises and legally disposed.
7. Conform to all applicable codes for demolition of items and systems, safety of adjacent systems, dust control, legal run-off control, disposal and all items necessary to complete the work completely.
8. Demolition shall be done in a manner so as not to damage adjacent work and not affect the operation of systems to remain in use. Any item to remain that is damaged by the contractor shall be replaced and/or repaired at the contractor's expense.
9. Demolition and cutting shall be done in a manner which does not deform or apply loads to the existing framing and equipment of the building to remain.
10. All walls, ceilings, floors, etc., being disturbed by the work shall be returned to finished conditions to match existing by the contractor and contractor shall do his own cutting and patching as necessary under his contract.
11. The contractor shall maintain existing services to and in the existing area as required.
12. The existing systems to remain are to be supported as required until the modified elements are installed and supported.
13. If necessary, the contractor shall provide temporary services in the existing areas.
14. Existing slabs shall be saw-cut in a manner that does not cause the steel framing or the rebar supporting the slab to be cut. Contractor shall field verify slab thickness and rebar spacing.
15. Existing slabs shall be core drilled at reentrant corners of new floor openings to prevent over cutting.
16. The demolished systems shall be reduced to pieces of a weight, and transported across the remaining structure in a manner, such that the remaining structure is not overstressed.
17. The electrical contractor shall disconnect and remove electric service to all mechanical equipment being removed as a result of the renovation.
18. Equipment and devices shall be removed complete including hangers, supports, controls, conduit, wire, pipes, ductwork, etc. Wiring shall be disconnected at circuit breakers, removed and breakers marked "spare."
19. All open ended piping and ductwork that is to remain shall be capped and property secured.
20. Any existing pipes, ductwork, conduit, low voltage control, wiring and/or electrical and mechanical devices being disturbed by the work shall be reworked by this contractor as required to return to its former existing operating condition.
21. Any pipes or ductwork, or control wiring, or tubing feeding through devices or equipment being relocated, reworked, or abandoned and serving other devices, and/or equipment shall be maintained in working condition.
22. Mechanical contractor shall remove and reclaim any refrigerant in existing systems prior to demolition of any equipment according to federal requirements.
23. All asbestos removal will be handled by the owner and is not a part of this work.
24. Use of explosives shall not be permitted.
25. Existing architectural, mechanical and electrical equipment and systems shall be protected from damage resulting from demolition.
26. Contractor shall submit a proposed deconstruction sequence to the owner and architect for review prior to commencement of work.

EXCAVATING/BACKFILLING

1. The contractor shall familiarize himself with the survey and the geotechnical investigation report before starting construction. All underground work shall be in accordance with the recommendations of the geotechnical report except where noted otherwise on drawings or specifications.
2. All building pad preparation and patching shall follow the recommendations of the geotechnical report and the structural drawings and architectural drawings (uno).
3. All objectionable materials encountered are to be removed from excavated areas of the site per the geotechnical report.
4. If unstable subgrade sectors cannot be stabilized by excavation and re-compaction, then crushed stone or similar coarse aggregate materials shall be rolled into the subgrade until a firm subgrade reaction is achieved.
5. The geotechnical engineer shall determine on site or off site imported material that can be used for engineered fill. All fill material shall be approved by the geotechnical engineer.
6. The proposed engineered fill materials are to be placed in lifts not exceeding eight (8) inches in loose measured thickness. Each lift is to be compacted as follows:
   - A. Slab on grade: Minimum of 95 percent maximum density by ASTM D698.
7. All fill materials shall be free of organic contaminations and other deleterious matter.
8. For back fill against basement walls, retaining walls, footings, etc., place in 8 inch thick layers, with each lift compacted at near optimum moisture content, until a minimum in place density of 95 percent of the maximum density as determined by ASTM D698 is achieved.
9. All soil surrounding and under footing shall be protected from frost action and freezing during the course of construction.
10. Notify structural engineer of any unusual soil conditions that are in variance with the geotechnical report.

CONCRETE PADS

1. This contractor shall provide to the general contractor, dimensions for the concrete foundations or bases under all equipment that rests on floors in mechanical equipment rooms or outside on grade. He shall follow drawings and/or manufacturer's literature with regard to design and construction of same. In the absence of more specific information, either on drawings or manufacturer's literature, the bases shall be level, shall have a minimum height above finished floor of 4 inches and extend 3 inches beyond the skids, feet or bed plate of the item of equipment.
2. Concrete pedestals and/or saddles for support of piping and/or tanks shall be designed to withstand stresses to which they may be subjected and to distribute properly the load and impact over building areas.
3. Concrete shall be a of mix producing compressive strength after 28 days of 3,000 psi minimum reinforcing and bolts shall be provided as required and work shall be done in accordance with applicable articles covering concrete work.
4. Space approximately 1 inch thick between bottom of equipment and top of concrete foundation or base which remains after shimming, shall be filled completely with grouting. Grout shall be made up with sand and cement designed for the purpose which does not shrink on setting up. Exposed surface of grouting shall be finished to make a neat appearance.

ABBREVIATIONS

A	AMPS
ACU	AIR CONDITIONING UNIT
AD	AREA DRAIN
ADDL	ADDITIONAL
AF	AIRFOIL BLADE FAN
AFC	ABOVE FINISH COUNTER
AFF	ABOVE FINISH FLOOR
AFG	ABOVE FINISH GRADE
AHU	AIR HANDLING UNIT
AP	ACCESS PANEL
ARCH	ARCHITECTURAL
B	BOILER
BLDG	BUILDING
BOT	BOTTOM
BT	BATHTUB
C	CONDENSER
CB	CATCH BASIN
CC	COOLING COIL
CD	CEILING DIFFUSER
CFH	CUBIC FEET PER HOUR
CFM	CUBIC FEET PER MINUTE
CH	CHILLER
CI	CAST IRON
CJ	CONTROL JOINT
CL (C)	CENTERLINE
CLG	CEILING
CO	CLEANOUT
COL	COLUMN
CONST	CONSTRUCTION
CONT	CONTINUOUS
CONTR	CONTRACTOR
CONV	CONVECTOR
CS	CLINIC SINK
CT	COOLING TOWER
CTX	CONNECT TO EXISTING
CU	CONDENSING UNIT
CUH	CABINET UNIT HEATER
D	DAMPER
DCBP	DOUBLE CHECK BACKFLOW PREVENTER
DCDA	DOUBLE CHECK DETECTOR ASSEMBLY
DE	DISHWASHER EXHAUST
DET	DETAIL
DF	DRINKING FOUNTAIN
DIA (Ø)	DIAMETER
DN	DOWN
DS	DOWN SPOUT
DSW	DISCONNECT SWITCH
DT	DRAIN TILE
DWDI	DOUBLE WIDTH DOUBLE INLET FAN
DWG	DRAWING
DWH	DOMESTIC WATER HEATER
EA	EXHAUST AIR
EBB	ELECTRIC BASEBOARD
EC	ELECTRICAL CONTRACTOR
EF	EXHAUST FAN
EG	EXHAUST GRILLE
EH	ELECTRIC HEATER
EJ	EXPANSION JOINT
EL	ELEVATION
ELEC	ELECTRICAL
ELEV	ELEVATOR
EM	EMERGENCY
EQ	EQUIPMENT
ETR	EXISTING TO REMAIN
EUH	ELECTRIC UNIT HEATER
EWC	ELECTRIC WATER COOLER
EWH	ELECTRIC WATER HEATER
EXIST (E)	EXISTING
EXP	EXPANSION
F	FURNACE
FA	FIRE ALARM
FBD	FACE & BYPASS DAMPER
FC	FORWARD CURVE BLADE FAN
FCO	FLOOR CLEAN-OUT
FCU	FAN COIL UNIT
FD	FLOOR DRAIN / FIRE DAMPER
FFE	FINISH FLOOR ELEVATION
FPB	FAN POWER BOX
FPC	FIRE PROTECTION CONTRACTOR
FS	FLOW SWITCH
FT	FIN TUBE RADIATION / FEET
G	GRILLE (EXHAUST, RETURN, OR TRANSFER)
GA	GAUGE
GALV	GALVANIZED
GC	GENERAL CONTRACTOR
GE	GENERAL EXHAUST
GND	GROUND
GRE	GRAVITY ROOF EXHAUSTER
GUH	GAS UNIT HEATER
GWH	GAS WATER HEATER
H	HUMIDIFIER
HB	HOSE BIBB
HC	HEATING COIL
HE	HOOD EXHAUST
HOA	HAND-OFF-AUTOMATIC
HORIZ	HORIZONTAL
HP	HORSEPOWER
HVAC	HEATING, VENTILATION, AIR CONDITIONING
HX	HEAT EXCHANGER
IE	INVERT ELEVATION
JB	JUNCTION BOX
KEC	KITCHEN EQUIPMENT CONTRACTOR
KHE	KITCHEN HOOD EXHAUST
KVA	KILOVOLT AMPERE
KW	KILOWATT
L	LOUVER
LAV / L	LAVATORY
LT	LAUNDRY TUB
LTG	LIGHTING
MAU	MAKE-UP AIR UNIT
MAX	MAXIMUM
MB	MOP BASIN
MBH	1,000 BTUH
MC	MECHANICAL CONTRACTOR
MECH	MECHANICAL
MFR	MANUFACTURER
MH	MANHOLE
MIN	MINIMUM
MTD	MOUNTED
MXB	MIXING BOX
N	NEW
NEC	NATIONAL ELECTRIC CODE
NF	NON FUSED
NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
NIC	NOT IN CONTRACT
NTS	NOT TO SCALE
OA	OUTSIDE AIR
OAI	OUTSIDE AIR INTAKE
OC	ON CENTER
OD	OVERFLOW DRAIN
P	PUMP
PC	PLUMBING CONTRACTOR
PH (ø)1 PHASE	
PNL	PANEL
PRE	POWER ROOF EXHAUSTER
PRV	PRESSURE REDUCING VALVE
PSF	POUNDS/SQUARE FOOT
PSI	POUNDS/SQUARE INCH
PVC	POLYVINYL CHLORIDE
RA	RETURN AIR
RAD	RADIUS
RD	ROOF DRAIN
RE	RETURN EXHAUST
REF	RETURN EXHAUST FAN
REQ'D	REQUIRED
RF	RETURN FAN
RG	RETURN GRILLE
RHC	REHEAT COIL
RPBP	REDUCED PRESSURE BACKFLOW PREVENTER ASSEMBLY
RPZ	REDUCED PRESSURE ZONE
RTU	ROOF TOP UNIT
SA	SUPPLY AIR
SECT	SECTION
SG	SUPPLY GRILLE
SF	SUPPLY FAN
SH	SHOWER
SK	SINK
SOVB	SHUT-OFF VALVE BOX
SQ	SQUARE
SS	SERVICE SINK
STL	STEEL
STRUCT	STRUCTURAL
SW	SAFE WASTE
SWSI	SINGLE WIDTH SINGLE INLET FAN
SYM	SYMMETRICAL
TA	TRANSFER AIR
TC	TEMPERATURE CONTROL
TCC	TEMPERATURE CONTROL CONTRACTOR
TD	TRENCH DRAIN
TG	TRANSFER GRILLE
TPV	TRAP PRIMER VALVE
TS	TAMPER SWITCH
TYP OR T/	TYPICAL
UH	UNIT HEATER
UL	UNDERWRITER'S LABORATORY
UNO	UNLESS NOTED OTHERWISE
UR	URINAL
UV	UNIT VENTILATOR
V	VOLTS
VAV	VARIABLE AIR VOLUME BOX
VD	VOLUME DAMPER
VERT	VERTICAL
VT	VITRIFIED TILE
VTR	VENT THRU ROOF
W	WATTS
WI	WITH
WC	WATER CLOSET
WCO	WALL CLEANOUT
WH	WALL HYDRANT
WP	WEATHERPROOF
YH	YARD HYDRANT
XFMR	TRANSFORMER

MECHANICAL LEGEND

CD	CONDENSATE DRAIN PIPING
CW	DOMESTIC COLD WATER PIPING
HW	DOMESTIC HOT WATER PIPING
RHW	DOMESTIC RECIRCULATING HOT WATER PIPING
HW (120°)	DOMESTIC HOT WATER PIPING - 120° F
RHW (120°)	DOMESTIC RECIRCULATING HOT WATER PIPING - 120° F
HW (140°)	DOMESTIC HOT WATER PIPING - 140° F
RHW (140°)	DOMESTIC RECIRCULATING HOT WATER PIPING - 140° F
G	GAS PIPING
GW	GREASE WASTE
GV	GREASE WASTE VENT
RL	REFRIGERANT LIQUID
RS	REFRIGERANT SUCTION
FCW	FILTERED COLD WATER PIPING
SROCW	SOFT REVERSE OSMOSIS WATER PIPING
RO	REVERSE OSMOSIS WATER PIPING
SAN	SANITARY SEWER PIPING
SCW	SOFT COLD WATER PIPING
SHW	SOFT HOT WATER PIPING
SHW(120°)	SOFT HOT WATER PIPING
SHW(140°)	SOFT HOT WATER PIPING
V	VENT PIPING
	PIPING ABOVE GRADE/FLOOR
	PIPING BELOW GRADE/FLOOR
	EXISTING PIPING TO BE REMOVED
	EXISTING PIPING TO REMAIN
	2-WAY MODULATING VALVE
	3-WAY MODULATING VALVE
	AUTOMATIC CONTROL VALVE
	BALANCE VALVE
	CHECK VALVE
	MODULATING VALVE
	PRESSURE REGULATING / REDUCING VALVE
	RELIEF VALVE
	SHUTOFF VALVE
	PRESSURE GAUGE W/COCK
	STRAINER
	THERMOMETER
	PIPE UNION
M	UTILITY METER
	DIRECTION OF DOWNWARD PITCH
	DIRECTION OF FLOW
DTS	DUCT TEMPERATURE SENSOR
	NEW DUCTWORK
(E)	EXISTING DUCTWORK TO REMAIN
MFSD	MOTORIZED FIRE/SMOKE DAMPER
FD	FIRE DAMPER
SD	SMOKE DAMPER
MOD	MOTOR OPERATED DAMPER
BDD	BACKDRAFT DAMPER

ROOMS OR AREAS SHOWN SHADED ON THIS PLAN, WITH THE FOLLOWING SHADING TYPE ARE DEDICATED FOR A SPECIFIC USE. EXAMPLES INCLUDE ELECTRICAL ROOMS, TECHNOLOGY/DATA CLOSETS, EXIT STAIRWELLS, AND ELEVATOR EQUIPMENT ROOMS. UNDER NO CIRCUMSTANCES SHALL PIPING, DUCTWORK, OR EQUIPMENT BE INSTALLED IN OR ROUTED THROUGH THESE ROOMS OR AREAS EXCEPT FOR BRANCH PIPING OR DUCTWORK SPECIFICALLY SERVING THE ROOM OR AREA. DEDICATED SPACE SHALL EXTEND VERTICALLY FROM FLOOR TO STRUCTURAL CEILING.

60°	DUCT-TYPE SMOKE DETECTOR WITH REMOTE TEST STATION AND AUXILIARY RELAY FURNISHED AND WIRED BY ELECTRICAL CONTRACTOR; INSTALLED IN DUCTWORK BY MECHANICAL CONTRACTOR PER CODE. COORDINATE EXACT LOCATION WITH MECHANICAL CONTRACTOR AND MANUFACTURER. PROVIDE CONDUIT AND WIRING NECESSARY TO SHUT DOWN HVAC UNIT UPON ACTIVATION OF SMOKE DETECTOR.
T2	TEMPERATURE CONTROL 120V JUNCTION BOX FOR LOW VOLTAGE CONTROL TRANSFORMER. POWER WIRING, BOX, AND 120V POWER BY ELECTRICAL CONTRACTOR. TEMPERATURE CONTROL CONTRACTOR SHALL PROVIDE TRANSFORMER AND LOW VOLTAGE POWER AND WIRING
CO2	CARBON DIOXIDE DETECTOR
CO	CARBON MONOXIDE DETECTOR
H RTU-1	HUMIDISTAT
T RTU-1	THERMOSTAT
S	SENSOR
TS	TEMPERATURE SENSOR
HS	HUMIDITY SENSOR
	POINT OF CONNECTION
XX / XX	EQUIPMENT TAG
CUH-1	CABINET UNIT HEATER DESIGNATION CUH-1 = SEE SCHEDULE
UH-1	UNIT HEATER DESIGNATION UH-1 = SEE SCHEDULE
	SUPPLY OR OUTDOOR AIR DUCT
	RETURN OR RELIEF DUCT
	EXHAUST DUCT
	DUCT LINING
SA-1	DUCT SOUND ATENUATOR SA-1 = SEE SCHEDULE
	DUCT UP
	DUCT DOWN
	AIRFLOW DIRECTION
UC	3/4" DOOR UNDERCUT
	DUCT RISE
	TRANSFER AIR
	DIFFUSER WITH BLANK -OFF SECTION
	CENTERLINE

BARNES&NOBLE

PROJECT DESIGNER:

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CO. LICENSE NUMBER: 42050
EXPIRES: 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

These plans are an instrument of service and the property of the Architect. Infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:

MECHANICAL
NOTES & LEGENDS

Drawing Number:

M-100

MECHANICAL INFORMATION IS BASED ON SITE OBSERVATION AND EXISTING DRAWINGS. THE GENERAL CONTRACTOR SHALL FIELD VERIFY THE EXISTING CONDITIONS FOR ACCURACY. NO ADDITIONAL COMPENSATION SHALL BE ALLOWED FOR REVISIONS TO THE DESIGN TO CORRECT FOR EXISTING CONDITIONS NOT AS INDICATED ON THE PLANS.

PLUMBING DEMOLITION NOTES

A. IN EVERY INSTANCE OF DEMOLITION AND/OR REMODELING, THE P.C. SHALL FIGURE A COMPLETE JOB AS NONE OTHER SHALL BE ACCEPTED.

B. THE DRAWINGS ARE TO BE USED ONLY AS A GUIDELINE FOR DEMOLITION. THE P.C. MUST VISIT THE SITE PRIOR TO BIDDING TO VERIFY ALL WORK REQUIRED FOR A COMPLETE JOB AND INCLUDE THE COST OF SUCH WORK IN HIS BID.

C. THE P.C. SHALL BE RESPONSIBLE TO REMOVE ALL EXISTING EQUIPMENT AND RELATED PIPING, INSULATION AND HANGERS, ETC.. ALL PIPING FROM DEMOLISHED EQUIPMENT AND/OR FIXTURES SHALL BE REMOVED. ALL EXPOSED UNUSED PIPING SHALL BE REMOVED. PIPING THAT PASSES THROUGH THE SLAB OR WALL SHALL BE CUT BELOW OR FLUSH WITH THE FINISHED FLOOR OR WALL. PATCHING SHALL BE BY G.C.

D. THE P.C. SHALL MAINTAIN SERVICES TO ALL EXISTING EQUIPMENT TO REMAIN, OR ANY OTHER EQUIPMENT UNAFFECTED BY DEMOLITION, BASED ON THE PHASING PLAN. THIS REQUIREMENT INCLUDES BUT IS NOT LIMITED TO THE FOLLOWING: WATER SERVICE, GAS SERVICE, SANITARY SEWERS AND STORM SEWERS.

E. ANY EXISTING PIPING AND OR MECHANICAL DEVICES BEING DISTURBED BY THE WORK SHALL BE REWORKED BY THIS CONTRACTOR AS REQUIRED TO RETURN IT TO ITS FORMER EXISTING OPERATING CONDITION.

F. ANY PIPING FEEDING THROUGH EQUIPMENT BEING RELOCATED, REWORKED, OR ABANDONED AND SERVING OTHER AREAS SHALL BE MAINTAINED AS REQUIRED.

G. THE P.C. SHALL DISCONNECT AND REMOVE WATER SERVICE AND GAS SERVICE TO ALL MECHANICAL AND STORE EQUIPMENT BEING REMOVED AS A RESULT OF THE REMODELING.

H. EXISTING MATERIALS SHALL BE TURNED OVER TO THE OWNER. IF NOT REQUIRED BY OWNER, THE PLUMBING CONTRACTOR SHALL REMOVE THESE MATERIALS FROM THE PREMISES.

I. EXISTING EQUIPMENT, PIPING OR OTHER PLUMBING RELATED EQUIPMENT THAT CONTAIN PCB'S, MERCURY, OR OTHER HAZARDOUS WASTE SHALL BE DISPOSED OF PROPERLY PER ALL EPA AND OTHER APPLICABLE GUIDELINES.

J. LOTS AND STREETS SHALL BE MAINTAINED FREE OF DIRT AND DEBRIS DURING CONSTRUCTION.

K. PLUMBING CONTRACTOR TO CONFIRM ALL EXISTING CONDITIONS PRIOR TO COMMENCEMENT OF DEMOLITION - REPORT ANY DISCREPANCIES TO THE ARCHITECT IMMEDIATELY.

L. PLUMBING CONTRACTOR TO COORDINATE ALL DEMOLITION WITH ALL OTHER TRADES PRIOR TO COMMENCEMENT OF DEMOLITION.

MECHANICAL DEMOLITION NOTES

A. IN EVERY INSTANCE OF DEMOLITION AND/OR REMODELING, THE M.C. SHALL FIGURE A COMPLETE JOB AS NONE OTHER SHALL BE ACCEPTED.

B. THE DRAWINGS ARE TO BE USED ONLY AS A GUIDELINE FOR DEMOLITION. THE M.C. MUST VISIT THE SITE PRIOR TO BIDDING TO VERIFY ALL WORK REQUIRED FOR A COMPLETE JOB AND INCLUDE THE COST OF SUCH WORK IN HIS BID.

C. EXISTING MATERIALS SHALL BE TURNED OVER TO THE OWNER. IF NOT REQUIRED BY OWNER, THE MECHANICAL CONTRACTOR SHALL REMOVE THESE MATERIALS FROM THE PREMISES.

D. VERIFY ALL DEMOLITION WORK REQUIRED WITH THE LANDLORD'S REPRESENTATIVE, PRIOR TO BID.

E. VERIFY ANY WORK REQUIRED TO PROTECT FIRE ALARM/SAFETY SYSTEM WIRING AND DEVICES, SPRINKLER PIPING, LANDLORD DUCTWORK, ETC. DURING DEMOLITION AND NEW CONSTRUCTION.

F. CONTRACTOR SHALL MAINTAIN EXISTING SERVICES TO AND IN THE EXISTING AREA AS REQUIRED BY OWNER. IF NECESSARY, THE CONTRACTOR SHALL PROVIDE TEMPORARY SERVICES IN THE EXISTING AREAS.

G. THE CONTRACTOR SHALL VISIT THE SITE & BECOME FAMILIAR WITH ALL EXISTING SYSTEMS & MATERIALS WHICH MUST BE REMOVED TO ACCOMMODATE NEW CONSTRUCTION. REMOVE ALL AIR TERMINALS, EXHAUST FANS, DUCTWORK, CONTROLS, HANGERS, INSERTS, GRILLES, ETC., UNLESS INDICATED TO REMAIN. PATCH ALL EXISTING WALLS OR FLOORS WHERE DUCT OR PIPE PENETRATIONS HAVE BEEN REMOVED TO MATCH EXISTING, UNLESS SPECIFICALLY NOTED TO BE PATCHED BY OTHERS. LEGALLY DISPOSE OF ALL EQUIPMENT, DUCTWORK, PIPING, ETC., WHICH IS REMOVED, EXCEPT ANY ITEMS SPECIFICALLY DESIGNATED BY OWNER TO BE SALVAGED. REMOVE & PLACE DESIGNATED SALVAGED EQUIPMENT IN A LOCATION DIRECTED BY THE OWNER.

H. REMOVE MECHANICAL EQUIPMENT AS INDICATED, NOT OTHERWISE INDICATED TO REMAIN, INCLUDING ALL PIPING, CONTROLS & MOUNTING BRACKETS. CAP CONCEALED PIPING BEYOND EXISTING FINISHED SURFACES. PATCH & SEAL ALL PENETRATIONS TO MATCH EXISTING.

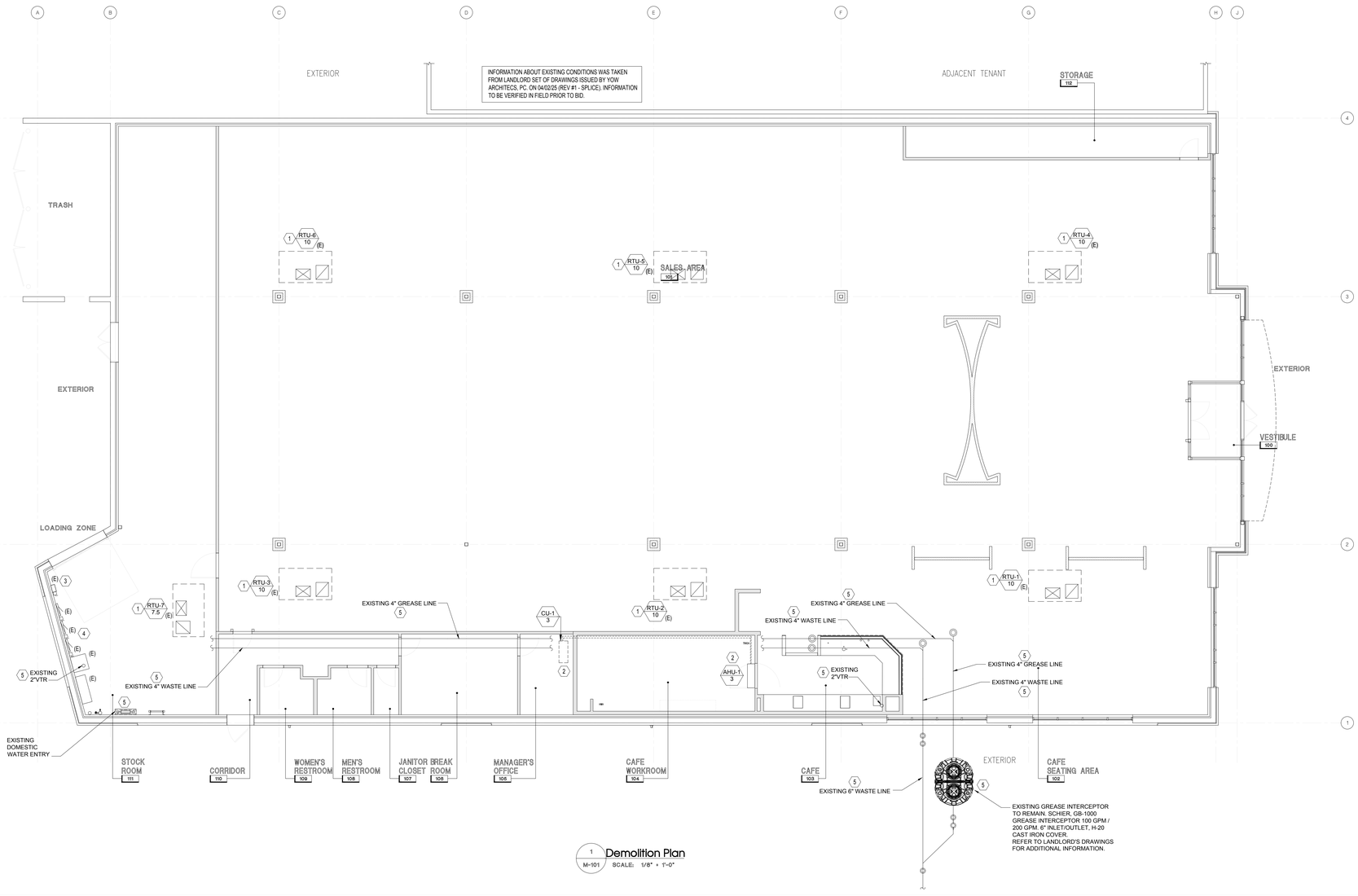
I. EXISTING EQUIPMENT, PIPING OR OTHER PLUMBING RELATED EQUIPMENT THAT CONTAIN PCB'S, MERCURY, OR OTHER HAZARDOUS WASTE SHALL BE DISPOSED OF PROPERLY PER ALL EPA AND OTHER APPLICABLE GUIDELINES. CONTRACTOR SHALL NOTIFY ARCHITECT IF ASBESTOS IS FOUND.

J. LOTS AND STREETS SHALL BE MAINTAINED FREE OF DIRT AND DEBRIS DURING CONSTRUCTION.

K. MECHANICAL CONTRACTOR TO CONFIRM ALL EXISTING CONDITIONS PRIOR TO COMMENCEMENT OF DEMOLITION - REPORT ANY DISCREPANCIES TO THE ARCHITECT IMMEDIATELY.

CODED NOTES: (#)

1. EXISTING ROOFTOP UNIT TO REMAIN.
2. EXISTING SPLIT SYSTEM TO REMAIN.
3. EXISTING ELETRIC UNIT HEATER TO REMAIN.
4. EXISTING ELECTRICAL EQUIPMENT TO REMAIN.
5. EXISTING PLUMBING SYSTEM TO REMAIN.



1 M-101 Demolition Plan
SCALE: 1/8" = 1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
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KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

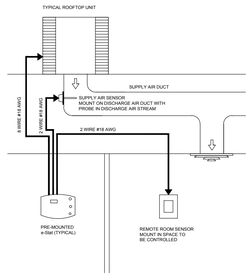
These plans are an instrument of service and the property of the Architect. Infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

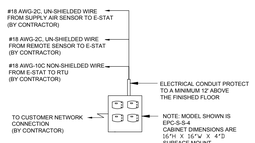
Professional Seal:

Drawing Description:
DEMOLITION PLAN

Drawing Number:
M-101



TYPICAL E-STAT WIRING SCHEMATIC
(FOR REFERENCE ONLY)



TYPICAL E-STAT PRE MOUNTED
ON CABINET
(FOR REFERENCE ONLY)

GENERAL NOTES:

A. THESE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL EXTENT OF THE WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COORDINATION AND PROPER INSTALLATION OF ALL MECHANICAL SYSTEMS. THE CONTRACTOR SHALL PROVIDE ALL NECESSARY OFFSETS AND FITTINGS WHICH MAY BE REQUIRED DUE TO SPACE CONSTRAINTS OR OTHER CONDITIONS.

B. THE CONTRACTOR SHALL PROVIDE ALL MISCELLANEOUS SUPPORTING STEEL, ETC. FOR THE PROPER INSTALLATION OF ALL MECHANICAL SYSTEMS.

C. THE CONTRACTOR SHALL COORDINATE FLOOR, WALL AND ROOF PENETRATIONS, LOUVER SIZES, ETC. WITH GENERAL TRADES.

D. THE CONTRACTOR SHALL VERIFY ALL CLEARANCES PRIOR TO FABRICATION OF ANY WORK.

E. THE CONTRACTOR SHALL COORDINATE THE LOCATION OF CEILING GRILLES, REGISTERS AND DIFFUSERS WITH THE ARCHITECTURAL REFLECTED CEILING PLANS.

F. DUCTWORK AND PIPING SHALL NOT BE LOCATED OVER THE TOP OF ANY ELECTRICAL PANELS OR EQUIPMENT.

G. THE CONTRACTOR SHALL COORDINATE AND PROVIDE ACCESS DOORS IN HARD CEILINGS FOR ALL EQUIPMENT WHICH REQUIRES ACCESS, SUCH AS: FIRE AND SMOKE DAMPERS, SMOKE DETECTORS, BALANCING DAMPERS, VAV BOXES, ETC.

H. CONTRACTOR SHALL FIELD VERIFY SLAB ON GRADE FLOOR CONSTRUCTION TYPE PRIOR TO CUTTING. UNDER NO CIRCUMSTANCES SHALL THE CONTRACTOR CUT A STRUCTURAL FLOOR SLAB THICKER THAN FOUR (4") INCHES WITHOUT PRIOR WRITTEN APPROVAL FROM ENGINEER OF RECORD. NOTIFY ENGINEER OF RECORD OF ANY SLAB THICKNESS GREATER THAN FOUR (4") INCHES PRIOR TO PROCEEDING WITH ANY SAW CUTTING.

CONTRACTOR SHALL FIELD VERIFY ALL EXISTING FIELD CONDITIONS PRIOR TO BEGINNING WORK. HOLD ALL NEW DUCTWORK AS TIGHT TO STRUCTURE ABOVE AS POSSIBLE.

MAINTAIN MINIMUM 10'-0" FROM ALL BUILDING INTAKES AND EXHAUST.

CONTRACTOR SHALL MAINTAIN ALL REQUIRED SERVICE CLEARANCES ON EQUIPMENT PER MANUFACTURER INSTALLATION GUIDELINES AND RECOMMENDATIONS. CONTACT ENGINEER OF RECORD IMMEDIATELY WITH ANY CONFLICTS.

COORDINATE ALL THERMOSTAT REQUIREMENTS WITH LIGHTSTAT.

THESE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL EXTENT OF THE WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COORDINATION AND PROPER INSTALLATION OF ALL MECHANICAL SYSTEMS. THE CONTRACTOR SHALL PROVIDE ALL NECESSARY OFFSETS AND FITTINGS WHICH MAY BE REQUIRED DUE TO SPACE CONSTRAINTS OR OTHER CONDITIONS

MECHANICAL INFORMATION IS BASED ON SITE OBSERVATION AND EXISTING DRAWINGS. THE GENERAL CONTRACTOR SHALL FIELD VERIFY THE EXISTING CONDITIONS FOR ACCURACY. NO ADDITIONAL COMPENSATION SHALL BE ALLOWED FOR REVISIONS TO THE DESIGN TO CORRECT FOR EXISTING CONDITIONS NOT AS INDICATED ON THE PLANS.

DESIGN OF THE NON-DUCTED RETURN AIR PLENUM CEILING IS BASED ON THE REQUIREMENT THAT ALL MATERIALS IN THE RETURN AIR PLENUM CEILING ARE ALL LABELED AS PLENUM RATED AND MEET THE FLAME & SMOKE SPREAD REQUIREMENTS FOR PLENUM INSTALLATION AND THE APPROVAL OF A RETURN AIR PLENUM FROM THE LANDLORD. IF THE ABOVE CONDITIONS CANNOT BE MET, THIS CONTRACTOR SHALL STATE THE AMOUNT TO BE ADDED TO HIS BASE BID FOR A TOTALLY DUCTED RETURN AIR SYSTEM.

CERTIFIED AIR TEST AND BALANCE REPORT TO BE FURNISHED AS PART OF CLOSE OUT PACKAGE.

CODED NOTES:

1. EXISTING ROOFTOP UNIT TO REMAIN.
2. PROVIDE NEW REMOTE TEMPERATURE SENSOR AT 96" A.F.F. SENSOR FURNISHED BY LIGHTSTAT. COORDINATE FINAL LOCATION WITH OWNER.
3. PROVIDE NEW THERMOSTAT ON LIGHTSTAT PANEL. THERMOSTAT FURNISHED BY LIGHTSTAT. COORDINATE FINAL LOCATION WITH OWNER.
4. 10" DIA. EXHAUST DUCT UP THROUGH ROOF. PROVIDE WITH ROOF CURB AND CAP. VERIFY ROUTING AND ROOF PENETRATION LOCATION IN FIELD PRIOR TO BID. COORDINATE ROOF WORK WITH LANDLORD'S ROOFING CONTRACTOR.
5. PROVIDE SMOKE DETECTOR LOCATED IN SUPPLY AND RETURN AIR DUCT TO SHUT DOWN UNIT UNDER ALARM. DETECTORS SHALL BE FURNISHED AND WIRED BY ELECTRICAL CONTRACTOR, INSTALLED BY MECHANICAL. ALL WIRING SHALL BE IN CONDUIT PER N.E.C.
6. EXISTING AIR HANDLING UNIT TO REMAIN. VERIFY EXACT LOCATION IN FIELD PRIOR TO BID.
7. CARBON MONOXIDE DETECTOR TO BE MOUNTED NEAR BY FIRST SUPPLY DIFFUSER. VERIFY EXACT MOUNTED LOCATION IN FIELD PRIOR TO BID.
8. RETURN AIR GRILLE TO BE MOUNTED AS HIGH AS POSSIBLE ABOVE CEILING.
9. EXISTING UNIT HEATER TO REMAIN. VERIFY EXACT LOCATION IN FIELD PRIOR TO BID.
10. 3,200 CFM FULL SIZE RETURN WITH WIRE MESH SCREEN.
11. 3/4" UNDERCUT DOOR
12. 1" COPPER CONDENSATE FROM AIR HANDLING UNIT TO MOP SINK. MATERIALS TO CONFORM TO NFPA90A. PIPE TO PITCH 1/4" PER FOOT.
13. EXISTING REFRIGERANT PIPING FROM EXISTING CONDENSING UNIT TO EXISTING AIR HANDLING UNIT. SIZE AND INSULATE PIPE PER MANUFACTURER'S RECOMMENDATIONS. NOTIFY ARCHITECT IMMEDIATELY IF LENGTH OF RUN EXCEEDS MANUFACTURERS RECOMMENDATIONS.
14. 2,400 CFM FULL SIZE RETURN WITH WIRE MESH SCREEN.
15. EXISTING CONDENSING UNIT LOCATED ON ROOF. VERIFY EXACT LOCATION IN FIELD PRIOR TO BID.

WORK RESPONSIBILITY

TENANT'S MECHANICAL CONTRACTOR SHALL FURNISH AND INSTALL THE FOLLOWING:

1. THIS CONTRACTOR SHALL FURNISH AND INSTALL ONE (1) NEW CEILING UNIT HEATER, THREE (3) NEW EXHAUST FANS, DUCTWORK, DIFFUSERS, REGISTERS, HANGERS, PROGRAMMABLE THERMOSTATS, SMOKE DETECTOR TEST STATIONS, AND ALL ITEMS REQUIRED TO PRODUCE A COMPLETE AND OPERABLE HVAC SYSTEM.

2. DUCTWORK: THE MECHANICAL CONTRACTOR IS TO FURNISH AND INSTALL, IN COMPLIANCE WITH THE MOST RECENT SMACNA STANDARDS FOR LOW AND MEDIUM PRESSURE, NEW DUCTWORK, INSULATION, FLEX DUCT, GRILLES, REGISTERS, DIFFUSERS, VOLUME DAMPERS, FIRE DAMPERS, SMOKE DETECTORS, SECONDARY CONDENSATE DRAIN, ETC. NECESSARY TO RENDER THE SYSTEM OPERATIONAL AS DESCRIBED IN THESE PLANS AND SPECIFICATIONS AND AS REQUIRED BY THE LANDLORD, LOCAL AND STATE CODES.

3. ALL DUCTWORK SHALL BE HUNG AS HIGH AS POSSIBLE TO MAINTAIN ARCHITECTURAL CEILING HEIGHT REQUIREMENTS.

4. ALL OUTSIDE AIR AND UNEXPOSED DUCTWORK WITHIN BUILDING, EXCEPT WHERE ACOUSTICALLY LINED, SHALL HAVE 3 INCH IN CLIMATE ZONES 1-4 AND 4" IN CLIMATE ZONES 5-8, FIBERGLASS DUCT WRAP INSULATION WITH FSK FACING EQUIVALENT TO JOHNS MANVILLE "MICROLITE EQ TYPE 75" (INSTALLED "R VALUE" = 8.3 FOR 3" AND 12 FOR FOR 4").

5. FIRE DAMPERS MUST BE INSTALLED AT ALL LOCATIONS WHERE DUCTWORK PENETRATES A FIRE RATED WALL. PROVIDE ACCESS DOORS AS REQUIRED. FIRE DAMPERS TO BE OF THE TYPE APPROVED BY THE AGENCIES HAVING JURISDICTION.

6. WHEN NEW DUCTWORK CONFLICTS WITH EXISTING DUCTWORK, PIPING, ETC., NEW DUCTWORK SHALL BE SET UP OR DOWN AS REQUIRED.

7. PROVIDE VOLUME DAMPERS ON ALL NEW SUPPLY AIR DUCT SPLITS AND TAPS.

8. POWER WIRING:

A. THE ELECTRICAL CONTRACTOR IS TO FURNISH AND INSTALL ALL EQUIPMENT AND MATERIAL REQUIRED TO PROVIDE POWER TO THE ROOFTOP UNITS FROM THE TENANT'S POWER SUPPLY.

B. THE ELECTRICAL CONTRACTOR IS TO FURNISH AND INSTALL A DISCONNECT SWITCH ON THE ROOFTOP UNITS.

C. THE ELECTRICAL CONTRACTOR IS TO PROVIDE POWER WIRING TO THE EXHAUST FANS.

9. CONTROL WIRING AND CONTROLS: THE MECHANICAL CONTRACTOR IS TO FURNISH AND INSTALL ALL NECESSARY WIRING (IN CONDUIT IF REQUIRED) AND CONTROLS REQUIRED TO PROVIDE A COMPLETE AND OPERATING SYSTEM.

10. WHEN THE SMOKE DETECTOR, IN SUPPLY & RETURN DUCTWORK, ACTIVATES IT SHALL SHUT DOWN THE ROOFTOP UNIT.

11. REFER TO ARCHITECTURAL DRAWINGS FOR EXACT LOCATION OF ALL CEILING DIFFUSERS AND REGISTERS.

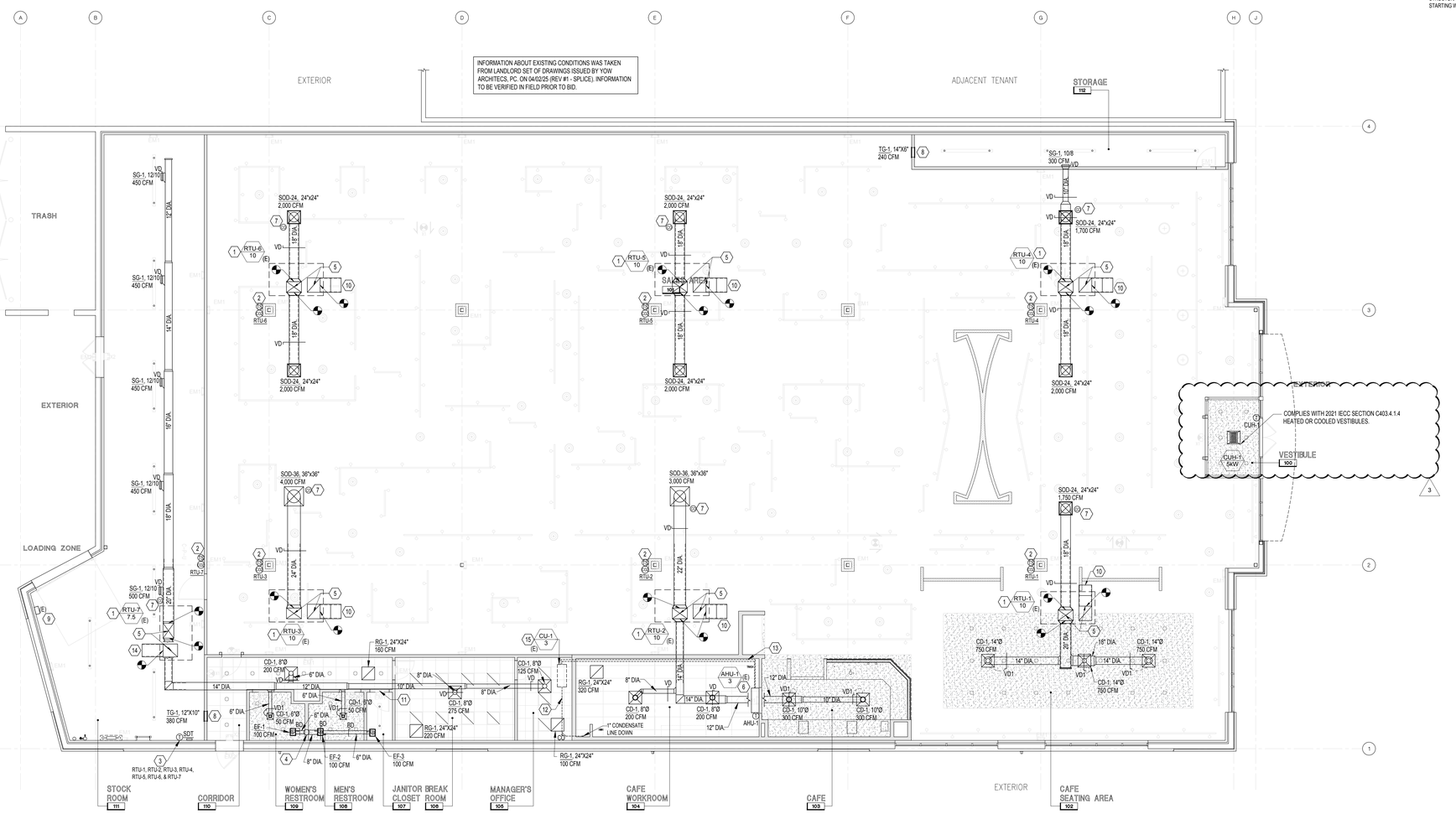
12. THE MECHANICAL CONTRACTOR IS TO FURNISH AND INSTALL A CONDENSATE DRAIN SYSTEM PER THE MANUFACTURERS RECOMMENDATIONS AND THE LANDLORD'S REQUIREMENTS. INTERIOR CONDENSATE DRAIN LINE SHALL BE INSULATED WITH 1/2" THICK ARMAFLEX. PITCH 1/4" PER FOOT.

13. TENANT'S MECHANICAL CONTRACTOR IS RESPONSIBLE FOR COMPLYING WITH ALL THE LANDLORD'S SPECIFICATIONS FOR THIS INSTALLATION.

14. THIS CONTRACTOR IS TO HIRE LANDLORD'S SPECIFIED CONTRACTOR FOR ALL ROOF PENETRATIONS.

15. THE MECHANICAL CONTRACTOR, AS PART OF THEIR WORK, IS TO STENCIL THE TENANT NAME AND SPACE NUMBER ON ALL ROOFTOP EQUIPMENT WITH 4" HIGH BLOCK LETTERS (2" HIGH BLOCK LETTERS ON ALL SMALL EQUIPMENT WHERE 4" WILL NOT FIT) IN A COLOR APPROVED BY THE BUILDING MANAGER / CONSTRUCTION INDIVIDUAL.

16. IF STRUCTURAL DRAWINGS FOR HVAC EQUIPMENT SUPPORTS ARE NOT ALREADY INCORPORATED INTO THIS SET OF PLANS AND SPECIFICATIONS, THE MECHANICAL CONTRACTOR, AT THEIR OWN COST AND EXPENSE, AND AS PART OF THE BID TO THE G.C., IS TO HIRE A STRUCTURAL ENGINEER TO DESIGN THE SUPPORTS FOR THE NEW HVAC UNITS AND A STRUCTURAL SUBCONTRACTOR TO FURNISH AND INSTALL SUCH HANGERS / SUPPORTS, COLUMNS, BEAMS, BRACING, ETC. TO HANG FROM THE STRUCTURAL AND / OR SUPPORT AT THE ROOF TOP FOR ALL NEW HVAC EQUIPMENT. G.C. TO SUBMIT AS REQUIRED ALL STRUCTURAL SHOP DRAWINGS TO THE LANDLORD'S ARCHITECT, AS REQUIRED, FOR APPROVAL, PRIOR TO STARTING WORK.



1 Mechanical Plan
M-102 SCALE: 1/8" = 1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

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BUILDING VALUE SINCE 1994

ARCHITECT:
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NEW JERSEY LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES: 09-30-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3
08-08-25	BD COMMENTS	4

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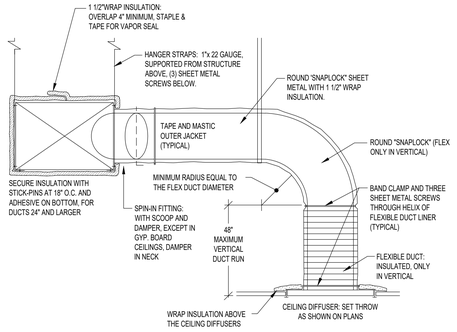
Professional Seal:

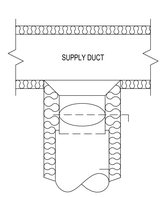
Drawing Description:
HVAC PLAN

Drawing Number:
M-102



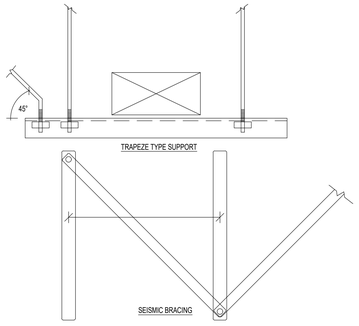
CEILING DIFFUSER CONNECTION DETAIL

N.T.S.



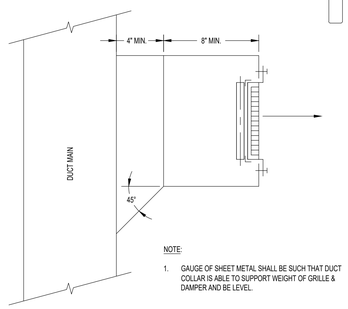
ROUND DUCT TAKE-OFF DETAIL

N.T.S.



DUCT SUPPORT DETAILS

N.T.S.

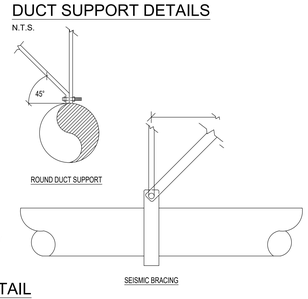


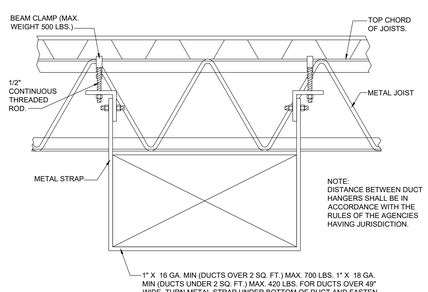
SUPPLY REGISTER DUCT CONNECTION DETAIL

N.T.S.

NOTE:

1. GAUGE OF SHEET METAL SHALL BE SUCH THAT DUCT COLLAR IS ABLE TO SUPPORT WEIGHT OF GRILLE & DAMPER AND BE LEVEL.

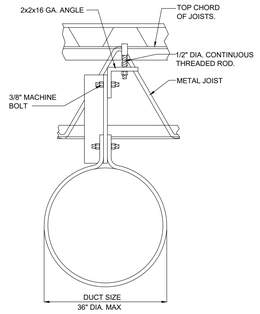




DUCTWORK SUPPORT DETAIL

N.T.S.

MAX. SIDE INCHES	RECTANGULAR DUCTS MIN. GALV. SHT. GAUGE	ALUMINUM MIN. B & S GAUGE
THROUGH 12	26 (0.022 IN.)	24 (0.020 IN.)
13 THROUGH 30	24 (0.028 IN.)	22 (0.025 IN.)
31 THROUGH 54	22 (0.034 IN.)	20 (0.032 IN.)
55 THROUGH 84	20 (0.040 IN.)	18 (0.040 IN.)
OVER 84	18 (0.052 IN.)	16 (0.051 IN.)

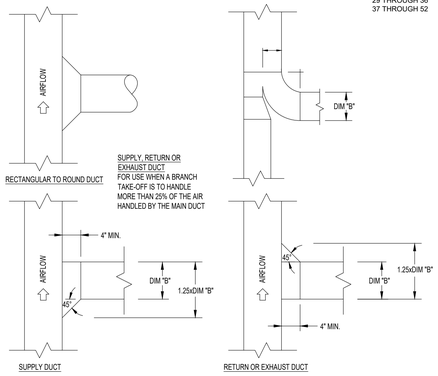


MAX WEIGHT PER FOOT = 16 LBS.

ROUND DUCT SUPPORT DETAIL

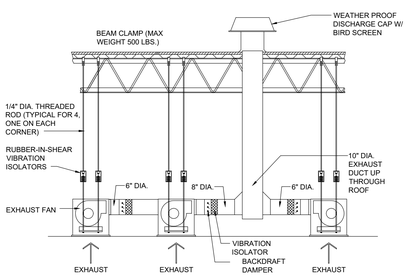
N.T.S.

DIAMETER INCHES	SPIRAL SEAM DUCT STEEL MIN. GALV. SHT. GAUGE	ROUND DUCTS LONGITUDINAL SEAM DUCT STEEL MIN. GALV. SHT. GAUGE	FITTINGS STEEL MIN. GALV. SHT. GAUGE
THROUGH 12	26 (0.019 IN.)	26 (0.022 IN.)	26 (0.022 IN.)
13 THROUGH 18	26 (0.022 IN.)	24 (0.028 IN.)	24 (0.028 IN.)
19 THROUGH 28	24 (0.028 IN.)	22 (0.034 IN.)	22 (0.034 IN.)
29 THROUGH 36	22 (0.034 IN.)	20 (0.040 IN.)	20 (0.040 IN.)
37 THROUGH 52	20 (0.040 IN.)	18 (0.052 IN.)	18 (0.052 IN.)



DUCT BRANCH TAKE-OFF DETAILS

N.T.S.



EXHAUST FANS DETAIL

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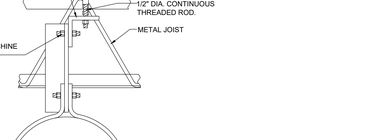
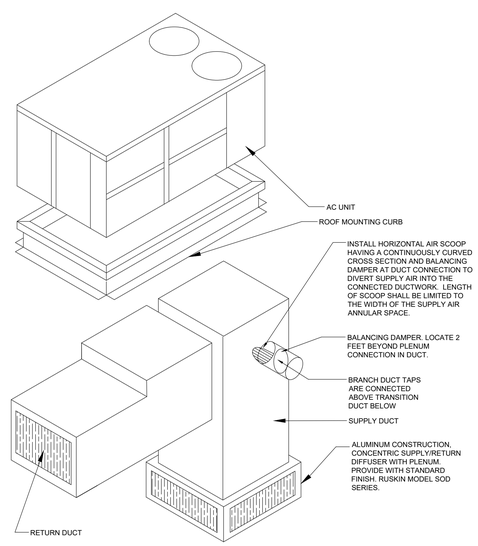


REMOTE VOLUME DAMPER DETAIL

N.T.S.



CONCENTRIC DIFFUSER DETAIL

N.T.S.

HEAT GAIN CALCULATIONS

	ITEM	ROOM NUMBER: / NAME					
	1	AREA (SQ. FT.)	17,806	DESIGN CONDITIONS		DRY BULB	WET BULB
	2	CEILING HEIGHT (FT.)	30		OUTSIDE	90 F	58 F
	3	VOLUME (CU. FT.)	534,180		INSIDE	78 F	60% RH
SENSIBLE GAINS			FACTOR	QUANTITY	BTUH	QUANTITY	BTUH
ABOVE CEILING	4	ROOF	5.16				
	5	WALL					
	6	LIGHTS	3.41				
	7						
	8	SUB-TOTAL (4 THRU 7)					
BELOW CEILING	9	ROOF	5.16	17,806	91,879		
	10	WALL	4.47	16,382	73,154		
	11	GLASS	58.40	741	43,271		
	12	PEOPLE	250	267	66,773		
	13	INFILTRATION					
	14	LIGHTS	3.41	17,800	60,751		
	15	OUTSIDE AIR	10.34	5,400	55,809		
	16	EQUIPMENT	138	160	20,717		
	17	SUB-TOTAL (9 THRU 16)			412,353		
LATENT GAINS							
BELOW CEILING	18	PEOPLE	200	267	53,418		
	19	INFILTRATION					
	20	EQUIPMENT					
	21	OUTSIDE AIR	-27.59	5,400	-149,013		
	22	SUB-TOTAL (18 THRU 21)			-95,595		
ROOM CONDITION							
BELOW CEILING	23	TOTAL LOAD (17 + 22)			316,758		
	24	SENSIBLE HEAT FACTOR (1 - (22 / 17))			1.23		
	25	SUPPLY AIR TEMP. DIFF.					
	26	SUPPLY AIR CFM ((17 / (1.08 X 25))			27,000		
	27	CFM PER SQ. FT. ((26 / 1)			1.52		
	28	AIR CHANGES PER HOUR ((26 X 60) / 3)			3.0		
	29	ROOM GRAND TOTAL (8 + 17 + 22)			316,758		
	30	AVG. ROOM LOAD BTUH PER SQ. FT. (29 / 1)			18		
	31	TOTAL TENANT AREA (1)				17,806	
	32	TENANT GRAND TOTAL LOAD (29)					316,758
	33	AVG. TENANT LOAD BTUH PER SQ. FT. (32 - 31)					18
	34	AVG. TENANT CFM PER SQ. FT. (26 / 31)					1.52

HEAT LOSS CALCULATIONS

	ITEM	ROOM NUMBER: / NAME					
	1	AREA (SQ. FT.)	17,806	DESIGN CONDITIONS		DRY BULB	
	2	CEILING HEIGHT (FT.)	30		OUTSIDE	-1 F	
	3	VOLUME (CU. FT.)	534,180		INSIDE	70 F	
EXTERIOR LOSSES			FACTOR	QUANTITY	BTUH	QUANTITY	BTUH
DAY CYCLE	4	ROOF	7.11	17,806	126,601		
	5	WALL	9.24	16,382	151,419		
	6	GLASS	45.60	741	33,718		
	7	INFILTRATION					
	8	OUTSIDE AIR	76.79	5,400	414,655		
	9	SUB-TOTAL (4 THRU 8)			726,393		
NIGHT CYCLE	10	ROOF	7.11	17,806	126,601		
	11	WALL	9.24	16,382	151,419		
	12	GLASS	45.60	741	33,718		
	13	OUTSIDE AIR	76.79				
	14	SUB-TOTAL (10 THRU 13)			311,738		
INTERIOR GAINS							
DAY CYCLE	16	LIGHTS 50% CREDIT	3.41	8,900	30,349		
	16	OTHER					
	17	SUB-TOTAL (15 + 16)			30,349		
NET HEAT LOSS							
	18	ROOM LOAD (9 - 17)			696,044		
DAY CYCLE	19	AVG. LOAD PER SQ. FT. (18 / 1)			39		
	20	TOTAL TENANT AREA (1)				17,806	
	21	TENANT GRAND TOTAL LOAD (18)					696,044
	22	AVG. TENANT LOAD BTUH PER SQ. FT. (21 / 20)					39
NIGHT CYCLE	23	AVG. ROOM LOAD BTUH PER SQ. FT. (14 / 1)					17.6
	24	TENANT GRAND TOTAL LOAD (14)					311,738
	25	AVG. TENANT LOAD BTUH PER SQ. FT. (24 / 20)					17.6

VENTILATION CALCULATIONS PER 2021 INTERNATIONAL MECHANICAL CODE

Description	Area (Ft2) Az	People Outdoor Air Rate CFM/person Table 403.3 Rp	Area Outdoor Airflow Rate CFM/Ft2 Table 403.3 Ra	Default Occupant Density Per Table 403.3 (People/ 1000 ft2) O	Area Outdoor Air Ra*Az	Zone Population O * Az/1000 Pz	Occupant Outdoor Air Rp*Pz	Breathing Zone Outdoor Air Vbz= RpPz + RaAz	Zone Air Distribution Effectiveness Ez	Zone Outdoor Air Req'd Voz = Vbz/Ez	Zone Outdoor Air Provided	Supply Air Design Vpz	Outdoor Air Percent Zp = Voz/Vpz
Sales	12634	7.50	0.12	15	1516	190	1421	2937	0.80	3672	4100	20500	20%
Storage	340	N/R	0.12	N/R	41			41	0.80	51	60	300	20%
Stockroom	1893	N/R	0.12	N/R	227			227	0.80	284	460	2300	20%
Breakroom	278	N/R	N/R	N/R					0.80		55	275	20%
Café Sitting	1371	7.50	0.12	15	165	21	154	319	0.80	398	440	2200	20%
Café	346	7.50	0.18	10	62	3	26	88	0.80	110	120	600	20%
Café Workroom	426	N/R	0.12	N/R	51			51	0.80	64	80	400	20%
Office	131	5.00	0.06	5.00	8	1	3	11	0.80	14	25	125	20%
Corridor	219	N/R	0.06	N/R	13			13	0.80	16	40	200	20%
Toilet Room	168	N/R	N/R	N/R					0.80		20	100	20%
Totals	17,806				2,083		1,605	3,688		4,610	5,400	27,000	20%

EXISTING ROOFTOP UNIT SCHEDULE

MARK	MANUFACTURER	MODEL	NOM. TONS	AIRFLOW ACFM	AIRFLOW DIRECTION	OA AIR CFM	EXT S.P. IN W.C.	COOLING EAT DB/WB	COOLING TOTAL MBH	HEATING FUEL SOURCE	HEATING INPUT MBH	HEATING OUTPUT MBH	ELECTRICAL VOLTAGE	ELECTRICAL PHASE	ELECTRICAL MCA AMPS	ELECTRICAL MOCP AMPS	APPROX. WEIGHT LBS	ACCESSORIES 1	2	3	4	5	6	7	8	9	10
RTU-1	LENNOX	LGT120H5E	10	4,000	DOWNFLOW	800	2.6	80/67	120.6	NAT GAS	180	144	460	3	25	30	1,618	X	X	X	X	X	X	X		X	X
RTU-2	LENNOX	LGT120H5E	10	4,000	DOWNFLOW	800	2.6	80/67	120.6	NAT GAS	180	144	460	3	25	30	1,618	X	X	X	X	X	X	X		X	X
RTU-3	LENNOX	LGT120H5E	10	4,000	DOWNFLOW	800	2.6	80/67	120.6	NAT GAS	180	144	460	3	25	30	1,618	X	X	X	X	X	X	X		X	X
RTU-4	LENNOX	LGT120H5E	10	4,000	DOWNFLOW	800	2.6	80/67	120.6	NAT GAS	180	144	460	3	25	30	1,618	X	X	X	X	X	X	X		X	X
RTU-5	LENNOX	LGT120H5E	10	4,000	DOWNFLOW	800	2.6	80/67	120.6	NAT GAS	180	144	460	3	25	30	1,618	X	X	X	X	X	X	X		X	X
RTU-6	LENNOX	LGT120H5E	10	4,000	DOWNFLOW	800	2.6	80/67	120.6	NAT GAS	180	144	460	3	25	30	1,618	X	X	X	X	X	X	X		X	X
RTU-7	LENNOX	LGH092H4M1Y	7.5	3,000	DOWNFLOW	600	2.6	80/67	93	NAT GAS	180	144	460	3	23	25	1,426	X	X	X	X	X	X	X		X	X

ACCESSORIES:
1) ENTHALPY CONTROLLED OUTDOOR AIR ECONOMIZER
2) POWER EXHAUST FAN
3) SYSTEM SENSOR SMOKE DETECTOR AND REMOTE KEY OPERATED TEST STATION
4) NON FUSED DISCONNECT SWITCH AND 115V GFI CONVENIENCE RECEPTACLE WIRED TO A SEPARATE CIRCUIT
5) CO2 SENSORS AND CONTROLS WITH LOW VOLTAGE WIRE
6) FACTORY PROVIDED CONDENSATE DRAIN PAN OVERFLOW SWITCH.
7) COMPLETE CONTROL PACKAGE AS SUPPLIED BY LENNOX NATIONAL ACCOUNTS
8) HUMIDITROL PACKAGE WITH HOT GAS RE-HEAT AND HUMIDITY CONTROLS
9) HUMIDISTAT AND LOW VOLTAGE WIRE
10) THERMOSTAT WITH 300FT OF LOW VOLTAGE WIRE

NOTES (APPLICABLE TO ALL UNITS):
1) COORDINATE INSTALLATION WITH OWNER'S CONSTRUCTION MANAGER AND WITH OTHER TRADES.
2) COORDINATE UNIT WITH STRUCTURE.
3) SHIM UNIT AND CURB LEVEL FOR PROPER CONDENSATE DRAINAGE.
4) PROVIDE FLEXIBLE CONNECTORS ON SUPPLY AND RETURN AIR DUCT CONNECTIONS. TRANSITION TO DUCT SIZES SHOWN ON PLAN.
5) INSTALL ALL ACCESSORIES FURNISHED WITH THE UNIT PER THE MANUFACTURER'S INSTRUCTIONS.
6) SET MINIMUM OUTSIDE AIR AS SPECIFIED ABOVE. OUTSIDE AIR DAMPER SHALL FULLY CLOSE ON UNIT SHUTDOWN.
7) RTU CURBS ARE TO BE ORDED FOR EACH RTU
8) ACOUSTICALLY LINED DUCTWORK DROPS
9) CONTROLS SHALL INCLUDE TEMPORARY THERMOSTATS, AND REMOTE TEST/RESET SWITCHES

EXISTING UNIT HEATER SCHEDULE - ELECTRIC HEAT

MARK	MANUFACTURER	MODEL	TYPE	AIRFLOW ACFM	HEATING INPUT SOURCE	HEATING OUTPUT (BTUH)	ELECTRICAL VOLTAGE	ELECTRICAL PHASE	APPROX. WEIGHT (LBS)	NOTES
EUH-1	QMARK	MUH05-81	SUSPENDED	350	ELEC.	5,000	17,000	208	3	27	1,2,3

NOTES:
1) PROVIDE WITH FACTORY MOUNTED AND WIRED DISCONNECT SWITCH.
2) PROVIDE WITH FACTORY TRANSFORMER AND CONTROLLER WITH REMOTE THERMOSTAT.
3) PROVIDE FACTORY BRACKET FOR SUSPENDED INSTALLATION.
4) PROVIDE WITH REQUIRED FACTORY ACCESSORIES FOR RECESSED INSTALLATION.
5) PROVIDE WITH INTEGRAL THERMOSTAT AND DISCONNECT SWITCH.

EXISTING SPLIT SYSTEM SCHEDULE

MARK INDOOR	MARK OUTDOOR	MANUFACTURER	INDOOR MODEL	OUTDOOR MODEL	NOM. TONS	AIRFLOW ACFM	COOLING SOURCE	COOLING TOTAL BTUH	ELECTRICAL VOLTAGE	ELECTRICAL PHASE	ELECTRICAL MCA AMPS	ELECTRICAL MOCP AMPS	APPROX. WEIGHT LBS	ACCESSORIES 1	2	3	4	5
CU-1	AHU-1	MITSUBISHI	PKA-A36KA7	PUY-A36NKA7	3.0	920	COMPRESSOR	24,000	208	1	28	41	211	X	X		X	

ACCESSORIES:
1) FACTORY MOUNTED AND WIRED DISCONNECT SWITCH.
2) FACTORY PROVIDED CONDENSER COIL HAIL GUARDS.
3) FACTORY FURNISHED REMOTE TEMPERATURE SENSOR.
4) FACTORY PROVIDED DRAIN PAN LEVEL SENSOR.

NOTES (APPLICABLE TO ALL UNITS):
1) COORDINATE INSTALLATION WITH OWNER'S CONSTRUCTION MANAGER AND WITH OTHER TRADES.
2) COORDINATE UNIT WITH STRUCTURE.
3) INSTALL UNIT LEVEL FOR PROPER CONDENSATE DRAINAGE.
4) PROVIDE FLEXIBLE CONNECTORS ON SUPPLY AND RETURN AIR DUCT CONNECTIONS. TRANSITION TO DUCT SIZES SHOWN ON PLAN.
5) INSTALL ALL ACCESSORIES FURNISHED WITH THE UNIT PER THE MANUFACTURER'S INSTRUCTIONS.

NEW EXHAUST FAN SCHEDULE

TAG	LOCATION	SERVICE	CFM	SP " W.C.	RPM	DRIVE	TYPE	WHEEL DIA.	WATTS	VOLTAGE PH. / HZ.	MANUFACTURER MODEL NO.	REMARKS
EF-1	TOILET ROOM	EXHAUST	100	0.25"	1,550	DIRECT	BI	-	79	120/1/60	PENNBARRY Z5H	1,3,4
EF-2	TOILET ROOM	EXHAUST	100	0.25"	1,550	DIRECT	BI	-	79	120/1/60	PENNBARRY Z5H	1,3,4
EF-3	JANITOR'S ROOM	EXHAUST	100	0.125"	1,550	DIRECT	BI	-	79	120/1/60	PENNBARRY Z5H	1,2,3

NOTES:
1) FAN SHALL BE MOUNTED IN CEILING WITH VIBRATION ISOLATORS.
2) FAN SHALL BE CONTROLLED BY LIGHT SWITCH.
3) FAN SHALL BE PROVIDED WITH LOW LEAKAGE BACKDRAFT DAMPER.
4) FAN SHALL BE CONTROLLED BY THE TOILET ROOM LIGHT SWITCH

NEW SMOKE DETECTOR TEST STATION

SDT	REMOTE TEST STATION WITH AUDIBLE AND VISUAL ALARM FOR DUCT SMOKE DETECTOR MANUFACTURED BY SIMPLEX MODEL 4098-9842.

NEW VOLUME DAMPER CABLE CONTROLS

TYPE	DESCRIPTION
"VD1"	THE VOLUME DAMPER SHALL BE ADJUSTABLE FROM THE FACE OF THE DIFFUSER BY USE OF THE BOWDEN CABLE CONTROL SYSTEM (#270-275 CONTROLLER) MANUFACTURED BY YOUNG REGULATOR COMPANY OR APPROVED EQUIVALENT. DAMPER MUST BE INSTALLED WITHIN 30 FEET FROM THE FACE OF THE DIFFUSER. COORDINATE INSTALLATION WITH GENERAL CONTRACTOR. YOUNG REGULATOR (440) 232-9700.

NEW GRILLE AND DIFFUSER SCHEDULE

MARK	MANUFACTURER	MODEL	FRAME OR BOARDER TYPE	MODULE SIZE	DAMPER MODEL NO.	FINISH	REMARKS
CD-1	TITUS	OMNI	LAY-IN	SEE PLANS	AG-100	MATCH CEILING CONDITION	1,2
RG-1	TITUS	350 RL	LAY-IN	SEE PLANS	AG-15		
SOD-1	RUSKIN	SOD	LAY-IN	SEE PLANS	AG-15		
TG-1	TITUS	350 RL	SURFACE	SEE PLANS	-		1
SG-1	TITUS	300FS	DUCT MOUNTED	SEE PLANS	AG-15		1

REMARKS:
1. ACCEPTABLE MANUFACTURERS: ANEMOSTAT, KRUEGER, NAILOR, PRICE, TITUS, OR TUTTLE AND BAILEY.
2. ALL CEILING DIFFUSERS ARE 4-WAY THROW UNLESS INDICATED OTHERWISE ON PLAN.

NEW CABINET UNIT HEATER SCHEDULE(ELECTRIC)

TAG	MANUFACTURER MODEL NO.	TYPE	AIRFLOW CFM	MOTOR RPM	CAPACITY (KW)	VOLTS-PH-Hz	MCA	MOCP	REMARKS:
CUH-1 (6kW)	Q-MARK CDF-557	RECESSED	300	1,400	5	277-1-60	18	25	1,2,3

NOTES:
1) ACCEPTABLE MANUFACTURERS: BERKO,BRASH, CHROMALOX, INDEECO, MARKEL, MARLEY, Q-MARK,TRANE, OR VULCAN.
2) PROVIDE WITH INTEGRAL/REMOTE THERMOSTAT. REMOTE THERMOSTAT TO SHUT DOWN UNIT WHEN OUTSIDE TEMPERATURE IS GREATER THAN 45F.
3) PROVIDE WITH RECESSED TRIM KIT.

3

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

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NJ LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER 42050
EXPIRES 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3
08-08-25	BD COMMENTS	4

These plans are an instrument of service and the property of the Architect, Infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:
MECHANICAL
DETAILS
AND SCHEDULES

Drawing Number:
M-201

Section 200500 - General Requirements

A. General
1. Specifications are applicable to all contractors and/or subcontractors for all mechanical systems in Divisions 01, 20, 21, 22, and 23.
2. This contractor is also referred to the architectural, structural, electrical and all other drawings and specifications pertinent to this project and fully coordinate with all other trades, owner and architect requirements. All of the above mentioned drawings and specifications are considered a part of the contract documents.
3. Conform to all Instructions to Bidders, general and special conditions of contract as specified by architect and/or owner.
4. Refer to "Alternate Proposals" for possible changes affecting the extent of this section of work.
5. Before submitting a bid, each contractor is requested to visit the job site to familiarize themselves with construction condition, check facilities and conditions and make all necessary observations and measurements. Note conditions under which work is to be performed and take all items into consideration in bid. No consideration will be given for his failure to do so.
6. Systems are to be complete and workable in all respects, placed in operation and properly adjusted.
7. Each contractor shall provide for his own clean-up, removal and legal disposal of all rubbish daily.
8. Each contractor shall protect his work, his existing and adjacent property against weather.
9. Each contractor shall protect his work, materials, apparatus and fixtures from damage. Any work damaged by failure to provide protection required, shall be removed and replaced with new material at the contractor's expense.
10. Each contractor must confirm all utility company requirements and connection points in field, prior to starting work. Each contractor shall include cost of utility companies work in their bid.
11. Each contractor must confirm size, location and materials at point of tie in connections in the field prior to rough-in of new work.
12. Arrange for and obtain owner's and insurance representative's permission for any service shutdowns.
13. Each contractor shall be solely responsible for construction means, methods, sequences of construction and the safety of workmen.
14. No piping, ductwork, wiring, etc., shall be installed or routed above or below electrical panels and equipment, through elevator equipment rooms or elevator shafts or stairways unless these items serve these areas only.
15. All contractors shall coordinate with the electrical contractor and obtain a written approval identifying the electrical characteristics of all mechanical equipment prior to ordering of equipment. No additional payment will be made for lack of contractor coordination of electrical characteristics.
16. Each contractor shall include modifying existing conditions to complete the project. During construction the contractors may uncover an existing condition that will have to be modified. Any such work which comes under the jurisdiction of this contractor shall be done by this contractor without extra cost to the owner and project.
17. Work related to the existing building shall be coordinated to minimize interference or interruption of normal building use by the owner. Refer to architectural plans for phasing requirements.
18. Ceiling grid systems shall not be supported from ductwork, heating or plumbing lines or any other utility lines, and vice versa. Each utility and the ceiling grid system shall be a separate installation and each shall be independently supported from the building structure - concrete, steel or masonry. Where interferences occur, in order to support ductwork, piping, ceiling grid systems, etc., trapeze type hangers or supports shall be employed which shall be located so as not to interfere with access to such mechanical equipment as valves, regulators, mixing boxes, fire dampers, etc.

B. Work Coordination and Scope
1. Each contractor under this division shall familiarize himself with the work to be done under other divisions of this specification and their related drawings and shall so coordinate and schedule his work as not to cause delays or interference with the work of others. Such coordination and scheduling shall accomplish the installation of mechanical and plumbing equipment and piping with a minimum of cutting through masonry and other adjustments.
2. Work included under this division shall consist of furnishing all materials, supplies, equipment, tools, transportation and facilities and performing all labor and services necessary for the complete installation of the mechanical systems of plumbing, fire protection, heating, ventilating, air conditioning, and specialty systems.
3. The contractor under this division shall report discrepancies in the work of others which affect his work. Any changes made necessary by failure or neglect to report such discrepancies shall be made by and at the expense of the contractor of this division. Obtain written instructions for changes necessary to accommodate work of others.
4. The contractor under this division shall be responsible for proper size and location of anchors, chases, recesses, opening, etc., required for the proper installation of his work.
5. The division of responsibility under separate mechanical, fire protection and plumbing contracts for tie-in points shall be as follows:
   a. The plumbing contractor shall provide domestic water and gas to within five feet (5'-0") of equipment connection furnished by the mechanical or electrical contractor, final connection by mechanical or electrical contractor. On the water lines, the plumbing contractor shall provide the shut-off valve, check valve, backflow preventor and pressure regulator. On the gas lines, the plumbing contractor shall provide the shut-off valve and pressure regulator.
   b. Plumbing contractor shall run the gas, water, sanitary and storm to 5'-0" outside the building or to points as noted on the drawings.
   c. Fire protection, plumbing and mechanical contractor shall provide sleeves to the general contractor for placement in floors, walls, etc. and coordinate such location. The plumbing contractor shall be responsible for flashing at vent roof terminals.
   d. The fire protection, plumbing and mechanical contractor shall check with the architectural drawings concerning the test borings to determine areas of rock which should be included in his excavation work. Failure to adjust for rock conditions shall not warrant cause for additional compensation.
   e. The plumbing contractor shall rough-in and connect all other fixtures and equipment where shown on the drawings but not previously mentioned. Provide with shut-off valves and p-traps with clean-out plug.
   f. The plumbing contractor shall provide gas, cold water and drain for the emergency generator and install valves, etc. Generator furnished by the electrical contractor.
   g. Unless responsibility to provide or furnish is otherwise stated on the electrical or mechanical drawings and electrical and mechanical specifications the contractor, under these divisions shall provide motors, special controls, disconnects, transformers, starters and relays as required for the proper operations of all equipment furnished under this division. All electrical equipment shall conform to requirements set forth under the electrical division and be suitable for operation on 60 cycle current available at the site.
   h. All motors 1/3 HP and smaller shall be single phase motors, 1/2 HP and larger, shall be three phase motors except where otherwise specified. Thermal overload protection for all motors shall be provided. Combination fused disconnect and magnetic line starters with auto-off-test switch shall be provided for all three-phase motors. Thermal overload relays shall be sized for 115 percent of full load motor current. For motors with VFD; motors shall be inverter duty motors that meets current "MG 1 Part 31" specifications. Motors to have a minimum of 20:1 turn down ratio. Motors over 20 Hp shall have shaft ground rings. The installation of all motors, starters and other electrical work under this mechanical division shall be done so as to conform with the National Electric Code. Each motor shall be of squirrel cage type, open-drip proof, normal starting torque, having ball bearings unless otherwise specified. For manufacturers that use PMAC motors, this contractor shall supply VFD's to operate motor.
6. Each contractor shall provide OSHA required small (Guard) system for all roof mounted equipment within 10'- 0" of roof edge where the roof edge does not have a 42" high parapet or higher.

C. Codes, Permits, Standards and Regulations
1. Contractors shall install work in full accordance with rules and regulations of all applicable codes (local, city, county, state, national codes, NFPA, OSHA, etc.), government regulations, utility company requirements, and applicable standards having jurisdiction over premises. This shall include safety requirements of the state department. Do not construe this as relieving contractor from compliance with any requirements of specifications which are in excess of code requirements and not in conflict therewith.
2. Contractors shall secure and pay for all fees, permits, and certificates of inspection incidental to this work required by foregoing authorities. Arrange for all required inspections and approvals.
3. Contractor shall be responsible for payments to all public utilities for work performed by them in connection with provision of service connections required under this division of specifications.
4. Deliver all permits and certificates to architect in duplicate.

D. Design Drawings
1. The design drawings, as submitted, are diagrammatic and are not intended to show exact location of equipment, piping and ductwork unless dimensions are given. Piping and ductwork are to be installed along the general plans shown on the drawings while conforming to actual building conditions. Each contractor shall confirm all dimensions by field measurement.
2. Before entering into a contract, the successful bidder may be required to submit satisfactory evidence to show that the manufacturer of all parts of the equipment offered have been regularly engaged in the manufacture of such equipment for three (3) years and have not less than three (3) installations of a similar type which have been in successful operation under conditions similar to those specified for not less than two (2) years.
3. All equipment, piping and material specified herein after as shown on the drawings shall be furnished and installed by the contractor, unless specifically indicated to the contrary. Installation shall comply with all required "Building Codes" and "Reference Standards."
4. If this contractor proposes to install equipment requiring space conditions other than those as specified and/or shown on the design drawings, or to rearrange the equipment, he shall assume full responsibility and submit drawings for the rearrangement of the space and shall obtain the full approval of the architect prior to start of any work.
5. The exact locations for fixtures, equipment and piping which is not covered by drawings shall be obtained from the architect or his representative in the field and the work shall be laid out accordingly.
6. Drawings and specifications are intended to supplement one another. Any materials or labor called for in one but not the other shall be furnished as if both were mentioned in the specifications and shown on the drawings.

E. Base Bid Equipment, Materials and Substitutions
1. All equipment and materials shall be new, free of defects and UL labeled.
2. Base bid manufacturers are included in the specification or listed in schedules on the drawings. All other manufacturers are considered substitution.
3. The name or make of any article, device, material, form of construction, fixture, etc., stated in this specification, whether or not the words "or approved equal" are used, shall be known as a "standard".
4. All cost shall be based on "standards" specified.
5. The equipment schedules on the drawings indicate manufacturer and their equipment model numbers that this design has been based on. Each contractor is required to bid upon the basis of design and furnish the makes specified.
6. Where more than one make or name is mentioned as being acceptable, it shall be understood that only the name or make referring to the manufacturers model numbers or sizes shall be considered the "Specified Standards." It shall be further understood that other makes and names, even though mentioned, have not been checked for detail and that their size and arrangement are the contractor's responsibility the same as a proposed substitute item. The use of other manufacturer's equipment that is listed as acceptable alternates that entails general trades, structural, mechanical, electrical, etc., revisions is this contractor's responsibility to provide revisions. Any additional cost of such changes shall be paid by the contractor submitting the acceptable alternates which necessitates changes in installing such submitted alternate equipment, even though such costs may be part of another division of work.
7. Bids concerning the use of substitute products must be accompanied by complete specifications and performance characteristic covering these products. Contractor shall provide all available test data and experience records which may be helpful to the architect in evaluating the quality and/or suitability of alternate products.
8. Contractor is also invited to bid on any other similar products the contractor desires to propose as substitutions, stating any difference in cost (add or deduct from base bid cost) for each proposed substitution on the substitution sheet. If the architect decides to accept any of the proposed substitutions, proper notations thereof shall be made in the written contract. Where several makes are mentioned in the specifications and the contractor fails to state that he prefers a particular make in his bid, the owner shall have the right to choose any of the makes mentioned without change in price. No consideration will be given to proposals for alternative products unless submitted with the original bids.
9. Substitutions are subject to the approval of the owner. If a substitution is submitted, it is the contractor's responsibility to evaluate it and certify that the substitution is equivalent in all respects to the base specifications.
10. If substitutions are approved, notify all other contractors, subcontractors, etc., affected by the substitution and fully coordinate with them. Any costs resulting from substitution, whether by this contractor or others, shall be the responsibility of and paid for by the substituting contractor. Approved shop drawings do not absolve this contractor from this responsibility.
11. All equipment shall be installed in full accordance with the manufacturer's data and installation instructions and service clearances. It is this contractor's responsibility to check and confirm these requirements prior to starting of any work.

F. Warranty
1. Fully warrant all materials, equipment and workmanship and the successful operation of all equipment and apparatus installed by this contractor for one (1) year from date of final acceptance.
2. Extend all manufacturers' warranties to owner; including five (5) year compressor and ten (10) year heat exchanger extended warranty on HVAC equipment to include material and labor.
3. Repair or replace without material and labor charge to the owner all items found defective during the warranty periods. In the case of replacement or repair due to failure within the warranty period, the warranty on that portion of the work shall be extended for a minimum period of one (1) year from the date of such replacement or repair.

G. Shop Drawing Submittals
1. Submit shop drawings for mechanical, plumbing, fire protection, and control systems; including but not limited to sheetmetal, plumbing fixtures and equipment with adequate details and scales to clearly show construction. Indicate the operating characteristics for each required item. Clearly identify each item on the submittal as to mark, location and use, using the same identification as provided on the construction documents.
2. Sheetmetal and fire protection shop drawings shall be fully dimensioned and coordinated based on field verified building dimensions and clearances and architectural ceiling layouts. Indicate structural systems, lighting, ductwork and piping at all critical locations.
3. Contractor shall review and indicate his approval of each shop drawing prior to submittal for review. Shop drawings will not be reviewed by the engineer unless the contractor's approval is noted. Do not start work or fabrication until shop drawings have been reviewed by the engineer and returned to the contractor.
4. Submittals will be reviewed only for general compliance with the contract documents and not for dimensions or quantities. The architect and engineer will make every effort to detect and correct errors, omissions, and inaccuracies in such drawings, but the failure to detect errors, omissions, and inaccuracies shall not relieve the contractor of responsibility for the proper and complete installation in accordance with the intent of the contract documents. The submittal review shall not relieve the contractor of responsibility for purchase of any item in full compliance with the contract documents or its complete and proper installation.
5. Where submittals vary from the contract requirements, the contractor shall clearly indicate on submittal or accompanying documents the nature and reason for the variations.
6. Each manufacturer or his representative must check the application of his equipment and certify at time of shop drawing submittal that the equipment specified has been properly applied and can be installed, serviced and maintained where indicated on the drawings. Advise engineer in writing with submittal drawings of any potential problems. The manufacturer shall be responsible for any changes that might be necessary because of physical characteristics of equipment that have not been called to the engineer's attention at the time of submittal.
7. Submit a minimum of one (1) print and an electronic "pdf" of shop drawings to the architect. The architect and engineer shall review and return a pdf. The contractor shall distribute copies as required to properly conduct the work, including requirements of the operating manual.

H. Record Drawings
1. Each contractor or subcontractor shall keep one (1) complete set of the contract drawings and equipment submittals on the job site on which he shall regularly record any deviations or changes from such contract drawings made during construction. All recording shall be done in color ink.
2. These drawings shall record the installed location of all concealed equipment, piping, electric service, sewers, wastes, vents, ducts, conduit, etc., by measure dimensions to each such item from column centerlines or readily identifiable and accessible walls or corners of the building. Plans also shall show invert elevation of sewers and top elevation of all other below-grade lines.
3. Record drawings shall be kept clean and undamaged and shall not be used for any purpose other than recording deviations from working drawings and exact locations of concealed work.
4. After the project is completed, these drawings shall be scanned to an electronic "pdf" format and pdf and hard drawings shall be delivered to the architect in good condition, as a permanent record of the installation as actually constructed.

I. Supervision
1. The contractor shall have in charge of work at all times during construction a competent foreman or superintendent whose experience and background shall qualify him for the work to be performed under this division. Once assigned, the foreman or superintendent shall be retained until completion of the project and any consideration as to his removal on grounds of incompetence shall either be initiated by or referred to the architect for decision.

SECTION 200510 - BASIC MATERIALS AND METHODS

A. GENERAL
1. PROVIDE ALL MATERIALS, LABOR, EQUIPMENT, AND ACCESSORIES REQUIRED TO FURNISH AND INSTALL THE MECHANICAL ITEMS IDENTIFIED IN THIS SECTION.
2. THIS SECTION INCLUDES BASIC MECHANICAL MATERIALS AND METHODS TO COMPLEMENT OTHER DIVISION SECTIONS IN THIS SPECIFICATION AND REQUIREMENTS INDICATED ON THE MECHANICAL DRAWINGS.

B. INTERFERENCES
1. BEFORE INSTALLING ANY WORK, CONTRACTOR SHALL SEE THAT IT DOES NOT INTERFERE WITH CLEARANCE REQUIRED FOR FINISH ON BEAMS, COLUMNS, PILASTERS, WALLS, OR OTHER STRUCTURAL OR ARCHITECTURAL MEMBERS, AS SHOWN ON ARCHITECTURAL DRAWINGS. IF ANY WORK IS SO INSTALLED AND IT LATER DEVELOPS THAT ARCHITECTURAL DESIGN CANNOT BE FOLLOWED, CONTRACTOR SHALL, AT HIS OWN EXPENSE, MAKE SUCH CHANGES IN HIS WORK AS ARCHITECT MAY DIRECT TO PERMIT COMPLETION OF ARCHITECTURAL WORK IN ACCORDANCE WITH PLANS AND SPECIFICATIONS.
2. INSTALL ADDITIONAL OFFSETS ON PIPING OR DUCTWORK WHERE REQUIRED TO OBTAIN MAXIMUM HEADROOM OR TO AVOID CONFLICT WITH OTHER WORK WITHOUT ADDITIONAL COST TO OWNER.
3. REPORT ANY INTERFERENCES BETWEEN WORK UNDER THIS DIVISION AND THAT OF ANY OTHER CONTRACTORS TO ARCHITECT AS SOON AS THEY ARE DISCOVERED. ARCHITECT WILL DETERMINE WHICH EQUIPMENT SHALL BE RELOCATED, REGARDLESS OF WHICH WAS FIRST INSTALLED, AND HIS DECISION SHALL BE FINAL.

C. PROTECTION OF WORK AND PROPERTY
1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR SAFEGUARDING WORK, PROPERTY, AND FACILITIES AGAINST DAMAGE, BOTH HIS OWN AS WELL AS OTHERS WITH WHICH HE MAY COME INTO CONTACT IN THE PERFORMANCE OF HIS WORK.
2. STORED MATERIALS SHALL BE PROTECTED AGAINST DAMAGE FROM WEATHER. PIPE, AND DUCT OPENINGS SHALL BE CLOSED WITH CAPS OR PLUGS DURING INSTALLATION. ALL FIXTURES AND EQUIPMENT SHALL BE COVERED AND PROTECTED AGAINST DAMAGE. ANY MATERIALS OR EQUIPMENT DAMAGED AT ANY STAGE IN THE CONSTRUCTION SHALL BE REPLACED OR REPAIRED. FINAL COMPLETION, ALL WORK SHALL BE IN A CLEAN AND UNBLEMISHED CONDITION.
3. DURING CONSTRUCTION, ALL RETURN AIR DUCTWORK AND TRANSFER AIR OPENINGS SERVING NEW AND EXISTING AIR HANDLING EQUIPMENT AND/OR ADJACENT TENANT SPACES SHALL BE PROTECTED. OPENINGS WHICH NEED TO REMAIN ACTIVE SHALL BE COVERED AND PROTECTED WITH MERV 8 FILTRATION MEDIA; OPENINGS WHICH CAN REMAIN INACTIVE DURING CONSTRUCTION SHALL BE COVERED WITH PLASTIC SHEATHING AND SEALED AIR TIGHT. FILTER MEDIA SHALL BE REPLACED REGULARLY AS REQUIRED DURING CONSTRUCTION IN ORDER TO ENSURE ADEQUATE AIRFLOW THROUGH ALL REQUIRED ACTIVE OPENINGS. IN ADDITION, AT THE END OF EACH PHASE OF CONSTRUCTION AND AT THE END OF THE CONSTRUCTION PROJECT, ALL FILTRATION MEDIA WITHIN EACH PIECE OF EQUIPMENT SERVING THE SPACE SHALL BE REPLACED.

D. EXCAVATION AND BACKFILL
1. PERFORM ALL EXCAVATION AND BACKFILL REQUIRED FOR INSTALLATION OF BELOW-GRADE PIPING AND DUCTWORK.
2. EXCAVATE AS REQUIRED TO INSTALL PIPING AT REQUIRED DEPTH AND PITCH. PIPE TO BE LAID ON SAND BEDDING TO GIVE UNIFORM BEARING ALONG LENGTH OF PIPE (SAND INSIDE BUILDING AND INTERLOCKING AGGREGATE OUTSIDE BUILDING).
3. BACKFILL WITH BEDDING MATERIAL TO A MINIMUM OF 12" ABOVE TOP OF PIPE AND COMPACT. BALANCE OF BACKFILL IN OUTDOOR GRASS AREAS SHALL BE CLEAN EARTH UP TO 6" ABOVE SURROUNDING GRADES. BACKFILL BELOW FINISHED FLOORS SHALL BE SAND. BACKFILL OUTDOORS UNDER PAVING SHALL BE INTERLOCKING AGGREGATE AND SHALL BE COMPACTED IN MAXIMUM 10" LAYERS.
4. ALL OTHER EXCAVATIONS SHALL BE BACKFILLED WITH CLEAN EARTH, EXCLUDING RUBBISH AND BOULDERS. BACKFILL SHALL BE THOROUGHLY TAMPED AND PUDDLED.
5. PATCH FLOOR AND PAVING TO MATCH EXISTING ADJACENT SURFACES.
6. BACKFILLING SHALL NOT BE DONE UNTIL PIPE LINES ARE PROPERLY TESTED IN THE PRESENCE OF THE ARCHITECT AND/OR INSPECTION OF THE GOVERNMENT AGENCY HAVING JURISDICTION.
7. CONTROL TRENCH SOIL COMPACTION DURING CONSTRUCTION FOR COMPLIANCE WITH THE MAXIMUM DENSITY SPECIFIED FOR THE FOLLOWING AREAS:
   a. BUILDING SLABS, WALKWAYS, ROADWAYS, OR PUBLIC THOROUGH-FARES; COMPACT TOP 12" OF SUBGRADE AND EACH LAYER OF BACKFILL FOR FILL MATERIAL AT 95 PERCENT DENSITY FOR COHESIONLESS SOILS, AND 90 PERCENT DENSITY FOR COHESIVE SOIL MATERIAL. TESTS TO BE PERFORMED BY AN INDEPENDENT TESTING SERVICE, WITH THE COMPLIANCE REPORT SUBMITTED TO THE ARCHITECT.
8. PIPE SHALL NOT BE LAID IN WATER. FURNISH ALL PUMPING EQUIPMENT, POWER, TEMPORARY CONNECTIONS, ETC., AND DO ALL PUMPING NECESSARY TO REMOVE GROUND OR CASUAL WATER.
9. WHERE TRENCHES CROSS ROADS, WALKS, OR PUBLIC THOROUGHFARES, PROVIDE SUITABLE BARRICADES AND BRIDGES ADEQUATELY PROTECTED BY SIGNS OR RED FLAGS DURING DAY AND LIGHTS AS NIGHT.
10. REPAVE ALL STREETS OR SIDEWALKS DISTURBED AT THIS CONTRACTOR'S EXPENSE TO RECOMMENDATIONS, PROCEDURES AND SATISFACTION OF ARCHITECT AND AUTHORITIES HAVING JURISDICTION.

E. SUPPORTS AND HANGERS
1. HANGERS AND SUPPORTS ARE TO BE PROVIDED TO PROPERLY SUPPORT, SECURE AND ALIGN PIPING AND TO MEET FIELD CONDITIONS AND AS MANUFACTURED BY GRINNELL, MICHIGAN HANGER OR CADDY.
2. ALL HANGERS, BRACKETS, CLAMPS, ETC., SHALL BE OF STANDARD WEIGHT STEEL. PERFORATED STRAP HANGERS SHALL NOT BE USED IN ANY WORK. WHEN TWO OR MORE PIPES ARE RUN PARALLEL, THEY MAY BE SUPPORTED ON UNISTRUT-TYPE TRAPEZE HANGERS. OTHER HANGERS FOR PIPE 3" IN SIZE AND SMALLER SHALL BE CLEVIS. FOR PIPE TRANSPORTING MEDIUM ABOVE 150 DEGREES F AND 4" IN SIZE AND ABOVE, USE PIPE ROLL. EACH HANGER IS TO BE SIZED TO INCLUDE PIPE INSULATION SADDLE FOR PROTECTION.
3. WHERE BUILDING SERVICE LINES ENTER OR LEAVE BUILDING SUCH AS WATER, SEWER, GAS, ETC., AND ARE INSTALLED ON FILLED EARTH, PROVIDE CONTINUOUS SUPPORT ON A REINFORCED CONCRETE BEAM FURNISHED AND INSTALLED UNDER THIS DIVISION. SUPPORT BEAM ON BUILDING AND WITH VERTICAL SUPPORT DOWN TO FOUNDATION FOOTING AND ON UNDISTURBED EARTH AT OTHER END. GAS MAIN SHALL ENTER BUILDING ABOVE GRADE.
4. ALL VERTICAL PIPING PASSING THROUGH FLOORS SHALL BE SUPPORTED AT THE FLOOR BY A RISER CLAMP.
5. ISOLATE ALL COPPER LINES FORM FERROUS HANGERS OR SUPPORTS BY USING FOIL FILLER OR VINYL TAPE.
6. SPACING TO COMPLY WITH ASHRAE STANDARDS AND CODE REQUIREMENTS.

F. PIPE SLEEVES, FLOOR AND CEILING PLATES
1. ALL PIPES PASSING THROUGH FLOORS OR MASONRY WALLS SHALL BE PROVIDED WITH MACHINE-CUT SCHEDULE 40 PIPE STEEL SLEEVES. THE SLEEVES SHALL BE SO SIZED TO ALLOW AT LEAST 1/4" CLEARANCE BETWEEN THE INSIDE SLEEVE WALL AND THE PIPE OR INSULATION SURFACE. SHEET METAL SLEEVES SHALL NOT BE USED IN THIS WORK. PIPE SLEEVES ARE TO EXTEND 2" ABOVE FINISHED FLOOR AND SEALED. PIPE SLEEVES ARE TO BE FULL WALL THICKNESS AND SEALED.
2. UNUSED SLEEVES SHALL BE PLUGGED AND FINISHED TO MATCH ADJOINING SURFACE.

G. ESCUTCHEONS
1. FIT ALL PIPE PASSING THROUGH WALLS, FLOORS OR CEILINGS IN FINISHED ROOMS WITH CHROME PLATED ESCUTCHEONS. WHERE SURFACE IS TO RECEIVE A PAINT FINISH, MAKE ESCUTCHEONS PRIME PAINTED; OTHERWISE, MAKE ESCUTCHEONS NICKEL OR CHROME PLATED. WHERE PIPING IS INSULATED, FIT ESCUTCHEONS OUTSIDE INSULATION.

H. PIPE IDENTIFICATION AND TAGS
1. IDENTIFY EACH PIPE, VALVE AND CONTROLS IN EQUIPMENT ROOMS, ABOVE ACCESSIBLE CEILINGS AND IN ACCESSIBLE SHAFTS.
2. COLOR CODE IDENTIFICATION BANDS OR MARKER BACKGROUNDS TO IDENTIFY CONTENTS OF PIPE WITH INITIALS AND DIRECTION OF FLOW LOCATED NEAR EACH VALVE AND FITTING, ON BOTH SIDES OF PIPE PASSING THROUGH WALLS AND ON LONG RUNS AT NOT OVER 20'-0" INTERVALS.
3. AT PLACE WHERE PIPE IS TO HAVE MARKING, COVERED PIPE SHALL BE PROPERLY PRIMED WITH CLEAR LACQUER. AFTER MARKING IS APPLIED, COAT WITH LACQUER. APPLY MARKING ADJACENT TO VALVES AND EQUIPMENT AT MAJOR CHANGES IN DIRECTIONS, WHERE PIPES PASS THROUGH WALLS OR FLOORS.
4. EACH PIECE OF EQUIPMENT SHALL BE IDENTIFIED BY A NUMBER, TOGETHER WITH A BRIEF DESCRIPTION OF ITS PURPOSE, E.G. "AIR HANDLING UNIT - EAST LOBBY." IDENTIFICATION SHALL BE EMBOSSED OR ENGRAVED PLASTIC OR STAMPED BRASS STRIPS FIRMLY ATTACHED TO THE EQUIPMENT OR ADJACENT WALL AT A VERY OBVIOUS LOCATION. THE LETTERING FOR SUCH STRIPS SHALL BE NOT LESS THAN 1/2" HIGH.
5. ALL VALVES SHALL BE PROVIDED WITH BRASS NUMBERED TAGS ATTACHED TO HANDLE WITH A BRASS CHAIN OR RING. WIRING OF TAGS WILL NOT BE ACCEPTABLE. AT THE COMPLETION OF THE WORK, A REPRODUCIBLE VALVE SCHEDULE SHALL BE PROVIDED. THREE (3) COPIES OF THIS SHALL BE MOUNTED IN METAL, GLASS COVERED FRAMES WHERE REQUESTED BY THE ARCHITECT. THE SCHEDULE SHALL GIVE A DESCRIPTION OF THE LINE OR EQUIPMENT CONTROLLED; THE NORMAL POSITION, EMERGENCY AND/OR SHUTDOWN POSITION AND LOCATION GIVEN EITHER BY DESCRIPTION OR DIAGRAM.
6. ALL CONTROLS, STARTERS, SWITCHES, ETC, SHALL BE IDENTIFIED BY EMBOSSED STENCIL OR ENGRAVED PLATE AS TO PURPOSE AND/OR EQUIPMENT CONTROLLED. CONTROL WIRING SHALL BE IDENTIFIED WITH PROGRAM NUMBER AND DEVICE IT SERVICES.

I. ACCESS PANELS
1. EACH CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING ALL REQUIRED ACCESS PANELS NECESSARY FOR HIS WORK. THIS INCLUDES ANY ACCESS PANELS REQUIRED FOR HVAC, PLUMBING AND FIRE PROTECTION. EACH CONTRACTOR SHALL ALSO PROVIDE ACCESS PANELS FOR ANY EXISTING CONDITIONS AS REQUIRED.
2. REFER TO ARCHITECTURAL DRAWINGS AND SPECIFICATIONS FOR TYPE OF ACCESS PANEL AND COORDINATE LOCATIONS PRIOR TO ANY WORK.
3. CONTRACTOR SHALL MARK LAY-IN CEILING TILES, IN A METHOD APPROVED BY THE ARCHITECT, WHERE ACCESS IS REQUIRED TO SUCH MECHANICAL, PLUMBING, AND FIRE PROTECTION EQUIPMENT, VALVES, REGULATORS, MIXING BOXES, FIRE DAMPER, ETC.

J. NOISE AND VIBRATION ISOLATION
1. FURNISH AND INSTALL VIBRATION ISOLATING MOUNTINGS TO ISOLATE FROM THE STRUCTURE, BY MEANS OF RESILIENT VIBRATION AND NOISE ISOLATORS, ALL MECHANICAL EQUIPMENT OVER 1 HP HAVING ROTATING OR RECIPROCATING PARTS. ISOLATORS SHALL BE SUPPLIED BY A SINGLE SOURCE, AND SHALL BE GUARANTEED BY THE MANUFACTURER TO PROVIDE ISOLATION EFFICIENCIES IN ACCORDANCE WITH THIS SPECIFICATION. SELECTION SHALL BE BASED ON EQUIPMENT PURPOSED, POWER DISSIPATED, FREQUENCY, WEIGHT DISTRIBUTION AND NATURE OF THE BUILDING STRUCTURE. MOUNTINGS SHALL BE DESIGNED TO PERMIT ATTACHMENT TO THE EQUIPMENT BASE OR PAD AND TO THE STRUCTURE AND SHALL BE SELECTED FOR UNIFORM DEFLECTION ALLOWING FOR UNEQUAL WEIGHT DISTRIBUTION.
2. SELECTION SHALL BE MADE BY THE MANUFACTURER OF THE MOUNTINGS TO PROVIDE A TRANSMISSIBILITY NOT EXCEEDING 16 PERCENT. THIS CONTRACTOR SHALL PROVIDE INERTIA PADS FOR EQUIPMENT WHERE CALLED FOR ON DRAWINGS OR RECOMMENDED BY THE MANUFACTURER OF THE MOUNTINGS. THESE SHALL CONSIST OF REINFORCED CONCRETE PADS OF SUITABLE SHAPE, OF WEIGHT 1-1/2 TIMES THE WEIGHT OF THE EQUIPMENT AND PROVIDED WITH WELD PLATES OR CHANNELS AT THE CORNERS TO WHICH THE MOUNTINGS MAY BE SECURED.
3. VIBRATION OR NOISE CREATED IN ANY PART OF THE BUILDING BY THE OPERATION OF ANY EQUIPMENT FURNISHED AND/OR INSTALLED UNDER THIS CONTRACT WILL BE PROHIBITED, AND THIS CONTRACTOR SHALL TAKE ALL PRECAUTIONS BY ISOLATING THE VARIOUS ITEMS OF EQUIPMENT, PIPE AND SHEET METAL WORK FORM THE BUILDING STRUCTURE. THE MAJOR ITEMS OF EQUIPMENT SHALL BE ISOLATED AS CALLED FOR ON THE PLANS AND SPECIFIED HEREIN. THE MINOR ITEMS SHALL BE HELD THE RESPONSIBILITY OF THIS CONTRACTOR.
4. MECHANICAL EQUIPMENT NOT INTERNALLY ISOLATED BY THE MANUFACTURER SHALL BE ISOLATED AS FOLLOWS:
   a. CONNECTIONS FROM PUMP OUTLET AND DISCHARGE NOZZLES TO PIPING SHALL BE MADE WITH FLEXIBLE CONNECTORS.
   b. ISOLATE EACH BASE MOUNTED PUMP FROM THE PIPING SYSTEMS BY USE OF APPROPRIATE SIZE CORRUGATED BELLOWS, TYPE 347 STAINLESS STEEL COUPLINGS WITH CONTROL RODS AS MANUFACTURED BY KEFLEX MFG. OR FLEXONICS FOR 300 PSIG DESIGN PRESSURE AT 800 DEGREES F. DESIGN TEMPERATURE.
   c. CENTRIFUGAL FANS, AIR CONDITIONING AND/OR HEATING AND VENTILATING UNITS UP TO 3"STATIC PRESSURE ON GRADE SHALL BE MOUNTED ON PRECOMPRESSED MOLDED FIBERGLASS, RUBBER-IN-SHEAR, OR STEEL SPRING ISOLATORS. IF THE DRIVE MOTOR IS NOT SUPPORTED DIRECTLY ON THE FAN, BOTH UNITS SHALL BE MOUNTED ON AN INTEGRAL STRUCTURAL STEEL BASE SUPPLIED BY THE ISOLATOR MANUFACTURER, OR SUFFICIENT RIGIDITY TO MAINTAIN ALIGNMENT BETWEEN THE FAN AND THE DRIVE MOTOR. THE BASE SHALL, IN TURN BE MOUNTED ON PRECOMPRESSED MOLDED FIBERGLASS, RUBBER-IN-SHEAR, OR STEEL SPRING ISOLATORS.

      THE FANS' ISOLATORS SHALL PROVIDE ISOLATION EFFICIENCIES AS FOLLOWS:

FAN SPEED OVER 700 RPM	95 PERCENT
FAN SPEED BETWEEN 450 AND 700 RPM	90 PERCENT
FAN SPEED BELOW 450 RPM, FAN WHEEL OVER 48" DIAMETER	80 PERCENT
FAN SPEED BELOW 450 RPM, FAN WHEEL UNDER 48" DIAMETER	NOISE ISOLATION ONLY

   d. CENTRIFUGAL FANS, AIR CONDITIONING AND/OR HEATING AND VENTILATING UNITS UP TO 3" STATIC PRESSURE ABOVE GRADE SHALL BE MOUNTED ON STEEL SPRING VIBRATION ISOLATORS IN COMBINATION WITH PRECOMPRESSED MOLDED FIBERGLASS NOISE ISOLATION PADS. IF THE DRIVE MOTOR IS NOT SUPPORTED DIRECTLY ON THE FAN, BOTH UNITS SHALL BE MOUNTED ON AN INTEGRAL CONCRETE INERTIA BASE, SUPPLIED BY THE ISOLATOR MANUFACTURER, OF SUFFICIENT RIGIDITY TO MAINTAIN ALIGNMENT BETWEEN THE FAN AND ISOLATORS IN COMBINATION WITH PRECOMPRESSED MOLDED FIBERGLASS NOISE ISOLATION PADS. SEE "C" ABOVE FOR ISOLATION EFFICIENCIES.
   e. CENTRIFUGAL FANS, AIR CONDITIONING AND/OR HEATING AND VENTILATING UNITS UP TO 3" STATIC PRESSURE CEILING SUSPENDED SHALL BE MOUNTED ON A SUITABLE PLATFORM AND THE PLATFORM IN TURN, SUSPENDED BY THREADED RODS FROM THE OVERHEAD STRUCTURE. RESILIENT HANGERS INCORPORATING STEEL SPRINGS AND PRECOMPRESSED MOLDED INSERTS SHALL BE INCORPORATED INTO EACH SUPPORTING ROD. SEE "C" ABOVE FOR ISOLATION EFFICIENCIES.
   f. CENTRIFUGAL FANS, AIR CONDITIONING AND/OR HEATING AND VENTILATING UNITS OVER 3" STATIC PRESSURE SHALL BE MOUNTED ON A REINFORCED CONCRETE INERTIA BLOCK WHOSE WEIGHT IS EQUAL TO OR GREATER THAN THAT OF THE SUPPORTED EQUIPMENT. THE CONCRETE SHALL BE POURED INTO A WELDED STEEL CHANNEL FRAME WITH WELDED-IN REINFORCING BARS AND PRELOCATED EQUIPMENT ANCHOR BOLTS, ALL SUPPLIED BY THE ISOLATOR MANUFACTURER. THE INERTIA BLOCK SHALL BE MOUNTED ON STEEL SPRING VIBRATION ISOLATORS IN COMBINATION WITH PRECOMPRESSED MOLDED FIBERGLASS NOISE ISOLATION PADS.

      SEE "C" ABOVE FOR ISOLATION EFFICIENCIES.
   g. PUMPS UP TO 5 HP SHALL BE MOUNTED ON RAILS, INCORPORATING PRECOMPRESSED MOLDED FIBERGLASS, RUBBER-IN-SHEAR, OR STEEL SPRING ISOLATORS. THE ISOLATORS SHALL PROVIDE 90 PERCENT ISOLATION EFFICIENCY.
5. PIPING AND DUCTWORK SHALL BE SUPPORTED INDEPENDENTLY OF THE MECHANICAL EQUIPMENT AND SHALL BE ISOLATED AS FOLLOWS:
   a. ALL SUSPENDED PIPING IN THE MECHANICAL EQUIPMENT AND AIR HANDLING ROOMS SHALL BE SUPPORTED FROM THE OVERHEAD STRUCTURE BY THREADED RODS INCORPORATING RESILIENT HANGERS. THE RESILIENT HANGERS SHALL CONTAIN STEEL SPRINGS AND PRECOMPRESSED MOLDED FIBERGLASS INSERTS, DESIGNED FOR STATIC DEFLECTIONS OF BETWEEN 1" AND 1-3/4" UNDER OPERATING CONDITIONS.
   b. ALL FLOOR SUPPORTED PIPING AND PIPE HANGERS IN THE MECHANICAL EQUIPMENT ROOMS SHALL BE MOUNTED ON STEEL SPRING VIBRATION ISOLATORS IN COMBINATION WITH PRECOMPRESSED MOLDED FI

1. CURB SHALL BE 18 GAUGE GALVANIZED STEEL WITH CONTINUOUS WELDED SEAMS, WOOD NAILER, COUNTERFLASHING, R-8 MINIMUM AND LINER INSULATION. TOP OF CURB SHALL BE A MINIMUM SIZE AS SHOWN IN DETAIL ON DRAWINGS, BUT NOT LESS THAN 14" ABOVE THE HIGH POINT OF ROOF WHERE CURB ATTACHES.
2. PROVIDE CURB FOR ALL ROOF PENETRATIONS OF DUCTS AND PIPING.
3. ALL CUTTING AND PATCHING OF EXISTING ROOF SHALL BE BY THE OWNER'S ROOFING CONTRACTOR AND PAID FOR BY THE MECHANICAL CONTRACTOR.
4. CURB SHALL BE INSTALLED WITH TOP LEVEL. CURB BASE TO MATCH ROOF PITCH.

SECTION 200523 - PIPING AND VALVES

A. GENERAL
1. FURNISH ALL MATERIAL, LABOR, EQUIPMENT, AND ACCESSORIES AS REQUIRED TO INSTALL COMPLETE FIRE PROTECTION, PLUMBING, AND HVAC PIPING SYSTEMS AS INDICATED ON DRAWINGS AND IN THESE SPECIFICATIONS.
2. INSTALL IN FULL ACCORDANCE WITH LOCAL CODE REQUIREMENTS, SEE OTHER SPECIFICATION SECTION FOR ADDITIONAL REQUIREMENTS AND INSTALL IN ACCORDANCE TO MANUFACTURER'S RECOMMENDATIONS AND REQUIREMENTS.

B. CONNECTIONS TO EQUIPMENT FURNISHED BY OTHERS
1. PROVIDE VALVED WATER AND/OR GAS CONNECTION FOR EQUIPMENT FURNISHED BY OTHER CONTRACTORS OR OWNER.
2. INCLUDE ACCESSORIES REQUIRED BY CODE, DRAWINGS AND MANUFACTURER'S INSTALLATION INSTRUCTIONS.
3. FULLY COORDINATE WITH LAB EQUIPMENT, POOL EQUIPMENT, KITCHEN EQUIPMENT, AND LAUNDRY EQUIPMENT SUPPLIERS AND CONFIRM ALL ROUGH-IN REQUIREMENTS PRIOR TO STARTING WORK.

C. INSTALLATION
1. ALL PIPING SHALL BE INSTALLED PARALLEL WITH OR PERPENDICULAR TO THE BUILDING WALLS. ALL VERTICAL RISERS SHALL BE INSTALLED PLUMB AND STRAIGHT. ALL PIPING ABOVE ACCESSIBLE CEILINGS SHALL BE INSTALLED AS HIGH AS POSSIBLE AND AT HEIGHT TO ALLOW SUFFICIENT SPACE FOR CEILING PANEL REMOVAL.
2. ALL PIPING SHALL BE INSTALLED WITH PITCH IN THE DIRECTION OF FLOW OF NOT LESS THAN 1" IN FORTY FEET, EXCEPT AS OTHERWISE SHOWN. IT MUST BE POSSIBLE TO DRAIN EVERY PORTION OF THE PIPING SYSTEM.
3. RUN LINES AS DIRECT AS POSSIBLE AND AVOID UNNECESSARY OFFSETS. HOWEVER, IF OFFSETS ARE REQUIRED IN ORDER TO OBTAIN MAXIMUM HEADROOM OR TO AVOID CONFLICT WITH OTHER WORK, THEY SHALL BE MADE AS REQUIRED OR AS REQUESTED BY THE ARCHITECT WITHOUT ADDITION COST TO THE OWNER. THE ARCHITECT RESERVES THE RIGHT TO MAKE MINOR CHANGES IN THE LOCATION OF PIPING AND EQUIPMENT DURING THE ROUGHING-IN, WITHOUT ADDITIONAL COST TO THE OWNER. ALL CHANGES PROPOSED BY OTHERS SHALL BE APPROVED BY THE ARCHITECT
4. LINES SHALL BE CUT ACCURATELY TO MEASUREMENT AT THE SITE AND WORKED INTO PLACE WITHOUT SPRINGING OR FORCING. SUFFICIENT OFFSETS, PIPE LOOPS OR EXPANSION JOINTS BETWEEN ANCHOR POINTS SHALL BE PROVIDED AS NEEDED, WHETHER OR NOT SHOWN, TO LIMIT STRESSES AND CONTROL MOVEMENT OF LINES SUBJECT TO THE THERMAL EXPANSION.
5. BEFORE ANY PIPING IS INSTALLED, IT SHALL BE UP-ENDED AND POUNDED TO REMOVE ANY FOREIGN MATTER PRESENT, AND SHALL BE SWABBED, IF NECESSARY, FOR THOROUGH CLEANING. AFTER INSTALLATION AND BEFORE FINAL CONNECTIONS MADE, ALL PIPING SYSTEM SHALL BE FLUSHED WITH A MATERIAL THAT IS NOT INJURIOUS TO EITHER PIPE OR EQUIPMENT. (SEE ALSO "TESTS AND ADJUSTMENTS.")
6. PIPE TO BE THREADED SHALL BE CUT SQUARE AND FULL THREADED WITH CLEAN-CUT TAPERED THREADS AND SHALL BE REAMED AFTER THREADING. THREADED CONNECTIONS SHALL BE MADE WITH PIPE THREAD COMPOUND APPLIED TO THE WALL THREADS ONLY.
7. THE EDGES OF PIPE TO BE WELDED SHALL BE MACHINE BEVELED WHEREVER POSSIBLE. BEFORE WELDING, THE SURFACES SHALL BE THOROUGHLY CLEANED. THE PIPING SHALL BE CAREFULLY ALIGNED. NO METAL SHALL PROJECT WITHIN THE PIPE. MITERED JOINTS ARE PROHIBITED. ONLY FACTORY FORMED FITTINGS SHALL BE USED. ELBOWS SHALL BE LONG RADIUS TYPE. FLANGES SHALL BE WELDING NECK TYPE. MITERING OF THE PIPE TO FORM ELBOWS OR THE NOTCHING OF STRAIGHT RUNS TO FORM THE TEE CONNECTION WILL NOT BE PERMITTED.
8. UNIONS OR COMPANION FLANGES SHALL BE INSTALLED IN ALL CONNECTIONS TO EQUIPMENT, AUTOMATIC VALVES, ETC., AS NECESSARY TO PERMIT REMOVAL OF EQUIPMENT AND SPECIALTIES FOR SERVICING, REPAIRING OR CLEANING. IT SHALL BE POSSIBLE TO REMOVE ANY PIECE OF EQUIPMENT BY REMOVING ONLY ONE OR TWO SECTIONS OF PIPING.
9. VALVES SHALL BE PROVIDED IN SUITABLE LOCATIONS AT EACH ITEM OF EQUIPMENT, BRANCH CIRCUIT, RISER, OR SECTION OF PIPING AS INDICATED OR REQUIRED FOR PROPER AND SAFE OPERATION OF THE SYSTEM AND TO FACILITATE MAINTENANCE AND/OR REMOVAL OF ALL EQUIPMENT AND APPARATUS. ON HORIZONTAL PIPE RUNS, INSTALL ALL VALVE STEMS VERTICALLY UP WHERE POSSIBLE AND IN NO CASE SHALL THE STEMS BE TURNED MORE THAN 90 DEGREES FROM THE VERTICALLY UP POSITION.
10. DRAIN VALVES SHALL BE PROVIDED AT ALL LOW POINTS, TRAPPED SECTION, AND ON THE EQUIPMENT SIDE OF ALL BRANCH VALVES TO PERMIT DRAINING OF ALL PARTS OF ALL LIQUID PIPING SYSTEMS. DRAIN VALVES SHALL HAVE THREADED HOSE ENDS WITH CAP AND CHAIN. DRAIN PIPING SHALL BE PROVIDED FROM PUMP GLANDS, RELIEF VALVES, ETC., TO SPILL AT THE FLOOR OVER FLOOR DRAINS OR OTHER ACCEPTABLE DISCHARGE POINTS. THE DRAIN LINE SHALL TERMINATE WITH PLAIN, UNTHREADED END WITH A MINIMUM 2" AIR GAP AT FLOOR DRAIN.
11. TAPS (HALF COUPLINGS OR TEES) SHALL BE PROVIDED AS NECESSARY TO PERMIT THE INSTALLATION OF TEMPERATURE CONTROL INSTRUMENTS, THERMOMETERS, PRESSURE GAUGES, AIR VENTS, ETC.
12. CONNECTIONS BETWEEN COPPER PIPING AND SCREWED FERROUS EQUIPMENT CONNECTIONS OR SCREWED FERROUS PIPING SYSTEMS SHALL BE MADE AS FOLLOWS:
    a. FOR STATIONARY NON-ROTATING, NON-VIBRATING EQUIPMENT CONNECTIONS: DIELECTRIC UNIONS.
    b. FOR ROTATING OR VIBRATING EQUIPMENT CONNECTION: CAST BRASS ADAPTER AND BRONZE FLANGES WITH DIELECTRIC SEPARATION OF FLANGES AND BOLTS.
    c. CONNECTIONS BETWEEN COPPER PIPING AND FERROUS EQUIPMENT FLANGES OR FLANGED FERROUS PIPING SYSTEMS SHALL BE MADE USING BRONZE COMPANION FLANGE WITH DIELECTRIC SEPARATION OF FLANGES AND BOLTS.
    d. BRASS OR BRONZE VALVES IN FERROUS PIPING WILL NOT REQUIRE DIELECTRIC SEPARATION.
    e. NIPPLES BETWEEN COPPER PIPING AND EQUIPMENT OR FIXTURE CONNECTION FITTINGS SHALL BE BRASS, NOT GALVANIZED STEEL.
13. ALL PRESSURE PIPING SYSTEMS SHALL BE INSTALLED TO CONFORM TO THE REQUIREMENTS OF THE LOCAL AHJ OR STATE'S PRESSURE PIPING SYSTEM CODE.
14. ALL EXCAVATIONS FOR INSTALLATION OF PIPE SHALL BE OPEN TRENCH WORK AND SHALL BE KEPT OPEN UNTIL PIPING HAS BEEN INSPECTED, TESTED, AND ACCEPTED.
15. ALL PIPING PASSING THRU CAST-IN PLACE CONCRETE CONSTRUCTION SHALL BE SLEEVED TO PROVIDE A MINIMUM OF 1/2"ANNULAR SPACE AROUND ENTIRE PIPE TO BE SLEEVED. SPACE BETWEEN SLEEVE AND PIPES IN FOUNDATION WALLS SHALL BE TIGHTLY CAULKED OR MECHANICAL SEAL TO GIVE A WATERPROOF PENETRATION.
16. ANY PIPING RESTING ON OR COMING IN CONTACT WITH BUILDING STRUCTURE SHALL BE INSULATED AT THAT POINT TO PREVENT TELEGRAPHING OF SOUND.
17. METAL PIPING LAID IN CORROSIVE FILL SHALL BE ENCASED IN CONCRETE OR IN SPLIT TILE.
18. ALL SEWERS 14 FEET -0" BELOW FINISH GRADE SHALL BE ENCASED IN CONCRETE.
19. THREADED JOINTS SHALL CONFORM TO AMERICAN TAPER PIPE THREAD ASA-B2.1-1960. ALL BURRS SHALL BE REMOVED, PIPE ENDS SHALL BE REAMED OR FILED TO SIZE OF BORE AND ALL CHIPS REMOVED. PIPE CEMENT SHALL BE USED ONLY ON MALE THREADS.
20. UNIONS SHALL HAVE METAL SEATS FOR DRAINAGE SYSTEMS AND METAL TO METAL GROUND SEATS ON WATER SYSTEM.
21. FURNISH AND INSTALL VALVE IN BRANCHES TO SILL COCKS, TOILET ROOMS AND OTHER FIXTURE GROUPS. PLUMBING FIXTURES SHALL HAVE WHEEL OR SCREWDRIVER STOPS AS SPECIFIED.
22. ALL PIPING SHALL BE RIGIDLY SUPPORTED AND SHALL NOT BE LOOSE OR SHAKY.

D. SANITARY, WASTE, VENT AND STORM SEWERS
1. INSTALL SEWERS, STACKS, VENTS, DRAINS, ETC., AS INDICATED ON THE DRAWINGS.
2. ALL DRAINAGE AND VENT PIPING SHALL BE CONSTRUCTED AND RUN AS DIRECT AS POSSIBLE, PROTECTED FROM CONTACT WITH SLAG OR CINDERS AND WHEREVER PRACTICABLE, SHALL BE LOCATED SO AS TO BE ACCESSIBLE FOR INSPECTION. THE ACTUAL RUNS AND LOCATIONS OF DRAINS, SOIL WASTE, AND LEADER PIPING SHALL BE INSTALLED AS TO MEET WITH THE VARIOUS CONDITIONS AS IN THE BUILDING AND ANY WORK NECESSARY TO CONCEAL PIPES OR CLEAR PIPES OF OTHER TRADES SHALL BE DONE AS DIRECTED BY THE ARCHITECT.
3. SEWERS TO BE PITCHED A MINIMUM OF 1/4" PER FOOT FOR 3" SIZES AND UNDER AND 1/8" PER FOOT FOR 4" SIZES AND LARGER OR TO SLOPE AS INDICATED ON DRAWINGS. KITCHEN SANITARY WASTE SHALL BE SLOPED ½" PER FOOT FOR ALL PIPE SIZES.
4. ALL PIPING SHALL BE CORRECTLY ALIGNED BEFORE JOINS ARE MADE. ALL CHANGES OF DIRECTION IN DRAINAGE AND VENT PIPING SHALL BE MADE BY MEANS OF "Y" BRANCHES AND 1/8, 1/8 OR 1/16 BENDS. NO LINES SHALL BE RUN WITH UNNECESSARY BENDS OR OFFSETS AND WHERE CHANGES IN DIRECTION ARE UNAVOIDABLE; THEY SHALL BE MADE BY USE OF PROPER FITTINGS. SINGLE AND DOUBLE SANITARY TEES, 1/4 BENDS AND 1/8 BENDS MAY BE USED IN VERTICAL SECTIONS WHEN DIRECTION OF FLOW IS FROM HORIZONTAL TO VERTICAL. CHANGES IN DIRECTION AND BRANCH CONNECTIONS SHALL BE MADE WITH APPROVED DRAINAGE FITTINGS COMPATIBLE WITH THE PIPING SYSTEM MATERIAL IN WHICH IT IS INSTALLED.
5. INSTALL CLEANOUTS AT BASE OF EACH VERTICAL WASTE AND RAINWATER STACK, EACH CHANGE IN A DIRECTION OF PIPING GREATER THAN 45 DEGREES, WITHIN FIVE FEET (5'-0") OF MAIN SEWER AFTER EXITING THE BUILDING, OR AS SHOWN ON DRAWINGS. CLEANOUTS ON UNDERGROUND LINES SHALL EXTEND UP FLUSH WITH FINISHED FLOOR OR GRADE. PROVIDE CLEANOUTS NOT OVER 50'- 0" ON CENTER ALONG STRAIGHT RUNS. CLEANOUTS SHALL BE SIZE OF PIPE TO WHICH IT IS INSTALLED UP TO 6" IN DIAMETER. PIPE OVER 6" IN DIAMETER SHALL HAVE A 6" CLEANOUT.
6. VENT TERMINALS SHALL BE TERMINATED AT LEAST 18" ABOVE ROOF. EACH VENT TERMINAL SHALL BE MADE WATER TIGHT WITH THE ROOF BY USING SHEET COPPER (8 OUNCES PSF) WITH BASE NOT LESS THAN 16" DIAMETER AND COLLAR FULL HEIGHT OF PIPE OR RUBBER BOOT PIPE FLUSHING. WHERE VENTS ARE 4" OR LARGER, FLASHING MAY BE TURNED OVER INTO TOP OF PIPE WITHOUT GAP. FURNISH FLASHING TO GENERAL CONTRACTOR FOR BUILDING INTO ROOFING MATERIAL.
7. ALL FIXTURES AND SANITARY DRAINS SHALL BE VENTED AS INDICATED ON DRAWINGS AND IN ACCORDANCE WITH CODE. VENT PIPES, WHERE NOT VERTICAL SHALL HAVE CONTINUOUS SLOPE UP TO VENT THROUGH ROOF.
8. OPENINGS IN PIPES SHALL BE PROPERLY PLUGGED WHEN WORK IS NOT IN PROGRESS
9. ROOF DRAINS SHALL BE PROVIDED WITH A FLUSHING RING AND A 30" X 30" X 12 OZ./S.F. -COPPER SHEET ASTM B152/B, 152 M FLASHING PROPERLY FASTENED TO THE FLASHING RING.
10. SEWERS SHALL BE LAID WITH FULL LENGTH OF EACH SECTION RESTING ON A SOLID BED. WHERE NECESSARY TO OBTAIN A FIRM SUPPORT, THE PIPE SHALL BE BEDDED ON SELECT MATERIAL AND THOROUGHLY TAMPED. AS PIPE IS LAID, CARE SHALL BE EXERCISED TO KEEP INTERIOR OF PIPE CLEAR OF FOREIGN MATTER. WHERE TRENCHING FOR PIPE IS EXCESSIVELY WIDE, THE CONTRACTOR SHALL AT HIS OWN EXPENSE, EMBED THE PIPE IN CONCRETE TO SUPPORT THE ADDED LOAD OF BACKFILLING.
11. PIPE SCHEDULE:
    a. BELOW GRADE INSIDE BUILDING
        1) SERVICE WEIGHT - CAST IRON PIPE ASTM A-74-82 WITH ASTM C-564-80 NEOPRENE COMPRESSION JOINTS OR NO-HUB CISPI WITH CLAMPS. ALL KITCHEN SANITARY SHALL BE CAST IRON ONLY.
        2) PVC-DWV SCH. 40 SOLID CORE PIPE, ASTM D-1785 WITH ASTM D-2665 DWV SOLVENT WELD SOCKET FITTINGS.
    b. ABOVE GRADE AND VENT MATERIAL SHALL BE AS FOLLOWS:
        1) NO-HUB CAST IRON PIPE CISPI 1-301-78.
        2) PVC-DWV SCH. 40 SOLID CORE PIPE, ASTM D-1785 WITH ASTM D-2665 DWV SOLVENT WELD SOCKET FITTINGS.
        3) 1-1/4" AND SMALLER, SCH. 40 GALVANIZED STEEL PIPE ASTM A-53/A53M, TYPE E WITH SCREWED FITTINGS ASME B-16.4, CLASS 125.
    c. SITE BELOW GRADE SEWERS
        1) NO-HUB CAST IRON PIPE CISPI 1-301-78.
        2) PVC-DWV SCH. 40 SOLID CORE PIPE. ASTM D-1785 WITH ASTM D-2665 DWV SOLVENT WELD SOCKET FITTINGS.
        3) UP TO 15" - PVC PIPE, ASTM D-3034 SDR 35 WITH ASTM F477 GASKET JOINTS.
        4) 18" AND OVER - REINFORCED CONCRETE PIPE (RCP) ASTM C 76-83 WITH ASTM C 443-79 RUBBER GASKET JOINTS.
    d. FORCED MAIN BELOW GRADE
        1) SOFT COPPER TUBE, TYPE L; COPPER PRESSURE FITTINGS AND SOLDERED JOINTS FOR 4" AND SMALLER PIPE.
        2) DUCTILE IRON AWWA C 110/A21.10, MECHANICAL JOINT PIPING, FITTINGS AWWA C153/A21.53, AND MECHANICAL JOINTS AWWA C111/A21.11.
        3) HARD COPPER TUBE, TYPE L; COPPER PRESSURE FITTINGS; AND SOLDERED JOINTS FOR 5" AND LARGER PIPE.
    e. FORCED MAIN ABOVE GRADE
        1) HARD COPPER TUBE, TYPE L; COPPER PRESSURE FITTINGS AND SOLDER JOINTS.
        2) GALVANIZED STEEL PIPE, PRESSURE FITTINGS AND THREADED JOINTS.
    f. EXPANSION JOINTS AND DEFLECTION FITTINGS
        1) DUCTILE-IRON, FLEXIBLE EXPANSION JOINTS; AWWA C110 OR AWWA C153 WITH TWO GASKETED BALL-JOINT SECTIONS AND ONE OR MORE GASKETED SLEEVE SECTIONS RATED FOR 250 PSI MINIMUM.
        2) DUCTILE-IRON EXPANSION JOINTS; THREE-PIECE ASSEMBLY OF TELESCOPING SLEEVE WITH GASKETS AND RESTRAINED-TYPE, BALL-AND-SPIGOT END SECTIONS; AWWA C110 OR AWWA C153; RATED FOR 250 PSIG MINIMUM.
        3) DUCTILE-IRON DEFLECTION FITTINGS; COMPOUND COUPLING FITTING WITH BALL JOINT, FLEXING SECTION, GASKETS, AND RESTRAINED JOINT ENDS; AWWA C110 OR AWWA C153; RATED FOR 250 PSI MINIMUM AND UP TO 15 DEGREES OF DEFLECTION.
12. PVC PIPING SHALL NOT BE INSTALLED UNLESS PERMITTED BY CODE AND SHALL NOT BE INSTALLED IN RETURN AIR PLENUMS.

E. DOMESTIC WATER PIPING
1. INSTALL DOMESTIC WATER PIPING AS INDICATED ON DRAWINGS. INCLUDE ALL FITTINGS, VALVES, HANGERS, AND OTHER ACCESSORIES INCLUDING WATER METER AND BACKFLOW PREVENTER. EXTEND DOMESTIC WATER PIPING TO ALL FIXTURES AND EQUIPMENT REQUIRED FOR COMPLETE INSTALLATION.
2. INCLUDE UNIONS, OR OTHER DISCONNECT MEANS, STOPS OR VALVES FOR ISOLATION OF FIXTURES AND EQUIPMENT. VALVES TO BE FULLY COMPATIBLE WITH PIPING FOR SERVICE INTENDED AS MANUFACTURED BY NIBCO, CRANE OR MILWAUKEE. INCLUDE HOSE OR DRAIN VALVES AT LOW POINTS WHERE FIXTURES CANNOT BE USED FOR DRAINAGE.
3. INSTALL SHOCK ABSORBERS AT EACH QUICK CLOSING FIXTURE AND WHERE REQUIRED TO PREVENT WATER HAMMER AS MANUFACTURED BY J.R. SMITH, SIOUX CHIEF OR ZURN. ABSORBERS SHALL BE INSTALLED IN VERTICAL UPRIGHT POSITION.
4. HANGERS ON INSULATED PIPE TO BE OUTSIDE OF INSULATION, SIZED ACCORDINGLY WITH A SUFFICIENT SADDLE TO PROTECT INSULATION AS MANUFACTURED BY GRINNELL OR MICHIGAN.
5. PIPE SCHEDULE:
    a. BELOW GRADE OUTSIDE BUILDING (2-1/2" AND LESS).
        1) SOFT COPPER ASTM B88, TYPE "K" WITHOUT JOINTS.
        2) PEX TUBE AND FITTINGS WITH STAINLESS STEEL CRIMP RINGS. NO JOINTS FOR BELOW GRADE INSTALLATION.
        3) PP, SDR11 SOCKET FITTINGS; AND FUSION-WELD JOINTS.
        4) PIPING MAY BE ROUTED INSIDE A LARGER CONDUIT SO PIPE CAN BE REMOVED WITHOUT EXCAVATION.
        5) IF MAXIMUM LENGTH OF PIPE AVAILABLE IS NOT LONG ENOUGH AND FITTING NEEDS TO BE INSTALLED UNDERGROUND, ALL FITTINGS SHALL BE STAINLESS STEEL MECHANICAL FITTINGS WRAPPED WITH POLYETHYLENE ENCASEMENT AND ENCASED IN CONCRETE.
    b. BELOW GRADE OUTSIDE BUILDING (2-1/2" AND LARGER)
        1) DUCTILE IRON CEMENT LINED IRON CEMENT LINED PIPE, AWWA H3-C1.2 DUCTILE FITTINGS, AND COMPRESSION OR MECHANICAL JOINT.
        2) PE, AWWA PIPE; PE, AWWA FITTINGS, AND HEAT-FUSION JOINTS.
        3) CPVC, SCHEDULE 40; SOCKET FITTINGS AND SOLVENT CEMENT JOINTS.
    c. BELOW GRADE INSIDE BUILDING (2-1/2" AND LESS).
        1) SOFT COPPER ASTM B88, TYPE "K" WITHOUT JOINTS.
        2) PEX TUBE AND FITTINGS WITH STAINLESS STEEL CRIMP RINGS. NO JOINTS FOR BELOW GRADE INSTALLATION.
        3) CPVC, SCHEDULE 40; SOCKET FITTINGS AND SOLVENT CEMENT JOINTS.
        4) PIPING MAY BE ROUTED INSIDE A LARGER CONDUIT SO PIPE CAN BE REMOVED WITHOUT EXCAVATION.
    d. BELOW GRADE INSIDE BUILDING (2-1/2" AND LARGER)
        1) SOFT COPPER ASTM B88, TYPE "K" WITHOUT JOINTS.
        2) CPVC, SCHEDULE 40; SOCKET FITTINGS AND SOLVENT CEMENT JOINTS.
        3) PP, SDR 11 SOCKET FITTINGS; AND FUSION-WELDING JOINTS.
    e. ABOVE GRADE (2" AND LESS)
        1) TYPE "L" HARD COPPER ASTM B 88-832 WITH WROUGHT COPPER FITTINGS ASTM B 16.22 1980 AND NON-LEAD OR ANTIMONY SOLDER JOINTS.
        2) CPVC, SCHEDULE 40; SOCKET FITTINGS AND SOLVENT CEMENT JOINTS.
        3) PEX TUBE AND FITTINGS WITH STAINLESS STEEL CRIMP RINGS.
    f. ABOVE GRADE (2-1/2" - 4")
        1) TYPE "L" HARD COPPER ASTM B 88-832 WITH WROUGHT COPPER FITTINGS ASTM B 16.22 1980 AND NON-LEAD OR ANTIMONY SOLDER JOINTS.
        2) CPVC, SCHEDULE 40; SOCKET FITTINGS AND SOLVENT CEMENT JOINTS.
        3) PEX TUBE AND FITTINGS WITH STAINLESS STEEL CRIMP RINGS.
        4) PP, SDR 11 SOCKET FITTINGS; AND FUSION-WELDING JOINTS.
    g. ABOVE GRADE (4" AND LARGER)
        1) GALVANIZED STEEL PIPE, GALVANIZED C.1 OR M.1 FITTINGS OR STAINLESS STEEL PIPE AND FITTINGS, AND SCREWED OR MECHANICAL JOINT.
        2) TYPE "L" HARD COPPER ASTM B 88-832 WITH WROUGHT COPPER FITTINGS ASTM B 16.22 1980 AND NON-LEAD OR ANTIMONY SOLDER JOINTS.
        3) CPVC, SCHEDULE 40; SOCKET FITTINGS AND SOLVENT CEMENT JOINTS.
        4) STAINLESS-STEEL SCHEDULE 10 PIPE, GROOVED-JOINT FITTINGS, AND GROOVED JOINTS.
6. FLUSH, VENT AND SANITIZE ALL WATER PIPING WITH CHLORINE AS REQUIRED PER AWWA, LOCAL BUILDING DEPARTMENT AND HEALTH DEPARTMENT CODES.
7. DOMESTIC HOT AND COLD WATER PIPING UNDER CONCRETE FLOOR TO BE COVERED WITH SAND SO THAT PIPING WILL NOT BECOME EMBEDDED IN THE FLOOR SLAB.
8. ALL PIPING UNDER CONCRETE FLOOR SHALL BE TYPE "K" SOFT COPPER, CONTINUOUS. SPLICES OR FITTINGS WILL NOT BE ALLOWED.
9. EXTREME CAUTION MUST BE TAKEN SO THAT NO COPPER PIPING AND INSULATION UNDER CONCRETE FLOORS BECOMES CRUSHED, CUT, SPLIT OR DEFORMED DURING THE POURING OF THE FLOOR SLAB.
10. ALLOW 1-1/4" PER 100 FEET OF LENGTH FOR EXPANSION IN DOMESTIC HOT WATER LINES.
11. ALL PIPING IN RETURN AIR CEILING PLENUMS OR WALLS SHALL BE PLENUM RATED MATERIALS.

F. GAS PIPING
1. INSTALL GAS PIPING IN ACCORDANCE TO THE LATEST VERSION OF THE NATIONAL FUEL AND GAS CODE, NFPA AND LOCAL GAS COMPANIES' REQUIREMENTS AND STATE AND LOCAL CODES.
2. INCLUDE METER, REGULATORS, VALVES AND CONNECT TO ALL GAS USING EQUIPMENT.
3. EQUIPMENT CONNECTIONS AT EACH UNIT SHALL INCLUDE GAS COCK, UNION, DIRT LEG, AND REDUCER TO UNIT CONNECTION SIZE. FOR ABOVE LOW PRESSURE GAS SYSTEMS, PROVIDE PRESSURE REDUCING VALVE AT EQUIPMENT OR LOW PRESSURE BRANCHES.
4. CONSTRUCT CONCRETE BASE TO BELOW FROST LINE FOR LARGE METER INSTALLATION.
5. PIPE SCHEDULE:
    a. BELOW GRADE, OUTSIDE BUILDING (<60 PSI)
        1) POLYETHYLENE PLASTIC ASTM D-2513 WITH STAB COUPLINGS OR FUSION WELD JOINTS.
        2) BLACK STEEL SCHEDULE 40 PIPE WITH WROUGHT-STEEL FITTINGS AND WELDED JOINTS, OR MECHANICAL COUPLINGS. COAT PIPE AND FITTINGS WITH PROTECTIVE COATING FOR STEEL PIPE. INSTALL CATHODIC PROTECTION ANODE ON SERVICE LINE.
    b. BELOW GRADE, OUTSIDE BUILDING (≥ 60 PSI)
        1) POLYETHYLENE PLASTIC ASTM D-2513 WITH STAB COUPLINGS OR FUSION WELD JOINTS.
        2) SCHEDULE 40 BLACK STEEL COATED AND WRAPPED WITH WELDED BLACK STEEL FITTINGS. INSTALL CATHODIC PROTECTION ANODE ON SERVICE LINE.
    c. ABOVE GRADE, LOW PRESSURE (≤ 2 PSI)
        1) SCHEDULE 40 SEAMLESS BLACK STEEL PIPE, BEVELED ENDS.
            a) 2" AND SMALLER - SCREWED FITTINGS, WROUGHT IRON.
            b) 2-1/2" AND LARGER - WELDED FITTINGS, BLACK STEEL.
    d. ABOVE GRADE, MEDIUM PRESSURE (2 - 10 PSI)
        1) SCHEDULE 40 BLACK STEEL WITH WELDED BLACK STEEL FITTINGS.
    e. UNDERGROUND, BELOW BUILDING
        1) BLACK STEEL SCHEDULE 40 PIPE WITH WROUGHT-STEEL FITTINGS AND WELDED JOINTS. PIPE IN CONTAINMENT CONDUIT WHICH IS STEEL PIPE WITH WROUGHT-STEEL FITTINGS AND WELDED JOINTS COATED WITH PROTECTIVE COATING FOR STEEL PIPE. CONDUIT TO BE VENTED TO ATMOSPHERE AT BOTH ENDS. INSTALL CATHODIC PROTECTION ANODE ON CONDUIT.
    f. VALVES SHALL NOT BE LOCATED ABOVE CEILING SPACES USED AS A RETURN AIR PLENUM.
    g. EXTERIOR EXPOSED BARE STEEL PIPE SHALL BE PAINTED WITH A PRIMER COAT AND TWO (2) COATS OR RUST INHIBITIVE PAINT, COLOR AS SELECTED BY ARCHITECT.
    h. ALL WELDING SHALL BE PERFORMED BY STATE CERTIFIED WELDERS.
    i. ALL PIPING IN NON-ACCESSIBLE SPACES SHALL HAVE WELDED JOINTS.

G. CONDENSATE DRAIN PIPING
1. TRAP SHALL BE INSTALLED NEAR EQUIPMENT IF NOT INTEGRAL WITH EQUIPMENT. INSTALL PIPING AT A UNIFORM SLOPE OF 1" IN FORTY FEET DOWNWARD INDIRECTION TO DRAIN.
2. PIPE SCHEDULE:
    a. PIPING OF ALL SIZES SHALL BE TYPE L HARD COPPER PIPE WITH BRASS OR COPPER FITTINGS AND SOLDERED JOINT.
    b. PIPING OF ALL SIZE FOR MECHANICAL ROOMS, ROOF AND NON-RETURN AIR PLENUM RATED CEILING SPACE SHALL BE TYPE PVC, SCHEDULE 40 WITH SOLVENT WELD SOCKET FITTINGS.

H. GENERAL HYDRONIC AND DOMESTIC VALVES AND STRAINERS
1. BALL VALVES 2" AND SMALLER SHALL BE 600# WOG, 150# SWP, TWO-PIECE, FULL PORT CAST BRONZE OR FORGED BRASS BODY, CHROME PLATED BRASS BALL, REPLACEABLE "TEFLON" OR "TFE" SEATS AND SEALS, BLOWOUT-PROOF STEM, VINYL-COVERED STEEL HANDLE AND HAVE THREADED ENDS. VALVES SHALL BE HAMMOND 8901, MILWAUKEE BA-125, STOCKHAM S-207, NIBCO T-585, OR APOLLO 77-100.
2. GATE VALVES 2" AND SMALLER SHALL BE 150# W.S.P., BRONZE, SCREWED PATTERN WITH RISING STEM, UNION BONNET, SOLID WEDGE DISC. VALVES SHALL CONFORM TO MSS SP-80. VALVES SHALL BE CRANE 431-UB, HAMMOND IR-629, LUNKENHEIMER 3151, JENKINS 47-U, STOCKHAM B-120, MILWAUKEE 1151, OR NIBCO T-135.
3. GATE VALVES 2-1/2" AND LARGER SHALL BE 125# W.S.P., CAST IRON, OUTSIDE SCREW AND YOKE, FLANGED PATTERN WITH BOLTED BONNET, RISING STEM, BRONZE SEATS, STEM AND DISC FACES. VALVES SHALL CONFORM TO MSS SP-70. VALVES SHALL BE CRANE 465-1/2, HAMMOND IR 1140, LUNKENHEIMER 1430, JENKINS 651-A, STOCKHAM G-623, MILWAUKEE F2885, OR NIBCO F-617-0.
4. GLOBE VALVES 2" AND SMALLER SHALL BE 150# W.S.P., BRONZE, SCREWED PATTERN WITH RISING STEM, AND UNION BONNET, AND ANSI 420-S STAINLESS STEEL TAPERED PLUG AND SEAT. VALVES SHALL CONFORM TO MSS SP-80. VALVES SHALL BE CRANE 14-1/2P, HAMMOND B-433, JENKINS 546-P, STOCKHAM B-29, MILWAUKEE 591 A, OR NIBCO T-235.
5. GLOBE VALVES 2-1/2" AND LARGER SHALL BE 125# W.S.P., CAST IRON, OUTSIDE SCREW AND YOKE, FLANGED PATTERN, WITH BOLTED BONNET, RISING STEM, BRONZE SEAT, STEM, AND DISC. VALVES SHALL CONFORM TO MSS SP-85 SPECIFICATION A-126, CLASS B. VALVES SHALL BE CRANE 351, HAMMOND IR 116, LUNKENHEIMER 1123, STOCKHAM G-512, MILWAUKEE F2981, OR NIBCO F-718-B.
6. CHECK VALVES 2" AND SMALLER SHALL BE 150# W.S.P. BRONZE, SWING CHECK, BRONZE SEAT, SCREWED PATTERN. VALVES SHALL CONFORM TO MSS SP 80, TYPE 4. VALVES SHALL BE CRANE 141, HAMMOND IB-946, MILWAUKEE 510, NIBCO T-43, STOCKHAM B-331B, LUNKENHEIMER 230, OR JENKINS 352.
7. CHECK VALVES 2-1/2" AND LARGER SHALL BE 125# W.S.P., CAST IRON, FLANGED PATTERN WITH ALL BRONZE TRIM. VALVES SHALL CONFORM TO MSS SP-71. VALVES SHALL BE CRANE 373, HAMMOND IR-1124, LUNKENHEIMER 1790, JENKINS 624, STOCKHAM G-931, MILWAUKEE F2974, OR NIBCO F-918-B.
8. BUTTERFLY VALVES (ALL SIZES) SHALL BE LUG STYLE DUCTILE IRON BODY AND EXTENDED NECK, DESIGNED FOR BUBBLE TIGHT SHUT-OFF AT 150 PSI. DISCS SHALL BE BRONZE OR ALUMINUM BRONZE AND ATTACHED TO STAINLESS STEEL STEM. SEAT MATERIAL SHALL BE EPDM (EPT) SUITABLE TO 275 DEGREES F. STANDARD VALVE MUST BE CAPABLE OF 200 PSI BI-DIRECTIONAL DIFFERENTIAL PRESSURE (DEAD-END SERVICE). MANUFACTURER MUST CERTIFY VALVES (SIZES 2" THRU 16") THAT SHALL BE CAPABLE OF PROVIDING BUBBLE TIGHT SEAL AT 200 PSI WHEN USED FOR END OF LINE SERVICE WITHOUT THE NEED OF A FLANGE ON THE DOWN STREAM SIDE. VALVES 18" AND LARGER MUST BE CAPABLE OF 150 PSI BI-DIRECTIONAL DIFFERENTIAL PRESSURE AND 150 PSI END OF LINE SERVICE AS STATED ABOVE. VALVES THRU 6" SHALL HAVE 10 POSITION LEVERLOCK HANDLE. VALVES 8" AND LARGER SHALL HAVE GEAR OPERATORS WITH HAND WHEELS. VALVES SHALL CONFORM TO MSS SP-67 AND MSS SP-25. VALVES SHALL BE CRANE 44 BXZ, DESURIK, MILWAUKEE ML 133E, NIBCO LD 2000, OR STOCKHAM LD-7.
9. ECCENTRIC PLUG (BALANCE) VALVES 2" AND SMALLER, SHALL BE 175#, CAST IRON BODY, BRONZE TRIM, SCREWED PATTERN WITH OPEN POSITION STOP. VALVES TO BE DEZURIK FIGURE 425 OR DRESSER WITH FIGURE 487 MEMORY STOP AND RESILIENT PLUG SEAL RATED AT 250 DEGREES F.
10. BALANCE VALES 2" AND SMALLER SHALL BE 125#, BRONZE BODY SCREWED PATTERN BALL TYPE CIRCUIT SETTER VAL

d. DEFECTIVE WORK OR MATERIALS SHALL BE REPLACED AND INSPECTION AND TESTS REPEATED WITHIN THREE DAYS.

13. CERTIFICATES OF APPROVAL OF SATISFACTORY COMPLETION AND FINAL INSPECTION SHALL BE OBTAINED BY THE PLUMBING CONTRACTOR. ONE COPY OF EACH APPROVAL SHALL BE GIVEN TO THE ARCHITECT.
14. DAMAGES WHICH RESULT FROM BREAKAGE OR FAULTY INSTALLATION SHALL BE THE RESPONSIBILITY OF THE PLUMBING CONTRACTOR.
15. AFTER THE SYSTEM HAS BEEN IN SERVICE FOR A TWO-WEEK PERIOD AND AGAIN BEFORE THE SYSTEM IS TURNED OVER TO THE OWNER, ALL DIRT POCKETS, TRAPS, AND STRAINERS SHALL BE CLEANED, REMOVED, AND REINSTALLED.

D. WATER USING EQUIPMENT: ALL WATER USING EQUIPMENT, SUCH AS BUT NOT LIMITED TO, COOLING COILS AND CONVERTORS, SHALL BE BALANCED TO OBTAIN THE REQUIRED WATER PRESSURE DROP AND FLOW. THIS CONTRACTOR SHALL LIST THE FLOW RATE AND REQUIRED PRESSURE DROP AND THE OBSERVED PRESSURE DROP FOR EACH PIECE OF EQUIPMENT.

E. AIR HANDLING EQUIPMENT: FOR EACH PIECE OF AIR HANDLING EQUIPMENT, THIS CONTRACTOR SHALL LIST THE DATA OF THE FAN, MOTOR AND DRIVE AND SHALL OBTAIN BY MEASUREMENT AND FURNISH TO THE ARCHITECT/ENGINEER THE FAN SPEED, MOTOR VOLTAGE, OPERATING AMPS, FOR CFM AND STATIC PRESSURE AS DETERMINED FROM THE MANUFACTURER'S FAN CURVES. THIS CONTRACTOR SHALL ALSO DETERMINE THE FAN CFM BY MEANS OF A VELOCITY TRAVERSE WHICH SHALL BE TAKEN A MINIMUM OF THREE FAN DIAMETERS FROM FAN OUTLET. BEFORE RUNNING ANY TESTS, THE CONTRACTOR SHALL HAVE INSTALLED ALL THE COMPONENTS OF THE SYSTEM AND SHALL ENSURE THE CLEANLINESS OF THE FILTERS.

F. DIFFUSERS, REGISTERS, GRILLES: AFTER COMPLETION OF THE AIR DISTRIBUTION SYSTEMS AND FINAL ADJUSTMENTS, THE CONTRACTOR SHALL ADJUST ALL DAMPERS AND AIR SUPPLY, RETURN AND EXHAUST OUTLETS SO THAT EACH OUTLET HANDLES ITS PROPER QUANTITY OF AIR. SUPPLY REGISTERS AND DIFFUSERS SHALL BE ADJUSTED TO PROVIDE FOR THE PROPER THROW AND A UNIFORM DISTRIBUTION PATTERN.

1. FOR SUPPLY, RETURN AND EXHAUST AIR OUTLETS, THE VELOCITY SHALL BE MEASURED WITH A HEATED WIRE RESISTANCE TYPE ANEMOMETER HELD 1" FROM THE FACE OF THE OUTLETS; THE AIR VELOCITY SHALL BE THE AVERAGE OF VELOCITY READINGS TAKEN AT POINTS NO MORE THAN 6" APART. THE AREA SHALL BE THE NET CORE AREA OF THE OUTLET.
2. TEST READINGS SHALL BE TAKEN FOR EACH REGISTER, GRILLE AND DIFFUSER. FOR EACH OF THESE UNITS, OBTAIN AND FURNISH INFORMATION ON MANUFACTURER, TESTING EQUIPMENT USED, PROCEDURE FOLLOWED, LOCATION, SIZE, AVERAGE, VELOCITY, GROSS AND NET CORE AREAS, OBSERVED CFM AND SPECIFIED CFM. SEPARATE TABULATIONS SHALL BE FURNISHED FOR EACH MANUFACTURER, EACH SYSTEM AND EACH TYPE OF REGISTER, GRILLE AND DIFFUSER.

G. MIXING DAMPERS: MIXING DAMPERS SHALL BE ADJUSTED ON THE BASIS OF THE TEMPERATURE OF THE MIXED OUTSIDE AND RETURN AIRSTREAMS.

1. THE MINIMUM FRESH AIR DAMPER POSITION SHALL BE SET TO OBTAIN A MIXED AIR TEMPERATURE DETERMINED FROM THE FOLLOWING EQUATION:

   MIXED AIR TEMPERATURE =

   (OUTSIDE AIR TEMP.) X (MIN. PERCENT OUTSIDE AIR) +

   (RETURN AIR TEMP.) X (1 - MIN. PERCENT OUTSIDE AIR)
2. PERCENTAGE OF AIR QUANTITY SHALL BE EXPRESSED AS A FRACTION OF THE TOTAL AIR SUPPLY.

H. HOLES IN DUCTS AND CASINGS USED FOR STATIC PRESSURE AND VELOCITY READINGS SHALL BE PROVIDED WITH REMOVABLE CLOSURES.

I. DURING THE TESTING PERIOD, THIS CONTRACTOR SHALL MAINTAIN ON THE JOB A COMPETENT INDIVIDUAL THOROUGHLY FAMILIAR WITH ALL PHASES OF AIR CONDITIONING, INCLUDING REFRIGERATION, TEMPERATURE CONTROL AND DISTRIBUTION, FOR AS LONG A PERIOD AS MAY B REQUIRED TO THOROUGHLY ADJUST ALL OF THE SYSTEMS AND TO DEMONSTRATE TO THE ARCHITECT THAT THEY ARE FUNCTIONING PROPERLY.

J. THE TESTING AND BALANCING ENGINEER SHALL, AS PART OF HIS WORK, PERFORM A "SPOT" RE-CHECK BALANCING CONDITIONS BETWEEN 30 TO 90 DAYS AFTER BOTH SUMMER AND WINTER BALANCING OPERATIONS AT WHICH TIME A REPRESENTATIVE OF THE TEMPERATURE CONTROL MANUFACTURER CAPABLE OF PERFORMING ADJUSTMENTS TO HIS SYSTEM SHALL ACCOMPANY THE BALANCING ENGINEER. THIS OPERATION SHALL INCLUDE A CHECK OF SPACE TEMPERATURE, CALIBRATION OF CONTROLS, PUMP AND FAN PERFORMANCE AND THE NECESSARY ADJUSTMENTS THERETO.

SECTION 200700 - INSULATION

A. GENERAL

1. FURNISH ALL MATERIAL, LABOR AND EQUIPMENT AS REQUIRED TO INSTALL COMPLETE PLUMBING AND HVAC INSULATION AS INDICATED ON MECHANICAL DRAWINGS AND IN THESE SPECIFICATIONS.
2. INSTALL IN FULL ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.

B. SCOPE: THIS CONTRACTOR SHALL FURNISH AND INSTALL ALL INSULATION NECESSARY TO THE PROJECT AND IN ACCORDANCE WITH THE FOLLOWING REQUIREMENTS. ALL INSULATION AND ACCESSORIES USED IN AN AIR PLENUM SPACE, AND ALL DUCT COVERING AND LINING, REGARDLESS OF PHYSICAL LOCATION, SHALL HAVE A COMPOSITE (INSULATION, JACKET, AND ADHESIVE) FIRE AND SMOKE HAZARD RATING AS TESTED UNDER PROCEDURE ASTM E-84, NFPA 255 AND UL 723, NOT EXCEEDING A FLAME SPREAD 25 AND SMOKE DEVELOPED 50. ALL OTHER AREAS SHALL HAVE INSULATING MATERIALS AND ACCESSORIES ON PIPES AND VESSELS RATED AT A FLAME SPREAD 25 AND SMOKE DEVELOPED 150 AS TESTED BY THE SAME PROCEDURE. ALL CALCIUM SILICATE SHALL BE ASBESTOS FREE.

C. WORKMANSHIP:

1. ALL INSULATION SHALL BE INSTALLED OVER CLEAN, DRY SURFACES. INSULATION MUST BE DRY AND IN GOOD CONDITION. WET OR DAMAGED INSULATION WILL NOT BE ACCEPTABLE. NO INSULATION SHALL BE APPLIED PRIOR TO PRESSURE TEST COMPLETION OF THE RESPECTIVE PIPING AND/OR DUCT SYSTEM.
2. ALL PIPE INSULATION SHALL BE INSTALLED WITH JOINTS BUTTED FIRMLY TOGETHER. ALL VALVES AND FITTINGS SHALL BE INSULATED USING MITERED SECTIONS OF INSULATION EQUAL IN DENSITY AND THICKNESS TO THE ADJOINING INSULATION, OR WITH AN INSULATION CEMENT EQUAL IN THICKNESS TO THE ADJOINING INSULATION OR PREMOLDED INSULATED FITTINGS. THE INSULATION APPLIED TO THE VALVES AND FITTINGS SHALL BE COVERED WITH THE SAME TYPE OF COVERING AS USED ON THE PIPE INSULATION. NO STAPLES.
3. ALL INSULATION ENDS SHALL BE TAPERED AND SEALED REGARDLESS OF SERVICES.
4. ALL INSULATED, EXPOSED PIPING 8'-0" AND BELOW TO THE FINISHED FLOOR SHALL INCLUDE A 0.020" THICK VINYL JACKET. THIS JACKET IS IN ADDITION TO THE NORMAL FINISH FOR THE RESPECTIVE SERVICE.
5. RIGID DUCT INSULATION SHALL BE IMPALED OVER WELDED PINS AND SECURED WITH WHITE INSULATION CAPS. ALL SEAMS SHALL BE FIRMLY BUTTED AND SEALED WITH WHITE PRESSURE SENSITIVE VAPOR BARRIER TAPE. NO STAPLES.
6. WRAP AROUND DUCT INSULATION SHALL BE APPLIED WITH ALL JOINTS BUTTED FIRMLY TOGETHER. INSULATION SHALL BE CEMENTED TO THE SURFACE WITH FIREPROOF ADHESIVE APPLIED IN 6" WIDE STRIPS ON 12" CENTERS. ALL JOINTS IN THE INSULATION COVERING SHALL BE SEALED WITH ADHESIVE. WHERE DUCTS ARE OVER 24" WIDE, THE DUCTWRAP SHALL BE ADDITIONALLY SECURED TO BOTTOM OF RECTANGULAR OR OVAL DUCTS WITH MECHANICAL FASTENERS ON 16" CENTERS TO PREVENT SAGGING. VAPOR BARRIER SHALL BE LEGIBLY PRINTED BY THE MANUFACTURER TO SHOW NOMINAL THICKNESS AND TYPE OF INSULATION. ALUMINUM CORNER ANGLES SHALL BE USED TO PREVENT OVER COMPRESSING INSULATION DURING INSTALLATION.
7. DUCTLINER INSULATION SHALL BE APPLIED WITH JOINTS PRECOATED WITH ADHESIVE AND BUTTED FIRMLY TOGETHER. LINING SHALL BE CEMENTED TO DUCTWORK WITH A MINIMUM OF 75 PERCENT COVERAGE OF FIRE RESISTANT ADHESIVE. MECHANICAL FASTENERS ON 16" CENTERS AND ADHESIVE SHALL BE USED WHEN DUCT WIDTH EXCEEDS 12" OR WHEN DUCT HEIGHT EXCEEDS 24".
8. ALL DUCTWORK IN THE MECHANICAL ROOMS IS TO BE CONSIDERED AS "EXPOSED DUCTWORK," I.E. SUPPLY, RETURN, RELIEF, AND OUTDOOR AIR.
9. ALL ROUND DIFFUSER DUCT DROPS CONNECTED TO LINED DUCTWORK SHALL BE INSULATED THE SAME AS "DUCTWORK" SCHEDULE NON-LINED.
10. ALL FLEXIBLE ELASTOMERIC INSULATION SHALL HAVE ALL FITTINGS, BUTT ENDS, AND SEAMS SEALED WITH VAPOR BARRIER ADHESIVE.
11. PROVIDE REMOVABLE INSULATION SECTIONS TO COVER PARTS OF EQUIPMENT WHICH MUST BE OPENED PERIODICALLY FOR MAINTENANCE INCLUDING METAL VESSEL COVERS, FASTENERS, FLANGES, CHILLED WATER BOXES, FRAMES AND ACCESSORIES.
12. REPAIR ALL DAMAGED SECTIONS OF THE EXISTING PIPING AND MECHANICAL INSULATION DAMAGED DURING THIS CONSTRUCTION PERIOD. USE INSULATION OF SAME THICKNESS AS EXISTING INSULATION. INSTALL NEW JACKET LAPPING AND SEAL OVER EXISTING.
13. REPLACE DAMAGED INSULATION WHICH CANNOT BE REPAIRED SATISFACTORILY, INCLUDING UNITS WITH VAPOR BARRIER DAMAGE AND MOISTURE SATURATED UNITS.

D. PLUMBING INSULATION (AS MANUFACTURED BY OWENS CORNING, KNAUF OR SCHULLER)

1. INSULATE ALL ABOVE-GRADE HOT WATER, HOT WATER RETURN AND COLD WATER PIPING WITH 1" THICK MOLDED FIBERGLASS HAVING AN ALL SERVICE JACKET.
2. INSULATE ALL ABOVE-GRADE, HORIZONTAL AIR CONDITIONING CONDENSATE FLOOR DRAINS AND WASTE LINES, OVERFLOW ROOF DRAINS AND PIPING, ROOF DRAINS AND PIPING AND ROOF DRAIN SUMPS WITH 1" THICK MOLDED FIBERGLASS HAVING AN ALL SERVICE JACKET.
3. INCLUDE INSULATION OF FITTINGS AND VALVES. KEEP VAPOR BARRIERS INTACT. APPLY PER MANUFACTURER'S RECOMMENDATIONS.
4. INSULATE ALL EXPOSED WASTE AND WATER SUPPLY PIPING UNDER LAVATORY WITH SAFETY COVERS PER ADA REQUIREMENTS (AS MANUFACTURED BY PLUMBEREX SPECIALTY PRODUCTS, MCGUIRE OR TRUEBRO).

E. HVAC INSULATION (AS MANUFACTURED BY OWENS CORNING, KNAUF)

1. ALL INSULATION TO BE APPLIED IN FULL ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS AND COMPLY WITH 25/50 FLAME AND SMOKE HAZARD RATINGS PER ASTM E-84, NFPA 255 AND UL 723.
2. INSULATE ALL SUPPLY, RETURN, OUTSIDE AND EXHAUST AIR DUCTS WITH 3/4" THICK LINED INSULATION OR LESS TO NONE WITH 1-1/2" THICK, 1.5 PCF, R-6, FOIL FACED REINFORCED KRAFT JACKET FIBERGLASS DUCT WRAP FULLY SECURED TO DUCT. LAP AND TAPE SEAMS AND SECURE TIGHTLY TO THE DUCTS WITH WIRE OR STICK PINS. EXPOSED DUCTWORK IN CONDITIONED SPACES WITHOUT CEILINGS SHALL NOT BE INSULATED, UNLESS OTHERWISE NOTED TO BE INSULATED. DUCTWORK IN CEILING PLENUM SPACE SHALL BE INSULATED.
3. INSULATE ALL SUPPLY, RETURN, OUTSIDE AND EXHAUST AIR DUCTS LINED OR NOT LINED LOCATED IN THE ATTIC SPACE WITH 3" THICK, 0.75 PCF NOMINAL DENSITY, R-10, FOIL FACED REINFORCED KRAFT JACKET, FIBERGLASS DUCT WRAP FULLY SECURED TO DUCT. LAP AND TAPE SEAMS AND SECURE TIGHTLY TO THE DUCTS WITH WIRE OR STICK PINS.
4. INSULATE ALL SUPPLY, RETURN, OUTSIDE AND EXHAUST AIR DUCTS LINED OR NOT LINED EXPOSED TO WEATHER OUTSIDE WITH 3" THICK MINERAL FIBER BOARD, 3 PCF NOMINAL DENSITY, R-13. PROVIDE A VENTURE CLAD INSULATION JACKETING, COLOR AS SELECTED BY ARCHITECT, FIELD-APPLIED JACKET.
5. INSULATE ALL AIR CONDITIONING CONDENSATE DRAIN PIPING WITH 1" THICK MOLDED FIBERGLASS INSULATION, C = 0.22.
6. INSULATE EXHAUST DUCTS FOR KITCHEN HOODS WITH TWO (2) LAYERS OF FIRE RATED 1" THICK INSULATING WOOL RATED FOR 1000 DEGREES F OR FIREMASTER DUCTWRAP AS MANUFACTURED BY 3M FOR A MINIMUM OF 2 HR RATING. PROVIDE REMOVABLE SECTIONS AT DUCT CLEANOUT PANELS.
7. INSULATE ALL SINGLE WALL BOILER BREECHING INSIDE THE BUILDING WITH A 3" TICK CALCIUM SILICATE WITH FIELD APPLIED 8-1/2 OZ. GLASS CLOTH LAGGING WRAP.
8. INSULATE ALL SINGLE WALL HOT WATER FLUES INSIDE THE BUILDING WITH A 2" THICK CALCIUM SILICATE WITH FIELD APPLIED 8-1/2 OZ. GLASS CLOTH LAGGING WRAP.
9. INSULATE ALL HOT WATER TANKS, CONVERTERS, COLD WATER STORAGE TANKS, CHILLED WATER PUMPS AND CHILLED WATER AIR SEPARATORS WITH A 2" THICK FIBERGLASS SERVICE BOARD 6 PCF NOMINAL DENSITY, R-10, WITH FIELD APPLIED 8-1/2 OZ. GLASS CLOTH LAGGING WRAP. DEVICES MAY BE INSULATED WITH 2" THICK FLEXIBLE ELASTOMERIC INSULATION, C = 0.24. ALL COLD WATER DEVICES SHALL HAVE A VAPOR SEAL.
10. INSULATE ALL REFRIGERANT SUCTION AND HOT GAS LINES WITH 1" ELASTOMERIC FOAM INSULATION, C = 0.24 WITH JOINTS AND SEAMS SEALED VAPOR TIGHT. INSULATION OUTSIDE SHALL BE PAINTED WITH TWO COATS OF PROTECTIVE COATINGS PER MANUFACTURER FOR PROTECTION TO WEATHER (AS MANUFACTURED BY AEROFLEX, ARMOCELL, OR K-FLEX).

SECTION 211000 - FIRE PROTECTION SYSTEMS

A. GENERAL

1. FURNISH ALL LABOR, MATERIALS AND EQUIPMENT AS REQUIRED TO INSTALL A COMPLETE FIRE PROTECTION SYSTEM FOR PROJECT.
2. FIELD-VERIFY SIZES AND LOCATION OF EXISTING SPRINKLER PIPING BEFORE FABRICATION OF NEW.
3. THIS CONTRACTOR SHALL BE RESPONSIBLE FOR THE REMOVAL AND REINSTALLATION OF EXISTING CEILING TILES, AS REQUIRED, FOR THE INSTALLATION OF WORK SHOWN IN AREAS WHERE EXISTING CEILINGS ARE TO REMAIN. SEE ARCHITECTURAL DRAWINGS FOR AREAS WHERE EXISTING CEILINGS ARE TO REMAIN.
4. THIS REMOVAL AND REINSTALLATION OF EXISTING LAY-IN CEILING TILES SHALL BE THE RESPONSIBILITY OF THE FIRE PROTECTION CONTRACTOR (UNDER THE SUPERVISION OF THE GENERAL CONTRACTOR) AS REQUIRED TO PERFORM HIS WORK. ANY DAMAGE TO EXISTING CEILING TILES OR SUPPORTS SHALL BE THE RESPONSIBILITY OF THE GENERAL CONTRACTOR. CEILING TILES MAY BE LEFT OUT OF THE CEILING AREAS UNDER CONSTRUCTION ONLY IF STORED IN AREAS AS DIRECTED BY THE OWNER SO AS NOT TO HINDER THE DAILY OPERATIONS OF THE BUILDING OCCUPANTIONS.
5. THIS CONTRACTOR SHALL MODIFY AND RELOCATE SPRINKLER PIPING AND PROVIDE NEW SPRINKLER PIPING AND HEADS, AS REQUIRED, TO ACCOMMODATE NEW MECHANICAL WORK IN FULL COMPLIANCE WITH NFPA 13. THIS CONTRACTOR SHALL ALSO PERFORM HYDRAULIC CALCULATIONS FOR SPRINKLER PIPING IN THE REMODELED AREAS IN ACCORDANCE WITH NFPA 13.

B. DESIGN BASIS

1. DESIGN BASIS FOR SYSTEM SHALL BE PER NFPA 13 (LATEST EDITION) BUILDING CODE REQUIREMENTS, LOCAL WATER DEPARTMENT, LOCAL FIRE DEPARTMENT, STATE FIRE MARSHAL, LOCAL CODE, AND OWNER AND OWNER'S FIRE INSURANCE UNDERWRITER REQUIREMENTS.
2. SYSTEM SHALL BE HYDRAULICALLY CALCULATED AS REQUIRED BY CODE.
3. PIPE SIZES INDICATED ON DRAWING ARE APPROXIMATE AND SHALL BE VERIFIED PER THE CONTRACTOR'S HYDRAULIC CALCULATIONS.

C. DRAWINGS AND CALCULATIONS

1. CONTRACTOR SHALL PREPARE SUBMITTAL DRAWINGS AND HYDRAULIC CALCULATIONS WITH A 10% FACTOR OF SAFETY FOR BUILDING IN ACCORDANCE WITH OWNER'S INSURANCE COMPANY BUILDING DEPARTMENT, AND LOCAL FIRE AUTHORITY REQUIREMENTS, TENANT'S REQUIREMENTS FOR DESIGN DENSITY, WHICHEVER IS MOST STRINGENT.
2. CONTRACTOR SHALL PERFORM A FLOW TEST DATA ON WATER MAIN AND SUBMIT DATA WITH CALCULATIONS.
3. IT IS THE FIRE PROTECTION CONTRACTOR'S RESPONSIBILITY TO VERIFY EACH TENANT'S DESIGN DENSITY WITH AGREED UPON LEASE DOCUMENTATION AND THAT TENANT'S PROTOTYPE OR INSURANCE UNDERWRITERS REQUIREMENTS.
4. PROVIDE WET STANDPIPE SYSTEM FOR PROJECT IN ACCORDANCE WITH NFPA 14 REQUIREMENTS.
5. CONTRACTOR AND DESIGNER SHALL BE STATE CERTIFIED.
6. COORDINATE LAYOUT AND INSTALLATION OF SPRINKLERS WITH DUCTWORK AND EQUIPMENT ABOVE CEILINGS AND OTHER CONSTRUCTION THAT PENETRATES CEILINGS, INCLUDING BUT NOT LIMITED TO LIGHT FIXTURES, SPEAKERS, HVAC EQUIPMENT, DOORS AND PARTITION ASSEMBLIES. NO SPRINKLER PIPING SHALL BE ROUTED BENEATH EQUIPMENT ABOVE ANY CEILINGS THAT MUST BE DROPPED DIRECTLY DOWN FOR SERVICE, REPAIR, OR REPLACEMENT.
7. EXAMINE AREAS AND CONDITIONS UNDER WHICH FIRE PROTECTION MATERIALS AND PRODUCTS ARE TO BE INSTALLED. DO NOT PROCEED WITH WORK UNTIL UNSATISFACTORY CONDITIONS HAVE BEEN CORRECTED IN MANNER ACCEPTABLE TO INSTALLER. SCHEDULE ROUGH-IN INSTALLATIONS WITH INSTALLATIONS OF OTHER BUILDING COMPONENTS.
8. SHOP DRAWINGS REVIEW DOES NOT RELIEVE FIRE PROTECTION CONTRACTOR FROM RESPONSIBILITY TO MEET EACH TENANT'S REQUIREMENTS FOR SPRINKLER COVERAGE.
9. FIRE PROTECTION CONTRACTOR IS RESPONSIBLE FOR VERIFYING ANY HIGH PILE STORAGE REQUIREMENTS OF FUTURE TENANTS AND PROVIDING AN INCOMING SPRINKLER SERVICE SIZE AND RISERS TO MEET THE REQUIREMENTS FOR ADEQUATE SPRINKLER COVERAGE.

D. PIPING

1. ALL PIPING SHALL BE INSTALLED IN ACCORDANCE WITH NFPA 13, 14 (LATEST EDITION) AND LOCAL CODE REQUIREMENTS .
2. FIRE PROTECTION PIPING SHALL BE AS FOLLOWS:
   a. BELOW-GRADE OUTSIDE BUILDING - DUCTILE IRON, CEMENT LINED. CLASS OF PIPE AS DIRECTED BY LOCAL WATER PURVEYOR WITH MECHANICAL OR PUSH-ON TYPE JOINTS.
   b. INSIDE BUILDING - PIPE AND TUBING SHALL BE STEEL OR COPPER IN ACCORDANCE WITH NFPA REQUIREMENTS.
   c. PIPING SHALL MATCH EXISTING BUILDING STANDARDS.
   d. CONTRACTOR SHALL ARRANGE WITH OWNER AND INSURANCE UNDERWRITER PRIOR TO SHUT DOWN OF EXISTING SYSTEMS.
   e. FLUSH ALL PIPINGS UPON COMPLETION OF PROJECT AND TEST PER NFPA REQUIREMENTS.
   f. NO PIPING SHALL BE INSTALLED AT LOCATIONS SUBJECT TO FREEZING.
3. EXCAVATION AND BACKFILL - SEE SECTION 200510, BASIC MATERIALS AND METHODS.

E. SPRINKLER HEADS

1. SPRINKLER HEADS SHALL BE UL LISTED, MATCH EXISTING BUILDING STANDARDS AND BE MANUFACTURED BY CENTRAL, STAR OR VIKING.
2. SPRINKLER HEADS SHALL BE AS FOLLOWS:
   a. AREAS WITH EXPOSED STRUCTURE
3. UPRIGHT - ROUGH BRASS.
   b. AREAS WITH CEILINGS
4. RECESSED PENDENT - CHROME PLATED WITH MATCHING TWO (2) PIECE, FLUSH ESCUTCHEON.
5. CONCEALED - BRASS FINISH WITH OFF-WHITE CEILING COVER PLATE.
6. SIDEWALL - CHROME PLATED WITH OFF-WHITE, TWO (2) PIECE, SEMI-RECESSED ESCUTCHEON.
7. INSTALL CONCEALED HEADS WITH WHITE FLUSH MOUNTED COVER PLATE IN (SALES AREA).
8. INSTALL HIGHER TEMPERATURE SPRINKLER HEADS WHERE REQUIRED BY CODE OR APPLICATION.
9. SPRINKLER HEADS SHALL BE LOCATED IN THE CENTER OF CEILING TILES OR THE CENTER OF AN AREA OF A 24" X 24" TILE SECTION. SEE ARCHITECTURAL REFLECTED CEILING PLANS.
10. SUBMIT SAMPLES OF SPRINKLER HEADS TO ARCHITECT PRIOR TO FABRICATION OF ANY PIPING.
11. INSTALL INSPECTOR'S TEST CONNECTION WITH VALVE AND TERMINATE DRAIN THROUGH EXTERIOR WALL WITH TEXT FITTING AND SPLASH BLOCK.

F. VALVES

1. INSTALL ALL VALVES AS REQUIRED BY NFPA 13, UL OR FM LISTED AND AS MANUFACTURED BY GRINNELL, HAMMOND OR MILWAUKEE.
2. ALL SHUT-OFF VALVES SHALL BE FITTED WITH TAMPER SWITCHES BY FIRE PROTECTION CONTRACTOR AND WIRED BY ELECTRICAL CONTRACTOR. TAMPER SWITCHES SHALL BE AS MANUFACTURED BY NOTIFIER, POTTER OR VIKING.
3. INSTALL FLOW SWITCH IN RISER AS MANUFACTURED BY NOTIFIER, POTTER OR VIKING AND WIRED BY ELECTRICAL CONTRACTOR.
4. INSTALL UL LISTED ALARM CHECK VALVE WITH ALL REQUIRED TRIM, INCLUDING WATER MOTOR ALARM BELL AND DRAINS AS MANUFACTURED BY CENTRAL STAR OR VIKING.
5. INSTALL WALL MOUNTED INDICATOR VALVE AS MANUFACTURED BY POTTER ROEMER, CROKER OR ELKHART AND APPROVED BY LOCAL AUTHORITIES.
6. INSTALL DOUBLE CHECK DETECTOR ASSEMBLY BACKFLOW PREVENTER, AS REQUIRED BY LOCAL WATER PURVEYOR AND AS MANUFACTURED BY WATTS, ZURN OR CONBRACO.

G. FIRE DEPARTMENT CONNECTION

1. WALL TYPE SIAMESE CONNECTIONS SHOULD BE POLISHED CAST BRASS WALL TYPE WITH WALL MOUNTED TWO-WAY CONNECTIONS. CONNECTION SIZES SHALL BE 4" OUTLET AND TWO - 2-1/2" FEMALE INLETS, HAVING NH (STORZ) STANDARD THREADS FOR THE CONNECTION SIZE INDICATED AS SPECIFIED IN NFPA . PROVIDE THE QUANTITY, SIZE AND TYPE OF CONNECTIONS AS REQUIRED BY THE LOCAL FIRE DEPARTMENT. EACH INLET SHALL HAVE A CLAPPER VALVE AND PLUG AND CHAIN. EACH INLET SHALL HAVE AN ESCUTCHEON OF CAST BRASS, FINISH TO MATCH CONNECTIONS, WITH WORDS "AUTO SPKR - FIRE DEPT. CONNECTION" IN RAISED LETTERS.
2. MANUFACTURER: SUBJECT TO COMPLIANCE WITH REQUIREMENTS, PROVIDE WALL TYPE SIAMESE CONNECTION OF ONE OF THE FOLLOWING:
   a. POTTER-ROEMER
   b. GUARDIAN FIRE EQUIPMENT, INC.
   c. CROKER
   d. ELKHART

H. EXTRA MATERIALS

1. VALVE WRENCHES: FURNISH TO OWNER, 2 VALVE WRENCHES FOR EACH TYPE OF SPRINKLER HEAD INSTALLED.
2. SPRINKLER HEADS AND CABINETS: FURNISH 2 EXTRA SPRINKLER HEADS OF EACH STYLE INCLUDED IN THE PROJECT. FURNISH EACH STYLE WITH ITS OWN SPRINKLER HEAD CABINET AND SPECIAL WRENCHES.
3. OBTAIN RECEIPT FROM OWNER THAT EXTRA STOCK HAS BEEN RECEIVED AND GIVE ARCHITECT A COPY OF THIS RECEIPT.

SECTION 224000 - PLUMBING FIXTURES AND EQUIPMENT

A. GENERAL

1. FURNISH ALL FIXTURES AND EQUIPMENT INDICATED AND SCHEDULED ON DRAWINGS, COMPLETE WITH ALL ACCESSORIES, CONTROLS, ETC., AS REQUIRED.
2. PROVIDE FACTORY-FABRICATED FIXTURES OF TYPE, STYLE AND MATERIAL INDICATED. FOR EACH TYPE FIXTURE, PROVIDE FIXTURE MANUFACTURER'S STANDARD TRIM, CARRIER, SEATS AND VALVES AS SHOWN BY THEIR PUBLISHED PRODUCT INFORMATION AND INDICATED IN THE PLUMBING FIXTURES SCHEDULE; EITHER AS DESIGNED AND CONSTRUCTED OR AS RECOMMENDED BY MANUFACTURER AND AS REQUIRED FOR COMPLETE INSTALLATION. WHERE MORE THAN ONE TYPE IS INDICATED, SELECTION IS INSTALLER'S OPTION, BUT ALL FIXTURES OF SAME TYPE MUST BE FURNISHED BY SINGLE MANUFACTURER. WHERE TYPE IS NOT OTHERWISE INDICATED, PROVIDE FIXTURES COMPLYING WITH GOVERNING REGULATIONS.
3. WHERE FITTINGS, TRIM AND ACCESSORIES ARE EXPOSED OR SEMI-EXPOSED, PROVIDE BRIGHT CHROME-PLATED OR POLISHED STAINLESS STEEL UNITS. PROVIDE COPPER OR BRASS WHERE NOT EXPOSED.
4. WATER OUTLETS: AT LOCATIONS WHERE WATER IS SUPPLIED (BY MANUAL, AUTOMATIC OR REMOTE CONTROL), PROVIDE COMMERCIAL QUALITY FAUCETS, VALVES OR DISPENSING DEVICES OF TYPE AND SIZE INDICATED AND AS REQUIRED TO OPERATE AS INDICATED. INCLUDE MANUAL SHUT-OFF VALVES AND CONNECTING STEM PIPES TO PERMIT OUTLET SERVICING WITHOUT SHUT-DOWN OF WATER SUPPLY PIPING SYSTEMS.
5. VACUUM BREAKERS: PROVIDE WITH FLUSH VALVES WHERE REQUIRED BY GOVERNING REGULATIONS, INCLUDING LOCATIONS WHERE WATER OUTLETS ARE EQUIPPED FOR HOSE ATTACHMENT.
6. WATER HAMMER ARRESTORS: PROVIDE WATER HAMMER ARRESTORS WHERE SHOWN ON THE DRAWINGS AND AS REQUIRED TO PREVENT WATER HAMMER AND EXCESSIVE VIBRATION IN THE DOMESTIC WATER SYSTEM. ARRESTORS TO BE OF SIZE INDICATED OR AS RECOMMENDED BY THE MANUFACTURER.
7. P-TRAPS: INCLUDE REMOVABLE P-TRAPS (WITH CLEAN OUT PLUG) WHERE DRAINS ARE INDICATED FOR DIRECT CONNECTION TO DRAINAGE SYSTEM.
8. CARRIERS: PROVIDE CAST IRON SUPPORTS FOR FIXTURES OF EITHER GRAPHITIC GRAY IRON, DUCTILE IRON OR MALLEABLE IRON AS INDICATED.
9. FIXTURE BOLT CAPS: PROVIDE MANUFACTURER'S STANDARD EXPOSED FIXTURE BOLT CAPS FINISHED TO MATCH FIXTURE FINISH.
10. ESCUTCHEONS: WHERE FIXTURE SUPPLIES AND DRAINS PENETRATE WALLS IN EXPOSED LOCATIONS, PROVIDE CHROME-PLATED SHEET STEEL ESCUTCHEONS WITH FRICTION SLIPS.
11. AERATORS: PROVIDE AERATORS OF TYPES APPROVED BY HEALTH DEPARTMENT HAVING JURISDICTION.
12. COMPLY WITH ADDITIONAL FIXTURE REQUIREMENTS CONTAINED IN FIXTURE SCHEDULE ON DRAWINGS.

B. BACKFLOW PREVENTER (IF REQUIRED)

1. PROVIDE REDUCED PRESSURE BACKFLOW PREVENTER CONSISTING OF ASSEMBLY INCLUDING ABUTTING SHUTOFF VALVES ON INLET AND OUTLET, AND DISCHARGE FUNNEL. BACKFLOW PREVENTER SHALL INCLUDE A MINIMUM OF FOUR (4) TEST COCKS AND PRESSURE-DIFFERENTIAL RELIEF VALVE LOCATED BETWEEN TWO (2) POSITIVE SEATING CHECK VALVES. BACKFLOW PREVENTER AND SHUTOFF VALVES SHALL BE THE SAME SIZE AS THE UPSTREAM PIPE.
2. BACKFLOW PREVENTERS SIZES 2" AND SMALLER SHALL HAVE NPT CONNECTIONS, BE OF BRONZE BODY CONSTRUCTION WITH BRONZE BALL TYPE SHUT-OFF VALVES AS SPECIFIED IN SECTION 15100 AND TEST COCK AND BRONZE BODY RELIEF VALVES WITH STAINLESS STEEL TRIM.
3. BACKFLOW PREVENTERS SIZES 2-1/2" AND LARGER SHALL HAVE FLANGED CONNECTIONS, BE OF EPOXY COATED CAST OR DUCTILE IRON BODY CONSTRUCTION WITH BRONZE TRIM, RESILIENT SEAT OS&Y SHUT-OFF GATE VALVES, EPOXY COATED RELIEF VALVE WITH STAINLESS STEEL TRIM AND BRONZE BODY-BALL VALVE TEST COCKS.
4. COMPLETE BACKFLOW PREVENTER ASSEMBLY SHALL BE RATED TO 150 PSI WORKING PRESSURE AND WATER TEMPERATURE RANGE FROM 32° F TO 140° F.
5. PROVIDE EACH BACKFLOW PREVENTER WITH A DRAIN FUNNEL FURNISHED BY THE MANUFACTURER. EXTEND DRAIN FROM FUNNEL TO NEAREST FLOOR DRAIN.
6. BACKFLOW DEVICES MUST MEET ASSE STANDARDS 1013, 1015 AND 1020 AND SHALL BE TESTED AT THE TIME OF INSTALLATION BY A PERSON CERTIFIED BY THE OHIO DEPARTMENT OF HEALTH. THE PLUMBING CONTRACTOR SHALL PAY FOR ALL COSTS ASSOCIATED WITH THIS TEST.
7. MANUFACTURER: SUBJECT TO COMPLIANCE WITH REQUIREMENTS, PROVIDE BACKFLOW PREVENTERS OF ONE OF THE FOLLOWING:
   a. CLA-VAL COMPANY
   b. CONBRACO INDUSTRIES, INC.
   c. FEBCO SALES, INC., SUB. OF CHARLES M. BAILEY CO., INC.
   d. HERSEY PRODUCTS, INC.
   e. WATTS REGULATOR COMPANY

SECTION 230900 - INSTRUMENTATION AND CONTROLS

GENERAL

1. FURNISH AND INSTALL COMPLETE TEMPERATURE CONTROL FOR ALL HVAC SYSTEMS.
2. PROVIDE NEW CONTROL DEVICES INCLUDING THERMOSTATS, HUMIDISTATS, DAMPER OPERATORS, MOTORS, TEMPERATURE SENSORS, STAGING RELAYS, AND OTHER RELATED DEVICES FOR A COMPLETE OPERATIONAL SYSTEM PER THE OPERATING SEQUENCE AND INDUSTRY STANDARDS.
3. MOUNT ALL CONTROLS FURNISHED AS ACCESSORIES TO EQUIPMENT AND PROVIDE ALL CONTROL WIRING

FOOD SERVICE EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY.
1	SINK BOWL, WELD-IN, DUMP SINK	N. WASSERSTROM & SONS	F4480234	1
2	SINK BOWL, WELD-IN, HAND SINK	N. WASSERSTROM & SONS	F4480234	1
3C	BLODGETT OVEN	BLODGETT	CTB SGL	1
3D	RAPID COOK OVEN	MERRYCHEF USA	E2S HIGH CLASSIC	1
100.1	PANTRY FAUCET	T&S BRASS	KL45-4000-WH	2
100.2	HOT WATER DISPENSER	IN SINK ERATOR	C1300	1
103	POS SYSTEM	-	-	1
104	ESPRESSO MACHINE	MELITTA	CT8	2
108	GARBAGE CAN	RUBBERMAID COMMERCIAL PRODUCTS	FG354060BLA	2
111	TOUCH-FREE SOAP DISPENSER	GOJO INDUSTRIES, INC.	PURELL CS6	1
112	SURFACE-MOUNTED PAPER TOWEL DISPENSER	BOBRICK	B-2621	1
112.1	BLENDER RINSER	BLENDTEC	JRE-610	1
113	COFFEE GRINDER	GRINDMASTER-UNIC-CRATHCO	890BS	1
114	COFFEE BREWER	BUNN	53100.0100	1
115	MANIFOLD	-		1
116	GLASS FILLER	T&S BRASS	B-1210	1
117	AIRPOT	SERVICE IDEAS	ECALS22SS	2
118	SELF-SERVICE REFRIGERATED CASE	STRUCTURAL CONCEPTS	B3424	2
119	PASTRY CASE, CURVED GLASS	FEDERAL INDUSTRIES	SSRPF-5052	1
120	DROP-IN ICE BIN	KROWNE	D278	1
121	BLENDER	VITAMIX	36019-ABAB	2
123	DIPPER WELL	T&S BRASS	B-2282-01-F05	1
124	DOUBLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-48-HC	1
131	SINGLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-27-HC	2

WORKROOM EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY
1A	R.O.WATER FILTRATION SYSTEM MANIFOLD BOARD	-	-	1
1B	R.O.WATER FILTRATION SYSTEM - ACCUMULATOR TANK	-	-	1
1C	R.O.WATER FILTRATION SYSTEM - WATER SOFTNER	-	-	1
1D	R.O.WATER FILTRATION SYSTEM - BRINE	-	-	1
2B	REFRIGERATOR, REACH-IN, SINGLE DOOR	TRUE MGF.	STG1R-1S-HC	1
3B	FREEZER, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2F-2S-HC	2
4	WATER HEATER	-	-	1
5	MOP SINK	FIAT	MSB2424	1
5.1	SERVICE FAUCET	CHICAGO FAUCET	897-RCF	1
6	WALL SHELF	SUPPLIED BY BN	-	1
8	SOAP DISPENSER	BN SUPPLIED PURELL	TFX	1
9	HAND SINK - WALL MOUNTED	HAND SINK PARTS & ACCESSORIES	DH-17-NO FAU	1
14	C-FOLD PAPER TOWEL DISPENSER	BN SUPPLIED C-FOLD	-	1
15	WASTE CONTAINER	-	-	1
16	THREE COMPARTMENT SINK	AMTEKCO INDUSTRIES LTD.	D724-03-74	1
16.1	PRE-RINSE FAUCET & ADD ON FAUCET			1
16.4	LEVER WASTE	T&S BRASS	B-3950	3
125	ICE MAKER, CUBE-STYLE	ITV ICE MAKER	SPIKA MS 500	1
127	WIRE SHELVING	1880 HOSPITALITY	FF2472C	4
129	REFRIGERATOR, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2DT-2S	1
133	DISHWASHER, UNDERCOUNTER	JACKSON WWS	DISHSTAR HT	1

GENERAL NOTES: (PLUMBING)

A. THESE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL EXTENT OF WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COORDINATION AND PROPER INSTALLATION OF ALL PLUMBING SYSTEMS. REFERENCE FOOD SERVICE DRAWINGS WHEN INSTALLING FOR EXACT LOCATIONS AND HEIGHTS.

B. THE CONTRACTOR SHALL PROVIDE ALL MISCELLANEOUS SUPPORTING STEEL ETC. FOR THE PROPER INSTALLATION OF ALL PLUMBING SYSTEMS.

C. THE CONTRACTOR SHALL COORDINATE FLOOR, WALL AND ROOF PENETRATIONS WITH THE GENERAL CONTRACTOR.

D. PIPING SHALL NOT BE LOCATED IN ELECTRICAL ROOMS OR OVER THE TOP OF ANY ELECTRICAL PANELS OR EQUIPMENT.

E. CONTRACTOR SHALL FIELD VERIFY SLAB ON GRADE FLOOR CONSTRUCTION TYPE PRIOR TO CUTTING. UNDER NO CIRCUMSTANCES SHALL THE CONTRACTOR CUT A STRUCTURAL FLOOR SLAB THICKER THAN FOUR (4") INCHES WITHOUT PRIOR WRITTEN APPROVAL FROM ENGINEER OF RECORD. NOTIFY ENGINEER OF RECORD OF ANY SLAB THICKNESS GREATER THAN FOUR (4") INCHES PRIOR TO PROCEEDING WITH ANY SAW CUTTING.

CODED NOTES: (#)

1. CONNECT NEW 4" WASTE LINE TO EXISTING 4" WASTE MAIN. FIELD VERIFY SIZE, LOCATION, INVERT ELEVATION, DIRECTION OF FLOW, AND POINT OF CONNECTION TO SANITARY MAIN PRIOR TO ANY WORK.

2. NEW 4" GREASE LINE TO BE CONNECTED TO EXISTING 4" GREASE LINE. FIELD VERIFY SIZE, LOCATION, INVERT ELEVATION, DIRECTION OF FLOW, AND POINT OF CONNECTION TO SANITARY MAIN PRIOR TO ANY WORK.

3. NEW 2" VENT LINE TO EXISTING 2"VTR. VERIFY LOCATION AND POINT OF CONNECTION PRIOR TO BID.

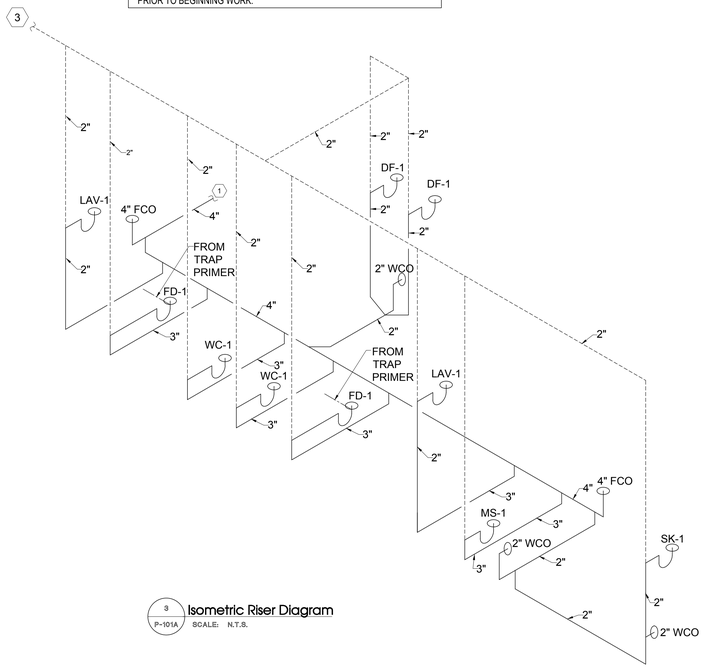
THESE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL EXTENT OF THE WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COORDINATION AND PROPER INSTALLATION OF ALL PLUMBING SYSTEMS. THE CONTRACTOR SHALL PROVIDE ALL NECESSARY OFFSETS AND FITTINGS WHICH MAY BE REQUIRED DUE TO SPACE CONSTRAINTS OR OTHER CONDITIONS

CONTRACTOR SHALL FIELD VERIFY ALL EXISTING FIELD CONDITIONS PRIOR TO BEGINNING WORK.

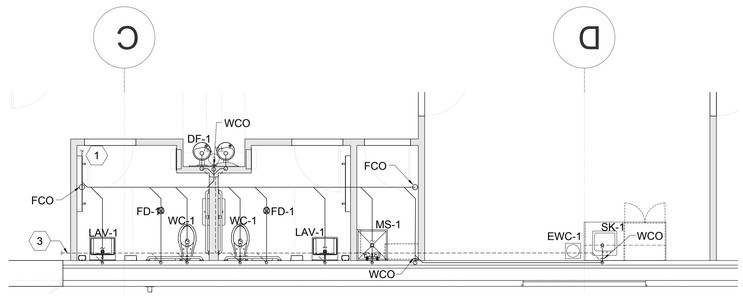
BUILDING DEPARTMENT NOTES: (PLUMBING)

A. THIS SET OF DRAWINGS COMPLY WITH 2021 IPC AND CO STATE AMENDMENTS TO 2021 IPC.
B. THIS SET OF DRAWINGS COMPLY WITH 2023 PPRBC
C. THE PROPER FIXTURE NUMBER AND THE MINIMUM NUMBER OF FIXTURES PER OCCUPANCY TYPE WAS CALCULATED PER 2021 IPC 403.1.
D. LAVATORIES COMPLY WITH 2021 IECC SECTION C404.5.1 BECAUSE THERE IS A RECIRCULATION LINE SPECIFIED ON THE HOT WATER PIPING. SEE P-102A.

3



3 P-101A Isometric Riser Diagram
SCALE: N.T.S.



2 P-101A Enlarge Sanitary Waste & Vent Plan
SCALE: 1/4" = 1'-0"

1

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J. COMMER
763 SUSQUEHANNA AVE
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Kenneth A. Fuller, P.E.
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CO LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES: 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3
08-08-25	BD COMMENTS	4

These plans are an instrument of service and the property of the Architect, Infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:
SANITARY WASTE
& VENT PLAN

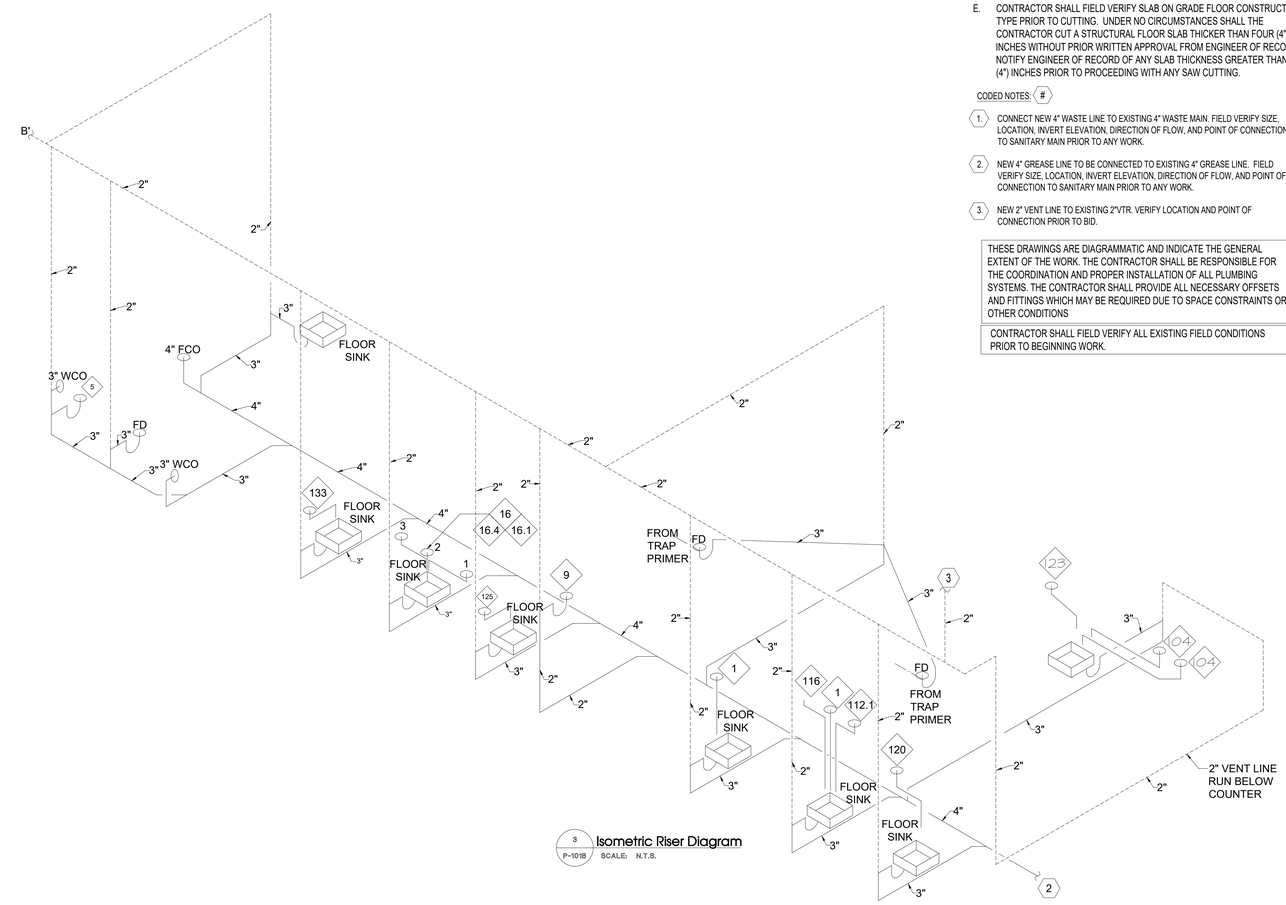
Drawing Number:
P-101A

FOOD SERVICE EQUIPMENT / UTILITIES SCHEDULE

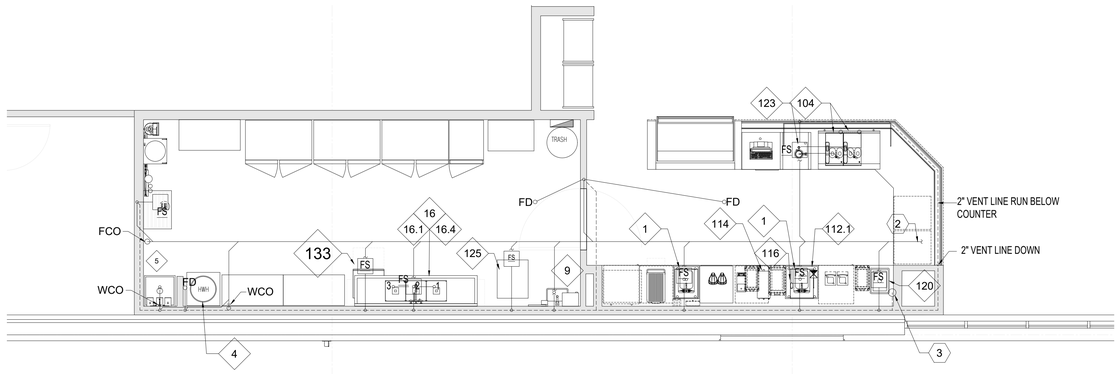
ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY.
1	SINK BOWL, WELD-IN, DUMP SINK	N. WASSERSTROM & SONS	F4480234	1
2	SINK BOWL, WELD-IN, HAND SINK	N. WASSERSTROM & SONS	F4480234	1
3C	BLODGETT OVEN	BLODGETT	CTB SGL	1
3D	RAPID COOK OVEN	MERRYCHEF USA	E2S HIGH CLASSIC	1
100.1	PANTRY FAUCET	T&S BRASS	KL45-4000-WH	2
100.2	HOT WATER DISPENSER	IN SINK ERATOR	C1300	1
103	POS SYSTEM	-	-	1
104	ESPRESSO MACHINE	MELITTA	CT8	2
108	GARBAGE CAN	RUBBERMAID COMMERCIAL PRODUCTS	FG354060BLA	2
111	TOUCH-FREE SOAP DISPENSER	GOJO INDUSTRIES, INC.	PURELL CS6	1
112	SURFACE-MOUNTED PAPER TOWEL DISPENSER	BOBRICK	B-2621	1
112.1	BLENDER RINSER	BLENDTEC	JRE-610	1
113	COFFEE GRINDER	GRINDMASTER-UNIC-CRATHCO	890BS	1
114	COFFEE BREWER	BUNN	53100.0100	1
115	MANIFOLD	-		1
116	GLASS FILLER	T&S BRASS	B-1210	1
117	AIRPOT	SERVICE IDEAS	ECALS22SS	2
118	SELF-SERVICE REFRIGERATED CASE	STRUCTURAL CONCEPTS	B3424	2
119	PASTRY CASE, CURVED GLASS	FEDERAL INDUSTRIES	SSRPF-5052	1
120	DROP-IN ICE BIN	KROWNE	D278	1
121	BLENDER	VITAMIX	36019-ABAB	2
123	DIPPER WELL	T&S BRASS	B-2282-01-F05	1
124	DOUBLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-48-HC	1
131	SINGLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-27-HC	2

WORKROOM EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY
1A	R.O.WATER FILTRATION SYSTEM MANIFOLD BOARD	-	-	1
1B	R.O.WATER FILTRATION SYSTEM - ACCUMULATOR TANK	-	-	1
1C	R.O.WATER FILTRATION SYSTEM - WATER SOFTNER	-	-	1
1D	R.O.WATER FILTRATION SYSTEM - BRINE	-	-	1
2B	REFRIGERATOR, REACH-IN, SINGLE DOOR	TRUE MGF.	STG1R-1S-HC	1
3B	FREEZER, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2F-2S-HC	2
4	WATER HEATER	-	-	1
5	MOP SINK	FIAT	MSB2424	1
5.1	SERVICE FAUCET	CHICAGO FAUCET	897-RCF	1
6	WALL SHELF	SUPPLIED BY BN	-	1
8	SOAP DISPENSER	BN SUPPLIED PURELL	TFX	1
9	HAND SINK - WALL MOUNTED	HAND SINK PARTS & ACCESSORIES	DH-17-NO FAU	1
14	C-FOLD PAPER TOWEL DISPENSER	BN SUPPLIED C-FOLD	-	1
15	WASTE CONTAINER	-	-	1
16	THREE COMPARTMENT SINK	AMTEKCO INDUSTRIES LTD.	D724-03-74	1
16.1	PRE-RINSE FAUCET & ADD ON FAUCET			1
16.4	LEVER WASTE	T&S BRASS	B-3950	3
125	ICE MAKER, CUBE-STYLE	ITV ICE MAKER	SPIKA MS 500	1
127	WIRE SHELVING	1880 HOSPITALITY	FF2472C	4
129	REFRIGERATOR, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2DT-2S	1
133	DISHWASHER, UNDERCOUNTER	JACKSON WWS	DISHSTAR HT	1



3 P-101B Isometric Riser Diagram
SCALE: N.T.S.



2 P-101B Cafe Sanitary Waste & Vent Plan
SCALE: 1/4" = 1'-0"

GENERAL NOTES: (PLUMBING)

A. THESE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL EXTENT OF WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COORDINATION AND PROPER INSTALLATION OF ALL PLUMBING SYSTEMS. REFERENCE FOOD SERVICE DRAWINGS WHEN INSTALLING FOR EXACT LOCATIONS AND HEIGHTS.

B. THE CONTRACTOR SHALL PROVIDE ALL MISCELLANEOUS SUPPORTING STEEL ETC. FOR THE PROPER INSTALLATION OF ALL PLUMBING SYSTEMS.

C. THE CONTRACTOR SHALL COORDINATE FLOOR, WALL AND ROOF PENETRATIONS WITH THE GENERAL CONTRACTOR.

D. PIPING SHALL NOT BE LOCATED IN ELECTRICAL ROOMS OR OVER THE TOP OF ANY ELECTRICAL PANELS OR EQUIPMENT.

E. CONTRACTOR SHALL FIELD VERIFY SLAB ON GRADE FLOOR CONSTRUCTION TYPE PRIOR TO CUTTING. UNDER NO CIRCUMSTANCES SHALL THE CONTRACTOR CUT A STRUCTURAL FLOOR SLAB THICKER THAN FOUR (4") INCHES WITHOUT PRIOR WRITTEN APPROVAL FROM ENGINEER OF RECORD. NOTIFY ENGINEER OF RECORD OF ANY SLAB THICKNESS GREATER THAN FOUR (4") INCHES PRIOR TO PROCEEDING WITH ANY SAW CUTTING.

CODED NOTES: #

1. CONNECT NEW 4" WASTE LINE TO EXISTING 4" WASTE MAIN. FIELD VERIFY SIZE, LOCATION, INVERT ELEVATION, DIRECTION OF FLOW, AND POINT OF CONNECTION TO SANITARY MAIN PRIOR TO ANY WORK.
2. NEW 4" GREASE LINE TO BE CONNECTED TO EXISTING 4" GREASE LINE. FIELD VERIFY SIZE, LOCATION, INVERT ELEVATION, DIRECTION OF FLOW, AND POINT OF CONNECTION TO SANITARY MAIN PRIOR TO ANY WORK.
3. NEW 2" VENT LINE TO EXISTING 2"VTR. VERIFY LOCATION AND POINT OF CONNECTION PRIOR TO BID.

THESE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL EXTENT OF THE WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COORDINATION AND PROPER INSTALLATION OF ALL PLUMBING SYSTEMS. THE CONTRACTOR SHALL PROVIDE ALL NECESSARY OFFSETS AND FITTINGS WHICH MAY BE REQUIRED DUE TO SPACE CONSTRAINTS OR OTHER CONDITIONS

CONTRACTOR SHALL FIELD VERIFY ALL EXISTING FIELD CONDITIONS PRIOR TO BEGINNING WORK.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J. COMMER
763 SUSQUEHANNA AVE
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STATE OF NEW JERSEY
LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES: 09-30-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

These plans are an instrument of service and the property of the Architect, Infringements will be prosecuted.

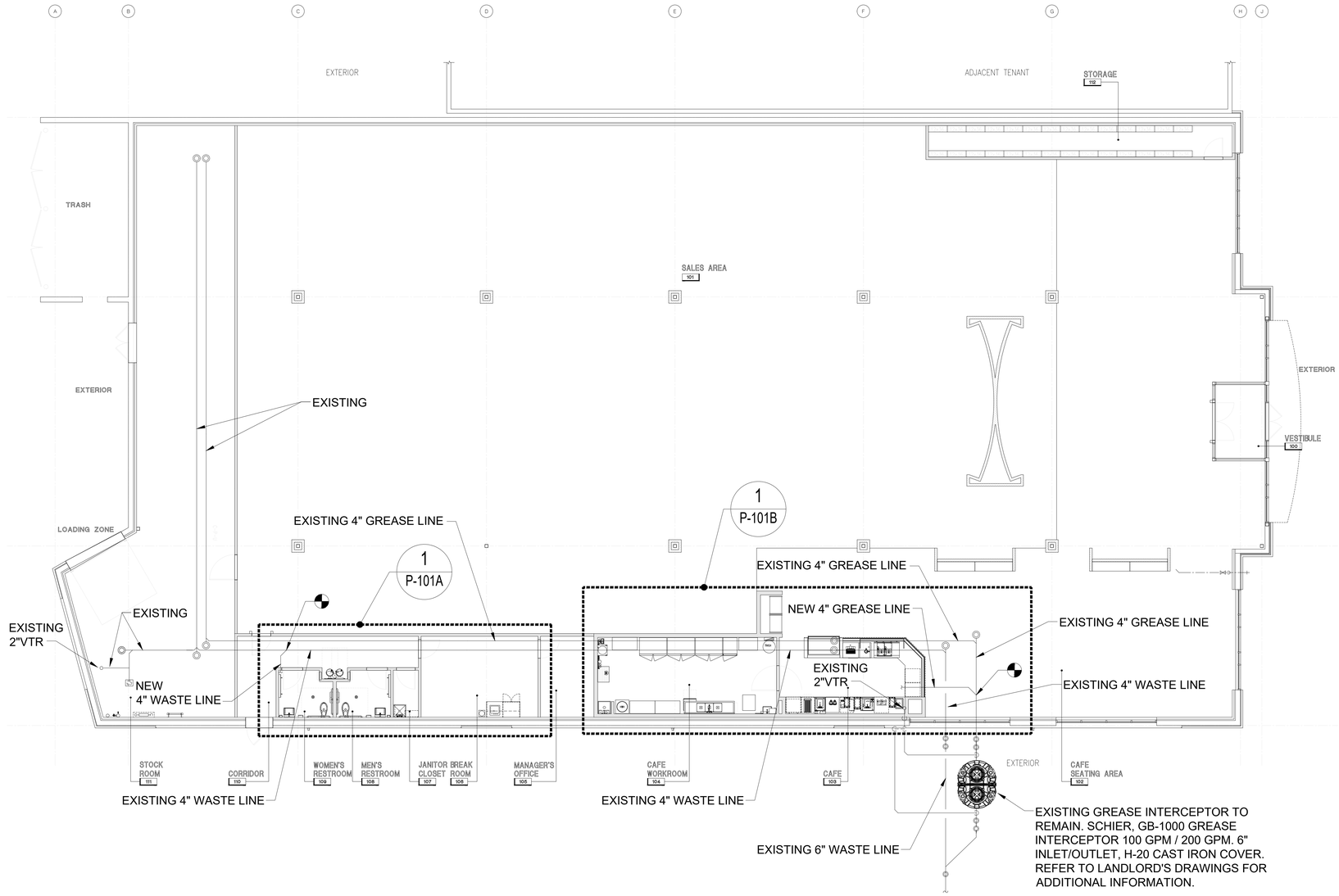
General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:
SANITARY WASTE
& VENT PLAN

Drawing Number:
P-101B

INFORMATION ABOUT EXISTING CONDITIONS WAS TAKEN FROM LANDLORD SET OF DRAWINGS ISSUED BY YOW ARCHITECS, PC. ON 04/02/25 (REV #1 - SPLICE). INFORMATION TO BE VERIFIED IN FIELD PRIOR TO BID.



1 P-101C Sanitary Waste & Vent Overall Plan
SCALE: 1/8" = 1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
WILLIAM J. COMMER
763 SUSQUEHANNA AVE
FRANKLIN LAKES, NJ 07417

Kenneth A. Fuller, P.E.
153 East Main St.
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kfullerpe@msn.com

STATE OF COLORADO LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES: 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

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General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:
SANITARY WASTE
& VENT PLAN

Drawing Number:
P-101C
3

FOOD SERVICE EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY.
1	SINK BOWL, WELD-IN, DUMP SINK	N. WASSERSTROM & SONS	F4480234	1
2	SINK BOWL, WELD-IN, HAND SINK	N. WASSERSTROM & SONS	F4480234	1
3C	BLODGETT OVEN	BLODGETT	CTB SGL	1
3D	RAPID COOK OVEN	MERRYCHEF USA	E2S HIGH CLASSIC	1
100.1	PANTRY FAUCET	T&S BRASS	KL45-4000-WH	2
100.2	HOT WATER DISPENSER	IN SINK ERATOR	C1300	1
103	POS SYSTEM	-	-	1
104	ESPRESSO MACHINE	MELITTA	CT8	2
108	GARBAGE CAN	RUBBERMAID COMMERCIAL PRODUCTS	FG354060BLA	2
111	TOUCH-FREE SOAP DISPENSER	GOJO INDUSTRIES, INC.	PURELL CS6	1
112	SURFACE-MOUNTED PAPER TOWEL DISPENSER	BOBRICK	B-2621	1
112.1	BLENDER RINSER	BLENDTEC	JRE-610	1
113	COFFEE GRINDER	GRINDMASTER-UNIC-CRATHCO	890BS	1
114	COFFEE BREWER	BUNN	53100.0100	1
115	MANIFOLD	-		1
116	GLASS FILLER	T&S BRASS	B-1210	1
117	AIRPOT	SERVICE IDEAS	ECALS22SS	2
118	SELF-SERVICE REFRIGERATED CASE	STRUCTURAL CONCEPTS	B3424	2
119	PASTRY CASE, CURVED GLASS	FEDERAL INDUSTRIES	SSRPF-5052	1
120	DROP-IN ICE BIN	KROWNE	D278	1
121	BLENDER	VITAMIX	36019-ABAB	2
123	DIPPER WELL	T&S BRASS	B-2282-01-F05	1
124	DOUBLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-48-HC	1
131	SINGLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-27-HC	2

WORKROOM EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY
1A	R.O.WATER FILTRATION SYSTEM MANIFOLD BOARD	-	-	1
1B	R.O.WATER FILTRATION SYSTEM - ACCUMULATOR TANK	-	-	1
1C	R.O.WATER FILTRATION SYSTEM - WATER SOFTNER	-	-	1
1D	R.O.WATER FILTRATION SYSTEM - BRINE	-	-	1
2B	REFRIGERATOR, REACH-IN, SINGLE DOOR	TRUE MGF.	STG1R-1S-HC	1
3B	FREEZER, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2F-2S-HC	2
4	WATER HEATER	-	-	1
5	MOP SINK	FIAT	MSB2424	1
5.1	SERVICE FAUCET	CHICAGO FAUCET	897-RCF	1
6	WALL SHELF	SUPPLIED BY BN	-	1
8	SOAP DISPENSER	BN SUPPLIED PURELL	TFX	1
9	HAND SINK - WALL MOUNTED	HAND SINK PARTS & ACCESSORIES	DH-17-NO FAU	1
14	C-FOLD PAPER TOWEL DISPENSER	BN SUPPLIED C-FOLD	-	1
15	WASTE CONTAINER	-	-	1
16	THREE COMPARTMENT SINK	AMTEKCO INDUSTRIES LTD.	D724-03-74	1
16.1	PRE-RINSE FAUCET & ADD ON FAUCET			1
16.4	LEVER WASTE	T&S BRASS	B-3950	3
125	ICE MAKER, CUBE-STYLE	ITV ICE MAKER	SPIKA MS 500	1
127	WIRE SHELVING	1880 HOSPITALITY	FF2472C	4
129	REFRIGERATOR, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2DT-2S	1
133	DISHWASHER, UNDERCOUNTER	JACKSON WWS	DISHSTAR HT	1

PIPE SIZING CALCULATIONS

TOILET/BREAK ROOM AREAS

FIXTURE	FIXTURE UNITS	NUMBER OF FIXTURES	TOTAL FIXTURE UNITS
WATER CLOSET (WC-1)	10.0	2	20.0
LAVATORY (LAV-1)	2.0	2	4.0
MOP SINK (MS-1)	3.0	1	3.0
BREAK ROOM SINK (SK-1)	1.4	1	1.4
DRINKING FOUNTAIN (DF-1)	0.25	1	0.25
ELECTRIC WATER COOLER (EWC-1)	0.25	1	0.25

CAFE AREA

FIXTURE	FIXTURE UNITS	NUMBER OF FIXTURES	TOTAL FIXTURE UNITS
HAND SINK	2.0	2	4.0
DUMP SINK	4.0	1	4.0
DIPPER WELL	4.0	1	4.0
HOT WATER DISPENSER	1.0	1	1.0
ESPRESSO MACHINE	1.0	1	1.0
COFFEE MACHINE	1.0	1	1.0
ICE MACHINE	0.5	1	0.5
3-BAY SINK	4.0	1	4.0
DISHWASHER	1.4	1	1.4
		TOTAL	49.8

WATER SUPPLY FIXTURE UNITS SHOWN ARE FROM 2021 I.P.C. TABLE E103.3(2).

PER 2021 I.P.C. TABLE E103.3(3) FOR SYSTEMS THAT ARE PREDOMINANTLY FLUSHOMETER VALVES 49.8 WATER SUPPLY FIXTURE UNITS EQUATES TO ROUGHLY 50.0 G.P.M.

PER 2021 I.P.C. FIGURE E103.3(3) FRICTION LOSS IN SMOOTH PIPE, 50.0 G.P.M. CAN BE PROVIDE WITHIN A 2" DIAMETER PIPE AT A VELOCITY OF 8 F.P.S.

WATER PIPE SIZE ANALYSIS
BARNES & NOBLE

PRESSURE CALCULATIONS	
DEVELOPED LENGTH	350 FT
120% DEVELOPED LENGTH (FITTINGS)	420 FT
ELEVATION OF HIGHEST FIXTURE	5.0 FT
PRESSURE AVAILABLE @ METER	60 PSI
TOTAL HEAD LOSS	2.2 PSI
METER LOSS (CHART A-1)	9.8 PSI
BACKFLOW PREVENTER LOSS	13.0 PSI
MINIMUM PRESSURE TO BE MAINTAINED	15.0 PSI
PRESSURE AVAILABLE TO OVERCOME PIPE FRICTION	20.0 PSI
PRESSURE AVAILABLE PER 100 FT OF PIPE	4.8 PSI

METER SIZE (INCHES):	1-1/2"
DISTRIBUTION SIZE (INCHES):	2
WASTE SIZE (INCHES):	4"

Water Supply Load Values - 2021 IPC table E103.3(2)
Table for Esitmating Demand - 2021 IPC Table E103.3(3)
Water Meters, Mains & Distribution Piping - 2021 IPC Table E201.1

INFORMATION ABOVE WAS TAKEN FROM LANDLORD'S DRAWINGS ISSUED BY MICHAEL ROYAL McSHEA ON 4/2/2025. INFORMATION TO BE VERIFIED IN FIELD PRIOR TO BID.

GENERAL NOTES: (PLUMBING)

A. THESE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL EXTENT OF WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COORDINATION AND PROPER INSTALLATION OF ALL PLUMBING SYSTEMS. REFERENCE FOOD SERVICE DRAWINGS WHEN INSTALLING FOR EXACT LOCATIONS AND HEIGHTS.

B. THE CONTRACTOR SHALL PROVIDE ALL MISCELLANEOUS SUPPORTING STEEL ETC. FOR THE PROPER INSTALLATION OF ALL PLUMBING SYSTEMS.

C. THE CONTRACTOR SHALL COORDINATE FLOOR, WALL AND ROOF PENETRATIONS WITH THE GENERAL CONTRACTOR.

D. PIPING SHALL NOT BE LOCATED IN ELECTRICAL ROOMS OR OVER THE TOP OF ANY ELECTRICAL PANELS OR EQUIPMENT.

E. CONTRACTOR SHALL FIELD VERIFY SLAB ON GRADE FLOOR CONSTRUCTION TYPE PRIOR TO CUTTING. UNDER NO CIRCUMSTANCES SHALL THE CONTRACTOR CUT A STRUCTURAL FLOOR SLAB THICKER THAN FOUR (4") INCHES WITHOUT PRIOR WRITTEN APPROVAL FROM ENGINEER OF RECORD. NOTIFY ENGINEER OF RECORD OF ANY SLAB THICKNESS GREATER THAN FOUR (4") INCHES PRIOR TO PROCEEDING WITH ANY SAW CUTTING.

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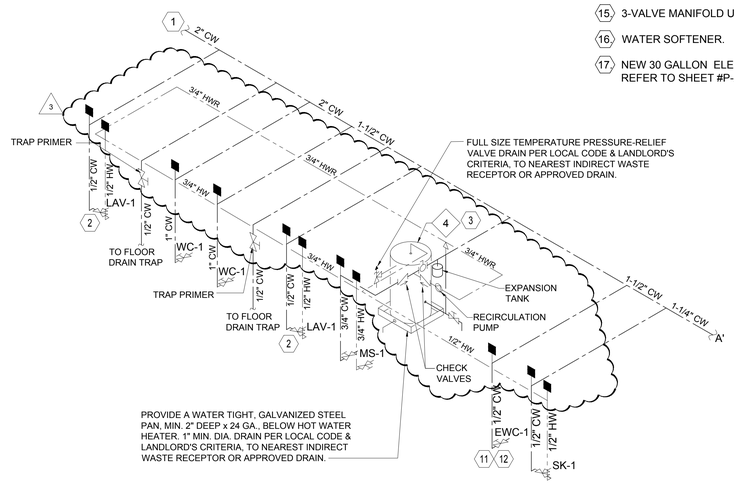
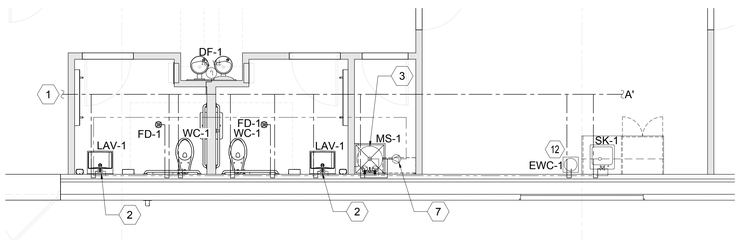
CONTRACTOR SHALL FIELD VERIFY ALL EXISTING FIELD CONDITIONS PRIOR TO BEGINNING WORK.

CODED NOTES: (#)

1. 2" COLD WATER LINE TO CONNECT TO EXISTING COLD WATER LINE. VERIFY LOCATION AND POINT OF CONNECTION IN FILED PRIOR TO BID.
2. THERMOSTATIC MIXING VALVE , WASTT'S SERIES LFMMV, ASPE 1017,1069 & 1070 APPROVED. MIXING VALVE TO BE SET FOR 110°F MAXIMUM UNLESS OTHERWISE NOTED ON PLANS.
3. NEW 20 GALLON ELECTRIC WATER HEATER TANK (EWH-2). REFER TO SHEET #P-201 FOR SPECIFICATIONS.
4. ROUTE 1/2" SROCW BELOW COUNTER AND CONNECT TO EQUIPMENT AS REQUIRED.
5. DROP 1/2" CW DOWN BELOW COUNTER AT THIS LOCATION.
6. 3/4" FCW TO ICE MAKER FILTER BOARD LOCATED TO THE RIGHT OF ICE CUBER. REFER TO GENERAL KITCHEN PLUMBING SCHEMATIC FOR MORE DETAIL.
7. NEW RE-CIRCULATION PUMP. REFER TO SHEET #P-201 FOR SPECIFICATIONS.
8. SOFTENED COLD WATER TO REVERSE OSMOSIS FILTER BOARD. REFER TO GENERAL KITCHEN PLUMBING SCHEMATIC FOR MORE DETAIL .
9. DROP 3/4" SROCW DOWN BELOW SCOUNTER AT THIS LOCATION. PROVIDE WATER SEALED PIPE SLEEVE AS REQUIRED. TRANSITION TO TYPE 'K' COOPER PIPE BELOW ROUTE PIPING BELOW COUNTEER TO BAR AREA AS SHOWN. NO FITTINGS FOR CHANGE OF DIRECTION SHALL BE INSTALLED IN PIPING BELOW COUNTER.
10. PROVIDE SROCW CONNECTION TO ITEM # 114. MOUNTED AT 46" AFF.
11. CONTRACTOR TO PROVIDE 1/2" COLD WATER LINE TO THE BOTTLEFREE WATER COOLER MANUFACTURED BY QUENCH. "QUENCH Q3" MODEL #Q3FS.
12. CONTRACTOR TO FURNISH AND INSTALL A SINGLE WATER BOX FOR WATER COOLER'S WATER SUPPLY PIPE. SINGLE WATER BOX TO BE MANUFACTURES BY WATER-TITE IPS CORPORATION , ITEM # 87967 MODEL # AB9700 OR EQUIVALENT.
13. PROVIDE SROCW CONNECTION TO ITEM # 116.
14. ROUTE 1/2" CW BELOW COUNTER AND CONNECT TO EQUIPMENT AS REQUIRED.
15. 3-VALVE MANIFOLD UNDER CABINET.
16. WATER SOFTENER.
17. NEW 30 GALLON ELECTRIC WATER HEATER TANK (EWH-1). REFER TO SHEET #P-201 FOR SPECIFICATIONS.



3 / P-102A ISOMETRIC RISER DIAGRAM
SCALE: N.T.S.



2 / P-102A Enlarge Domestic Water Supply Plan
SCALE: 1/4" = 1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
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EXPIRES 10-31-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3
08-08-25	BD COMMENTS	4

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Professional Seal:

Drawing Description:
DOMESTIC WATER PLAN

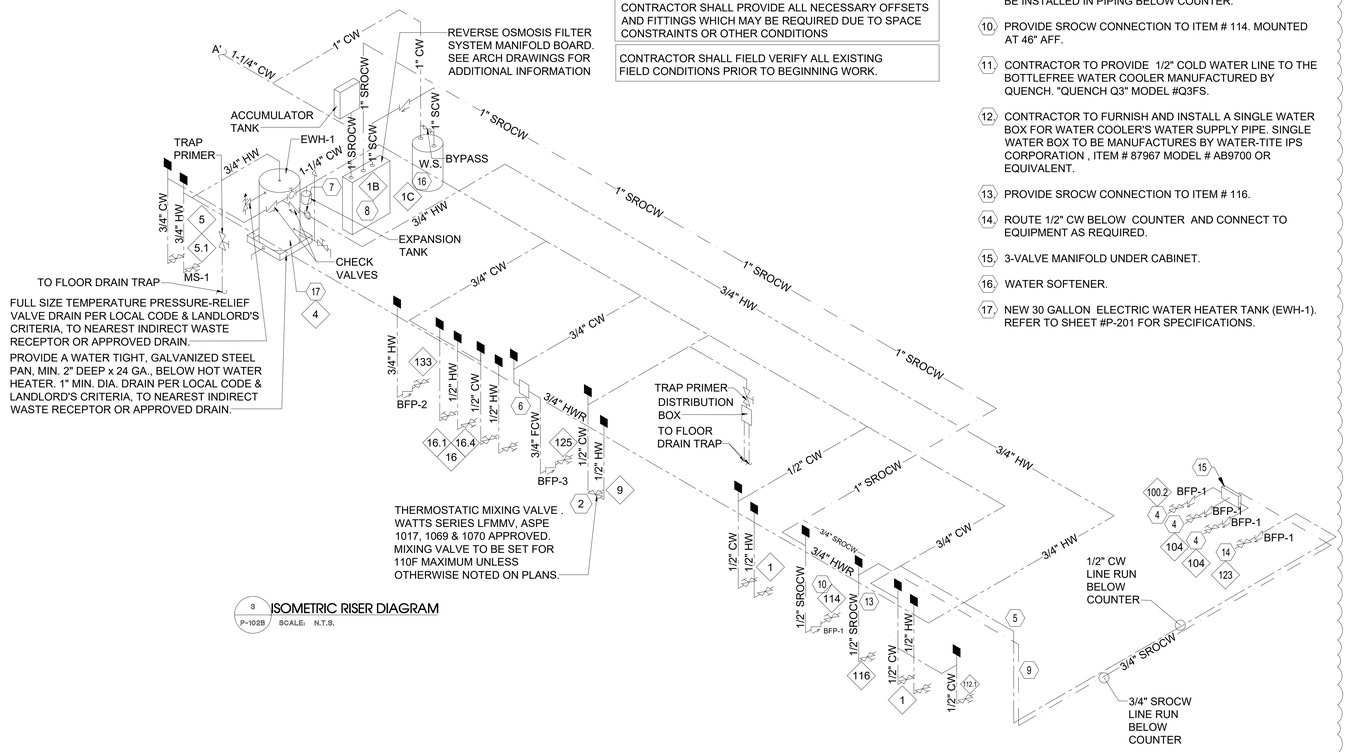
Drawing Number:
P-102A

FOOD SERVICE EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY.
1	SINK BOWL, WELD-IN, DUMP SINK	N. WASSERSTROM & SONS	F4480234	1
2	SINK BOWL, WELD-IN, HAND SINK	N. WASSERSTROM & SONS	F4480234	1
3C	BLODGETT OVEN	BLODGETT	CTB SGL	1
3D	RAPID COOK OVEN	MERRYCHEF USA	E2S HIGH CLASSIC	1
100.1	PANTRY FAUCET	T&S BRASS	KL45-4000-WH	2
100.2	HOT WATER DISPENSER	IN SINK ERATOR	C1300	1
103	POS SYSTEM	-	-	1
104	ESPRESSO MACHINE	MELITTA	CT8	2
108	GARBAGE CAN	RUBBERMAID COMMERCIAL PRODUCTS	FG354060BLA	2
111	TOUCH-FREE SOAP DISPENSER	GOJO INDUSTRIES, INC.	PURELL CS6	1
112	SURFACE-MOUNTED PAPER TOWEL DISPENSER	BOBRICK	B-2621	1
112.1	BLENDER RINSER	BLENDTEC	JRE-610	1
113	COFFEE GRINDER	GRINDMASTER-UNIC-CRATHCO	890BS	1
114	COFFEE BREWER	BUNN	53100.0100	1
115	MANIFOLD	-		1
116	GLASS FILLER	T&S BRASS	B-1210	1
117	AIRPOT	SERVICE IDEAS	ECALS22SS	2
118	SELF-SERVICE REFRIGERATED CASE	STRUCTURAL CONCEPTS	B3424	2
119	PASTRY CASE, CURVED GLASS	FEDERAL INDUSTRIES	SSRPF-5052	1
120	DROP-IN ICE BIN	KROWNE	D278	1
121	BLENDER	VITAMIX	36019-ABAB	2
123	DIPPER WELL	T&S BRASS	B-2282-01-F05	1
124	DOUBLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-48-HC	1
131	SINGLE UNDER COUNTER REFRIGERATOR	TRUE MGF.	TUC-27-HC	2

WORKROOM EQUIPMENT / UTILITIES SCHEDULE

ITEM#	DESCRIPTION	MANUFACTURER	MODEL#	QTY
1A	R.O.WATER FILTRATION SYSTEM MANIFOLD BOARD	-	-	1
1B	R.O.WATER FILTRATION SYSTEM - ACCUMULATOR TANK	-	-	1
1C	R.O.WATER FILTRATION SYSTEM - WATER SOFTNER	-	-	1
1D	R.O.WATER FILTRATION SYSTEM - BRINE	-	-	1
2B	REFRIGERATOR, REACH-IN, SINGLE DOOR	TRUE MGF.	STG1R-1S-HC	1
3B	FREEZER, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2F-2S-HC	2
4	WATER HEATER	-	-	1
5	MOP SINK	FIAT	MSB2424	1
5.1	SERVICE FAUCET	CHICAGO FAUCET	897-RCF	1
6	WALL SHELF	SUPPLIED BY BN	-	1
8	SOAP DISPENSER	BN SUPPLIED PURELL	TFX	1
9	HAND SINK - WALL MOUNTED	HAND SINK PARTS & ACCESSORIES	DH-17-NO FAU	1
14	C-FOLD PAPER TOWEL DISPENSER	BN SUPPLIED C-FOLD	-	1
15	WASTE CONTAINER	-	-	1
16	THREE COMPARTMENT SINK	AMTEKCO INDUSTRIES LTD.	D724-03-74	1
16.1	PRE-RINSE FAUCET & ADD ON FAUCET			1
16.4	LEVER WASTE	T&S BRASS	B-3950	3
125	ICE MAKER, CUBE-STYLE	ITV ICE MAKER	SPIKA MS 500	1
127	WIRE SHELVING	1880 HOSPITALITY	FF2472C	4
129	REFRIGERATOR, REACH-IN, DOUBLE DOOR	TRUE MGF.	STG2DT-2S	1
133	DISHWASHER, UNDERCOUNTER	JACKSON WWS	DISHSTAR HT	1



3 P-102B ISOMETRIC RISER DIAGRAM
SCALE: N.T.S.

2 P-102B Cafe Domestic Water Supply Plan
SCALE: 1/4" = 1'-0"

GENERAL NOTES: (PLUMBING)

A. THESE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL EXTENT OF WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COORDINATION AND PROPER INSTALLATION OF ALL PLUMBING SYSTEMS. REFERENCE FOOD SERVICE DRAWINGS WHEN INSTALLING FOR EXACT LOCATIONS AND HEIGHTS.

B. THE CONTRACTOR SHALL PROVIDE ALL MISCELLANEOUS SUPPORTING STEEL ETC. FOR THE PROPER INSTALLATION OF ALL PLUMBING SYSTEMS.

C. THE CONTRACTOR SHALL COORDINATE FLOOR, WALL AND ROOF PENETRATIONS WITH THE GENERAL CONTRACTOR.

D. PIPING SHALL NOT BE LOCATED IN ELECTRICAL ROOMS OR OVER THE TOP OF ANY ELECTRICAL PANELS OR EQUIPMENT.

E. CONTRACTOR SHALL FIELD VERIFY SLAB ON GRADE FLOOR CONSTRUCTION TYPE PRIOR TO CUTTING. UNDER NO CIRCUMSTANCES SHALL THE CONTRACTOR CUT A STRUCTURAL FLOOR SLAB THICKER THAN FOUR (4") INCHES WITHOUT PRIOR WRITTEN APPROVAL FROM ENGINEER OF RECORD. NOTIFY ENGINEER OF RECORD OF ANY SLAB THICKNESS GREATER THAN FOUR (4") INCHES PRIOR TO PROCEEDING WITH ANY SAW CUTTING.

THESE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL EXTENT OF THE WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COORDINATION AND PROPER INSTALLATION OF ALL PLUMBING SYSTEMS. THE CONTRACTOR SHALL PROVIDE ALL NECESSARY OFFSETS AND FITTINGS WHICH MAY BE REQUIRED DUE TO SPACE CONSTRAINTS OR OTHER CONDITIONS

CONTRACTOR SHALL FIELD VERIFY ALL EXISTING FIELD CONDITIONS PRIOR TO BEGINNING WORK.

CODED NOTES: (#)

1. 2" COLD WATER LINE TO CONNECT TO EXISTING COLD WATER LINE. VERIFY LOCATION AND POINT OF CONNECTION IN FILED PRIOR TO BID.
2. THERMOSTATIC MIXING VALVE . WASTTS SERIES LFMMV, ASPE 1017,1069 & 1070 APPROVED. MIXING VALVE TO BE SET FOR 110°F MAXIMUM UNLESS OTHERWISE NOTED ON PLANS.
3. NEW 20 GALLON ELECTRIC WATER HEATER TANK (EWH-2). REFER TO SHEET #P-201 FOR SPECIFICATIONS.
4. ROUTE 1/2" SROCW BELOW COUNTER AND CONNECT TO EQUIPMENT AS REQUIRED.
5. DROP 1/2" CW DOWN BELOW COUNTER AT THIS LOCATION.
6. 3/4" FCW TO ICE MAKER FILTER BOARD LOCATED TO THE RIGHT OF ICE CUBER. REFER TO GENERAL KITCHEN PLUMBING SCHEMATIC FOR MORE DETAIL.
7. NEW RE-CIRCULATION PUMP. REFER TO SHEET #P-201 FOR SPECIFICATIONS.
8. SOFTENED COLD WATER TO REVERSE OSMOSIS FILTER BOARD. REFER TO GENERAL KITCHEN PLUMBING SCHEMATIC FOR MORE DETAIL .
9. DROP 3/4" SROCW DOWN BELOW SCOUNTER AT THIS LOCATION. PROVIDE WATER SEALED PIPE SLEEVE AS REQUIRED. TRANSITION TO TYPE 'K' COOPER PIPE BELOW ROUTE PIPING BELOW COUNTEER TO BAR AREA AS SHOWN. NO FITTINGS FOR CHANGE OF DIRECTION SHALL BE INSTALLED IN PIPING BELOW COUNTER.
10. PROVIDE SROCW CONNECTION TO ITEM # 114. MOUNTED AT 46" AFF.
11. CONTRACTOR TO PROVIDE 1/2" COLD WATER LINE TO THE BOTTLEFREE WATER COOLER MANUFACTURED BY QUENCH. "QUENCH Q3" MODEL #Q3FS.
12. CONTRACTOR TO FURNISH AND INSTALL A SINGLE WATER BOX FOR WATER COOLER'S WATER SUPPLY PIPE. SINGLE WATER BOX TO BE MANUFACTURES BY WATER-TITE IPS CORPORATION , ITEM # 87967 MODEL # AB9700 OR EQUIVALENT.
13. PROVIDE SROCW CONNECTION TO ITEM # 116.
14. ROUTE 1/2" CW BELOW COUNTER AND CONNECT TO EQUIPMENT AS REQUIRED.
15. 3-VALVE MANIFOLD UNDER CABINET.
16. WATER SOFTENER.
17. NEW 30 GALLON ELECTRIC WATER HEATER TANK (EWH-1). REFER TO SHEET #P-201 FOR SPECIFICATIONS.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

ARCHITECT:
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3173 CINEMA POINT,
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80922

Project Number 33288
Store Number 3566

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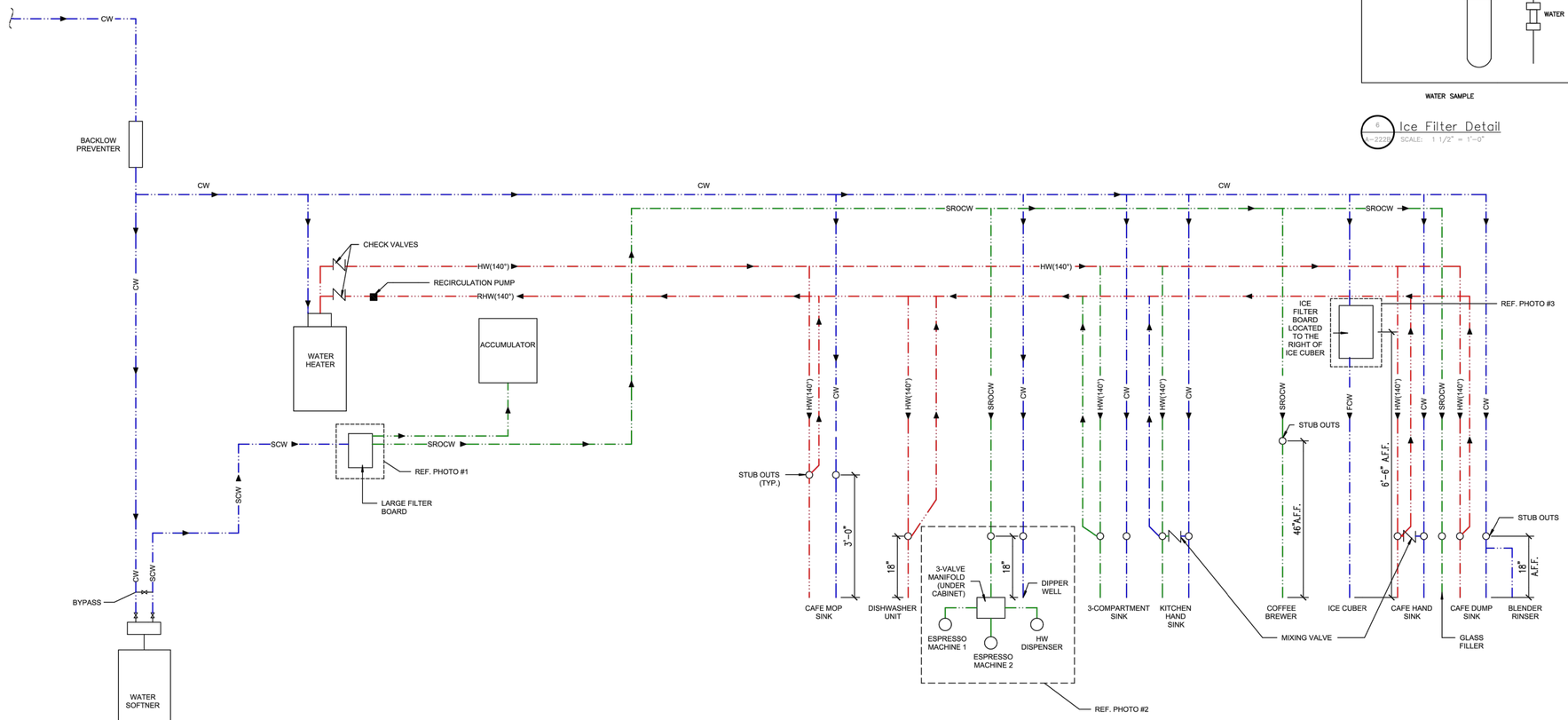
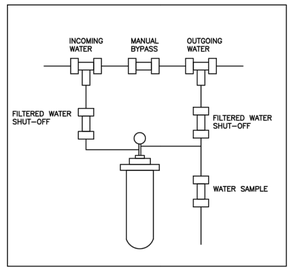
Professional Seal:

Drawing Description:
DOMESTIC WATER PLAN

Drawing Number:
P-102B



1 A-222B — Cafe & Cafe Workroom Plumbing Riser Schematic
SCALE: 3/4" = 1'-0"
GENERAL PLUMBING SCHEMATIC



WATER SAMPLE

6 A-222B — Ice Filter Detail
SCALE: 1 1/2" = 1'-0"

RED = HW
BLUE = CW or SCW
GREEN = SROCW

REFER TO PLUMBING ISOMETRIC FOR SIZING

THIS SCHEMATIC IS DIAGRAMMATIC AND INTENDED FOR PIPE ROUTING REFERENCE ONLY. INSTALLATION OF EQUIPMENT SHALL BE PER MANUFACTURERS GUIDELINES AND RECOMMENDATIONS

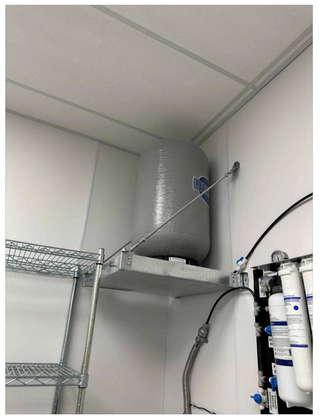
APPROVED BACK FLOW PREVENTING DEVICE IN LINE WITH THE DOMESTIC WATER SUPPLY TO ALL APPLIANCES REQUIRING SAME (COFFEE MAKER, ETC.)

INCOMING WATER PRESSURE MUST BE BETWEEN 50 - 55 P.S.I. PROVIDE A BID ALTERNATE FOR BOOSTER PUMP DESIGNED FOR APPROXIMATE FLOW OF 60 GPM, AND APPROXIMATE PRESSURE BOOST OF 30 PSI TO SERVE THE CAFE. PUMP SHALL HAVE A VFD DRIVE FOR VARIABLE FLOW AND BID ALTERNATE SHOULD INCLUDE COST FOR ALL REQUIRED ACCESSORIES AND CONTROLS. THRUSH BRAND MODEL PB-VS-40-30 OR EQUAL. ENGINEER TO SIZE PUMP BASED ON FLOW TEST DATA IF PRESSURE IS DEEMED IN ADEQUATE.

2 A-222B — Reference Photo #1
SCALE: 3/4" = 1'-0"
WATER FILTRATION BOARD IN WORK ROOM

3 A-222B — Reference Photo #2
SCALE: NTS
DIPPING STATION IN CAFE

4 A-222B — Reference Photo #3
SCALE: NTS
HAND SINK AND ICE MACHINE FILTER IN WORKROOM



5 A-222B — Reference Photo #4
SCALE: NTS
ACCUMULATOR TANK IN WORKROOM

1 P-102C — Cafe & Cafe Workroom Plumbing Riser Schematic
SCALE: N.T.S. (FOR ADDITIONAL INFORMATION REFER TO WITH SHEET #A-223B)

THIS PLUMBING RISER SCHEMATIC IS FOR REFERENCE ONLY. REFER TO PLUMBING DESIGN IN SHEET #P-102B, FOR WATER DISTRIBUTION, PIPE SIZES, AND CODE COMPLIANCE COMPONENTS.

BARNES&NOBLE

PROJECT DESIGNER:

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KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER 42050
EXPIRES 10-31-25

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80922

Project Number 33288
Store Number 3566

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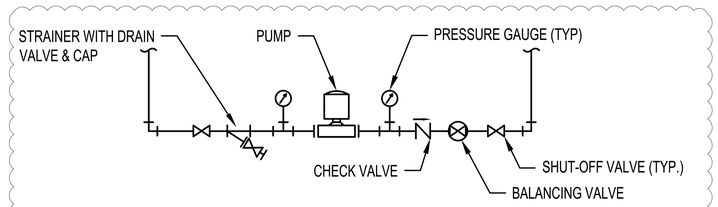
Professional Seal:

Drawing Description:
DOMESTIC WATER PLAN

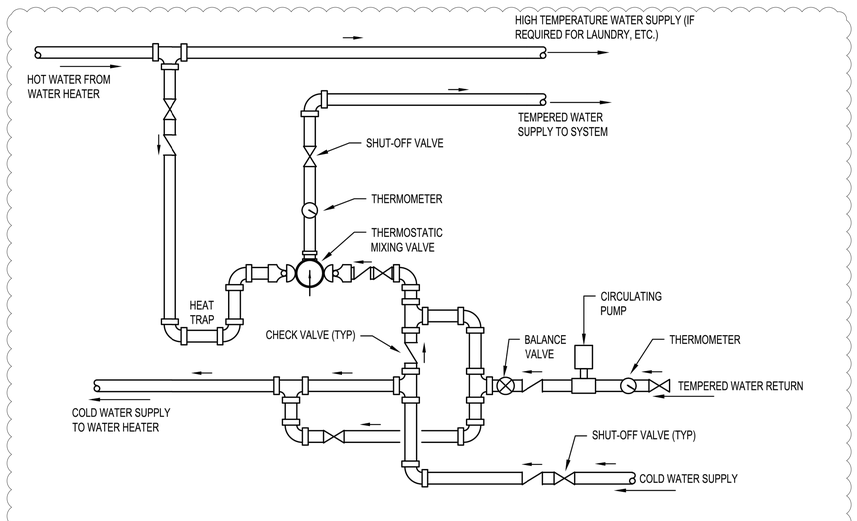
Drawing Number:
P-102C



IN-LINE PUMP PIPING DIAGRAM

N.T.S.

1



NOTE:

1. CONTRACTOR SHALL VERIFY EXACT PIPING REQUIREMENTS WITH MANUFACTURER.

THERMOSTATIC MIXING VALVE DETAIL WITH RECIRCULATING PUMP

N.T.S.

1

INFORMATION ABOUT EXISTING CONDITIONS WAS TAKEN FROM LANDLORD SET OF DRAWINGS ISSUED BY YOW ARCHITECS, PC. ON 04/02/25 (REV #1 - SPLICE). INFORMATION TO BE VERIFIED IN FIELD PRIOR TO BID.

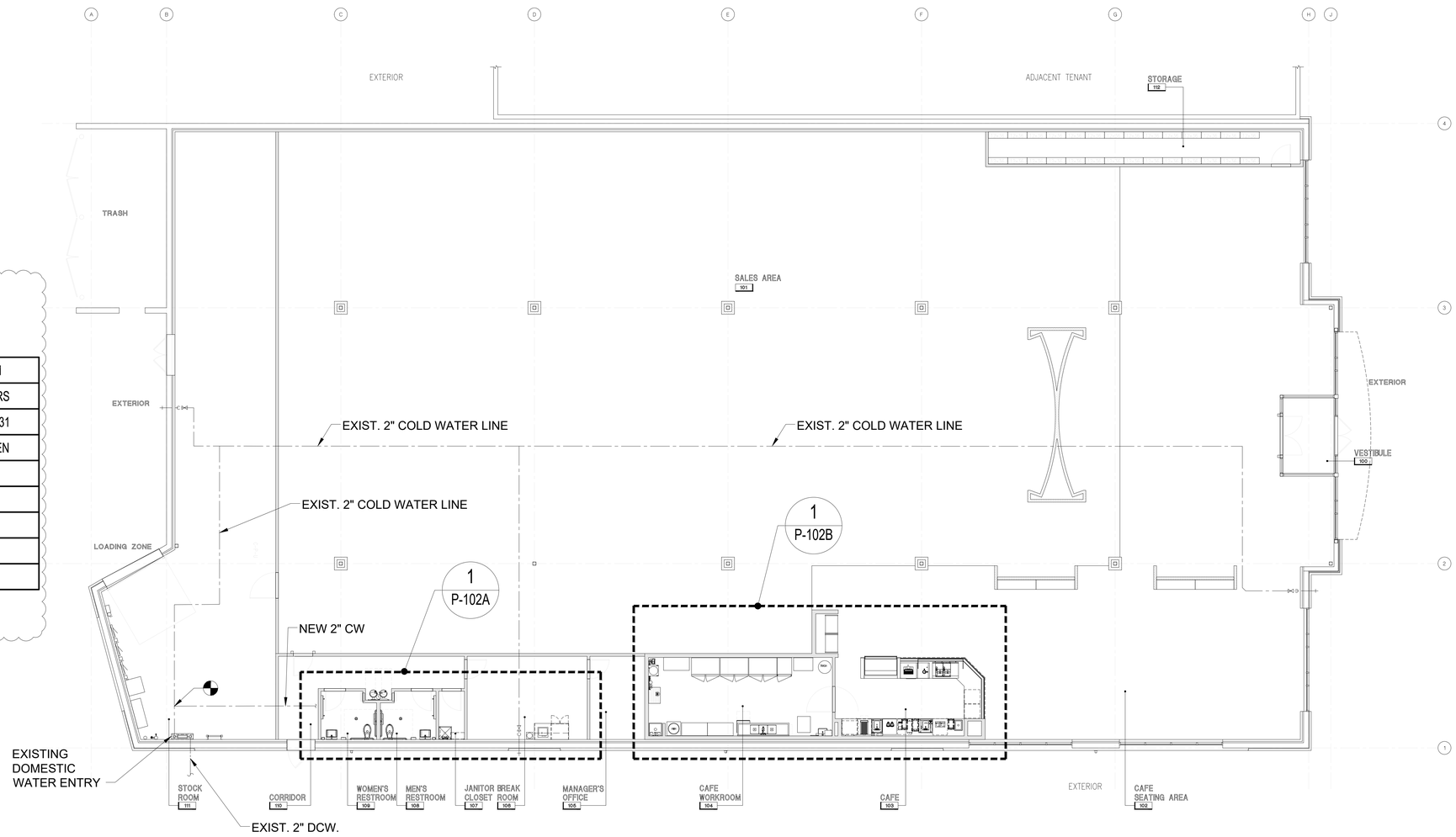
THERMOSTATIC MIXING VALVE SCHEDULE

MARK	TMV-1
MANUFACTURER	POWERS
MODEL No.	LFMM431
SERVICE	KITCHEN
GPM	14
PRESSURE DROP (PSI)	5
INLET TEMPERATURE (COLD WATER °F)	45
OUTLET TEMPERATURE (HOT WATER °F)	105
REMARKS	1

REMARKS:

1. ACCEPTABLE MANUFACTURERS: LAWLER, LEONARD, SYMMONS, WATTS.

1



1 P-102D Domestic Water Supply Overall Plan

SCALE: 1/8" = 1'-0"

BARNES&NOBLE

PROJECT DESIGNER:

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ARCHITECT:
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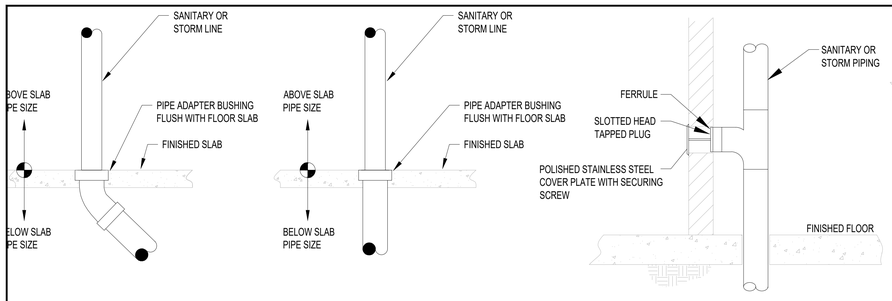
Professional Seal:

Drawing Description:
DOMESTIC WATER SUPPLY OVERALL PLAN

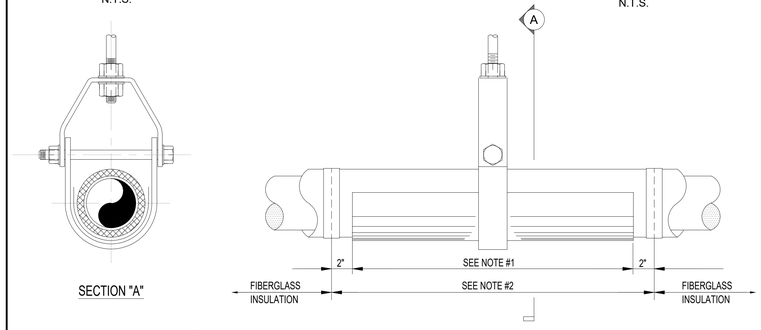
Drawing Number:
P-102D

1



PIPE SLAB PENETRATION DETAIL

N.T.S.



NOTES:

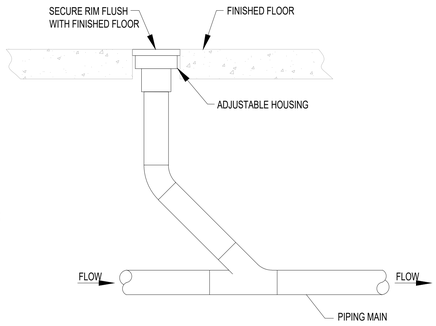
1. SADDLE FOR PIPE SIZES OF 3" OR SMALLER SHALL BE 18 GAUGE, 12" LONG. SADDLE FOR PIPE SIZES OF 4" OR LARGER SHALL BE 16 GAUGE., AND 3 TIMES PIPE DIAMETER LONG.
2. FIBERGLASS INSULATION FOR PIPES 2 1/2" IN DIAMETER AND SMALLER. HIGH DENSITY INSULATION INSERT FOR PIPES 3" IN DIAMETER AND LARGER.

INSULATED PIPE SADDLE DETAIL

N.T.S.

WALL CLEANOUT DETAIL

N.T.S.



NOTE:

1. CLEANOUT SHALL BE THE SAME SIZE AS THE PIPE TO WHICH IT IS CONNECTED UP TO 4". FOR PIPES LARGER THAN 4", THE MINIMUM SIZE OF THE CLEANOUT IS 4".

FLOOR CLEANOUT DETAIL

N.T.S.

LEGEND

	CONNECT TO EXISTING
CW	DOMESTIC COLD WATER PIPING
SCW	SOFTENED COLD WATER PIPING
FCW	FILTERED COLD WATER PIPING
SROCW	SOFTENED REVERSE OSMOSIS COLD WATER PIPING
SHW (120°)	DOMESTIC HOT WATER PIPING
RHW (120°)	RECIRCULATING HOT WATER (120°) PIPING
SHW (140°)	DOMESTIC HOT WATER (140°) PIPING
RHW (140°)	RECIRCULATING HOT WATER (140°) PIPING
V	VENT PIPING
SAN	SANITARY PIPING
GW	GREASE WASTE PIPING
D	EQUIPMENT INDIRECT DRAIN PIPING
COND.	CONDENSATE DRAIN PIPING
	SHUT-OFF VALVE
	PRESSURE REGULATING VALVE
	PIPE UNION
	BALANCING VALVE
	CHECK VALVE
	KITCHEN/BAR EQUIP. TAG
XX	CODED NOTE
XX / XX	EQUIPMENT TAG

AFF	ABOVE FINISHED FLOOR
BFP	BACKFLOW PREVENTER
CW	COLD WATER
SAN	SANITARY
ST	STORM PIPING
GW	GREASE WASTE
FP	FIRE PROTECTION PIPING
G	GAS PIPING
CO	CLEANOUT
EX. (E)	EXISTING
FD	FLOOR DRAIN
VTR	VENT THROUGH ROOF

PLUMBING FIXTURE SCHEDULE

FIXTURE NO.	FIXTURE DESCRIPTION	WASTE	VENT	C.W.	H.W.	
FD-1	FLOOR DRAIN	3"	2"	-	-	DUCO CAST IRON BODY WITH FLASHING COLLAR AND ADJUSTABLE ROUND NICKEL BRONZE STRAINER HEAD, TRAP PRIMER CONNECTION.
FS-1	FLOOR SINK	4"	2"	-	-	JOSAM #49324A (12"x12") ENAMELED CAST IRON FLOOR SINK W/INTERNAL DOME STRAINER, NO HUB OUTLET, AND 1/2 GRATE. SEE ARCHITECTURAL PLUMBING LOCATION PLAN.
MS-1	MOP SINK	3"	1-1/2"	1/2"	1/2"	FIAT MODEL #MSB2424 WITH CHICAGO FAUCET #897. SERVICE SINK FITTING WITH RIGID VACUUM BREAKER, 3/4" HOSE THREADED OTULET. PAIL HO AND WALL BRACE, FAUCET HOSE & CLAMP. MOUNT FAUCET 48" AFF. PROVIDE WITH A&J WASHROOM UJ12A MOP HOLDER.
WC-1	TOILET ROOM WATER CLOSET	4"	2"	1"	-	WALL MOUNTED, AMERICAN FLOWlse ADA RETROFIT ELONGATED FLUSHOMETER TOILET 2294.011EC. #736046-100 FLUSHOMETER 1.6 GPF,
LAV-1	TOILET ROOM H.C. LAVATORY	1-1/2"	1-1/2"	1/2"	1/2"	AMERICAN STANDARD DECORUM WALL-HUNG LAVATORY WITH EVERCLEAN MODEL 9024.001EC. AMERICAN STANDARD SERIN MONOBLOCK BATHRO FACUET MODEL 2064.131.002, POLISHED CHROME. THERMOSTATIC MIXING VALVE (#TLT10R) FOR SINGLE FAUCET (NOTE: THERMOSTATIC MIXING VA TO BE ASSE 1070 COMPLIANT). MOUNT UNDER PUBLIC HAND WASHING SINK/LAVATORY AND SET TO 105°F.
TP	TRAP PRIMER	-	-	1/2"	-	EQUAL TO PPP INC. MODEL OREGON #1 TRAP PRIMER VALVE.
SK-1	BREAK ROOM DROP-IN SINK	1-1/2"	1-1/2"	1/2"	1/2"	ELKAY #LRADQ252150 25" x 21-1/4" x 5" SINGLE BOWL DROP-IN ADA SINK, 18 GA STAINLESS STEEL, SELF-RIMMING BOWL. KOHLER TRITON BOWE 0.5 DECK MOUNTED BATHROOM FAUCET WITH WRISTBLADE HANDLES, MODEL K-400T70-5AKA, P-TRAP, SUPPLIES, STOPS, GRID DRAIN, (LESS SPRAY) W OPTIONAL 4" DECK PLATE AND MIXING VALVE.
FCO-1	FLOOR CLEANOUT	-	-	-	-	J.R. SMITH 4020 DUCO CAST IRON CLEANOUT WITH ADJUSTABLE SCORIATED TOP. COLOR AS SELECTED BY ARCHITECT.
WCO-1	WALL CLEANOUT	-	-	-	-	J.R. SMITH 4402 DUCO CAST IRON CLEANOUT WITH ADJUSTABLE SCORIATED TOP.
EWC-1	ELECTRIC WATER COOLER	3"	2"	1/2"	-	"QUENCH Q3" MODEL #Q3FS.
DF-1	DRINKING FOUNTAIN	3"	2"	1/2"	-	OASIS MMRSL RADII, BI-LEVEL MODULAR FOUNTAIN, ADA.

PLUMBING FIXTURE NOTES

- ALL SIZES SHOWN ABOVE ARE FIXTURE CONNECTION SIZES. USE INCREASER WHERE APPLICABLE SEE ISOMETRIC RISER DIAGRAMS.
- ALL FIXTURES DENOTED AS HANDICAPPED SHALL BE INSTALLED PER APPLICABLE HANDICAPPED CODE OR ANSI A117.1.
- HOT WATER SUPPLY PIPING FOR LAVATORIES SHALL INCLUDE A FLOW RESTRICTING DEVICE LIMITING THE FLOW OF 0.5 GPM MAXIMUM. HOT WATER SUPPLY TEMPERATURE SHALL NOT EXCEED 105°F.
- ALL SUBSTITUTIONS WILL BE REJECTED OUTRIGHT. THERE ARE NO EXCEPTIONS.

DOMESTIC WATER HEATER SCHEDULE (ELECTRIC)

MARK	EWH-1	EWH-2
MANUFACTURER	AO SMITH	AO SMITH
MODEL No.	DSE-30	DEL-20
TYPE	TANK	TANK
RECOVERY AT 80°F (GPH)	62	30
STORAGE (GAL)	30	20
No. OF ELEMENTS	1	1
HEATING KW	12.3	4
ELECTRICAL (VOLT/PH)	208 / 3 / 60	277 / 1 / 60
REMARKS	1, 2	1, 2

REMARKS:
1. RECOVERY RATE BASED ON 40° ENTERING WATER TEMPERATURE.
2. FURNISH AND INSTALL VACUUM RELIEF VALVE ON COLD WATER SUPPLY LINE ABOVE HIGHEST POINT IN THE TANK. INSTALL PER MANUFACTURER'S INSTRUCTIONS.
3. ALL MODELS MEET OR EXCEED THE THERMAL EFFICIENCY AND/OR STANDBY LOSS REQUIREMENTS OF THE U. S. DEPARTMENT OF ENERGY AND CURRENT EDITION ASHRAE/IES 90.1.

RECIRCULATING PUMP SCHEDULE

MARK	P-1
MANUFACTURER	BELL & GOSSETT
MODEL No.	NBF-25
SIZE	3/4"
TYPE	IN-LINE
SERVICE	HOT WATER
GPM	2
TOTAL HEAD (FT.)	16.5
NPSH	-
IMPELLER DIA. (")	2 1/2
WATTS	125
RPM	2950
ELECTRICAL (VOLT/PH)	115 / 1
OPER. WT. (LBS.)	10.4
REMARKS	1

REMARKS:
1. ACCEPTABLE MANUFACTURERS: ARMSTRONG, PACO, TACO, WEINMAN, PATTERSON, GUNDFOS.

BACK-FLOW PREVENTER

TAG	MAKE	MODEL	TYPE	ASSE Std	QUANTITY
BFP-1	ZURN	975XL2	REDUCED PRESSURE PRINCIPLE ASSEMBLY	1013	3
BFP-2	WATTS	LF800M4QT	ANTI-SIPHON VACUUM BREAKER	1020	1
BFP-3	WATTS	SD-3	DUAL CHECK WITH VENT	1022	1

GREASE INTERCEPTOR SIZING

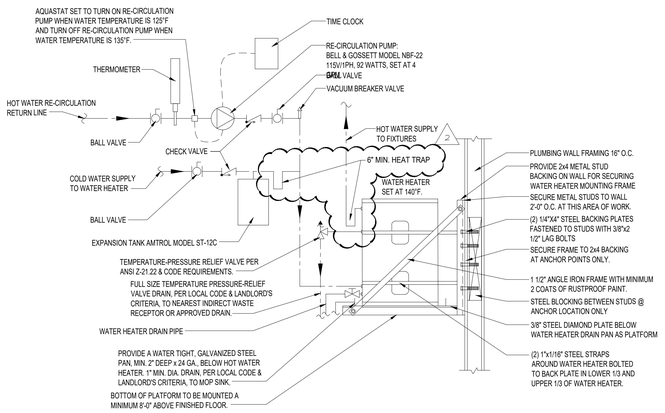
TAG	FIXTURE	QUANTITY	CUBIC OF FIXTURE (L X W X D)	GALLONS OF FIXTURE (1 GAL = 231 CUBIC INCHES)	ACTUAL DRAINAGE LOAD (=75% OF FIXTURE CAPACITY)	FLOW RATE=ACTUAL DRAINAGE LOAD / DRAINAGE PERIOD (1 OR 2 MINUTE DRAINAGE PERIOD)
16	THREE COMPARTMENT SINK	1	3(16"x18"x12") = 10,368 CUBIC INCHES	10,368 / 231 = 44.8 GALLONS	.75 x 44.8 GALLONS = 33.6 GALLONS	33.7 / 2 = 16.8 GPM FLOW RATE
4	HAND SINK - WALL MOUNTED	1	13.75"x9.88"x5.38" = 730.8 CUBIC INCHES	730.8 / 231 = 3.16 GALLONS	.75 x 3.16 GALLONS = 2.37 GALLONS	2.37 / 2 = 1.185 GPM FLOW RATE
5	MOP SINK	1	24"x24"x10" = 5,760 CUBIC INCHES	5,760 / 231 = 24.9 GALLONS	.75 x 24.9 GALLONS = 18.7 GALLONS	18.7 / 2 = 9.35 GPM FLOW RATE
1	SINK BOWL, WELD-IN, DUMP SINK	1	12"x16"x10" = 1,920 CUBIC INCHES	1,920 / 231 = 8.3 GALLONS	.75 x 8.3 GALLONS = 6.2 GALLONS	6.2 / 2 = 3.1 GPM FLOW RATE
1	SINK BOWL, WELD-IN, DUMP SINK	1	12"x16"x10" = 1,920 CUBIC INCHES	1,920 / 231 = 8.3 GALLONS	.75 x 8.3 GALLONS = 6.2 GALLONS	6.2 / 2 = 3.1 GPM FLOW RATE
25	ICE MAKER (FLOOR SINK)	1				= 1 GPM FLOW RATE
33	DISHWASHER, UNDERCOUNTER (FLOOR SINK)	1				= 1 GPM FLOW RATE
	FLOOR DRAIN (FD)	3				= 0 GPM FLOW RATE
20	DROP-IN ICE BIN (FLOOR SINK)	1				= 1 GPM FLOW RATE
	FLOOR SINK (FS)	1				= 1 GPM FLOW RATE
	TOTAL GPM					= 37.535 GPM

GREASE INTERCEPTOR IN THIS PROJECT IS EXISTING TO REMAIN.
THE EXISTING GREASE INTERCEPTOR WAS LOCATED OUTSIDE THE SPACE / BUILDING. REFER TO SHEETS # P-101A & # P-101B FOR EXACT LOCATION. THE EXISTING GREASE INTERCEPTOR IS A SCHIER, GB-1000 GREASE INTERCEPTOR 100 GPM / 200 GPM. 6" INLET/OUTLET, H-20 CAST IRON COVER. REFER TO LANDLORD'S DRAWINGS FOR ADDITIONAL INFORMATION.

2

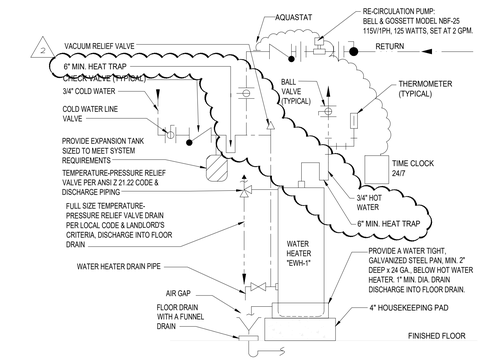
BUILDING DEPARTMENT NOTES:

1. THIS SET OF DRAWINGS COMPLY WITH CURRENTLY ADOPTED 2021 IECC AND 2023 PPRBC.
2. THE DEPARTMENT OF ENERGY RECOMMENDS A MINIMUM R-3 INSULATION FOR DOMESTIC HOT WATER PIPES. CONTRACTOR TO USE ARMFLEX OR APPROVED EQUIVALENT.
3. HEATED-WATER CIRCULATIING AND TEMPERATURE MAINTENANCE SYSTEMS TO INCLUDE THE HEAT TRACE TEMPERATURE MAINTENANCE SYSTEMS, CONTROLS FOR HOT WATER STORAGE, AUTOMATIC CONTROLS, TEMPERATURE SENSORS AND PUMPS TO PRODUCE A COMPLETE AND OPERABLE SYSTEM. MANUAL CONTROLS SHALL BE IN A LOCATION WITH READY ACCESS. REFER TO WATER HEATER DETAILS FOR ADDITIONAL INFORMATION.
4. A FINAL COMMISSIONING REPORT SHALL BE DELIVERED TO THE BUILDING OWNER PER SECTION C408.2.5 OF THE 2021 IECC.



WATER HEATER DETAIL (EWH-2) FOR RESTROOMS AND BREAKROOM

N.T.S.



WATER HEATER DETAIL (EWH-1) FOR CAFE AREA

N.T.S.

BARNES&NOBLE

PROJECT DESIGNER:

WJCA
BUILDING VALUE SINCE 1994

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NEW JERSEY LICENSED PROFESSIONAL ENGINEER
KENNETH A. FULLER, P.E.
CO. LICENSE NUMBER: 42050
EXPIRES: 09-30-25

3173 CINEMA POINT,
COLORADO SPRINGS, CO
80922

Project Number 33288
Store Number 3566

Revision Log:

Date	Description	No.
04-25-25	ISSUED FOR PERMIT, BID, LANDLORD & CLIENT REVIEW.	1
06-12-25	LL AND BD COMMENTS	2
07-23-25	BD COMMENTS	3

These plans are an instrument of service and the property of the Architect. Infringements will be prosecuted.

General Contractor to verify all conditions and dimensions at the premises. Discrepancies shall be reported to the Architect prior to the commencement of any work.

Professional Seal:

Drawing Description:

PLUBMING SCHEDULES, LEGEND & DETAILS

Drawing Number:

P-201

WJCA
BUILDING VALUE SINCE 1994

CLIENT'S NAME:

REVISION 1 **ISSUE DATE:** 06.12.2025

Project:	**Barnes & Noble COLORADO SPRINGS, CO**	Store No:	**3566**
Dwg. Date:	**06.12.2025**	Project No:	**33288**

☐ Architect Clarification ☐ MEP Clarification

☑ Building Department Comments ☑ Landlord Comments ☐ Owner's Changes

☐ Field Condition

☐ Other :

	Reference Document	Comments
1	A-000	Added bubbles in REV 1 drawing index column
2	A-001	Updated square footage, Updated occupancy group, Updated occupancy load table, Updated section 1005.3.2, Updated plumbing code analysis.
3	A-001A	No changes.
4	A-002	Updated Scope of work in Division 05 – Metals, Updated Scope of work in Division 26 – HVAC, Updated Tenant coordinator's details, Added Landlord notes.
5	A-003	Added occupants calculation for each area.
6	DM-101	Updated Demolition Key Notes no. 15.
7	A-101	Updated notes in details no. 3, Added key notes no. 28.
8	A-111	No changes.
9	A-121	No changes.
10	A-131	No changes.
11	A-151	No changes.
12	A-200	Updated turning area for restroom ADA from 5'-0" to 5'-7".
13	A-201	No changes.
14	A-210	Added accessible clearances per BD comment
15	A-220	No changes.

16	A-221	No changes.
17	A-222A	No changes.
18	A-222B	No changes.
19	A-223	No changes.
20	A-224	No changes.
21	A-225	No changes.
22	A-226A	No changes.
23	A-226B	No changes.
24	A-227	No changes.
25	A-228	No changes.
26	A-300	No changes.
26	A-400A	No changes.
27	A-400B	No changes.
28	A-410	Added BD notes.
29	A-412	No changes.
30	A-413	No changes.
31	A-414	No changes.
32	A-500	No changes.
33	A-510	No changes.
34	E-101	Added BD note referencing 2021 IECC
35	E-102	Added BD note referencing 2021 IECC
36	E-103	Added BD note referencing 2021 IECC. Changed convenience outlets in office to controlled type outlets. Added note.
37	E-104	Added BD note referencing 2021 IECC. Showed SCA and SCCR ratings for CUH-1.
38	E-105	No changes.
39	E-201	No changes.
40	E-202	Added BD note referencing 2021 IECC. Provided SCA calculation for panel ‘K’. Added ground size for transformer ‘T2’

41	E-202.1	No changes.
42	E-203	No changes.
43	E-301	No changes.
44	E-302	No changes.
45	M-100	No changes.
46	M-101	No changes.
47	M-102	No changes.
48	M-201	No changes.
49	M-301	No changes.
50	M-302	No changes.
51	M-303	No changes.
52	P-101A	Revised text size. Removed Sanitary Overall Plan to Sheet P-101D Added column lines to detail 2 Added Building Department Note Plumbing to sheet.
53	P-101B	Resized text. Resized viewports.
54	P-101C	Added Sanitary Waste & Vent Overall Plan
55	P-102A	Resized text. Added Pipe Sizing Calculations.
56	P-102B	Resized text. Resized viewports.
57	P-102C	No changes
58	P-102D	Added resized Domestic Water Supply Plan Added In-Line Pump Piping Detail. Added Thermostatic Mixing Valve Detail. Added Thermostatic Mixing Valve Schedule
59	P-201	Resized text. Rearranged sheet and moved three Details to P-101D.

WJCA
BUILDING VALUE SINCE 1994

CLIENT'S NAME:

REVISION 2 **ISSUE DATE:** 07.24.2025

Project:	**Barnes & Noble Colorado Springs, OH**	Store No:	**3566**
Dwg. Date:	**07.24.2025**	Project No:	**33288**

☐ Architect Clarification ☐ MEP Clarification

☑ Building Department Comments ☐ Landlord Comments ☐ Owner's Changes

☐ Field Condition

☐ Other :

	Reference Document	Comments
1	A-000	Added bubbles in REV. 2 drawing index column.
2	A-001	Updated occupant group. Removed void/unused space from occupant load table, included in storage SF. Updated section 1005.3.2 calculation based on new occupant load. Updated plumbing code analysis based on new occupant load.
3	A-001A	No changes.
4	A-002	No changes.
5	A-003	Updated storage area to match occupant load table on A-001.
6	DM-101	No changes.
7	A-101	No changes.
8	A-111	No changes.
9	A-121	No changes.
10	A-131	No changes.
11	A-151	No changes.
12	A-200	No changes.
13	A-201	No changes.
14	A-210	No changes.
15	A-220	No changes.

16	A-221	No changes.
17	A-222A	No changes.
18	A-222B	No changes.
19	A-223	No changes.
20	A-224	No changes.
21	A-225	No changes.
22	A-226A	No changes.
23	A-226B	No changes.
24	A-227	No changes.
25	A-228	No changes.
26	A-300	No changes.
27	A-400A	No changes.
28	A-400B	No changes.
29	A-410	No changes.
30	A-412	No changes.
31	A-413	No changes.
32	A-414	No changes.
33	A-500	Updated door schedule to call out panic hardware for both new and existing doors.
34	A-510	No changes.
35	E-101	No changes
36	E-102	Added CUH-1 to this sheet. It was already shown on sheet #E-104.
37	E-103	Added note to the utility receptacles.
38	E-104	No changes.
39	E-105	No changes.
40	E-201	No changes.
41	E-202	No changes.
42	E-202.1	No changes.
43	E-203	No changes.
44	E-301	No changes.

45	E-302	No changes.
46	M-100	No changes.
47	M-101	No changes.
48	M-102	No changes.
49	M-201	Eliminated clouds from Existing Rooftop Unit Schedule.
50	M-301	No changes.
51	M-302	No changes.
52	M-303	No changes.
53	P-101A	Added three (3) Building Department comments.
54	P-101B	No changes.
55	P-101C	No changes.
56	P-102A	Added table with service pressure, service size, meter size, pressure loss, etc. Adjusted hot water recirculation piping in Isometric Riser Diagram.
57	P-102B	No changes.
58	P-102C	No changes.
59	P-102D	No changes.
60	P-201	Added four (4) Building Department comments. Adjusted Water Heater details to add heat traps. Added remark #3 to Water Heater schedules about efficiency.

WJCA
BUILDING VALUE SINCE 1994

CLIENT'S NAME:

REVISION 3 **ISSUE DATE:** 08.8.2025

Project:	**Barnes & Noble Colorado Springs, OH**	Store No:	**3566**
Dwg. Date:	**08.8.2025**	Project No:	**33288**

☐ Architect Clarification ☐ MEP Clarification

☑ Building Department Comments ☐ Landlord Comments ☐ Owner's Changes

☐ Field Condition

☐ Other :

	Reference Document	Comments
1	A-000	Added bubbles in REV. 3 drawing index column.
2	A-001	No changes.
3	A-001A	No changes.
4	A-002	No changes.
5	A-003	No Changes
6	DM-101	No changes.
7	A-101	No changes.
8	A-111	No changes.
9	A-121	No changes.
10	A-131	No changes.
11	A-151	No changes.
12	A-200	No changes.
13	A-201	No changes.
14	A-210	No changes.
15	A-220	No changes.
16	A-221	No changes.
17	A-222A	No changes.

18	A-222B	No changes.
19	A-223	No changes.
20	A-224	No changes.
21	A-225	No changes.
22	A-226A	No changes.
23	A-226B	No changes.
24	A-227	No changes.
25	A-228	No changes.
26	A-300	No changes.
27	A-400A	No changes.
28	A-400B	No changes.
29	A-410	No changes.
30	A-412	No changes.
31	A-413	No changes.
32	A-414	No changes.
33	A-500	No changes
34	A-510	No changes.
35	E-101	No changes
36	E-102	No changes
37	E-103	No changes
38	E-104	No changes.
39	E-105	No changes.
40	E-201	No changes.
41	E-202	No changes.
42	E-202.1	No changes.
43	E-203	No changes.
44	E-301	No changes.
45	E-302	No changes.
46	M-100	No changes.
47	M-101	No changes.

48	M-102	Added note to vestibule.
49	M-201	Added note to vestibule heater.
50	M-301	No changes.
51	M-302	No changes.
52	M-303	No changes.
53	P-101A	Revised Note C to reflect 2021 IPC 403.1. Added note that the recirculation line on the hot water allowed the lavatories to comply with 2021 IPC Section C404.5.1.
54	P-101B	No changes.
55	P-101C	No changes.
56	P-102A	Added note on P-101A that the recirculation line on the hot water allowed the lavatories to comply with 2021 IPC Section C404.5.1.
57	P-102B	No changes.
58	P-102C	No changes.
59	P-102D	No changes.
60	P-201	No changes

TAKE-OFF MARKUPS - LEGEND

Source: takeoff_canon.json · one PDF annotation (Line / PolyLine) per wall on the plan sheet where it was measured · Subject = type · Contents = TYPE · LF · H · SF · DETAIL · id

Sample	Type	Walls	LF	Marked	No geometry	Walls without geometry (not marked): 0
	2	13	278.57	13	0	
	3	6	38.68	6	0	
	4	22	83.64	22	0	
	5	1	5.70	1	0	
	A	3	20.27	3	0	
	B	1	3.17	1	0	
	C	1	3.17	1	0	
	D	1	5.14	1	0	
	E1	3	27.88	3	0	
	E2	1	29.30	1	0	
	F	1	3.26	1	0	
	H1	2	42.15	2	0	
	H2A	1	12.17	1	0	
	H2B	1	1.95	1	0	
	H3A	1	4.08	1	0	
	H3B	1	6.17	1	0	
	I	4	4.52	4	0	
	OTHER FRAMING	14	137.31	14	0	
	TOTAL	**77**	**707.12**	**77**	**0**	